



Flight Operations Manual

Revision Number: 01

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Fast Reference Links	Revision 1 Highlights
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Taxi	Go-Around
Takeoff	Landing
Departure	After Landing
Enroute	Parking
Arrival	Post Flight/RON

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Qualifications/Limitations	Appendices
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FAST REFERENCE LINKS

The sections most often needed quickly. This page is not a substitute for the QRC/QRH.

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ADDITIONAL CHAPTERS

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International Operations	Electronic Flight Bag (EFB)
Training and Checking	Sabre/DECS
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APPENDICES

An appendix is used for material that is referenced by the manual but revised on its own cycle. Appendices issued at this revision are listed here and in the List of Effective Pages.

Appendix A — Delay Codes. The delay coding reference used for the delay report required by section 1c.3.3. See section 23.7 Delay Coding for the reporting policy.

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NORTHERN AIRWAYS

FLIGHT OPERATIONS MANUAL

Revision 1 — 06 AUG 26

Northern Airways, Inc. · Flight Operations

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Scope

This manual states **policy**. It applies to all Northern Airways mainline flight operations conducted under 14 CFR Part 121, and to all Northern Airways crewmembers, dispatchers, and management personnel in the performance of those operations.

The FOM answers "**am I permitted to?**" The Aircraft Operating Manual (AOM) answers "**how do I do it?**" Where a procedure is aircraft-specific, this manual does not restate it — it directs the reader to the applicable AOM.

Where a conflict exists, the more restrictive requirement governs, in this order of precedence:

1. 14 CFR and applicable foreign regulation
2. Northern Airways Operations Specifications (OpSpecs)
3. This manual (FOM)
4. Aircraft Operating Manual (AOM) / Quick Reference Card and Handbook (QRC/QRH)
5. Aircraft Flight Manual (AFM) limitations, where more restrictive than the above

Report any conflict to the Chief Pilot's office and to Flight Operations Publications. Do not resolve a conflict by informal interpretation.

Applicability to Subsidiary and Regional Operators

This manual applies to **Northern Airways mainline only**. It does not apply to Pioneer, Aloha, Aurora, or Corvus, each of which holds a separate certificate and publishes its own FOM. It does not apply to Cascadian, Embassy, or Cardinal when operating under their own certificates, including when operating in Northern Airways livery under a capacity purchase agreement.

Where a mainline crewmember interacts with a regional or subsidiary operation — jumpseat, through-flight, irregular operations, misconnect handling — the applicable section of this manual governs the mainline crewmember's conduct only.

How to Use This Manual

Navigation

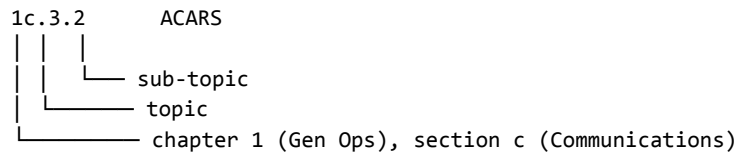
This manual is organized as a **matrix**, not as a linear sequence of chapters.

- **Chapters** are the top level and carry a number. Most correspond to a phase of flight; the remainder are standing subjects that do not belong to a single phase.
- **Sections** recur inside a chapter and carry a letter — General (g), Communications (c), Weather (w), Flight Planning (f), Maintenance (m), Security (s), Non-Normal/Emergency (n), Performance (p), Dangerous Goods/Cargo (d). A chapter with only one subject has no section letter.
- **Topics** and **sub-topics** are numbered beneath the section.

Find the phase you are in, then the kind of question you have. A question about talking to the ramp during boarding is in Preflight Communications, not in a single consolidated communications chapter.

Paragraph Numbering

Every numbered paragraph carries its full address:



So 1c.3.2 is Gen Ops, Communications, Dispatch, ACARS — and nothing else in the manual can share that number.

Pagination

Pages are numbered **by section**: 1c-12 is Gen Ops Communications, page 12. A chapter with no section letter numbers plainly — 25-8 is Electronic Flight Bag, page 8. A revision to Communications does not repaginate General.

Each page carries a two-line running header — chapter and section on the left with the page number on the right, then the current topic on the left with **Home** on the right — and a running footer with the revision date and the manual title.

Cross-References

Inside this manual, refer to material by its paragraph number and title:

See section [1w.4 Turbulence](#).

See also section [1g.5 Sterile Flightdeck Period](#).

Where the target has not yet been numbered, or where naming it is clearer, refer to it by chapter, section, and title:

See Preflight Maintenance, eAML.

The > path form — FOM> Gen Ops> Communications — is used **by other manuals referring into this one**, not within this manual. An AOM cites the FOM that way; the FOM does not cite itself that way.

Reserved Numbers

Chapter and section numbers that are not in use are marked **Reserved** rather than removed. Numbering is never compressed. A section deleted in a future revision becomes Reserved and its number is not reissued.

Conventions

Convention	Meaning
Bold text in quotes	Required verbatim phraseology. State it as written.

Convention	Meaning
<i>Italic</i>	Emphasis, or a term defined in the Glossary at first use.
Note:	Amplifying information. Not a requirement.
Caution:	Practice which, if not observed, may result in damage to equipment.
Warning:	Practice which, if not observed, may result in injury or loss of life.
Will / must	Mandatory. No crew discretion.
Should	Recommended. Not mandatory, but a deviation is expected to have an operational reason.
May / can	Permitted at the discretion of the individual named.
— End —	Closes a procedure block.
Revision bar ()	Left margin. Marks text changed since the previous revision.
— End —	

Revision Bars

A vertical bar in the left margin identifies material added or changed in the current revision. Bars are removed at the next revision. A bar is not applied for formatting, repagination, or typographical correction.

The Captain's Authority

Nothing in this manual limits the authority of the pilot in command under 14 CFR 91.3(a). The captain may deviate from any policy in this manual to the extent required to meet an emergency. Report any such deviation. See Reports and Forms, Reports to Dispatch.

Manual Library

Manual	Abbrev	Owner	Contains
Flight Operations Manual	FOM	Flight Operations	Policy, fleet-common. This manual.
Aircraft Operating Manual	AOM	Fleet Support Team	Aircraft-specific procedure, flows, checklists, limitations, performance. One per fleet.
Quick Reference Card / Handbook	QRC/Q RH	Fleet Support Team	Quick action items (QRC) and full non-normal and emergency checklists (QRH). One per fleet.

Manual	Abbrev	Owner	Contains
Flight Attendant Manual	FAM	Inflight	Cabin crew policy and procedure.
Dispatch Operations Manual	DOM	Integrated Operations Center	Dispatcher policy, operational control.
Airport Analysis / Runway Data	—	Flight Operations Engineering	Runway-specific weight and obstacle data.
Route Manual (Jeppesen)	—	Navigation Services	Charts, airport diagrams, procedures.
Minimum Equipment List	MEL	Maintenance / Fleet Support	Dispatch with inoperative equipment.
Configuration Deviation List	CDL	Maintenance / Fleet Support	Dispatch with missing external parts.
General Maintenance Manual	GMM	Technical Operations	Maintenance program.
— End —			

Aircraft Operating Manuals by Fleet

AOM	Aircraft
A32F	A319, A320, A321, A321neo
A33F	A330-900neo
73F	737-800, 737 MAX 8
757	757-200
767	767-300
777	777-200ER
E90	E190
— End —	

Note: Regional fleets (Q400, ERJ-145, E170, E175) are operated by Cascadian, Embassy, and Cardinal under their own certificates and are not covered by a Northern Airways AOM.

Revision Control

Responsibility

Flight Operations Publications is responsible for the format, distribution, and revision control of this manual. The content of each chapter is owned by the department named below. Fleet Support Teams own the AOMs, not this manual.

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1 Gen Ops	g c w m s n d p	Flight Standards
2 Flight Plan / Dispatch Release	—	Integrated Operations Center
3 Preflight	g c w f m s d n	Flight Standards
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17 Training and Checking	—	Flight Training & Standards
18 Passengers and Cargo	g d	Inflight / Flight Standards
20 Administration	—	Flight Operations Administration
21 Qualifications / Limitations	—	Flight Training & Standards
22 Threat & Error Management	—	Flight Safety
23 Reports and Forms	—	Flight Safety
24 Meteorology	—	Integrated Operations Center — Meteorology
25 Electronic Flight Bag (EFB)	—	Flight Operations Technology

Chapter	Sections	Content Owner
26 Sabre / DECS	—	Flight Operations Technology
27 Glossary	—	Flight Operations Publications
— End —		

Chapter 19 is **Reserved**.

Revision Cycle

This manual is revised on a scheduled cycle four times per year, effective the first day of January, April, July, and October. Interim revisions are issued as required for regulatory change or safety-of-flight matters.

Revision Numbering

Revisions are numbered sequentially from 1. Interim revisions carry a letter suffix (1A, 1B). The revision number and date appear in the footer of every page.

Acknowledgment

Each crewmember will acknowledge receipt of each revision in the Controlled Document Library within **10 days** of the effective date. An unacknowledged revision does not relieve a crewmember of compliance.

Suggestions and Discrepancies

Individuals discovering a discrepancy, or making a suggestion for improvement, will submit it through the Manual Feedback form in the Controlled Document Library. Include the navigation path, the revision number, and the proposed change. Every submission receives a written disposition.

Do not annotate a personal copy of this manual and rely on the annotation.

List of Effective Pages

Generated at each revision. Verify the LEP matches your copy before use.

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Revision History

Rev	Date	Summary
1	06 AUG 26	Original issue. Supersedes all prior Northern Airways flight operations policy documents.
— End —		

Summary of Changes — Revision 1

Original issue. Revision bars are not used in an original issue.

This manual supersedes the previous Northern Airways Flight Operations Manual in its entirety. Two structural changes are significant:

- **Aircraft-specific procedure has been removed.** Flows, expanded flows, and the Normal Procedures Checklist formerly carried in the FOM now reside solely in the applicable AOM. See A32F AOM> 2> Flow Guide.
- **Numbering has changed from a linear outline to the chapter/section matrix.** Prior cross-references by paragraph number are no longer valid.

Chapter 1: Gen Ops

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Fleet-common policy that applies across more than one phase of flight. Where a policy applies to a single phase only, it is found in that phase chapter.

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1g.1 Operational Priorities

Every decision is measured against four priorities, in this order:

1. **Safety**
2. **Regulatory compliance**
3. **Passenger comfort and service**
4. **Schedule and efficiency**

A lower priority is never traded against a higher one.

Safety. The safe conduct of the flight is the first consideration in every decision. Where a course of action is safe and another is safer, take the safer one.

Regulatory compliance. Operate within 14 CFR, the OpSpecs, and this manual. Where operating outside them is necessary for safety, the captain has that authority and will report its use.

Passenger comfort and service. Passengers have chosen Northern Airways. A smooth ride, an informed cabin, and a professional crew are part of the product.

Schedule and efficiency. On-time performance and fuel efficiency matter and are pursued — after the three priorities above are satisfied.

Note: No one at Northern Airways will ask a crew to invert this order. If someone does, report it to the Director of Operations and to Flight Safety.

1g.2 Standard Operating Procedures

Adherence

Standard operating procedures are complied with as written. SOPs exist so that two pilots who have never met can operate a complex aircraft together and each know what the other will do next. Deviation removes that predictability.

The captain may deviate from normal operations, **but only when unusual circumstances require a deviation in the interest of safety and as necessary to maintain operations**. A deviation for convenience, habit, or personal preference is intentional non-compliance. See section [22.9 Intentional Non-Compliance](#).

Structure

Document	Contains
This manual (FOM)	Policy. What is permitted and required.
AOM	Aircraft-specific procedure, flows, checklists, limitations, performance.

Document	Contains
QRC/QRH	Quick action items and non-normal/emergency checklists.
— End —	

Emergency and non-normal procedures are contained in the **QRC/QRH** and in the applicable AOM> 16> Non-Normals. They are not restated in this manual.

1g.3 PF and PM Duties

Assignment

The captain assigns pilot flying (PF) and pilot monitoring (PM) duties before each flight. The assignment does not change in flight except:

- On transfer of control under section [22.7 Applying the ABCs](#).
- When the captain assumes control per [1g.3 Captain Assumption of Control](#)
- During a non-normal, per the applicable QRC/QRH procedure

Duties

Pilot Flying	Pilot Monitoring
Controls flight path, thrust, and configuration	Monitors flight path, thrust, and configuration
Directs the automation level and mode	Verifies mode changes and annunciations
Calls for checklists and configuration changes	Accomplishes checklists and configuration changes
Makes the decision to go around, reject, or divert	Makes required callouts; may call for a go-around
Retains the aircraft during a non-normal	Handles radios, QRC/QRH, and coordination during a non-normal
— End —	

Captain Assumption of Control

The captain will assume control of the aircraft when:

- Conditions are below the first officer's authorized minimums
 - The situation requires the exercise of command authority
 - The first officer is incapacitated or unresponsive
 - In the captain's judgment, the safety of the flight requires it
- State **"My aircraft."** The other pilot responds **"Your aircraft."**

Transfer of Control

Every transfer of control is verbal, explicit, and acknowledged. There is no silent transfer.

Transferring	Accepting
"Your aircraft."	"My aircraft."
— End —	

1g.4 Flightdeck Management

Professional Environment

The flightdeck is a workplace. Conversation, activity, and demeanor are appropriate to that.

Personal Electronic Devices

Item	Policy
Company EFB	Authorized at all times. See Electronic Flight Bag (EFB).
Personal phone or tablet	Powered off or in airplane mode, stowed, from the sterile flightdeck period through the end of the sterile period on arrival.
Personal phone, above 10,000 ft	May be used for a purpose related to the operation of the flight only. Airplane mode.
Headphones for personal audio	Prohibited at any time in the flightdeck.
Camera, photography	Prohibited in flight. Prohibited on the ground where a passenger or a security area is in frame.
Laptop	Prohibited in the flightdeck.
— End —	

Warning: Use of a personal electronic device for any purpose unrelated to the operation of the flight is a distraction and is prohibited.

Exterior Lights

Light	Policy
Beacon	On before engine start, off after engines are shut down. A rotating beacon means the aircraft is live.
Navigation and logo	On from first engine start until shutdown at the destination.
Strobes	On entering a runway, off clear of the runway.

Light	Policy
Landing and runway turnoff	On below 10,000 feet and whenever conflicting traffic is a factor.
Wing and nacelle inspection	As required to inspect for ice. Do not leave on where the reflection degrades night vision.
— End —	

Before leaving the flightdeck after shutdown, turn off all unnecessary flightdeck lights and reduce display brightness. This increases display unit life, reduces flightdeck heat, and prevents screen burn-in.

Night Operations

Use the minimum flightdeck lighting that allows the instruments to be read, to preserve outside visibility. Both pilots keep a flashlight readily accessible.

Reading Material

Only material related to the operation of the flight is permitted in the flightdeck below FL250. Above FL250, in cruise, non-operational reading is permitted for one pilot at a time provided the other pilot is actively monitoring, and is discontinued immediately on any deviation, ATC communication, or non-normal.

Food and Beverage

Both pilots will not eat at the same time. Crew meals for the captain and first officer will be different where different meals are available. Hot liquids are not placed on the center pedestal or on any surface from which a spill would reach a control or a circuit breaker panel. Refer to the applicable AOM> 1> Flightdeck Discipline.

1g.5 Sterile Flightdeck Period

Source: 14 CFR 121.542

Definition

For the purpose of sterile flightdeck procedures, **critical phase of flight** means:

- All ground operations involving taxi, takeoff, and landing, and
- All other flight operations conducted **below 10,000 feet**, except cruise flight.

Requirement

During a critical phase of flight, all activities in the flightdeck not required for the safe operation of the aircraft are **prohibited**, including non-essential conversation.

Prohibited activities do **not** include required company operational communications — on/off times, gate information, or maintenance requirements.

During the sterile flightdeck period, flight attendants contact the flightdeck only to deal with safety-related information or situations. See section [1c.4 Flight Attendants](#).

Application

The sterile flightdeck is not in effect when the aircraft is not in motion with the parking brake set.

Phases

Phase	Sterile
Aircraft stopped, parking brake set	No
Pushback commenced through takeoff	Yes
Climb, below 10,000 ft AFE	Yes
Cruise	No
Descent, below 10,000 ft AFE	Yes
Landing through taxi-in	Yes
Aircraft parked, brake set	No
— End —	

Note: Northern Airways applies the sterile period from the **commencement of pushback**, not from the beginning of taxi. Pushback and engine start are the two highest communication-error phases in the LOSA data.

Announcement

The PM announces "**Sterile**" at the start of the sterile period in climb-out where it was interrupted, and "**Ten thousand, sterile**" on descent through 10,000 feet.

1g.6 Flow Guide

General

The Flow Guide is published in the applicable AOM> 2> Flow Guide. It states, in time order for a typical flight, the flow each pilot follows and how the two pilots' flows integrate. The Flow Guide contains abbreviated tasks, triggers, and flows only — it does not supply system or component detail. Detailed procedure is in the expanded flow chapters and in this manual.

Definitions

Trigger

An event that cues the pilot to begin an action or a flow.

Flow

A pattern of sequential items accomplished from memory, before the associated checklist is called.

Task

An item that is not sequential and may be completed whenever appropriate during the phase.

Abbreviated Flow

The Flow Guide depicts each flow in abbreviated form — the item list without the expanded detail. Each abbreviated flow item is linked to its expanded flow in the applicable phase-of-flight chapter of the AOM.

Expanded Flow

Full procedural guidance, including the flow diagram and system detail, is presented in the phase-of-flight chapters of the AOM. See Expanded Flows.

First Flight

Some checks are accomplished once, not on every sector. Where a flow item applies only to the first aircraft flight of the day or the first flight a crew flies on a given aircraft without interruption, the AOM marks it accordingly and the pilot follows that marking.

Format

The Flow Guide uses a consistent visual convention:

- **Flow triggers** are shaded and introduce the flow that follows.
- **Flow items** are marked with an arrow bullet (➤) and represent the typical sequence for the flight. A flow item printed in blue is linked to its expanded procedure.
- **Numbered flow items** are performed in the printed order.
- **Task triggers** are unshaded conditional statements; they give the crew latitude on when a task is accomplished, not whether it is accomplished.
- **Task items** are marked with a dash (—) and have no required order.
- **Callouts** are shaded distinctly from triggers. A callout in quotes and bold is required to be stated verbatim; a callout in quotes and not bold is suggested language, not a required script.

Note: The Flow Guide is a memory aid for a typical flight, not a substitute for the expanded procedure. Where a flow item and its expanded procedure appear to conflict, the expanded procedure governs.

1g.7 Checklist Discipline

General

Northern Airways uses a **flow-then-checklist** philosophy. The flow accomplishes the task, from memory, before the checklist is called. The checklist verifies that the flow was accomplished — it is a verification, not the procedure itself. See section [1g.6 Flow Guide and the applicable AOM> 2> Flow Guide](#).

Challenge and Response Concept

The normal procedures checklist incorporates the challenge and response concept.

Challenger

The person who reads the checklist and verbally issues the challenge. On the ground the first officer is the challenger, and in flight the PM is the challenger.

Responder

The person who verbally issues the response. On the ground the captain is the responder, and in flight the PF is the responder.

Normal Checklist Execution

Table 1.1 Normal Checklist Execution

Checklist	Call	Challenge	Respond
Before Start	Captain	First Officer	Captain and First Officer ‡
After Start	Captain	First Officer — Silent	First Officer — Silent
Before Takeoff	Captain	First Officer	Captain and First Officer ‡
After Takeoff	PF	PM — Silent	PM — Silent
Descent	PF	PM	PF and PM ‡
Before Landing	PF	PM	PF and PM ‡
Shutdown	Captain	First Officer	Captain
Secure	N/A	Read & Do	Read & Do
— End —			

Both pilots respond on Cross of Lorraine (‡) items.

Checklist Format

Checklists are printed in one of two formats:

- **Challenge and response.** The checklist title appears in white inside a black box. Each line pairs a challenge on the left with the required response on the right.
- **Silent read and do.** The checklist title appears in black inside a white box. These items are read and accomplished without a spoken challenge and response.

A dashed line within a checklist marks a break in the sequence — some action (a configuration change, for example) must occur before the checklist can continue past that line. A **Cross of Lorraine** (‡) printed after a response marks an item that requires a response from both pilots.

Table 1.2 Checklist Format Example — Before Start (Challenge and Response)

Challenge	Response
Test Panel	Tested
Avionics Panel	Checked
Hydraulic Panel	Checked
<i>(sequence break — flaps must be set before continuing)</i>	
Parking Brake	On/Off
Beacon	On
Altimeters	___ . ___ ‡
— End —	

A silent read and do checklist item, such as **Flaps**, prints its title in black inside a white box rather than white inside a black box, and the response (**Up**) is read and accomplished without a spoken challenge.

Execution

When

The Flow Guide indicates when to initiate each checklist. Flow items are accomplished before the checklist is called; a checklist is not initiated until the crew can give it full attention through to completion.

How

Both pilots are in their assigned seats for a checklist, except the Secure Checklist. A checklist item is not complete until every expanded step for that item is complete.

The **challenger**:

- reads every challenge aloud, except where the checklist is accomplished silently
 - challenges are never recited from memory
- ensures the correct response is verbalized and verifies the switch position or action is correct
- repeats the response where the Cross of Lorraine (‡) is printed

The **responder(s)**:

- verify the required switch position or action is correct
- verbalize the exact printed response

Note: Where an item has not been completed at the time it is challenged, the responding pilot(s) complete it — discussing it first if necessary — and then give the response. The checklist is not answered ahead of the aircraft's actual configuration.

Specific Responses

- **Altered from normal.** Where a Flight Ops Bulletin, MEL/CDL item, or inoperative equipment requires a switch or control to differ from its normal checklist position, the responder states the actual position and at least one pilot states the reason — for example, "ON due to MEL."
- **Not installed.** Where an item is not installed on the aircraft, the responder states "Not installed."
- **Blank line.** Where the printed response is blank, the responder states the actual condition — for example, "Runway two-seven, checked."
- **Or.** Where the response is a variable joined by "or," the responder states the actual position in use.
- **Slash.** A forward slash pairs two items in one line; the challenge and response on each side of the slash correspond to each other.
- **Checked.** "Checked" means every required step for that item has been accomplished.
- **As Required / Set.** The responder states or sets the actual required status of the item.

A checklist is not interrupted without cause. Where an interruption is unavoidable, the checklist is resumed **from the beginning**. The PM announces completion: "[Checklist name] complete."

Incomplete Checklist

A checklist that cannot be completed because of an unresolved item is not complete. Do not continue past the point the checklist protects.

1g.8 Automation Policy

Level of Automation

Select the level appropriate to the situation, and state it aloud. Northern Airways does not require the highest level of automation available.

Level	Use
Autopilot with LNAV/VNAV or managed modes	Normal cruise, high-workload arrivals, low visibility
Autopilot with selected modes	Vectors, ATC-driven changes, non-standard clearances
Flight director, manual flight	Proficiency, when workload permits
Raw data, manual flight	Proficiency, in benign conditions, with the other pilot's concurrence
— End —	

Reducing Automation

When the aircraft is not doing what is expected, **reduce** the level of automation. Do not attempt to diagnose an automation anomaly while the aircraft continues to fly under its control.

Warning: Uncertainty about what the automation is doing is an undesired aircraft state. Disconnect, fly the aircraft, then investigate.

Manual Flight Proficiency

Northern Airways encourages manual flight to maintain proficiency, in visual conditions, in low-workload environments, with the concurrence of both pilots, and outside RVSM airspace. Manual flight is not undertaken in:

- Low visibility operations
- RVSM airspace
- High-density terminal areas during peak periods
- Any phase where the other pilot has expressed a preference otherwise

Mode Awareness

Every mode change is announced by the pilot making it and verified by the other pilot from the FMA. An unannounced mode change is a monitoring failure by both pilots.

1g.9 Altitude Awareness

Assigned Altitude

Both pilots will confirm every altitude clearance. The PM reads back; the PF confirms the readback and the altitude set in the altitude window.

Altitude Alerting

- The PM announces **"One thousand to go"** approaching any assigned altitude.
- Both pilots verify the altitude window against the clearance.
- A discrepancy is resolved with ATC before the aircraft reaches the altitude.

Approaching an Altitude with a Restriction

Verify the restriction is loaded and armed, and cross-check the raw data. A crossing restriction met by the automation is verified, not assumed.

Altitude Deviation

Any deviation of 300 feet or more from an assigned altitude is reported through ASAP.

1g.10 Airspeed and Courses

- Airspeeds in this manual and in the AOMs are **indicated airspeed in knots** unless stated otherwise.
- Courses, headings, and radials are **magnetic** unless stated otherwise.
- Altitudes are **MSL** unless annotated AGL or AFE.
- Distances are **nautical miles** unless annotated statute or metric.
- Weights are **pounds** unless the aircraft is configured in kilograms, in which case refer to the applicable AOM> 20> Aircraft General.

1g.11 Altimetry

Transition

- **Transition altitude** is where altimeters are set to standard on climb.
- **Transition level** is where altimeters are set to local on descent.

In the United States, transition altitude is 18,000 feet and the transition level is FL180. Outside the United States, both vary by state and are published on the chart.

Set standard **climbing through** the transition altitude, and local **descending through** the transition level.

Cross-Check

Both pilots cross-check altimeters against each other and against the field elevation before departure.

The maximum permitted divergence between altimeters is **200 feet**. Where the divergence exceeds 200 feet, the aircraft is not dispatched until the discrepancy is resolved.

RVSM

RVSM airspace is **FL290 through FL410 inclusive**. RVSM operation requires the equipment and procedures published in OpSpecs.

Where an RVSM significant system becomes inoperative in RVSM airspace, advise ATC and request a clearance to exit RVSM airspace. See section [7g.1 General](#).

For further information on transition altitudes and levels outside the United States, see Departure International.

Setting

Both altimeters are set and cross-checked by both pilots at every transition. The PM announces the setting; the PF confirms.

QFE

Northern Airways does not operate using QFE. Where a QFE setting is issued, request QNH.

Metric Altimetry

Where metric altitude assignment is in use, see Enroute International and the applicable AOM for the metric display procedure.

Cold Temperature Altitude Correction

See Approach General, Cold Temperature Corrections.

1g.12 Crew at the Controls

Two Pilots at the Controls

Both pilots will be at the controls with seat belts fastened:

- From the commencement of pushback through 10,000 feet AFE on departure
- From 10,000 feet AFE on descent through the completion of taxi-in
- During any non-normal or emergency
- In turbulence, or when turbulence is anticipated
- During any period of reduced visibility on the ground

Physiological Break

One pilot may leave the flightdeck in cruise for a physiological need. **The flightdeck is never left with only one person in it** — see section [1s.2 Two-Person Flightdeck](#). Before departing the flightdeck:

1. Notify the remaining pilot and obtain concurrence.
2. Confirm the remaining pilot is fit and alert.
3. Confirm the aircraft is in a stable cruise condition with no non-normal in progress.
4. Brief and summon the designated flight attendant. The flight attendant enters the flightdeck and the door is secured **before** the pilot leaves. See section [1s.2 Two-Person Flightdeck for the full procedure](#).

The remaining pilot will wear the oxygen mask if the aircraft is above **FL250** while the other pilot is out of the flightdeck, regardless of who else is present. Refer to 14 CFR 121.333(c)(3) and the applicable AOM.

Seat Belts and Shoulder Harnesses

Shoulder harnesses are worn during takeoff and landing by both pilots (14 CFR 121.311(f)). Seat belts are worn at all times while at the controls.

1g.13 Flightdeck Door

See section [1s.2 Flightdeck Security](#).

1g.14 Smoking and Vaping

Smoking and the use of electronic smoking devices are prohibited aboard all Northern Airways aircraft, including in the flightdeck, at all times, by all persons. This is a company policy without exception.

1c.1 General

Radio communications are a critical link in the ATC system. The use of proper radio procedures enhances safety and is the mark of a professional pilot. Standard phraseology will be adhered to at all times.

1c.2 Radio Communication Guide

Proper radio procedure includes the following:

- Use the **full call sign** — company name and flight number — when communicating with ATC.
- If there is any doubt about a clearance, or about who a clearance is for, **ask for verification**. Do not act on a clearance you are not certain is yours, particularly when in an area of high terrain, or when the clearance includes heading, frequency, or route changes, or instructions to hold short of a runway.
- On initial contact, notify ATC of the assigned altitude and, if applicable, the assigned heading and airspeed. When on other than published routes, include the presently assigned routing.
- Acknowledge every frequency change, including the full call sign.
- Read back all runway hold short instructions, runway crossing clearances, takeoff clearances, and landing clearances, in full, including the runway.

Pilots will **confirm with one another** any clearance including headings, frequencies, routes, and hold short instructions, particularly where there is any question of whether the other pilot heard the information.

Similar Call Signs

Where a similar call sign is operating on frequency, both pilots will increase vigilance and will confirm every clearance. Advise ATC that a similar call sign is on frequency. Report the occurrence through ASAP.

1c.2.1 Clearance Readback

Acknowledge the following ATC clearances and instructions with the **full flight number** and a **verbatim** read back of the clearance:

General

- initial IFR clearance, and flight plan clearance limits or amendments
- instructions to initiate contact on a specific radio frequency

Taxi

- taxi clearance and runway assignment

- taxi clearance involving hold short clearances restricting runway or taxiway access
- taxi clearance to cross any runway surface

Takeoff

- line up and wait clearance
- takeoff clearance

Inflight

- clearances to change altitude
- instructions to report out of or reaching a specific altitude
- instructions and clearances specifying VOR radials, turns to headings, airspeeds, rates of descent, and runway information

Approach and Landing

- clearance to land
- instructions to reject a takeoff, abandon an approach, or execute a go-around

Advisory Information

ATC advisory information is read back at the flightdeck crew's discretion.

Warning: A hold short instruction, a runway crossing clearance, and a landing clearance are read back in full including the runway, every time. A partial readback is how a runway incursion begins.

Loss of Communication

Where two-way communication is lost, comply with 14 CFR 91.185 in domestic airspace and with the procedure published for the airspace elsewhere. See section [7c.4 Lost Communications](#).

1c.3 Dispatch

Source: 14 CFR 121.99

1c.3.1 General

The Company must be able to **immediately communicate with the aircraft at all times**. Flights will continuously monitor ACARS, SELCAL, and/or the appropriate VHF or HF frequency.

Note: VHF 121.5 does not transmit on SELCAL.

Listening Watch

ACARS is the primary means of maintaining the listening watch domestically. Foreign ACARS coverage is limited. Where ACARS is inoperative, or the aircraft is

in an area without ACARS coverage, establish a SELCAL or monitor the appropriate voice frequency.

1c.3.2 ACARS

ACARS is the primary means of two-way communication between the aircraft and Dispatch, via either an ACARS ground station or SATCOM.

Away from the gate, ACARS reports:

- normal transmission of **OOOI** — out, off, on, and in times
- enroute position reports
- in-range reports
- aircraft systems performance data
- flight-to-IOC and IOC-to-flight communications
- SELCAL used to establish dispatcher-to-captain voice communication

ACARS is intended **only** for Company operational communication.

Note: Non-operational messages are not authorized.

Verification

Inputs. Both the captain and first officer will verify the correct flight number, origin, and destination are entered **prior to departure**.

Received messages. Verify all messages for the correct flight number, date, and aircraft number.

TPS and load closeout. Besides verifying the received message, compare the TPS and the load closeout against the documents obtained from Operations for reasonableness — zero fuel weight, takeoff gross weight, and V-speeds. Compare all subsequent TPS and closeout messages against those previously received.

Inoperative ACARS. Refer to the MEL and the applicable division information.

1c.3.3 Delay Report

Report delays in minutes, by ACARS, or by voice where ACARS is not functioning. Delay codes are published in **Appendix A — Delay Codes**; the policy for assigning and splitting them is in section [23.7 Delay Coding](#).

Reporting the reason for a delay is essential for identifying delay problem areas. It is the **captain's responsibility** to judge the number of minutes to charge to each delay code that involves the flight crew, and to state the reason in the remarks where the code alone does not capture it.

Where the delay is coded **SD** — **Station Default** at departure because no code has been assigned, that is a station administrative placeholder, not a description of the cause. It does not relieve the crew of reporting what actually happened.

1c.3.4 VHF Radio and Telephone Communications

VHF Radio

When contacting the controlling dispatcher on the dispatch direct VHF network, state the **dispatch desk number** — listed on the flight plan — the flight number, and the VHF frequency.

"Northern dispatch, desk 17, Northern one eight five on one three zero decimal two five."

Transmitting the desk number and frequency eliminates confusion and expedites a response.

Telephone

Where the aircraft is on the ground and radio contact is impractical, contact the desk by telephone using the number printed on the release. See section [20.3 Contact Information](#).

1c.3.5 When to Contact Dispatch

The crew will contact dispatch:

- Whenever the flight cannot be conducted as released
- Before any diversion, where time permits
- On any change to destination, alternate, route, or fuel
- On any mechanical irregularity affecting the operation
- On any medical event, security event, or passenger disturbance
- On any emergency declaration, after the aircraft is under control
- Whenever the captain wants a second opinion

Note: The dispatcher shares operational control and is a resource, not an approval authority to be avoided. A dispatcher who is not told cannot help.

Method

ACARS is the primary method. Company radio and SATCOM are available. See section [20.3 Contact Information](#).

Where ACARS is inoperative and no additional manual company communication method is available. See Sabre/DECS FOS Outage Backup Plan.

1c.4 Flight Attendants

Crew Briefing

The captain will brief the flight attendants before each departure, or before the first departure of a duty period where the same crew continues together. See section [1c.4 Flight Attendants](#).

Flightdeck Contact During the Sterile Period

During the sterile flightdeck period, flight attendants contact the flightdeck **only** for safety-related information or situations.

A flight attendant who believes a safety-related condition exists will call the flightdeck, whether or not the sterile period is in effect.

Standard Notifications

Event	Communication
Cabin secure for departure	Flight attendant to flightdeck
Sterile period ended, climb	Flightdeck to cabin, per aircraft signal or interphone
Turbulence anticipated	Flightdeck to cabin, with expected duration
Ten minutes to landing	Flightdeck to cabin
Cabin secure for landing	Flight attendant to flightdeck
— End —	

1c.5 Passengers

1c.5.1 Routine PA

The captain, or the first officer at the captain's direction, makes a passenger announcement:

Table 1.3 Required Passenger Announcements

When	Content
After boarding is complete, before departure	Welcome, flight time, routing, expected ride, and anything already known about a delay
Whenever a delay of 10 minutes or more occurs or becomes known	What is known, what is not, and when the next update will come
Passing through 10,000 feet, when workload permits	Flight time remaining, expected ride, destination weather
Before an expected period of turbulence	That it is coming, roughly when, and how long

When	Content
On any significant change — diversion, extended holding, gate change	What changed and what happens next
On descent or arrival	Arrival time, destination weather, and connection guidance
— End —	

1c.5.2 Content

State what is known, what is not known, and **when the next update will come**. Give a time for the next update and meet it. An update that says nothing new but arrives when promised keeps the cabin calm; a promised update that does not arrive does the opposite.

Do not:

- speculate on a mechanical issue, on weather, or on ATC
- attribute blame to another department, to ATC, or to another agency
- promise a connection, a gate, or a departure time you do not control
- use technical language that sounds routine to a crew and alarming to a passenger

Say **why** where you can. "We are number fourteen for de-icing" tells a cabin more than "we are experiencing a short delay," and it is verifiable.

1c.5.3 Delivery

Speak slowly and slightly more deliberately than feels natural. Hold the handset close and steady. Check the volume of the first few words.

Where the announcement follows an abnormal event, deliver it early, briefly, and calmly. The tone carries more information to the cabin than the words do.

Note: A passenger who has just felt something unusual is listening for whether the person speaking sounds concerned. Content matters less than manner.

1c.5.4 After an Abnormal Event

Following a rejected takeoff, a go-around, severe turbulence, or a return to the gate, make an announcement **as soon as the aircraft is under control and the immediate workload is complete**.

Cover, briefly:

1. that the crew is aware of what happened
2. what happened, in plain language

3. that the aircraft is safe, if it is
4. what happens next, and when the next update will come

A go-around is the most commonly misread event in the cabin. Explain it — an approach that was discontinued and will be flown again — because passengers who are not told assume the worst.

1c.5.5 Irregular Operations

An informed cabin is a calm cabin. During an irregular operation, **more frequent and shorter announcements are better than one comprehensive announcement delivered late.**

Where the aircraft is held on the ground with passengers aboard, announcements are made at least every **30 minutes**, whether or not there is new information. See section [15g.5 Tarmac Delay](#).

Coordinate with the flight attendant in charge so the cabin crew is not answering questions the flightdeck is about to address.

1c.5.6 Medical Event

Where a medical event is in progress, do not describe the passenger, the condition, or the outcome. Announce only what the cabin needs — a request for medical assistance, or that the flight is diverting and why in general terms.

See section [18g.4 Special Categories](#).

1c.6 Ramp

Communication with the ground crew is via interphone. Standard exchanges are described in the applicable AOM> 4> Pushback/Start.

The flightdeck will confirm with the ground crew, before pushback:

- Nosewheel steering bypass pin installed and its removal confirmed after pushback
- Direction of push and the expected orientation
- Area clear
- All doors closed and all equipment clear

1c.7 Company to Flight Crew

1c.7.1 General

The Company communicates with crews in flight and on the ground through several channels. Each has a different urgency and a different acknowledgment requirement.

Channel	Carries	Acknowledgment
ACARS	Operational messages to and from Dispatch and the IOC	By reply where the message requests it
CCI	Operational alerts and Company messaging	Required for a read and acknowledge message
Comply365	Bulletins, manuals, and forms	By review before the trip
Dispatch VHF or SATCOM	Voice with the controlling dispatcher	By voice
— End —		

1c.7.2 Bulletins

Flight Operations Bulletins (FOB)

The Flight Operations Bulletin is the communication avenue used to consolidate general flight operations information. Bulletins are located in **Comply365** in the Flight Operations Bulletin collection.

A FOB carries information that applies between revisions of this manual. Where a FOB and this manual conflict, the FOB is the later document and governs until the next revision incorporates it.

Fleet-Specific Bulletins

The fleet-specific bulletin consolidates fleet-specific information and is located in Comply365 in the fleet's Fleet Specific Ops, Fleet Bulletins collection.

Review outstanding bulletins for the fleet before the first flight of a trip.

1c.7.3 Operational Alert

Where a disruptive event — weather, a security-related situation, or another event affecting the operation of the airline — occurs, operational messages may be sent by CCI.

Priorities and Authorization

CCI messages are categorized in three priorities. They follow the same logic as warnings, cautions, and notes respectively.

Table 1.4 CCI Message Priorities

Priority	Content	Authorized by
Read & Acknowledge — requires pilot acknowledgment, flight critical	Safety and/or security threat	SVP Flight Operations and IOC; MD Flight Operations Technical; MD Flight Line Operations
Medium — requires timely action, not flight critical	May cause operational delays; procedural reminders	Flight operations management
Low — requires routine action, not flight critical	Information only	Flight operations administration
— End —		

Procedure

When a read and acknowledge message is issued, a pop-up window appears in mobile CCI. The pop-up **restricts the use of all other CCI functions — including signing fit for duty — until the crewmember reads and acknowledges the message.**

Note: A read and acknowledge message does not prevent a crewmember from using alternate sign-in procedures, and does not restrict access to DECS.

Connectivity. Cellular or WiFi connectivity is required to receive read and acknowledge messages in CCI.

During ground movement. Where a read and acknowledge message is received during ground movement, stopping the aircraft to acknowledge the message may be necessary.

Caution: Do not acknowledge a CCI message while the aircraft is moving. Stop, or defer the acknowledgment until the aircraft is stopped. The message is not more urgent than taxiing the aircraft.

1c.8 Flightdeck to Ground Hand Signals

1c.8.1 General

Where the interphone is inoperative, where the ground crew disconnects, or where noise or a language barrier makes voice communication unreliable, standard marshalling signals are used.

Marshalling signals are a two-way exchange. The marshaller signals the crew; the crew acknowledges. An unacknowledged signal has not been received.

Before Accepting a Signal

- Confirm the signaller is the marshaller and is in view of both pilots where possible
- Confirm the signal is intended for this aircraft
- Where a signal is not understood, do not move. Set the parking brake and request clarification

Warning: Do not move the aircraft on a signal you cannot positively identify. A misread marshalling signal at the gate is a collision.

Wands

Wands are used at night and in reduced visibility. Illuminated wands may be used to form any signal that would otherwise be given by hand.

1c.8.2 Signals

The following are the standard signals. They are reproduced from the ICAO marshalling signal set and are common to every station Northern Airways serves.

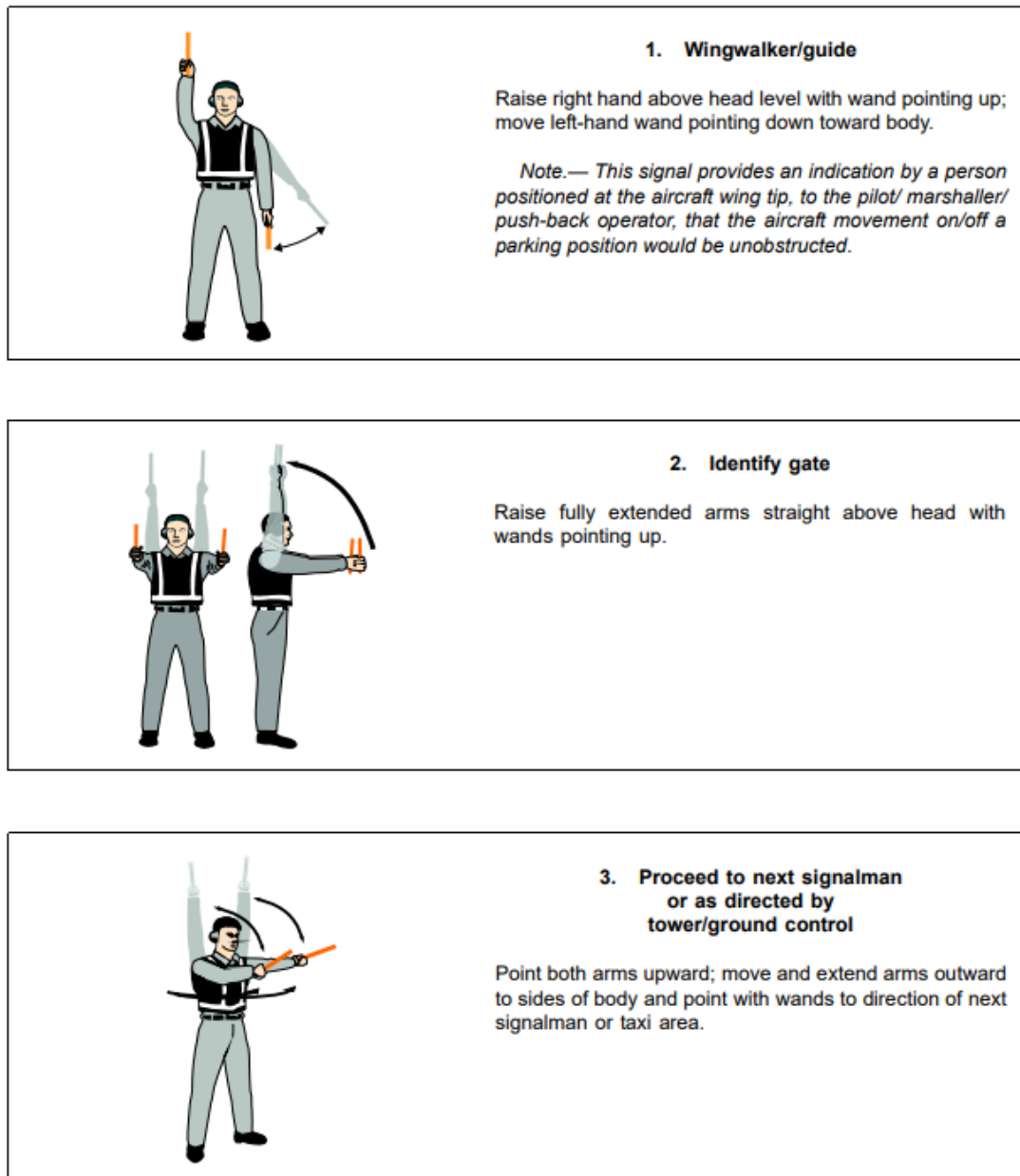


Figure 1.1 Marshalling Signals — Wingwalker, Identify Gate, Proceed to Next Signalman

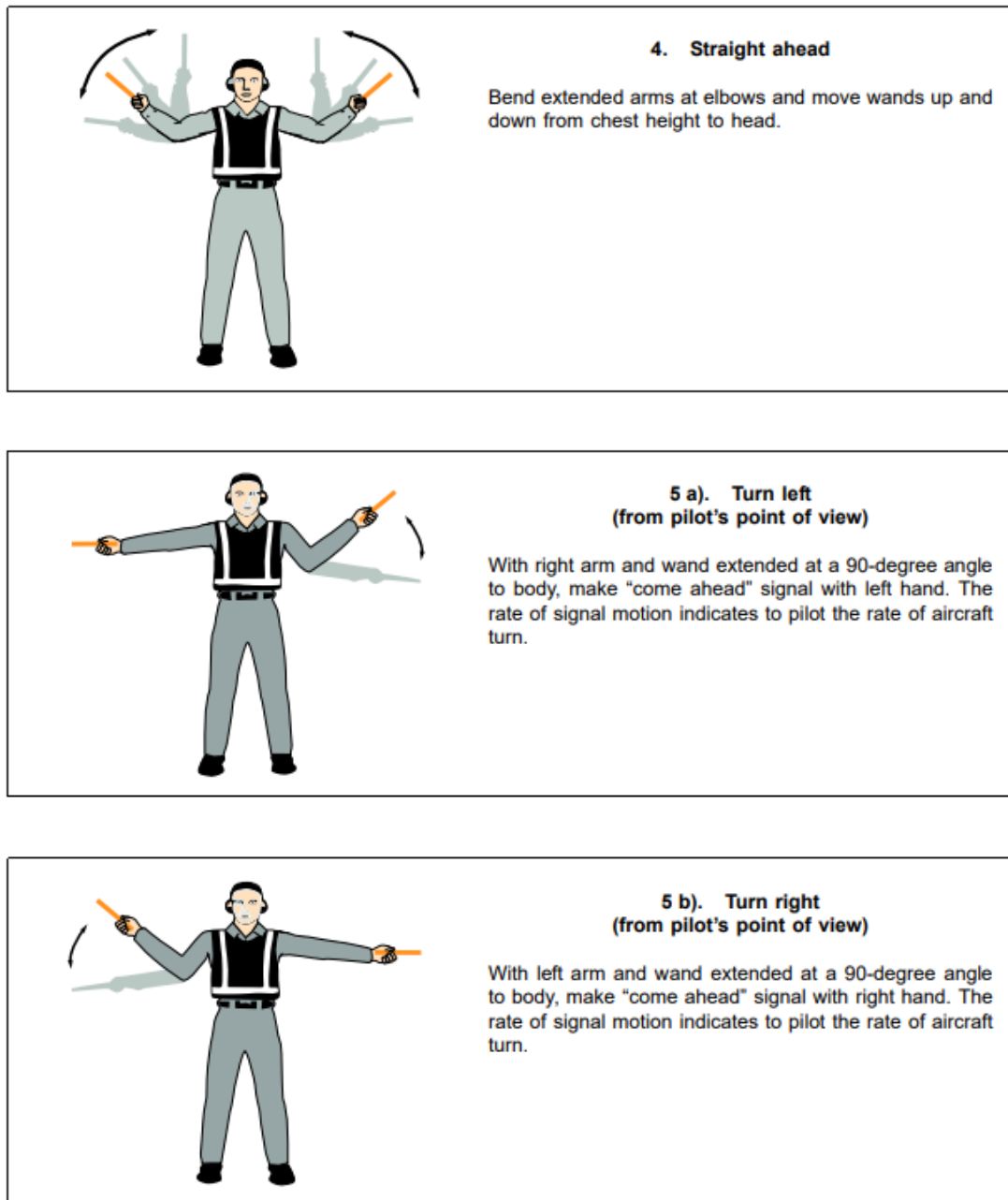


Figure 1.2 Marshalling Signals — Straight Ahead, Turn Left, Turn Right

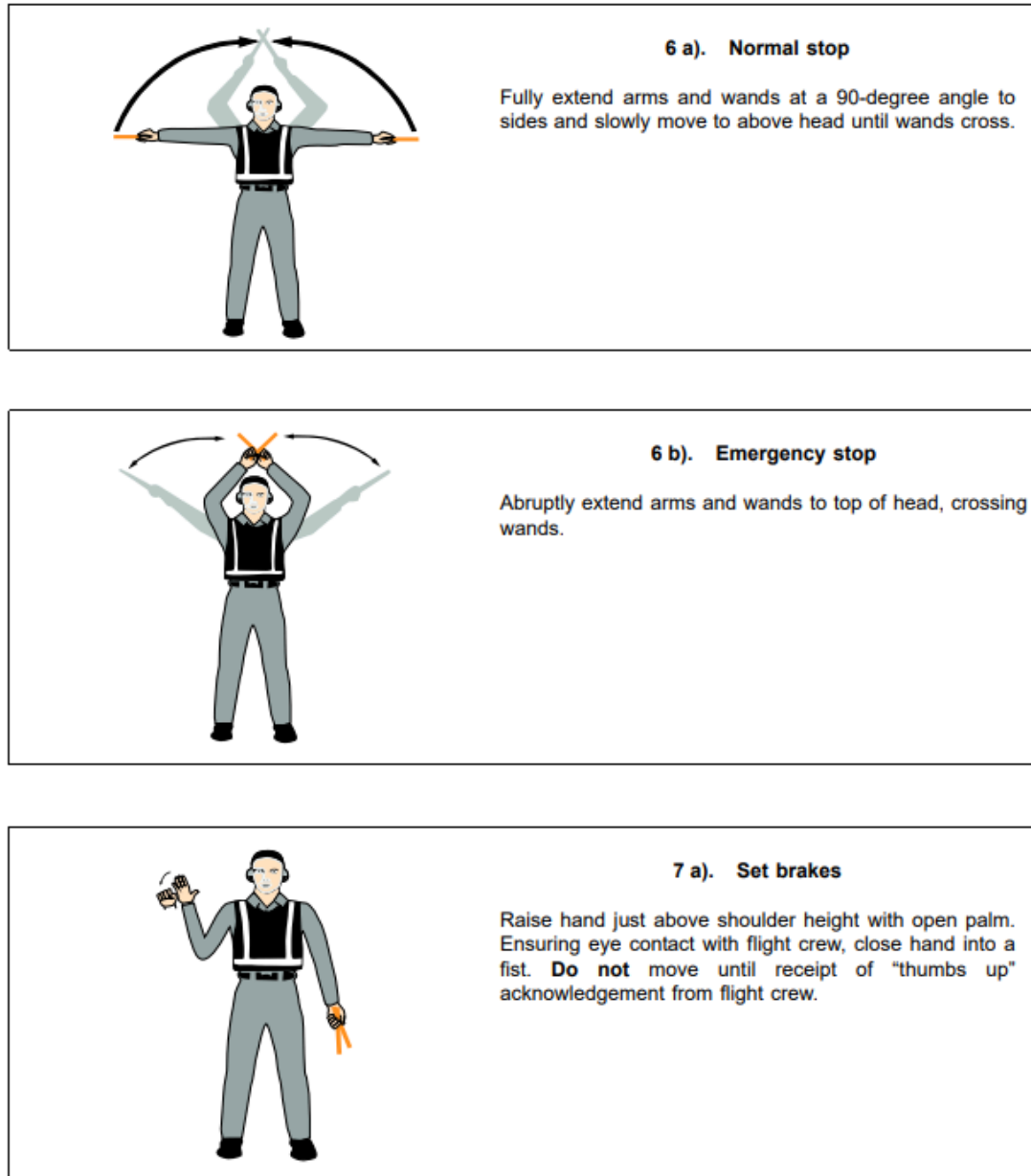


Figure 1.3 Marshalling Signals — Normal Stop, Emergency Stop, Set Brakes

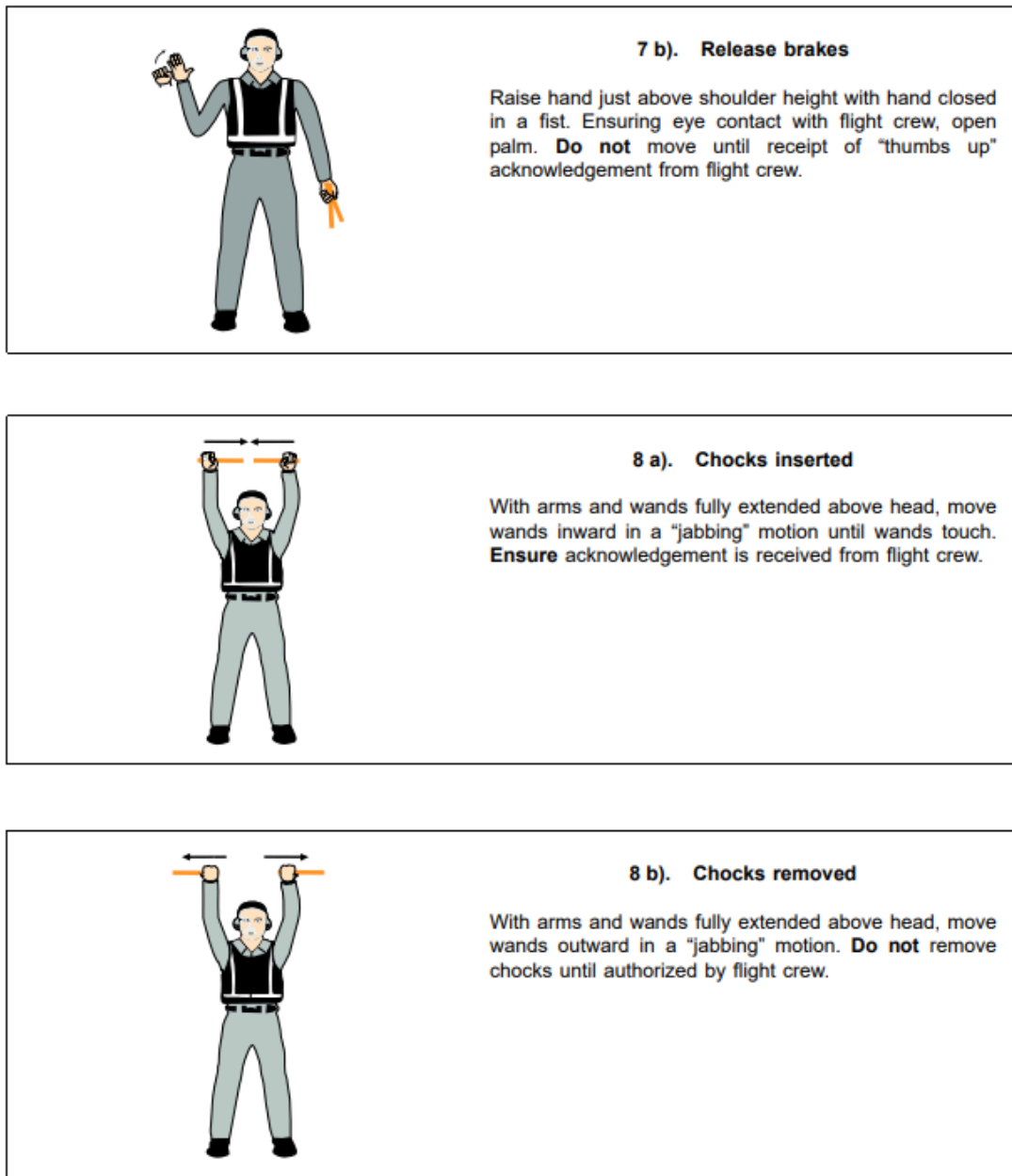


Figure 1.4 Marshalling Signals — Release Brakes, Chocks Inserted, Chocks Removed

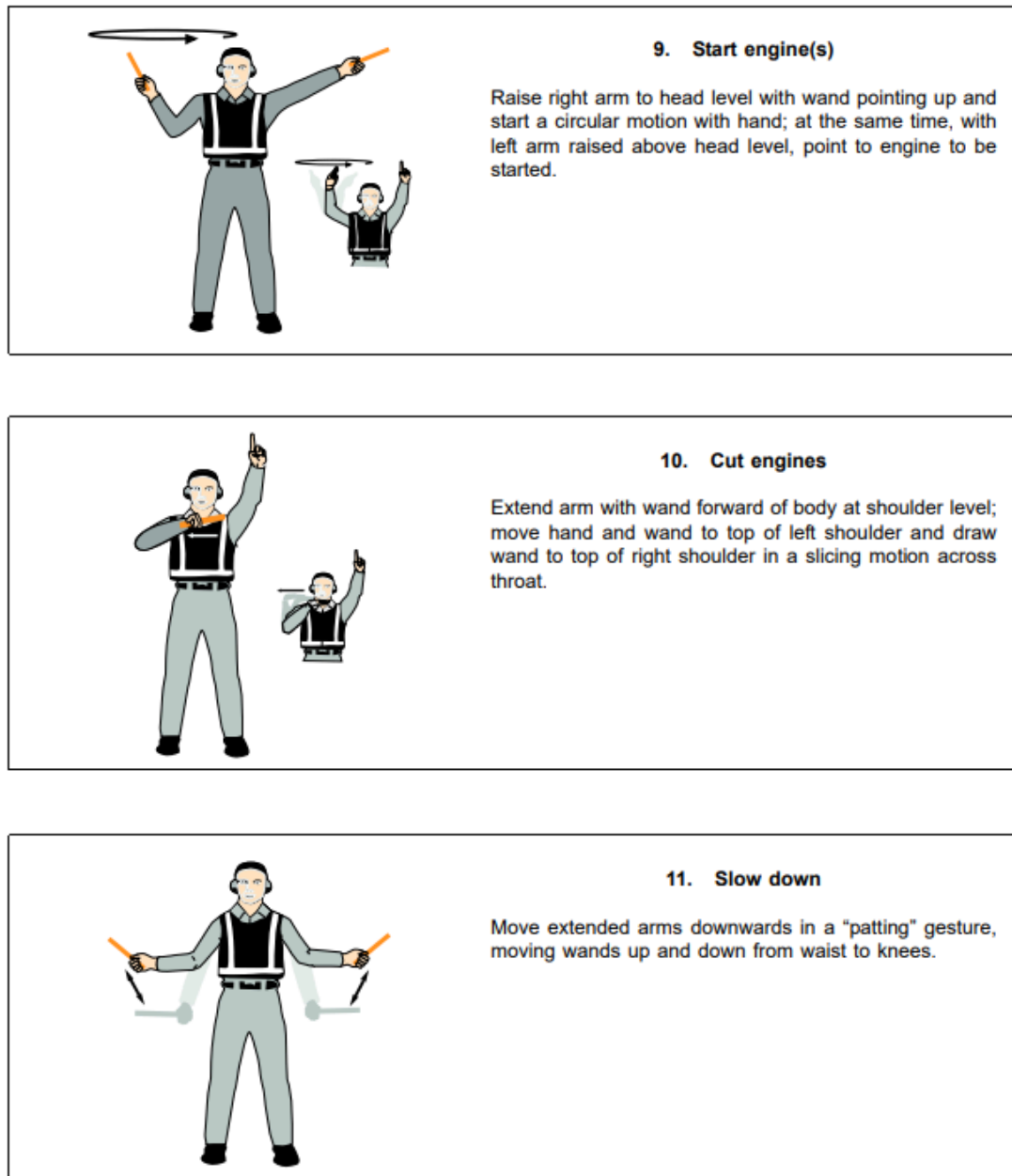


Figure 1.5 Marshalling Signals — Start Engine, Cut Engines, Slow Down

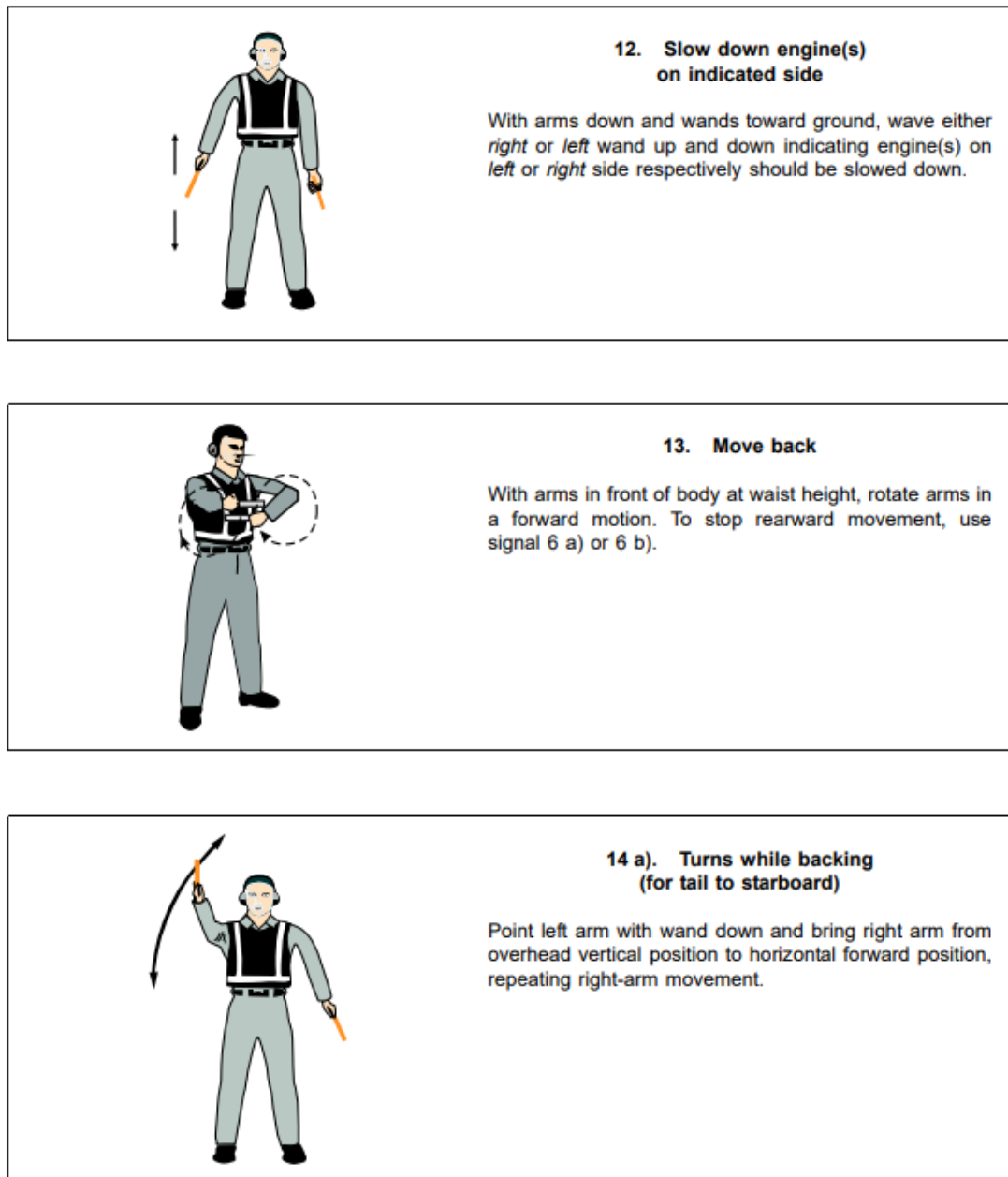


Figure 1.6 Marshalling Signals — Slow Down Engine, Move Back, Turns While Backing

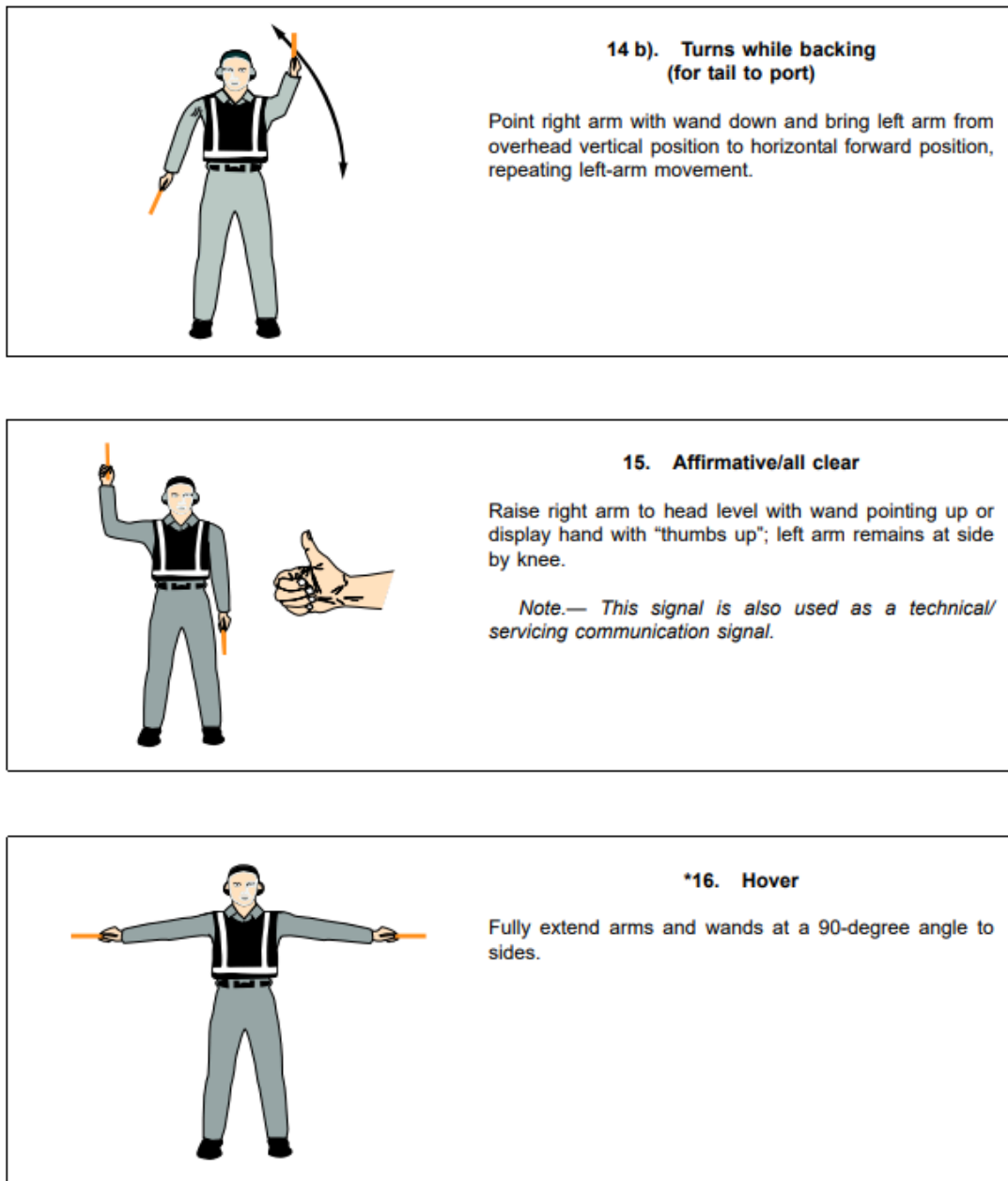


Figure 1.7 Marshalling Signals — Turns While Backing, Affirmative, Hover

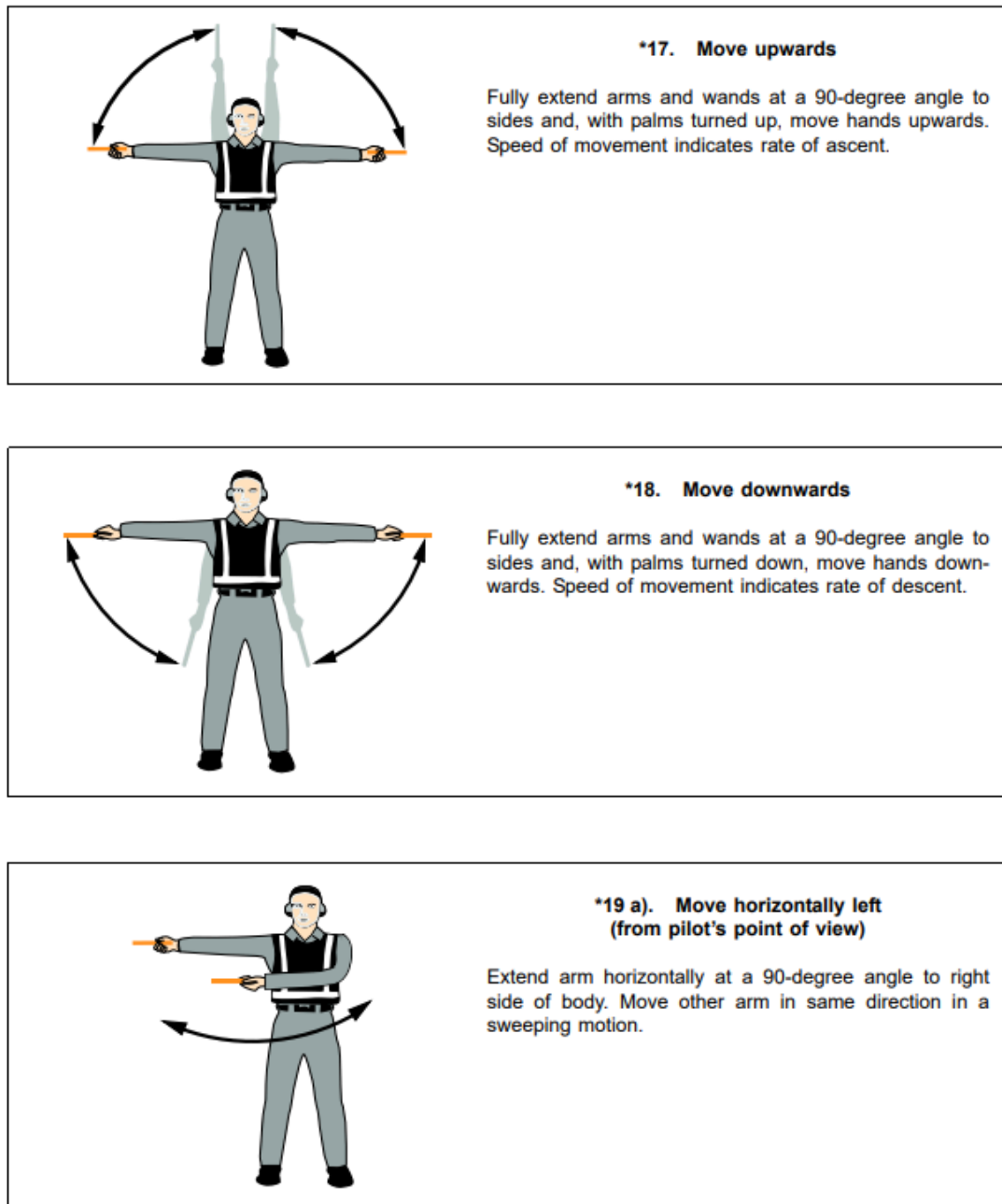


Figure 1.8 Marshalling Signals — Land, Fire, Hold Position

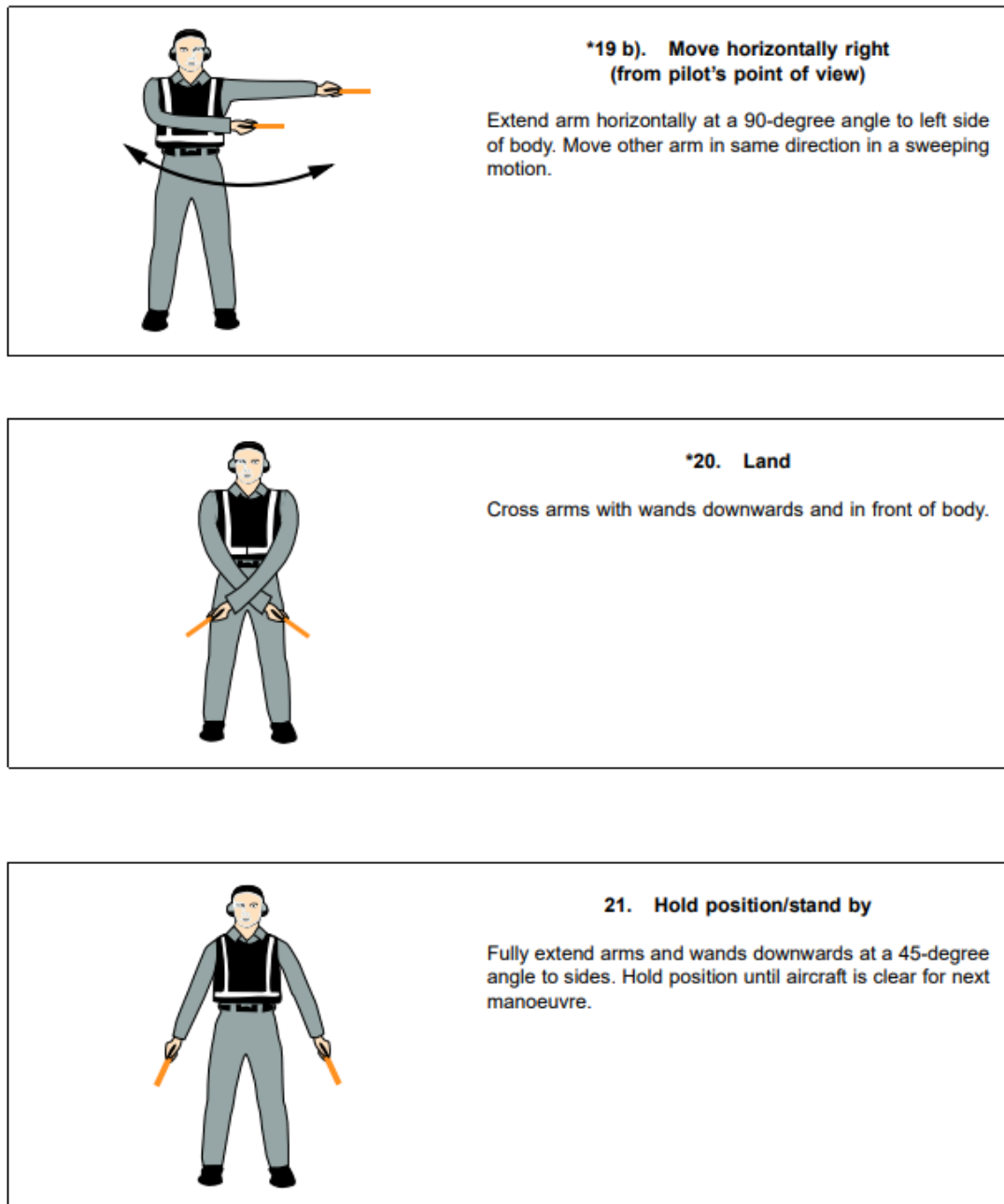


Figure 1.9 Marshalling Signals — Dispatch Aircraft, Do Not Touch Controls, Connect Ground Power

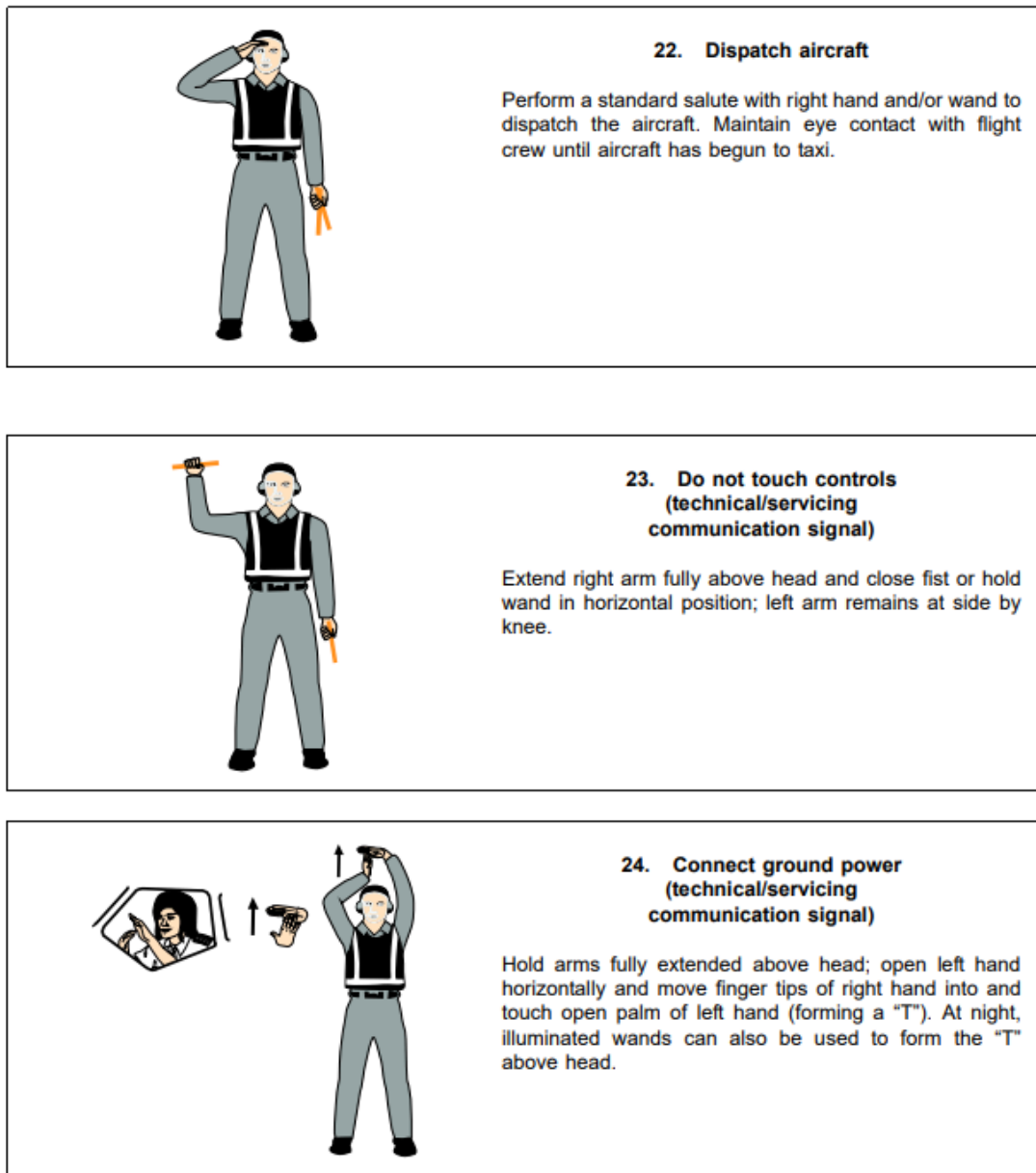


Figure 1.10 Marshalling Signals — Disconnect Power, Negative, Establish Communication

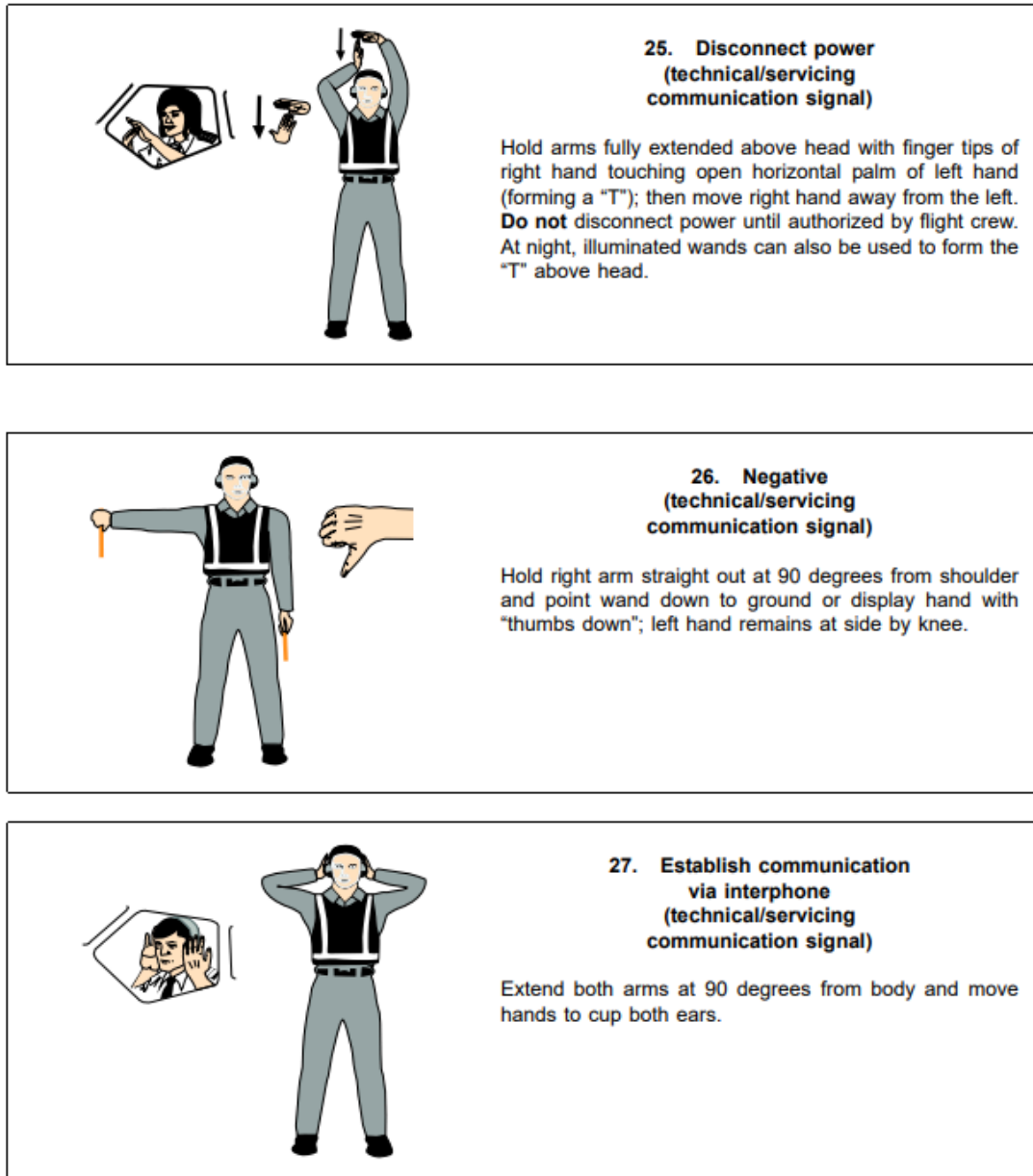


Figure 1.11 Marshalling Signals — Disconnect Power, Negative, Establish Communication via Interphone

1c.8.3 Signals Requiring Immediate Compliance

Signal	Crew Action
Emergency stop	Stop immediately. Set the parking brake.
Fire	Stop. Accomplish the applicable QRC/QRH procedure. Do not move the aircraft.
Hold position	Stop and hold. Do not resume without a further signal.

Signal	Crew Action
Do not touch controls	Do not move any flight control or move the aircraft. Ground personnel are in a hazardous position.
— End —	

For the aircraft-specific pushback and start procedure, see the applicable AOM> 4> Pushback/Start.

Weather products, sources, and interpretation are in See Meteorology. This section states operating policy.

1w.1 General

Weather is the most frequently encountered category of threat. It is anticipated in planning, briefed, and reassessed continuously.

1w.2 Windshear / Microburst

Avoidance

Windshear is avoided, not penetrated. Where windshear is reported, suspected, or indicated by the predictive windshear system, **do not take off and do not continue an approach.**

Indicators

Indicator	Action
Predictive windshear warning	Immediate escape maneuver per the AOM
Reactive windshear warning	Immediate escape maneuver per the AOM
LLWAS or TDWR alert	Do not depart. Do not continue the approach.
PIREP of ± 15 knots or more airspeed loss/gain	Do not depart. Do not continue the approach.
Convective activity within 3 NM of the departure or arrival path	Assess. Delay is the default.
Virga, blowing dust ring, localized strong outflow	Delay.
— End —	

Escape Maneuver

The escape maneuver is aircraft-specific. Refer to the applicable AOM> 17> Inflight Maneuvers and to the QRC/QRH. It is a memory item.

Delay

Delay the departure or the approach until the condition no longer exists. See section [1g.1 Operational Priorities](#).

1w.3 Thunderstorms

Avoidance

Avoid a thunderstorm cell laterally. Do not overfly a cell, and do not attempt to pass beneath one.

Avoidance distances increase with altitude, because the anvil and the associated hail and turbulence extend further from the core at height. The distances applicable to the fleet and the altitude band are published in the applicable AOM> 15> Adverse Weather and in Meteorology.

Give additional margin downwind of a cell, and treat a gap between two cells as closing unless both are being actively assessed on radar.

Radar

The radar is operated, tilted, and interpreted actively by the PM. A radar left in a fixed tilt is not being used. See the applicable AOM> 15> Adverse Weather.

Warning: Attenuation makes the far side of a strong cell appear clear. A radar shadow is not a gap.

Deviations

Request a deviation early. A deviation requested 100 NM out is routinely approved; one requested 20 NM out may not be. If a required deviation is not approved, advise ATC that the deviation is required and turn.

1w.4 Turbulence

Seat Belt Sign

The seat belt sign is turned **ON** before pushback, after the passengers have boarded, and remains on until the flight is clear of the sterile period and the ride is smooth. It is turned on any time turbulence is encountered or anticipated.

Anticipated Turbulence

Notify the cabin crew with the expected onset and duration. Where turbulence is anticipated to be moderate or greater, direct the flight attendants to be seated and secured.

Encountered Turbulence

Intensity	Action
Light	Seat belt sign on. Advise the cabin.

Intensity	Action
Moderate	Seat belt sign on. Flight attendants seated. Turbulence penetration speed. Advise ATC. PIREP.
Severe	Seat belt sign on. Everyone seated. Turbulence penetration speed. Autopilot per AOM. Do not chase altitude. Advise ATC. PIREP. ASAP report. Notify dispatch.
— End —	

See the applicable AOM> 15> Adverse Weather for turbulence penetration speed and autoflight configuration.

Reporting

PIREP every encounter of moderate or greater turbulence, and every encounter that differs materially from the forecast. Other crews depend on it.

1w.5 Icing

Ground Icing

See Preflight General, Deicing for the ground deicing and anti-icing program, holdover times, and the clean aircraft concept.

Severe Icing

If severe icing is encountered, flight crews will be vigilant for any decay in the handling characteristics of the aircraft. Conditions can deteriorate rapidly to a dangerous situation, and flight crews will take all steps necessary to exit severe icing as soon as is practicable.

Visual cues that may indicate severe icing:

- Unusually extensive ice accreted to the airframe in areas which are not normally observed to collect ice
- Extensive accumulation of ice on the upper surface of the wing aft of the protected area

FAA guidance states that landing the aircraft when freezing rain or freezing drizzle is encountered on approach would, in many cases, be the most expeditious method of exiting the conditions. In considering options, flight crews will also bear in mind:

- Request **priority handling** from ATC to facilitate a route or altitude change to exit the conditions
- A climb may or may not exit the icing; an inversion may make it worse

— Autopilot use in severe icing may mask control difficulty. Refer to the applicable AOM.

Warning: Severe icing is outside the aircraft's certification envelope. The airplane is not approved to operate in it. Exiting is not optional.

1m.1 General

Airworthiness is determined by Maintenance, not by the crew. The crew's obligations are to **observe**, **report accurately**, and **not to operate** an aircraft known to be unairworthy.

Contact Maintenance Control through dispatch. See section [20.3 Contact Information](#).

1m.2 eAML

The electronic Aircraft Maintenance Log is the legal record of aircraft discrepancies and their resolution.

Verify the eAML before accepting the aircraft. See section 3m.2 eAML.

Entry Types

Entry	Made by	Purpose
MECHANICAL DISCREPANCY	Crew	Records a condition affecting airworthiness or the operation. Requires a maintenance action.
MAINTENANCE ACTION	Maintenance	Records the corrective action or the deferral, and the authority for it.
INFO-ONLY	Crew	Relays information to Maintenance on a non-airworthiness matter. Requires no maintenance action.
— End —		

Entries Always Required

An eAML entry is made for each of the following, whether or not the aircraft appears affected:

- **Rejected takeoff**
- **Bird strike** — state the flap position
- **Lightning strike or static discharge**
- **Overweight landing**
- **Severe turbulence**
- **Volcanic ash encounter**
- **Smoke, odour, or fumes**

Review Before Acceptance

Review the **preceding 3 days** of eAML entries before accepting the aircraft, not only the open items. A pattern across several days is visible in the history and is not visible in the open list.

Marking	Meaning
*F	The item carries a follow-up action. Shown against the item on the release.
Yellow placard	A repetitive maintenance action is required for the item.
— End —	

Foreign Language Entries

An entry may be made in a language other than English by a person authorized to make it. The entry is **translated for the crew before departure**. Do not accept an aircraft carrying an entry the crew cannot read.

INFO-ONLY Entries

Acceptable INFO-ONLY subjects include:

- items not affecting airworthiness, such as cabin cosmetic damage
- refills and replacements
- required checks, such as an ETOPS verification
- transitory conditions and pilot-corrected items, such as a system reset
- insect or rodent control

Entering Mechanical Discrepancies

Enter every mechanical discrepancy, including one that resolves itself. Enter:

- what was observed, and by whom
- the phase of flight and the conditions
- any EICAS, ECAM, or annunciator message, quoted exactly
- the crew action taken and the result

Do not enter a diagnosis, a suspected cause, or a proposed corrective action.

Note: Recording a discrepancy is not a judgment as to its significance. Engineering determines significance from the record.

Deferrals

A deferral is applied under the MEL or CDL, or under other authority granted by tracking placard or engineering authorization. Every deferral cites the authority for it. See section [1m.3 MEL and CDL](#).

Inflight Verification Check

Where Maintenance requests an inflight operational check of a system, the request appears as a MECHANICAL DISCREPANCY entry. The captain responds with a new entry summarizing the check.

- Where the check is satisfactory, make an **INFO-ONLY** entry.
- Where the check is unsatisfactory, make a **MECHANICAL DISCREPANCY** entry.

Outage

Where the eAML is unavailable, use the paper log procedure published in the Controlled Document Library and advise Maintenance Control. See section 3m.2 eAML.

1m.3 MEL and CDL

A deferred item is dispatched under the MEL or CDL with the associated operational (O) and maintenance (M) procedures. The crew is responsible for knowing and complying with the (O) procedure for every open item.

Verify at preflight:

- Every MEL/CDL item on the release is placarded and matches the eAML
- Every (O) procedure is understood and any operational penalty is applied
- The item does not conflict with the planned operation — ETOPS, RVSM, CAT II/III, or an airport requiring the affected system

See Preflight Maintenance and, for ETOPS, See Preflight Maintenance, ETOPS.

1s.1 General

Security procedures are governed by 49 CFR Part 1544 and by the Northern Airways Aircraft Operator Standard Security Program (AOSSP). The AOSSP contains Sensitive Security Information and is issued separately to authorized personnel.

Warning: This manual does not restate AOSSP content. Do not discuss security procedures on an open frequency, in a public area, or on any unsecured system.

1s.1.1 Inflight Security Coordinator

The **captain is the Inflight Security Coordinator** for the flight. The captain has final authority over the security of the aircraft and everyone aboard it, and retains that authority whether or not a Federal Air Marshal is aboard.

Note: A Federal Air Marshal may or may not intervene in a passenger disturbance. Their presence does not transfer the security decision, and the crew does not plan on their intervention.

1s.1.2 Threat Levels

Northern Airways classifies an inflight security threat at one of four levels. The level determines the response.

Table 1.5 Inflight Security Threat Levels

Level	Situation
1	Suspicious behavior. A person acting in a way that warrants attention but has not interfered.
2	Interference. A person has interfered with a crewmember's duties, or has refused a lawful instruction.
3	Threat. A credible threat has been made against the aircraft, the crew, or the passengers.
4	Attempted or actual breach of the flightdeck, or an attempt to seize the aircraft.
— End —	

Level 1 and 2

Advise the flightdeck. The cabin crew manages the situation under the AOSSP. The captain decides whether to request law enforcement to meet the aircraft, and whether to divert. See section [18g.5 Denied Boarding and Removal](#).

Level 3

Advise the dispatcher and consider a diversion. Follow the AOSSP.

Level 4

An attempted breach of the flightdeck is the highest level and requires immediate action:

1. **Declare an emergency.**
2. **Squawk 7500.**
3. **Expect an intercept.** Standard interception procedures are in section [16n.2 Contingency Procedures](#).
4. **Land as soon as practical.**

Warning: Do not open the flightdeck door during a level 4 event for any reason, including a request that appears to come from a crewmember. The door stays closed until the aircraft is on the ground and the situation is resolved.

1s.1.3 Flightdeck Door Challenge

Where the flightdeck door is open and suspicious behavior is detected in the cabin or at the door, any crewmember will call out:

"Door! Door! Door!"

The call means the door is to be closed and locked immediately. It is acted on without discussion, by whoever is nearest. Confirmation of what was seen comes afterward.

1s.2 Flightdeck Security

The Door Stays Closed

The flightdeck door will be closed and locked from the time all external doors are closed before pushback until the aircraft is parked at the destination gate (14 CFR 121.587).

Opening the Door in Flight

The door is opened in flight only in accordance with the AOSSP procedure. The procedure requires positive identification, cabin surveillance, and coordination with the cabin crew. It is not modified, abbreviated, or skipped for convenience.

Returning to the Flightdeck

A pilot returning to the flightdeck uses the interphone until their identity is confirmed visually through the peephole. Identity is confirmed before the door is unlocked, not as it opens.

Suspect Device

Where a suspect device is found in flight, it is moved to the least risk bomb location published for the aircraft in the QRC/QRH, and the QRC/QRH procedure is accomplished. Do not attempt to open, disarm, or submerge the device.

Two-Person Flightdeck

Two persons will be on the flightdeck at all times during flight.

Where a pilot leaves the flightdeck for any reason, a **designated flight attendant takes their place** and remains until the pilot returns. The flightdeck is never occupied by one person.

Procedure

Before a pilot leaves:

1. The remaining pilot and the designated flight attendant are both briefed and ready.
2. The flight attendant enters the flightdeck **before** the pilot leaves.
3. The pilot leaves only once the flight attendant is inside and the door is secured behind them.

On return, the pilot is positively identified before the door is unlocked, and the flight attendant leaves only after the pilot is seated. See section [1s.2 Opening the Door in Flight](#).

The Designated Flight Attendant

The flight attendant occupying the flightdeck is briefed on the flightdeck door, the interphone, and the requirement not to touch any control. Their function is presence — a second set of eyes and the ability to open the door and summon help.

Warning: A pilot does not leave the flightdeck before the flight attendant is in position, however brief the absence is expected to be. The sequence is not reversed for convenience.

Denial of Entry

The pilot in command may deny flightdeck entry to any person, at any time, for any reason related to the security or safety of the flight.

1s.3 Jumpseat

Flightdeck jumpseat occupancy is governed by 14 CFR 121.547 and by the AOSSP. See section [1s.3 Jumpseat for the authorization](#), identification, and briefing requirements.

The captain has absolute authority over jumpseat occupancy and may decline any rider without stating a reason.

Every jumpseat rider will be briefed on:

- Seat, harness, and oxygen mask operation
- Emergency exit and evacuation
- Sterile flightdeck period
- The requirement not to interfere, and the expectation that the rider will speak up if they see something

1n.1 General

Non-normal and emergency procedures are contained in the **QRC/QRH** and in the applicable AOM> 16> Non-Normals. They are not restated here.

The **QRC** (Quick Reference Card) carries memory items and quick action items — the handful of steps accomplished from recall before any checklist is opened. The **QRH** (Quick Reference Handbook) carries the full non-normal and emergency checklists, including ECAM-driven procedures and their follow-up items. Identify which applies before acting: a memory item is flown first and read afterward; everything else is read and done.

This section states the policy that surrounds those procedures.

Priorities in a Non-Normal

1. **Aviate.** Someone flies the aircraft. Announce who.
2. **Navigate.** Know where you are and where the terrain is.
3. **Communicate.** ATC first, then the cabin, then the Company.
4. **Manage.** QRC/QRH, then decision, then execution.

Sequence

Except for memory items, and except where the aircraft is in an undesired state, no non-normal requires immediate action. Stabilize the aircraft and establish who is flying before beginning the QRC/QRH.

1n.2 Declaring an Emergency

When

Declare an emergency whenever the crew requires priority handling or when the safety of the flight is in question. The captain does not need permission, does not need to consult dispatch first, and does not need to be certain.

How

"Mayday, Mayday, Mayday" or "Pan-Pan, Pan-Pan, Pan-Pan", followed by call sign, nature of the emergency, intentions, altitude, position, souls on board, and fuel remaining in time.

Reporting

A declaration that proves unnecessary carries no penalty and no reporting burden beyond the reports required by Reports and Forms.

See section [1g.1 Operational Priorities](#).

After

Notify dispatch once the aircraft is under control. See Reports and Forms.

1n.3 Crew Incapacitation**Recognition**

Incapacitation may be obvious or subtle. Suspect it when a pilot:

- Fails to respond to two communications. See section [22.7.2 Two-Communication Rule](#).
- Responds inappropriately or incoherently
- Fails to make an expected callout or control input

Immediate Actions

1. **Assume control.** State "**My aircraft.**"
2. Engage the autopilot and stabilize.
3. Declare an emergency.
4. Secure the incapacitated pilot away from the controls. Summon a flight attendant.
5. Request a qualified pilot from the cabin if one is aboard.
6. Plan a landing at the nearest suitable airport.

See the applicable AOM> 16> Non-Normals and to the QRC/QRH.

Subtle Incapacitation

Subtle incapacitation is more dangerous than obvious incapacitation because it is not recognized. The two-communication rule exists specifically to detect it. Apply the rule literally — two communications without adequate response, take control.

1n.4 Flightdeck Fire Extinguisher

Halon extinguishers installed in the flightdeck are effective on **Class A, B, and C** fires — ordinary combustibles, flammable liquids, and energised electrical equipment.

Operation

The first action is to **break the safety wire** retaining the pin or lever. The extinguisher will not discharge until this is done.

Then aim at the base of the fire and discharge in short bursts, ventilating in accordance with the applicable QRC/QRH procedure.

Warning: Halon displaces oxygen. Where a Halon extinguisher is discharged in the flightdeck, don oxygen masks and select 100 percent before or

immediately after discharging. Accomplish the applicable QRC/QRH smoke procedure.

For a lithium battery fire, water or another non-alcoholic liquid is used to cool the cell after the fire is knocked down. Halon alone does not prevent thermal runaway. See the applicable QRH> Lithium Battery Fire.

1n.5 Reserved

1i.1 General

This section states standing policy applicable to a region, irrespective of the phase of flight. Phase-specific international procedure is in the phase chapter; the international operating requirements are in International Operations.

Divisions and the qualifications that go with them are in section [21.6.4 Division Qualification](#).

1i.2 South (Latin) America

Operations in Latin and South America share a set of characteristics that do not appear together anywhere else on the network: high terrain close to the airport, high field elevation, non-radar and procedural airspace, mixed language proficiency, and approach procedures that depend on RNP AR.

Latin Terrain Qualification is required to operate into terrain-challenged cities in the region. See section [21.6.5 Latin Terrain Qualification](#).

1i.2.1 Regional Considerations

Terrain

Terrain in the region is close, high, and frequently on more than one side of the airport. Obstacle clearance is assured only along the published procedure or the engine-out departure path. See section [1p.2.6 Takeoff Obstructions](#).

Brief the escape procedure for every departure and every approach, and know which way it turns before you need it.

Airspace and Communication

Expect procedural control, position reporting, and limited or no radar. Expect English proficiency to vary. Read back every clearance in full, and where a clearance is not fully understood, request it again rather than acting on the part that was clear.

Performance

Field elevation and temperature combine to make the climb limit the governing limit far more often than field length. Verify the TPS reflects the reported field temperature, not a forecast figure, before accepting the performance. See section [1p.2.4 Takeoff Thrust Ratings](#).

Weather

Convective activity develops quickly in the afternoon and evening, and terrain blocks radar returns. A radar shadow behind terrain is not a gap. See section [1w.3 Thunderstorms](#).

1i.2.2 Mexico

Altimetry

Altimeter settings in Mexico are issued in **hectopascals**. Set the units the aircraft expects and confirm the setting with the other pilot. See section [5g.6.2 Altimetry](#).

Mexico City

MMMX is a special qualification airport and requires Latin Terrain Qualification. See section [21.9.3 Designated Special Airports](#).

The field elevation places the aircraft at a true altitude well above the indicated figures crews are used to at sea level stations. Expect:

- climb-limited takeoff weights, frequently below the structural maximum
- high true airspeed for a given indicated speed, and correspondingly larger turn radii on the departure and the approach
- a longer landing roll for the same indicated approach speed

Terrain surrounds the basin. The published departure and engine-out procedures exist because of it, and a deviation from them is not covered by the obstacle analysis.

Documentation and Entry

Carry the documents required for entry and overflight. Customs, immigration, and agricultural requirements are stated in section [16g.4 Customs and Immigration](#).

Fuel

Fuel availability and quality vary by station. Where fuelling is conducted at a station not on the approved vendor list, Maintenance Control will direct a clear and bright test, and the result is recorded as an INFO entry in the eAML. See section [3g.5 Fuel](#).

Note: Canada and Mexico are domestic for qualification purposes unless the routing crosses the Gulf of Mexico or another qualifying overwater segment. Being domestic for qualification does not make an airport in Mexico ordinary — the special qualification and terrain requirements apply regardless.

1i.2.3 High-Elevation Andean Stations

Several stations south of Mexico share the Mexico City characteristics — runway elevation of 8,000 feet or more, terrain close on more than one side, and a

climb-limited rather than field-limited takeoff — without sharing its airspace or ATC environment.

At a high-elevation Andean station:

- verify the takeoff performance was computed for the field elevation and the actual reported temperature, not the forecast temperature used for planning
- brief the missed approach and the escape procedure before every approach — terrain rises on both sides of most approach corridors in the region
- expect procedural, non-radar ATC, with position reporting in place of radar vectors
- expect the true airspeed effect described in section [1i.2.2 Mexico to apply equally here](#)

Every station in this category requires Latin Terrain Qualification and is listed as a designated special airport. See section [21.9.3 Designated Special Airports](#).

1i.2.4 Weather

The Intertropical Convergence Zone produces year-round convective activity across the region, with the most severe activity typically in the afternoon and evening. Terrain blocks and distorts radar returns — a clear return behind high terrain is not evidence of a clear path.

June through November is the Atlantic and Eastern Pacific hurricane season. Where a tropical system affects a station on the routing, expect the alternate requirement and the fuel plan to be adjusted well before the system is close enough to affect the terminal forecast. See section [2.5 Alternates](#).

1i.2.5 Communication and Language

Where the region's controllers do not use English as a first language, phraseology may deviate from ICAO standard even where transmissions are in English. Confirm an ambiguous instruction rather than infer it from context, and expect to hold traffic information that a domestic controller would volunteer without being asked.

1i.3 Caribbean

Caribbean operations share Latin America's warm-weather convective pattern but differ from it in airport geography — island runways, frequently with water on one or more sides, limited or no suitable alternate within range on some routings, and, at several islands, United States preclearance facilities.

1i.3.1 Regional Considerations

Alternates

An island station may have no suitable alternate within range, or only one. Where the destination weather is marginal and no alternate exists within fuel range, the flight is planned to hold sufficient fuel to return to a mainland alternate. See section [2.5 Alternates](#).

Runway Environment

Many Caribbean runways are shorter than the mainline network average and are bounded by water at one or both ends. Verify landing performance against the actual runway length and the reported condition — a wet-runway correction on a short island runway removes a proportionally larger part of the margin than it does on a long mainland runway.

Weather

See section [1i.2.4 Weather](#). Hurricane season planning applies equally in the Caribbean, and a system approaching from the east frequently affects Caribbean stations before it affects the Gulf.

1i.3.2 United States Preclearance

At a station with a U.S. Customs and Border Protection preclearance facility, passengers clear U.S. immigration and customs before boarding, and the flight is treated as a domestic arrival at the U.S. destination.

Where preclearance is in effect:

- confirm with the gate agent that preclearance is complete before departure — a flight that departs before preclearance is finished arrives as an international flight regardless of what was planned
- the general declaration and crew documentation requirements of section [16g.4 Customs and Immigration still apply to the crew](#)

1i.3.3 Documentation and Entry

Entry requirements vary island to island more than they do across contiguous Latin America. Verify the current requirement for the specific station — do not assume a requirement from one Caribbean station applies at another.

1i.4 Europe (Transatlantic)

Transatlantic operations are flown under the North Atlantic organized track system for the oceanic segment. See section [16c.1 Oceanic Clearance](#) and section [2.2.11 Oceanic Track Message](#). This section addresses the differences that apply on the continental side, at the departure and arrival gateways.

1i.4.1 Regional Considerations

Altimetry and Units

European states set altimeters in **hectopascals** and use metric units for runway visual range and, at some stations, runway length. See section [5g.6 Foreign Airports for the conversions](#).

Noise and Curfews

Several European gateways enforce a noise curfew or a noise abatement procedure with a defined penalty for non-compliance. Review the noise abatement requirement for the arrival and departure airport during flight planning — a late arrival into a curfew station can require a hold, a diversion, or a curfew exception coordinated by the dispatcher in advance.

Airspace

RVSM and PBN requirements are more uniformly enforced across European airspace than in Latin America. Verify aircraft and crew qualification for the specific airspace to be flown, including any requirement unique to a state's upper or lower airspace. See section [16g.3.4 Airspace and Separation](#).

Slot Times

Many European airports operate under slot control. A significant delay may require a new slot from the dispatcher before departure — do not assume a late departure will be accepted without one.

1i.4.2 Winter Operations

Transatlantic winter operations combine the cold-weather items of section [9w.3 Cold Weather Operations with materially stronger headwinds on the westbound crossing](#). Review the fuel plan for a westbound winter crossing with particular attention to the alternate and contingency fuel margins — the wind forecast used for planning is more likely to be exceeded in winter than in summer.

1i.4.3 Documentation and Entry

Schengen-area entry requirements differ from non-Schengen European states. Verify the visa and documentation requirement for the specific country of arrival, not the region generally. See section [16g.2 Documentation](#).

1i.5 Pacific / Asia

Pacific and Asia operations are the network's longest-range segments and are flown almost entirely over water on ETOPS-approved routings. See section [7g.5 ETOPS](#).

1i.5.1 Regional Considerations

Extended Overwater

Verify ETOPS entry and exit points, the alternate airports along the route, and the fuel required to divert from the most critical point before departure. A Pacific routing frequently has fewer enroute alternate options than any other segment of the network — treat every one on the plan as a real diversion option, not a formality. See section [7g.5 ETOPS](#).

Weather

Typhoon season affects Western Pacific and Asian stations on a seasonal pattern distinct from the Atlantic hurricane season. Where a tropical system affects a station on the routing, expect the same alternate and fuel plan adjustments described in section [1i.2.4 Weather](#).

Fatigue and Time Zone Change

Pacific crossings involve the largest time zone changes in the network, frequently combined with flight time near the maximum for the pairing. Apply the fatigue reporting requirements of section 21.11 Fatigue Risk Management without discounting a symptom because the flight "always feels this way."

Airspace and Phraseology

Expect greater variation in controller English proficiency than in Europe, and expect some airspace in the region to use non-ICAO-standard procedures. Confirm an unclear instruction before acting on it. See section [16g.3.3 Phraseology](#).

1i.5.2 Documentation and Entry

Entry, health, and customs requirements vary significantly by country in this region and change more frequently than in the other regions covered by this section. Verify

the current requirement for the specific arrival station before departure rather than relying on the requirement from a previous trip. See section [16g.2 Documentation](#).

1d.1 Dangerous Goods/Hazardous Materials

1d.1.1 Definition and Classification

Dangerous goods are articles or substances capable of posing a risk to health, safety, property, or the environment when transported by air. They are classified by the hazard class assigned in the ICAO Technical Instructions and the IATA Dangerous Goods Regulations.

The terms **dangerous goods (DG)**, **hazardous materials (HAZMAT)**, and **restricted articles (RA)** are synonymous and interchangeable. The closeout uses RA; the notification uses DG; training material uses HAZMAT. They mean the same thing.

There are **nine** hazard classes. All nine are permissible on Northern Airways aircraft — only certain divisions within a class are forbidden. A class number alone therefore never tells you whether a shipment may be carried.

The Company Dangerous Goods program is published in the Controlled Document Library. It is the controlling document.

1d.1.2 Responsibilities

Party	Responsibility
Shipper	Classification, packing, marking, labeling, and documentation
Acceptance agent	Acceptance check against the shipper's declaration
Load agent	Loading, segregation, and securing per the loading instruction
Dispatcher	Verifying the shipment is permitted on the routing and aircraft
Captain	Acknowledging the notification and deciding whether to accept the load
— End —	

Ground personnel are responsible for **accepting, inspecting, loading, and notifying the captain** of dangerous goods.

Captain's Authority

The dangerous goods information in this manual does not transfer regulatory compliance responsibility to the captain. It is quick reference material for use where the captain questions the acceptability of a shipment.

The captain has the authority to **reject any dangerous goods shipment** where safety of flight is in question — incorrect or incomplete paperwork, improper packaging, a quantity or weight limit exceeded, a forbidden item, or a damaged shipment.

Contact the Pilot Dangerous Goods Hotline, or Cargo Operations Control. Where Cargo Operations Control is closed, or where immediate assistance is required, contact the dispatcher.

1d.1.3 Notification to Captain (NOTOC)

Source: 49 CFR 175.33

The captain is provided written notification of every dangerous goods shipment aboard before departure. The notification is known as the **NOTOC**.

The requirement is satisfied by either:

- a computer-generated **AUTO NOTOC** form, or
- a preprinted **OK-333** form, used as the backup where the AUTO NOTOC system is unavailable

The NOTOC states the proper shipping name, class and division, number of packages and net quantity, and the loading position of each item. It must be **readily available to the captain in flight**, in the event of an incident or emergency in which the shipment becomes a safety factor.

For package totals and bin locations quickly, use the **Dangerous Goods Summary** on the NOTOC header rather than reading the item list.

Note: An AUTO NOTOC is produced for lithium batteries even where a NOTOC is not required for them, so that the crew is aware they are aboard.

Do not depart without the NOTOC where dangerous goods are aboard. See section [3d.1 Notifications](#).

Cross-Check Against the Closeout

The load closeout states restricted articles as **R/A** by compartment, forward and aft. Where the closeout shows restricted articles aboard and no NOTOC has been received, **obtain the NOTOC before takeoff** — including where the closeout arrives after pushback. See section [4f.1 Weight](#).

1d.1.4 Carriage by Crewmembers and Passengers

Items forbidden in baggage are listed in the Dangerous Goods program. A crewmember who becomes aware of a forbidden item aboard will advise the dispatcher and have the item removed before departure.

In the Flightdeck

Restricted articles permitted as carry-on baggage are permitted in the flightdeck, **with the exception of dry ice** — which is not permitted in the flightdeck in any quantity.

Dry Ice

The maximum net weight of dry ice per package, where the package is not contained in a Unit Load Device, is **440 lb (200 kg)**.

Dry ice in passenger checked baggage is stated on the load closeout as **D/I**, by compartment.

1d.1.5 Inflight Incident

Where a dangerous goods incident occurs in flight, accomplish the applicable QRC/QRH procedure, advise ATC and the dispatcher, and record the event. See section [23.5.1 NTSB Notification](#).

The QRC/QRH is the reference for an emergency involving dangerous goods. This manual states policy; it does not carry the response procedure.

Note: Dangerous goods information is passed to ATC by the **IOC Duty Director**, not by the crew. Advise the dispatcher, state that dangerous goods are aboard, and continue flying the aircraft. The crew is not expected to read the NOTOC over the radio.

Carriage Onward

Where the aircraft continues with a new crew, the NOTOC passes to the oncoming captain as part of the crew change. See section [14d.1 Crew and Documentation](#).

1p.1 General**1p.1.1 Purpose**

The purpose of Gen Ops Performance is to provide information that is general in nature for the operation of the aircraft. It supplements the flight planning system and the Takeoff Performance and Weight & Balance System. Additional performance data from the FAA-approved Airplane Flight Manual (AFM), required by 14 CFR Part 121, is contained in the performance section of the applicable AOM.

This section does not list fleet-specific performance. It is the guide to understanding and standardizing the computation and entry of takeoff data across fleets.

1p.1.2 Interpolation

Except as listed below, interpolation is required when using performance tables. Interpolation by mental computation is sufficient.

Interpolation is **not** required:

— When ranges of values are shown on the table as the input parameter.

Example: a V-speeds table shows ranges for both altitude and temperature. For those parameters interpolation is not allowed. Takeoff gross weight is not shown as a range, so gross weight must be interpolated to determine the V-speeds.

— When the resulting value is conservative.

Example: determining maximum landing weight at a field elevation of 606 feet, it is acceptable to use the value for 1,000 feet rather than interpolate from sea level, provided this does not restrict the operation.

1p.1.3 Data

Not all performance data included is certified or required by regulation. Some is manufacturer advisory or recommended data.

1p.2 TPS General**1p.2.1 Takeoff Performance Data**

The Takeoff Performance System (TPS) is the primary source of takeoff data. For most operations the TPS provides all necessary data. In some situations TPS data is used together with the takeoff performance section of the applicable AOM.

Note: A TPS is based on the Planned Takeoff Weight in the system at the time it was requested. TPS data requested at different times can differ. Both pilots will confirm they are working from the same TPS.

1p.2.2 Takeoff Performance System (TPS)

TPS provides thrust settings, flap configurations, stabilizer trim settings, V-speeds, and airport analysis data for takeoff. It interfaces with the Load Planning System (LPS) for weight and balance, and with FlightKeys for the flight plan and fuel requirement.

The term TPS also refers to the flightdeck paperwork, or departure plan. It is based on the planned takeoff weight, the planned temperature and pressure altitude estimated for departure time, and the runways in use, up to five. TPS is used together with the load closeout to ensure compliance with takeoff requirements.

TPS data is organized into four sections:

— **HEADER**

Contains general information listing the flight number, date, aircraft number, station, and the planned weights.

— **THRUST/V-SPEED DATA**

Contains the takeoff data for most takeoffs. It shows thrust and V-speeds based on the planned takeoff weight.

— **AIRPORT ANALYSIS DATA**

Contains maximum weight information for the selected runways. It may be used to verify calculations shown in the thrust/V-speed section, or used alone together with the AOM to determine takeoff data. Because it shows data for a limited temperature range only, it cannot be used to determine the assumed temperature for a flex thrust takeoff.

— **WEIGHT AND BALANCE DATA**

Contains preliminary weight and balance data. See section [1p.5.1 TPS Weight and Balance Section](#).

1p.2.3 TPS Calculation Process

TPS has two modes of operation during planning. It first determines the maximum takeoff weight and passes it to the Load Planning System. After a planned load is established, it computes takeoff data for each runway based on the planned takeoff weight and generates the output to the flight crew.

1. **MTOW calculation.** TPS first calculates the Maximum Takeoff Weight for the primary runway — the first runway listed — from the climb, runway, and structural weight limits for the ambient conditions. Enroute and landing limited weights are also checked. This MTOW is internal to the load planning system and is not shown on the TPS. It is not the same MTOW that appears on the departure plan.
1. **PTOW calculation.** A Planned Takeoff Weight is generated from aircraft EOW, payload, and fuel information available from the load planning and flight planning systems.
1. **Check PTOW against MTOW.** TPS confirms PTOW does not exceed MTOW for the primary runway — the runway being planned for takeoff based on winds and other operational conditions. If PTOW exceeds MTOW for all flap configurations on the primary runway, the load agent must act: select another primary runway, select APU ON, use available headwind, or restrict payload.
1. **Flap selection.** TPS chooses the flap configuration that achieves the maximum assumed temperature. If no configuration achieves flex thrust, the fleet standard configuration is selected unless PTOW is too heavy for it, in which case the configuration that accommodates PTOW is chosen.

The flap configuration determined for the primary runway is used for all other runways. The configuration specified by TPS does not preclude the crew from requesting or using a different one.

On a secondary runway, an asterisk next to the flap configuration indicates that another configuration would be more advantageous for that runway. Where a secondary runway carrying an asterisk is used for departure and time permits, consider requesting new TPS data for that runway.

1p.2.4 Takeoff Thrust Ratings

TOGA (Maximum Takeoff Thrust)

TOGA is the maximum takeoff thrust for the pressure altitude and temperature. It is shown for both APU OFF and APU ON.

Note: Any reference to maximum thrust means TOGA thrust. Maximum continuous thrust is less than TOGA. Limitations for both are in the limitations section of the applicable AOM.

Engine bleed air normally supplies the packs during takeoff. Where necessary to accommodate payload, MTOW can be increased by using APU bleed air instead. The TPS shows this as APU ON or APU OFF.

For a takeoff using APU bleed, select APU bleed on. The engine bleeds do not need to be turned off — priority logic shuts them off automatically when the APU is running with APU bleed selected on.

Where TPS specifies APU ON and the APU is inoperative, the MEL contains a procedure to turn the packs off for takeoff. This gives equivalent performance, because the engines then supply no bleed air to the packs during takeoff.

Thrust Bump

Where thrust bump is required, the TPS header states it, and a banner appears in the airport notes:

```
*****
**                                     **
**      USE OF BUMP REQUIRED           **
**                                     **
*****
```

Thrust bump is used at high-elevation and high-temperature stations on fleets where it is installed — PHX and other hot-and-high domestic stations, not the Latin America network. Because it is required only at certain stations, follow the procedures designed to highlight it:

1. Review the TPS carefully during preflight.
2. During the departure briefing, after threats are reviewed, the PF briefs the takeoff performance. Where thrust bump is required, highlight it at that point.
3. The captain engages thrust bump as part of the taxi flow, after both engines are started, and confirms the bump annunciation.

Flex Takeoff Thrust

Flex takeoff thrust is a certified method of lowering takeoff thrust to reduce engine wear and extend engine life. It uses an assumed temperature to compute the thrust setting. The terms *standard thrust* and *flex thrust* are used interchangeably; a system message referring to standard thrust means flex thrust.

TPS uses the Assumed Takeoff Weight to determine a maximum temperature for each combination of runway and flap configuration that will accommodate the ATOW. That maximum is the assumed temperature, labelled **AT** on the TPS, and is shown for each runway where flex thrust is authorized.

Flex takeoff thrust is **not authorized**:

- When using improved performance.
- When the runway is contaminated by standing water, slush, snow, or ice.

- Where engine anti-ice is used and the thrust/V-speed section does not indicate ANTI-ICE ON.
- Where windshear is reported or expected.
- Where takeoff weight exceeds ATOW. A new TPS may be obtained which authorizes flex thrust at the takeoff weight.
- Where an airport operations advisory page requires maximum thrust.
- With a MEL or CDL item containing a takeoff weight correction, unless the takeoff data is automatically corrected by TPAS, or the item prohibits flex thrust.
- Whenever takeoff data is used from a CRC/MEL message.

Caution: Ensure no condition exists that prohibits flex thrust even though flex thrust appears on the TPS.

1p.2.5 TPS Weight Limit Data

Takeoff climb (CLIMB) and runway (RWY) limiting weights are computed for each airport and runway by Operations Engineering, loaded into TPS as tables, and used to determine maximum weight and maximum assumed temperature.

Maximum weights are computed from airport elevation, runway length, runway slope, takeoff obstructions, and temperature. Corrections are calculated for pressure altitude and wind. Takeoff performance requirements are generally based on the worst case of an engine failure just before V1. See section [1p.7 FAR Performance](#).

Takeoff will only be made from a runway for which airport analysis information is available. Where cleared for takeoff from a longer portion of a TPS-designated runway, data for the shorter portion — the intersection — may be used.

1p.2.6 Takeoff Obstructions

The obstructions used to determine maximum takeoff weight are based on a straight-out departure, or along the flight path of a special engine-out departure procedure where one is specified. Obstructions are analyzed only to the end of the takeoff acceleration segment.

Following an engine failure after V1, obstruction clearance is assured only along the extended runway centerline, or along the special engine-out departure path where one exists.

Warning: Where it becomes necessary or desirable to turn out of the departure corridor, the pilot must assure obstruction clearance.

1p.2.7 Improved Performance

Several FAA-approved methods are used to increase takeoff weight above basic performance levels. These are referred to as improved performance and are used to accommodate payload. The primary concept is *improved climb* — optimizing the V-speeds away from the normal speeds for that runway to increase the overall maximum takeoff weight. See section [1p.7.7 Improved Performance](#).

Where improved performance is required, **MAX-IMP** appears in the AT column on the runway line in the thrust/V-speed section. Where it is authorized, the limit weight — the lighter of the computed climb and runway limits — is shown in an IMPROVED PERFORMANCE field in the airport analysis section.

Only one maximum weight is shown per runway in the improved performance portion of the airport analysis section: the more limiting of the improved performance runway and climb limits.

Caution: Improved performance data in the airport analysis section is for reference only. It may be used only where it appears in the thrust/V-speed section for that runway. The thrust/V-speed section is the only source of improved performance speeds.

Limitations on the Use of Improved Performance

- Prohibited with a MEL or CDL item containing a takeoff weight correction, unless the data is automatically corrected by TPAS or the item permits improved performance.
- Engine anti-ice may be used only where the thrust/V-speed data is based on ANTI-ICE ON.
- Takeoff with a tailwind is permitted only where the thrust/V-speed data is based on that tailwind.
- Maximum (TOGA) thrust is required.
- V-speeds from the AOM performance section are not used. Improved performance V-speeds can only be generated by TPS.
- Where takeoff weight exceeds PTOW by more than **2,000 lb** and improved performance is required, a new TPS must be obtained. Takeoff weight may not exceed MTOW in any case.

Where takeoff weight is less than PTOW, the improved performance V-speeds may be used. Alternatively, where time permits, check the airport analysis section to determine whether the takeoff weight can be accommodated without improved

performance; if it can, the corresponding V-speeds may be taken from the AOM V-speed tables.

Where the temperature is hotter than planned, the thrust/V-speed V-speeds may still be used provided:

- the temperature is not more than 2°C above the planned temperature
- takeoff weight does not exceed PTOW by more than 2,000 lb, and
- takeoff weight does not exceed the improved performance maximum takeoff weight shown in the airport analysis section

Where the current temperature exceeds planned by more than 2°C, new takeoff data must be obtained by ACARS or voice.

1p.2.8 Reserved

1p.3 TPS Description

The TPS departure plan contains four sections — header, thrust/V-speed, airport analysis data, and weight and balance data.

1p.3.1 Header

1	WBD*82/08AUG/0004 JFK									
2	STA	PRES	ALT	FLT	AIRPL	DTE/TIME				
	JFK	13		82	803	08/0004Z				
3	*** A321NY WET ***									
4	TEMP	PTOW	ATOW	ZFW	FUEL	TXI	FUEL			
	23C	188.3	190.3	145.7	43.2P	0.6				
	***** THRUST / V-SPEED *****									
	TOW CG									
	XX.X									
5	F	S	GRN	DOT						
	167	215	239							
	RWY	CONF	APU	V1	VR	V2	THR	AT	MTOW	
6	22R	02*	OFF	141	154	155	FLEX	47C	222.7S	
7	22RX	02*	OFF	141	154	155	FLEX	47C	222.7S	
8	22L	02	OFF	136	147	149	TOGA	MAX-WT	196.9T	
9	31L	01*	OFF	160	170	171	FLEX	51C	222.7S	
10	31R	01	OFF	139	153	158	FLEX	37C	207.7T	

Figure 1.12 Takeoff Performance System — Header and Thrust/V-Speed Data

1. **Message identifier, flight, date, time, and station.**
2. **Pressure altitude.** A minus sign denotes below sea level.
3. **Aircraft, engine, and runway surface condition** — dry or wet. This line includes BUMP where thrust bump is required and the TPS is predicated on it.
4. **Planned temperature (OAT)** used for the TPS calculation. It is the estimated OAT for the planned departure time. Validate it against the current surface observation, the ATIS, or local ATC.
5. **PTOW** — planned takeoff weight, in thousands of pounds.
6. **ATOW** — assumed takeoff weight, in thousands of pounds. ATOW equals PTOW plus 2,000 lb and is used to calculate the assumed temperature. The assumed temperature is valid as long as the closeout takeoff weight does not

exceed ATOW. Where maximum thrust is required for all runways, ATOW is replaced by **STD N/A**.

7. **ZFW** — zero fuel weight, in thousands of pounds, including any planned ballast fuel. $ZFW = EOW + \text{payload} + \text{ballast fuel}$.
8. **ZFW CG** — zero fuel weight center of gravity, in percent MAC.
9. **FUEL** — in thousands of pounds. A **P** suffix indicates planned fuel and excludes ballast; an **A** suffix indicates actual fuel aboard as reported by the crew.
10. **TXI FUEL** — planned taxi burn from gate departure to the start of the takeoff roll.

1p.3.2 Thrust/V-Speed

The thrust/V-speed section carries the data used for most takeoffs.

Field	Contains
MAX N1 / EPR	TOGA thrust setting. Changes only with temperature and pressure altitude. Shown for APU OFF and APU ON.
TOW CG	Takeoff center of gravity in percent MAC
STAB	Takeoff stabilizer trim in units. N/U nose up, N/D nose down.
F / S / GRN DOT	Flap retract, slat retract, and green dot speeds
RWY	Runway designator. Up to five runways. The first is the primary runway. An intersection is identified in the airport notes.
CONF	Takeoff flap configuration. An asterisk on a secondary runway indicates another configuration would be more advantageous there.
APU	Takeoff APU configuration, OFF or ON
N1 / EPR	Takeoff thrust setting — flex where flex is planned, TOGA where TOGA is planned
V1 VR V2	Takeoff V-speeds for the pressure altitude, PTOW, and assumed temperature where flex is planned, or planned temperature where TOGA is planned
AT	Assumed (flex) temperature
MTOW	Maximum takeoff weight for the conditions shown, in thousands of pounds
— End —	

Note: Where the dispatcher manually adjusts MTOW — a D suffix — the V-speeds shown are **not** automatically corrected.

Assumed Temperature Messages

Where flex thrust is not authorized for a runway, one of the following appears in the AT field.

Table 1.6 Assumed Temperature Field Messages

Message	Meaning
MAX-WT	Weight is too heavy for flex thrust, or AT is within 2°C of OAT
MAX-IMP	Improved performance is required to accommodate PTOW. V-speeds are based on improved performance and APU ON.
MAX-PNL T	The dispatcher has manually adjusted MTOW for the primary runway. APU ON is selected automatically.
MAX-SPC L	TOGA thrust required by Company policy for the runway or flap setting — usually noise abatement or another special requirement
MAX-TCH K	A TOGA thrust check is required
— End —	

MTOW Limiting Condition

The letter suffix on MTOW identifies what limits it.

Table 1.7 MTOW Limiting Condition Suffixes

Suffix	MTOW is limited by
T	Climb or runway limits
S	Structural weight limit
L	Landing weight limit — maximum landing weight at the destination plus flight plan burn
A	Climb or runway limits for APU ON, and/or headwind or tailwind
D	Dispatcher has manually adjusted MTOW for the primary runway. V-speeds are not adjusted.
E	Dispatcher has limited MTOW for an enroute consideration — driftdown, depressurization, or fuel
X	Based on improved performance. May be preempted by S or L; verify against the assumed temperature field.
— End —	

Note: A heavier maximum weight may be available by changing the flap configuration or selecting APU ON.

Runway Messages

Where TPS cannot compute takeoff data for a runway, one of the following appears in place of the data.

Table 1.8 Runway Messages

Message	Meaning and action
PTOW EXCEEDS MTOW ALL FLAPS	PTOW is too heavy for every available flap configuration. The aircraft is too heavy for takeoff on this runway.
PTOW EXCEEDS MTOW - RQST NEW TPS	PTOW is too heavy for the selected configuration. Takeoff is possible on another. Request a new TPS, or compute from the airport analysis section — see section 1p.4.8.
FLAP N/A THIS RWY - RQST NEW TPS	TPS holds no data for that configuration on that runway. Request a new TPS, or compute manually.
WIND ADJM-CALL LOADS OR COMPUTE DATA	A wind adjustment applies to the primary runway. TPS computes only the primary and parallel runways. Contact the load agent, compute manually, or request an ACARS TPS.
PERF PNLT - CALL DISP OR COMPUTE DATA	The dispatcher has manually adjusted MTOW, usually for a contaminated runway. Flap configuration and APU ON are locked. Contact the dispatcher or compute manually.
— End —	

A parallel runway is one whose designation differs by one — where the primary is 17, runways 16 and 18 are parallel, including all left, center, right, and intersection variants.

Airport Notes

The airport notes contain remarks on the TPS data: the taxiway from which each intersection departure is measured, a reminder where a special departure procedure exists, the USE OF BUMP REQUIRED banner where applicable, and any SIGMET entered by the dispatcher.

Warning: Do not depart from an intersection for which no TPS line exists. An intersection takeoff computed from the full length is an unverified accelerate-stop distance.

1p.3.3 Airport Analysis Data

***** AIRPORT ANALYSIS DATA *****						
1	STRUCT WT LIMIT 222.7					
	CONF	APU	CLIMB LIMIT	TEMP C	RWY 31L	RWY 31R
2		OFF	207.7	23	222.7	207.7
3	01	OFF	207.4	25	222.4	207.4
4		ON	212.2	23	222.7	212.2
		ON	211.9	25	222.4	211.9
5	HDWND ADD / KT				410	368
6	TLWND SUB / KT				2030	1927
7	E/O ACCEL /AFL/ FT				1000	1000
	/MSL/ FT				1020	1020
8	PLANNED WIND KT				T1	T1
9	LENGTH - FT				14511	10000
10	SLOPE - PCT				.0	.0

Figure 1.13 Takeoff Performance System — Airport Analysis Data

1. **STRUCT WT LIMIT** — maximum structural takeoff weight.
2. **CONF** — flap configuration. All configurations analyzed for each runway in the thrust/V-speed section are shown, in ascending order, unless Dispatch has constrained the configuration, in which case that one is displayed first.
3. **APU** — APU configuration, OFF or ON.
4. **CLIMB LIMIT** — climb limited takeoff weight in thousands of pounds, based on TOGA thrust.
5. **TEMP C** — planned OAT. Planned temperature plus 2°C is also displayed, to allow for a last-minute temperature increase.

6. **RWY** — runway/obstacle limited takeoff weight in thousands of pounds, based on TOGA thrust. All zeros indicate the data for that runway and configuration has not been loaded into TPS.
7. **HDWND ADD / KT** — headwind correction in pounds per knot, applied to the runway limit.
8. **TLWND SUB / KT** — tailwind correction in pounds per knot. N/A means takeoff with a tailwind is not authorized on that runway. A correction of zero means the maximum tailwind component is authorized with no weight adjustment.
9. **E/O ACCEL** — engine-out acceleration height in feet, above field level and above mean sea level.
10. **IMPROVED PERFORMANCE** — improved performance maximum weight data. Only runway limit data is shown, as it contains both the runway/obstacle and climb limits. For reference only. PLANNED WIND shows the head or tailwind component, prefixed H or T.
11. **LENGTH - FT** — runway length available for takeoff.
12. **SLOPE - PCT** — runway slope. Negative is downhill, positive uphill.

Reading the Line

Compare the runway MTOW against the actual takeoff weight from the load closeout. Where the actual weight exceeds the MTOW shown, that runway is not available at that weight, regardless of the speeds printed alongside it.

Caution: V-speeds and thrust on a TPS line are computed for the assumed takeoff weight shown in the header, not for the actual weight. Where the actual weight differs by more than the fleet tolerance, request a new TPS.

1p.3.4 TPS Engine Anti-Ice Corrections

Where engine anti-ice is used for takeoff there is no reduction to the thrust setting, but there is a small reduction in takeoff thrust from the additional bleed air, and a corresponding small reduction in runway and climb limited weights. Wing anti-ice is not used for takeoff.

Temperature Colder than 15°C

The airport analysis section shows engine anti-ice corrections for the climb and runway limits. The climb and runway limit data itself is **not** corrected by TPS:

```

-----
LENGTH - FT      12000 12000
SLOPE - PCT       .0    -.1
-----
A/I ON SUB FROM  CLB   .9 RWY   .9

```

END

Temperature 10°C or Colder with Anti-Ice Selected On

Where the dispatcher has selected engine anti-ice on in TPS:

- the thrust/V-speed section indicates **ANTI-ICE ON**, and all data in that section includes the anti-ice corrections
- the airport analysis section shows the corrections as above, and its climb and runway limit data remains uncorrected

Where the TPS indicates ANTI-ICE ON and engine anti-ice is not used for takeoff, the TPS may still be used. Where maximum weight is desired, request a new TPS.

Flex Thrust with Anti-Ice

Where the thrust/V-speed section indicates ANTI-ICE ON, flex thrust may be used with engine anti-ice on. Otherwise use the airport analysis section to compute the climb and runway limits including the anti-ice corrections, and use TOGA thrust.

Caution: Flex thrust may not be used with engine anti-ice on unless the thrust/V-speed section indicates ANTI-ICE ON.

1p.3.5 Improved Performance on the TPS

Where improved performance data is available for a runway, the airport analysis section shows the improved performance maximum weights. The thrust/V-speed section shows improved performance data only where PTOW requires it.

Improved performance uses the same engine-out acceleration height as normal performance for that flap configuration.

Where a planned wind component applies to the primary runway, the improved performance weights for the primary and any parallel runway are based on that wind, and the wind is shown below the improved performance weights, prefixed H or T.

1p.3.6 ACARS TPS

A TPS may be obtained through an ACARS takeoff data request.

Note: Entering a flap configuration of 99 allows TPS to search for a configuration that permits flex thrust where one exists.

The ACARS TPS format is similar to the printed TPS except:

- data and airport notes are provided for a single runway only
- where a head or tailwind applies, weights and speeds are adjusted for the primary runway or a parallel runway on the original TPS; any other runway is not adjusted for wind unless the request includes it

- printer width requires additional lines for runway data and airport notes. Where maximum thrust is required, the maximum thrust message replaces the AT value on the second line.

ACARS TPS Messages

UNABLE TO COMPUTE TPS FOR FLAPS XX TEMP XX RWY XXXX is returned where:

- takeoff weight is too heavy for the flap configuration, runway, and temperature combination requested — try a different configuration, request a new ACARS TPS, or compute manually per section 1p.4.8
- the flap configuration requested is not available for that runway
- Dispatch has adjusted MTOW for a MEL, CDL, or contaminated runway condition. Only the configuration and runway shown on the original TPS are available; all others are locked out. Contact Dispatch where necessary.

FMS Uplink Verification

Verification that data has been uplinked to the FMS is provided by a printer message stating **FMC UPLINKED DATA COMPUTED AT XX C OAT**, the OAT being the planned temperature at initialization. The same message prints when new data is uplinked following a flightdeck request.

Uplinked Data

Where uplinked assumed temperature and V-speeds change from the original TPS due to a change of weight, that data may be used without requesting a new TPS, provided the quick reference guide does not require one.

1p.4 TPS Variations

1p.4.1 TPS Quick Reference Guide

Where a weight, temperature, or wind variation exists, use the applicable guide below. Where a combination exists, check **weight**, then **temperature**, then **wind**, in that order.

Do not exceed the structural or airport analysis limited weights. Maximum thrust may be required by a limitation — see section 1p.4.10.

Warning: Ensure the load closeout is received before takeoff. Every check below is made against closeout figures, not planned figures.

Weight Variation

Table 1.9 Weight Variation

TPS plans	Actual takeoff weight	Action
Flex thrust	Lighter than or equal to ATOW and MTOW	Use flex thrust. Use uplinked V-speeds or the thrust/V-speed section.
Flex thrust	Heavier than ATOW or MTOW	Request a new TPS, or apply the rules below.
Maximum thrust	Lighter than or equal to PTOW	Use TOGA thrust. Use uplinked V-speeds or the thrust/V-speed section.
Maximum thrust	Heavier than PTOW	Apply the rules below.
— End —		

Where a new TPS is not obtained:

- Where maximum thrust is planned, including improved performance, and takeoff weight does not exceed PTOW by more than 2,000 lb and is lighter than MTOW — use the thrust/V-speed data for that runway.
- Where takeoff weight exceeds ATOW and is lighter than MTOW, and improved performance does not apply — use TOGA thrust with V-speeds from the AOM takeoff data section.
- Where takeoff weight is heavier than the MTOW in the thrust/V-speed section — contact Dispatch where the TPS is enroute (E), landing (L), or dispatch (D) limited. Otherwise check the airport analysis section for a flap, APU, and temperature combination, corrected for wind and anti-ice, that allows the takeoff:
 - accommodated without improved performance — use TOGA thrust and AOM V-speeds
 - accommodated with improved performance — request a new TPS
 - not accommodated — contact Dispatch

Temperature Variation

Table 1.10 Temperature Variation

TPS plans	Current temperature	Action
Flex thrust	Colder than or equal to AT	Takeoff authorized using flex thrust. Use uplinked V-speeds or the thrust/V-speed section.
Flex thrust	Warmer than AT	Request new TPS data.

TPS plans	Current temperature	Action
Maximum thrust	Colder than or equal to planned temperature	Takeoff authorized using TOGA thrust. Use uplinked V-speeds or the thrust/V-speed section.
Maximum thrust	Warmer than planned by 2°C or less	Request a new TPS, or check the airport analysis section for a combination that allows the takeoff.
Maximum thrust	Warmer than planned by more than 2°C	Request new TPS data.
— End —		

Where improved performance is planned, the thrust/V-speed V-speeds may be used provided takeoff weight does not exceed PTOW by more than 2,000 lb and does not exceed the improved performance maximum takeoff weight in the airport analysis section.

Wind Variation

Where **WIND ADJM-CALL LOADS OR COMPUTE DATA** appears for the takeoff runway, request a TPS with the appropriate wind component, contact the load planner or dispatcher, or compute manually per section 1p.4.8.

Table 1.11 Wind Variation

TPS shows	Actual wind	Action
No wind, or a headwind less than actual	Headwind greater than or equal to the TPS wind	Takeoff authorized using the thrust/V-speed data
A headwind greater than actual	Headwind less than the TPS wind	Request a TPS with the updated headwind, or compute manually
No wind or a headwind	Tailwind	Request a TPS with the updated tailwind, or follow section 1p.4.7 Unplanned Tailwind Takeoff
A tailwind greater than or equal to actual	Tailwind less than or equal to the TPS wind	Takeoff authorized using the thrust/V-speed data
A tailwind less than actual	Tailwind greater than the TPS wind	Request a TPS with the updated tailwind, or compute manually
— End —		

A TPS with updated wind is requested by ACARS, or through Dispatch and load planning — the latter is required where improved performance applies.

1p.4.2 MEL and CDL Adjustments

Some MEL and CDL items require adjustment to the takeoff performance data. These are handled in one of three ways:

- the Takeoff Performance Adjustment System (TPAS)
- the Contaminated Runway Correction/MEL (CRC/MEL) program
- manual calculation

TPAS is used normally. Items with large adjustments require CRC/MEL. Manual calculation is the backup.

1p.4.3 Takeoff Performance Adjustment System (TPAS)

TPAS automatically adjusts takeoff data for MEL and CDL items entered by the dispatcher. This affects only how takeoff performance is computed. It does not change any other MEL or CDL procedure, and does not cover enroute or landing penalties.

Where the dispatcher enters an item in TPAS:

- all data for all runways on the TPS — thrust, weights, speeds, assumed temperature, and airport analysis — is adjusted
- maximum thrust is required only where the item specifically requires it. In that case TPS plans TOGA and shows MAX-PNLT or MAX-WT in the AT column.
- where maximum thrust is not required by the item but is needed to accommodate the planned takeoff weight, MAX-WT is shown
- normal thrust rating, flap configuration, and APU logic apply
- where more than one item is listed, the corrections are cumulative

A banner across the top of the TPS states **PERFORMANCE ADJUSTMENTS MADE BY TPAS**, and a **TPS PERFORMANCE ADJUSTMENTS** block below the header lists the items used and any dispatcher remarks. Only items carrying a takeoff performance correction are listed. This applies equally to a TPS obtained by ACARS.

Caution: A D suffix on MTOW means the dispatcher computed the maximum takeoff weight manually and the V-speeds are **not** adjusted. The crew makes any required V-speed adjustment. See section 1p.4.4.

1p.4.4 Contaminated Runway and Manual Performance Adjustment

The dispatcher can compute MTOW manually and enter it in TPS, shown by a D suffix. This is normally done for a contaminated runway, because TPS does not

automatically account for contamination, but may also be used to restrict MTOW for another reason.

The dispatcher cannot adjust V-speeds where MTOW is entered manually. The crew is responsible for any adjustment required. For a contaminated runway the dispatcher can provide V-speeds on the contaminated runway correction message; those speeds do not appear on the TPS.

A manually adjusted MTOW is valid only for the primary runway and for the conditions the dispatcher used. TPS therefore locks the thrust rating, flap configuration, and APU ON:

- the thrust/V-speed section displays the primary runway only. All other runways show **PERF PNLT - CALL DISP OR COMPUTE DATA**. Using another runway or configuration requires manual adjustment of the MTOW and V-speeds from the airport analysis section and the AOM V-speed tables.
- flap configuration is locked; the TPS configuration logic is bypassed
- maximum thrust is required, and MAX-PNLT is displayed for the primary runway
- TPS defaults to APU ON. APU OFF may be used where the adjusted MTOW is within the APU OFF runway and climb limits — see section 1p.4.8.
- ACARS takeoff data requests are limited to the primary runway and configuration. Any other request returns UNABLE TO COMPUTE TPS.

1p.4.5 Contaminated Runway Correction / MEL Program

To optimize takeoff data for MEL and CDL items carrying large performance penalties, the CRC/MEL program computes maximum takeoff weight and V-speeds. It handles MEL items, contaminated runways, or both.

It is used only for identified items. The MEL or CDL states in the operational procedure that performance weight adjustments are not provided through TPS and that Dispatch will provide a CRC/MEL correction message.

The program sits outside TPS and cannot update the TPS printout. The dispatcher runs it and provides corrected runway data for up to five runways, sent to a station printer or to the aircraft by ACARS. **V-speeds on a CRC/MEL message do not uplink to the FMS.**

The program uses the same logic as the TPS improved performance calculation, so the resulting V-speeds may be faster or slower than normal.

- Use maximum (TOGA) thrust and the V-speeds from the CRC/MEL message.

- The dispatcher normally selects APU OFF. Request APU ON where necessary to maximize weight.
- Where actual takeoff weight exceeds PTOW by more than 2,000 lb, or the temperature increases, request new data from Dispatch.

Caution: Actual takeoff weight cannot exceed the MTOW for the runway shown on the CRC/MEL message. The MTOW on that message does **not** observe a landing-limited MTOW — where the TPS shows an L suffix, that limit must also be observed.

CRC/MEL Message Fields

Field	Contains
Title, date, time	Message identification and time of calculation
FLT / NOSE	Flight number, date, departure station, and aircraft number
APU / ANTIICE	APU and engine anti-ice configuration assumed
TEMP / WIND / QNH	Temperature, wind component, and the QNH used to correct to pressure altitude
PTOW	Planned takeoff weight
CONTAM	Type and depth of contamination, in inches
MAX CROSSWIND	Maximum crosswind for the runway surface condition
MAX EPR / N1	Maximum thrust data. The same for all runways in the conditions specified.
RWY / CONF	Runway designator and flap configuration
V1 VR V2	V-speeds for the planned takeoff weight
MTOW	Maximum allowable takeoff weight for the runway. Does not consider a landing-limited MTOW.
EO_ACCEL	Engine-out acceleration height, AFL and MSL
OVER MTOW	PTOW is heavier than the MTOW for that runway and configuration
NOT AUTHD	The computation failed for that runway and configuration
— End —	

1p.4.6 Wind Adjustments

The load agent may issue a TPS with MTOW adjusted for a planned tailwind, or for up to half of the reported headwind. The captain, in coordination with Dispatch, may authorize the load planner to use the full headwind component as determined by the captain.

Where a wind adjustment is made:

- the thrust/V-speed section displays the primary runway and any parallel runway. All others show **WIND ADJM-CALL LOADS OR COMPUTE DATA**.
- an ACARS request for any runway other than the primary or a parallel runway will not include the wind adjustment unless the request states it
- the airport analysis runway limits remain based on zero wind. Apply the headwind or tailwind correction to adjust them.

1p.4.7 Unplanned Tailwind Takeoff

Where taking off with a tailwind and the TPS is based on a headwind or zero wind, use one of the following. Where improved performance is required, method 1 is the only option.

Method 1. Request a TPS with the updated tailwind by ACARS, or through Dispatch and load planning.

Method 2 — manual tailwind takeoff data.

- Check the airport analysis data to confirm the takeoff weight is not heavier than the maximum allowable. Multiply the tailwind correction by the tailwind component and subtract the result from the zero wind maximum weight for the runway.
- Use TOGA thrust.
- Use the uplinked TOGA V-speeds and correct V1 for the tailwind, or look up V-speeds in the AOM takeoff data section using the actual temperature and correct V1 for the tailwind.

The V1 tailwind correction is published by fleet in the applicable AOM. It is repeated here for the A320 family:

Table 1.12 V1 Correction for Tailwind — A320 Family

Aircraft	Engine	Sharklet	V1 correction for tailwind
A319	CFM	Yes	No correction
A319	CFM	No	Subtract 1 kt per 10 kt of tailwind
A319	IAE	No	Subtract 1 kt per 10 kt of tailwind
A320	CFM	No	Subtract 1 kt per 10 kt of tailwind
A320	IAE	No	Subtract 1 kt per 10 kt of tailwind
A321	IAE	Yes	No correction
A321	CFM56	No	Subtract 1 kt per 10 kt of tailwind
A321	IAE	No	Subtract 1 kt per 5 kt of tailwind
A321neo	LEAP-1A	Yes	Subtract 1 kt per 5 kt of tailwind

Aircraft	Engine	Sharklet	V1 correction for tailwind
— End —			

For every other fleet, use the correction published in the applicable AOM.

1p.4.8 Manual Takeoff Calculation

TPS locks a runway, flap configuration, and APU ON where performance adjustments have been made by the dispatcher or load agent. It may be desirable to compute takeoff performance for a different runway, configuration, APU OFF, or wind component.

The crew may use the airport analysis section together with the weight adjustments, thrust, and V-speeds in the AOM takeoff performance section and the MEL or CDL as applicable:

- determine the maximum weight from the climb and runway limits and apply any applicable corrections
- look up the V-speeds in the AOM takeoff performance section
- use maximum (TOGA) thrust. Where the TPS temperature is valid, use the TOGA thrust setting from the TPS; otherwise use the FMS.
- an assumed temperature cannot be determined, so flex thrust cannot be used

Note: Manual takeoff calculation cannot be used with improved performance, because the corresponding V-speeds cannot be determined. Where the actual takeoff weight requires improved performance, request a new TPS.

1p.4.9 Temporary Airport Analysis

Where a situation temporarily reduces the runway/obstacle limited takeoff weights — a partial runway closure, an obstruction near the extended centerline — a Temporary Airport Analysis is issued for the affected runway.

Where a TAA is in effect, the TPS states the affected runway, the TAA number, the date of issue, and the conditions. Limit weights for that runway are adjusted automatically. **No further adjustment is required.**

```
...RWY 28L DATA BASED ON TAA 26-081...
*****
JFK TMPRY ARPT ANALYS NR <26-081 PT 1 OF 1 ISSUED <07/16/26
WITH - RWY 10R/28L E 690 FT CLSD
THE FOLLOWING APPLIES -
RWY 10R/28L 10810 FEET AVBL FOR TAKEOFF AND LANDING
CHECK NOTAMS/FIELD CONDITIONS FOR APPLICABILITY OF TAA
***STATION FIELD REPORT MUST INDICATE THIS TAA WHEN IN EFFECT***
*****
```

Where a runway in use has been partially closed and no TAA has been issued for it, notify Dispatch as soon as possible.

1p.4.10 Maximum Thrust Required by a Limitation

Limitations Without a Takeoff Weight Penalty

- Windshear is reported or expected, including a windshear advisory.
- An airport operations advisory page requires maximum thrust.

Where the takeoff weight does not exceed ATOW and the actual temperature does not exceed the AT, no further check of maximum weight is necessary. Use TOGA thrust and the TPS V-speeds.

Limitations With a Takeoff Weight Penalty

- The runway is contaminated by standing water, slush, snow, or ice.
- Takeoff weight exceeds ATOW and a new TPS authorizing flex thrust at that weight has not been obtained.
- A MEL or CDL item contains a takeoff weight correction, unless corrected automatically by TPAS or the item prohibits flex thrust.
- Takeoff data is used from a CRC/MEL message.

The takeoff weight must not exceed the maximum takeoff weight from the airport analysis data incorporating the applicable penalties. Use TOGA thrust and maximum thrust V-speeds. Where the actual temperature is hotter than planned by more than 2°C and a weight penalty applies, new TPS data is required.

1p.4.11 Weight and/or Temperature Differs from Plan

Flex Thrust Planned

The flex thrust setting is generated from ATOW. Where TPS calls for flex thrust, check takeoff weight against **both** ATOW and MTOW, and check actual temperature against the AT.

Maximum Thrust Planned

TOGA thrust from TPS is based on planned temperature. Manual adjustment of the thrust setting is not required, because the FMS adjusts TOGA thrust for the actual temperature. However, the runway and climb limit weights must be rechecked as temperature rises, and takeoff weight must be checked against both MTOW and PTOW, because V-speeds are based on PTOW.

1p.4.12 Reserved

1p.5 Loading

The Load Planning System, within DECS, is the primary system for computing weight and balance and for confirming the aircraft is within weight and CG limits. It works with TPS and the flight planning system to coordinate weight and fuel loading.

The primary output to the flight crew is:

- the TPS weight and balance data
- the load closeout

Who the Load Agent Is

The **load agent**, also called the load planner, works in the Integrated Operations Center. The load agent builds the load plan, distributes the payload across the compartments to keep the aircraft within its weight and CG limits, issues the loading instruction, and produces the load closeout.

The load agent is **not** a ramp agent and is not at the aircraft. Ramp personnel load the aircraft against the instruction the load agent issues.

Where the crew needs a load plan changed — a wind adjustment, a different primary runway, a payload restriction — the request goes to the load agent, not to the ramp.

See section [1c.6 Ramp](#).

1p.5.1 TPS Weight and Balance Section

These are **planning** figures. They are not the load closeout. See section [1p.5.2 Load Closeout](#).

***** WEIGHT AND BALANCE DATA *****											
-----LOAD-----TOTALS-----LIMITS-----CMPT MAX--AS LDED--											
1	EOW	109746	ZFW	145696	MZFW	166669		F1	9800		2900
2	PSGR WT	30580	FUEL	43242P	***	STD	***	A1	10200		2500
3	CGO WT	5400	RMP	188938	MRMP	222666					
4	BALLAST	0	TXI	596							
5			TOW	188342	MTOW	222666					
6			CNFIG	F 20	C 0	W 0	Y 176	CAP	196		
			PSGRS	F 20	C 0	W 0	Y 119	TOT	139		
7	CRT ADDRESS	L036	AGENT	LATOYA NÚÑEZ			PHONE	614-454-5685			

Figure 1.14 TPS Weight and Balance Data

1. **Weight breakdown of the planned load, in pounds.** EOW is the empty operating weight — aircraft empty weight plus operating items, including operating crewmembers on the crew manifest with bags, buffet equipment, cargo containers, cabin equipment, residual fuel, engine oil, potable water, and

chemical fluids. Deadheading crewmembers are not included. PSGR WT is passenger weight, including carry-on and infants under two, and flightdeck (W) and cabin (X) jumpseat riders. CGO WT is checked baggage, mail, and freight combined. BALLAST is the weight of ballast fuel.

2. **Weight and fuel totals.** ZFW is EOW plus payload. FUEL is total fuel — a **P** suffix is planned fuel excluding ballast, an **A** suffix is actual fuel aboard as reported by the crew.
3. **Limits and fuel distribution.** MZFW is maximum zero fuel weight. The distribution code states how the fuel is loaded.
4. **Ramp and taxi.** RMP is ramp weight against MRMP, its maximum. TXI is the planned taxi burn.
5. **Takeoff weight.** TOW equals RMP less TXI, shown against MTOW.
6. **Configuration and passengers.** CNFIG states the seating configuration by class with total capacity. PSGRS states the passengers planned in each class.
7. **Load agent.** Printer address, name, and telephone number.

Table 1.13 Fuel Distribution Codes

Code	Meaning
STD	Standard distribution — wing tanks filled first, then center
005 – 010	Fuel in the center tank, to the nearest 500 lb, where the wing tanks are not full
WTO	Wing tanks only
MCX	No fuel in the center tank
MEL	CG limits curtailed against a second pump failure
NAT	No fuel in the additional center tanks
MCA	No fuel in the second additional center tank
— End —	

Where a code other than the above appears, contact the load agent.

Compartment Limits

CMPT MAX states the weight limit for each cargo compartment. AS LDED states the weight planned into it. No AS LDED figure may exceed its CMPT MAX.

Note: An operating exception crewmember is not reflected as a W rider even where they occupy the flightdeck jumpseat.

Verification

Verify against the TPS, then verify the load closeout against these figures once it is issued:

- ZFW, TOW, and ramp weight are each below their stated maximum
- no compartment AS LDED figure exceeds its CMPT MAX
- the PTOW here matches the weight the thrust and V-speeds were computed for

Caution: This block states structural maximums. It does not state the performance-limited weight for the runway. A weight within every structural limit here may still exceed the runway MTOW on the TPS line.

1p.5.2 Load Closeout

The load closeout is required **before takeoff**. Pushback and taxi may commence without it; the closeout is normally received during taxi. It may be printed, displayed electronically, or read over a voice radio call. **In all cases both the flight number and the aircraft number must be provided.**

For the crew procedure on receipt, the brief, and the independent verification, see section [4f.1 Weight](#).

Table 1.14 Load Closeout Fields

Field	Contains
RVSN / time	Closeout revision number and time of calculation
Flight / route / nose	Flight number, city pair, and aircraft number prefixed N
TOW	Takeoff weight = ZFW + FOB
FOB	Fuel on board — actual (A) or planned (P) total fuel, less planned taxi burn
ZFW	Zero fuel weight = EOW + payload. Excludes ballast fuel. Use this value to initialize onboard systems.
STAB	Stabilizer trim setting for takeoff, in units. N/U nose up, N/D nose down.
L/A	Location and count of live animals in the hold, forward and aft
R/A	Whether restricted articles are loaded, forward and aft
W/B	Location and number of wheelchair batteries
D/I	Location and number of checked bags containing dry ice
ZFW CG	Zero fuel weight center of gravity, percent MAC
TOW CG	Takeoff center of gravity, percent MAC
PSGR	Total passenger count — adults and children, excluding infants — with W and X riders

Field	Contains
LAP	Infants not occupying a seat
CREW	Crewmembers, flightdeck and cabin
TSOB	Total souls on board — passengers, W and X riders, infants, and crew
CHILD	Child count, also included in the passenger total
PSGR WGT	Total passenger weight
CGO WGT	Total cargo weight
EOW	Empty operating weight
SECOK	Security check complete
— End —	

Note: Where a fuel figure is entered that differs from planned by 4,000 lb or more, an ACARS message is generated. If the figure is re-entered it is accepted even where it is in error. The message exists to prompt the load agent to verify the fuel load.

Crew Verification

Both pilots verify the closeout against the aircraft limits and against the TPS:

- ZFW, TOW, and CG are within limits
- the stabilizer trim setting matches what is set
- TOW does not exceed the MTOW for the runway in use
- TOW is within tolerance of the weight the TPS assumed
- the passenger count matches the flight attendant report
- TSOB is correct — this is the figure passed to ATC in an emergency

Warning: Do not depart on figures that have not been verified against the aircraft limits. A closeout that does not match the aircraft is not a paperwork error — it is an unverified center of gravity.

1p.5.3 Manual System

Where the flight planning system is unavailable, complete ACARS messages cannot be sent. For the description and procedures see section [26.3 FOS Outage Backup Plan](#).

1p.5.4 CG Restrictions with a Low Passenger Count

Certain seating distributions with a low passenger count can drive the takeoff or zero fuel weight CG out of limits. Where this occurs the load planner applies an exception code restricting seating in certain coach rows — either the first four or the first eight.

The load planner will:

- turn off the automatic radio closeout
- instruct the gate agent which rows must remain empty
- advise the flight crew by free text appended to the load closeout, in the form:

DUE TO TOW AND/OR ZFW CG LIMITATIONS, THE FIRST (4 or 8) ROWS OF COACH MUST REMAIN EMPTY FOR THE DURATION OF THE FLIGHT.

Confirm the flight attendant in charge is aware of the restriction, and that it applies for **takeoff, enroute, and landing**.

1p.6 Landing Performance

A landing performance calculation is completed for every flight automatically, based on planned conditions. It does not account for conditions existing at the time of arrival, including changes in weather, runway condition, or aircraft weight. Results are not visible to the flight crew; the dispatcher is alerted to invalid performance.

The flight crew determines whether conditions at the time of arrival remain within the standard assessment criteria used for that calculation. Where any criterion is exceeded, the crew independently calculates landing performance using the applicable landing performance tool before landing.

Standard assessment criteria vary by fleet and are listed on the reverse of the normal procedures checklist and in the applicable AOM.

Landing performance is computed on the **Land App**, on all fleets. Where the Land App is unavailable, use the **ACARS Landing Report**.

Takeoff performance is not computed on the EFB. See section [1p.2.2 Takeoff Performance System \(TPS\)](#).

1p.7 FAR Performance

This section states the regulatory basis for the takeoff and landing performance the TPS computes. Fleet-specific data is in the applicable AOM.

1p.7.1 Introduction

Transport category performance is certified under 14 CFR Part 25 and operated under 14 CFR Part 121 Subpart I. The certified data assumes a trained crew flying the aircraft to the AFM procedure, in the configuration analyzed, on a surface in the condition assumed.

1p.7.2 Takeoff Performance

Takeoff weight is limited by the most restrictive of:

Limit	Requirement
Field length	The runway must accommodate the longest of the all-engine takeoff distance factored by 1.15, the accelerate-go distance, and the accelerate-stop distance
Climb	The aircraft must meet the certified one-engine-inoperative climb gradients in each takeoff segment
Obstacle	The net takeoff flight path must clear each obstacle in the departure area by 35 feet vertically
Tire and brake	Tire speed and brake energy limits must not be exceeded
Structural	Maximum structural takeoff weight
Landing	Maximum landing weight at the destination plus the planned burn
— End —	

1p.7.3 Takeoff Thrust

TOGA thrust is the certified maximum. Flex thrust reduces it using an assumed temperature, and is certified provided the aircraft still meets every applicable limit at the assumed temperature. Flex is not a performance reduction — the aircraft meets the same regulatory margins at the reduced thrust.

1p.7.4 Rejected Takeoff

The accelerate-stop distance assumes a reject initiated at V1 with maximum manual braking, spoilers, and reverse thrust where credited. V1 is a decision speed, not a decision point — the decision to reject is made before V1.

See section [5g.4 Rejected Takeoff](#).

1p.7.5 Obstacle Clearance

Obstacle clearance is assured along the extended runway centerline, or along a special engine-out departure procedure where one is published. It is not assured anywhere else. See section [1p.2.6 Takeoff Obstructions](#).

1p.7.6 High Altitude Takeoffs

At high elevation the climb limit becomes the governing limit more often than the field length limit, and the margin between V1 and VR narrows. Engine-out acceleration height is frequently raised above the standard 1,000 feet AFL to clear terrain, which lengthens the period at takeoff thrust.

1p.7.7 Improved Performance

Improved climb trades excess field length for climb capability by increasing V2 above the minimum. The higher V2 improves the one-engine-inoperative climb gradient, which raises the climb limited weight, at the cost of a longer takeoff run. It is used where the runway is long relative to the climb requirement.

See section [1p.2.7 Improved Performance for the operational limitations](#).

1p.7.8 Takeoffs on Wet and Contaminated Runways

A wet runway increases the accelerate-stop distance and reduces the runway limited weight. Contamination by standing water, slush, snow, or ice adds displacement and impingement drag and reduces braking further.

TPS computes wet runway data. It does **not** compute contaminated runway data — that is provided by the dispatcher through the CRC/MEL message. See section 1p.4.5.

Warning: Flex thrust is not authorized on a contaminated runway.

1p.7.9 Enroute

Enroute performance is limited by the net level-off altitude following an engine failure, and by the driftdown profile where terrain is a factor. See section [2.6 Engine-Out Driftdown](#).

1p.7.10 Approach and Landing

Landing weight is limited by the most restrictive of the structural maximum, the landing field length requirement, the approach climb limit, and the landing climb limit. The landing distance available must be at least the certified landing distance factored to the applicable requirement for the runway condition.

Chapter 2: Flight Plan/Dispatch Release

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- [2.3 Computer Flight Plan](#)
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- [2.5 Alternates](#)
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- [2.7 Weight and Balance](#)
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2.1 General

Source: 14 CFR 121.593, 121.597, 121.687

No flight will originate without a dispatch release signed by the dispatcher and accepted by the captain. The release, the computer flight plan, and the weather package together constitute the flight paperwork.

Both the captain and the dispatcher exercise operational control. See section [20.4 Operational Control](#).

Flight Numbering

A flight number is stated without leading zeros. A **stub** flight number — a segment operating under a parent flight number with a suffix — is stated in full, including the suffix, on every transmission and in every log entry.

Currency

Review the flight paperwork for currency before acceptance. Paperwork more than **1 hour** old at the time of departure requires an updated weather package.

Airports Without an Operating Control Tower

An airport may have no control tower, or a tower that is closed at the time of the operation. In either case there is **no ATC clearance issued on the ground and no traffic separation provided**.

Dispatch

A flight may be released to or from an airport without an operating control tower where:

- the airport is authorized in OpSpecs for the operation
- weather at the airport is at or above the minimums applicable to the approach to be flown, with no tower in operation
- a means of obtaining the local altimeter setting is available — an AWOS, ASOS, or a nearby reporting station within the distance published for the approach

Where the altimeter source is unavailable, the approach minimums are raised or the approach is not authorized. Confirm with the dispatcher before departure.

Departure

Obtain the IFR clearance by telephone or by radio from the controlling facility before taxi. State the expected departure time and comply with any void time issued.

Broadcast intentions on the common traffic advisory frequency throughout taxi, takeoff, and departure. Both pilots maintain a visual watch for traffic, vehicles, and animals on the movement area — nobody is checking the runway for you.

Arrival

Broadcast position and intentions on the CTAF from the initial approach fix onward. Do not assume the frequency is monitored by other traffic.

Cancel IFR only after landing, or where positively clear of terrain and traffic and in visual conditions. Cancelling early releases the airspace to other traffic.

Warning: At an airport without an operating control tower, a landing clearance does not exist. The runway is confirmed clear visually before landing, and the crew is responsible for separation from all other traffic.

Runway Condition

No field condition report is issued when the tower is closed. Where the surface condition cannot be established, plan on the most adverse condition reasonably expected and confirm the landing distance supports it. See section [10g.2 Contaminated Runway](#).

2.2 Dispatch Release

2.2.1 Structure of the Release Package

The release is issued as one document. Every page carries the same header — page number, release version, issue time in UTC, city pair, and flight number — and the same document identifier in the footer. A page bearing a different release version or document identifier belongs to a different release and is not used.

Table 2.1 Release Package Contents

Block	Contains	Paragraph
Flight summary	Type of operation, desk, fuel, route, on-time analysis, weights, driftdown	2.2.2
Fuel summary	Each component of required fuel, in pounds, with time and distance	2.2.3
On time analysis, weights, and driftdown	Scheduled versus planned times, weights, wind, and driftdown data	2.2.4
Equipment and remarks	MEL, CDL, NEF, SEL, restrictions, remarks	2.2.5
Filed flight plan	The ICAO flight plan as filed	2.2.6
Navigation log	Waypoint-by-waypoint course, wind, speed, distance, time, and burn	2.2.7
Time and fuel log	Estimated time and fuel remaining at each waypoint	2.2.8
Signature block	Dispatcher, crew, approach capability, captain's acceptance	2.2.9
Enroute winds	Temperature deviation and wind by waypoint at four flight levels	2.2.10
Track message	Oceanic organized track system, where the route uses it	2.2.11
Station field report	Departure station runway, taxiway, and facility status	2.2.12
Weather	ATIS, TAF, and METAR for each airport, and enroute advisories	2.2.13
NOTAMs	Airport and FIR NOTAMs, grouped by category	2.2.14
Charts	Route, significant weather, upper air data, vertical profile	2.2.15
Takeoff performance	The TPS for the departure station	2.2.16
— End —		

Review every block before accepting. A block that is missing, or that reads as empty where it should read NONE, is a transmission fault — request a retransmission rather than departing on a partial release.

2.2.2 Flight Summary

The first page states the flight and the plan in summary.

	PAGE 1	RELEASE 1.0	2026-AUG-08 00:04:01Z	KJFK-EGPH	NAC82
1	FLAG		DESK: FD01	PHONE: 800-	
2	- IFR	NAC82/7	803/N803NY	JFK	EDI ALTN GLA NCL
3	MIN T/O FUEL	042632	RLS FUEL	043242	
4	TOT BRN	35089	PLAN ARR FUEL	7557 01HR/23MIN	COST INDEX:
5	ROUTE:	KJFK DCT GREKI DCT JUDDS DCT MARTN DCT CBIRD DCT TAFFY N643B RIKAL NATV NETKI DCT REVNU DCT MOLAK DCT ELBOB DCT DUFFY DCT GOTNA DCT BLACA P600 TUNSO TUNSL E GPH			
6	ALTN RTE -	FL080 / GRICE3C	GRICE DCT STIRA DCT EGPF		
7	ALT2 RTE -	FL150 / TLA6C	TLA Y96 NATEB DCT EGNT		

Figure 2.1 Dispatch Release — Flight Summary

1. **Type of operation** — FLAG, DOMESTIC, or SUPPLEMENTAL — with the dispatch desk identifier and telephone number. Quote the desk on any call about the flight.
2. **IFR line** — flight number and leg count, registration, departure and destination, and each alternate.
3. **MIN T/O FUEL and RLS FUEL** restated for quick reference.
4. **TOT BRN, PLAN ARR FUEL, and COST INDEX** — total burn, the fuel and time expected on arrival, and the index the plan was optimized against.
5. **ROUTE** — the planned route in full. Compare it against the clearance received; where they differ, reload and verify before departure.
6. **ALTN RTE** — routing and planned level to the first alternate.
7. **ALT2 RTE** — routing and planned level to the second alternate.

2.2.3 Fuel Summary

The fuel summary states each component of the fuel required, in pounds, with the time and distance it buys.

1	PLAN ARR FUEL	007557	0123		

	ARPT	FUEL	TIME	DIST	
2	ENRT BRN EDI	035089	0605	2985	

3	ACF99	000014			
4	RSV	005932	0107		
5	MAF	000000	0000		
6	DISP ADD	000000	0000		
7	ALTN GLA	001611	0017	0056	
8	-ALT2 NCL	002621	0029	0114	

9	MIN T/O	042632			

10	DISP EXTRA	000000	0000		

11	TAXI JFK	000596	0020		
12	RLS FUEL JFK	043242			

13	TOTAL JFK	043242			

Figure 2.2 Dispatch Release — Fuel Summary

1. **PLAN ARR FUEL:** fuel and time expected on arrival at the destination
2. **ENRT BRN:** enroute burn, time, and distance to the destination
3. **ACF99:** Analyzed Contingency Fuel, computed under the OpSpec B343 program in lieu of fixed contingency fuel
4. **RSV:** reserve fuel and the time it represents
5. **MAF:** minimum additional fuel, where the route requires it
6. **DISP ADD:** fuel added by the dispatcher; the reason is stated in RMKS
7. **ALTN:** fuel, time, and distance to the first alternate
8. **ALT2:** fuel, time, and distance to the second alternate
9. **MIN T/O:** minimum takeoff fuel; the sum of the lines above
10. **DISP EXTRA:** discretionary fuel above the minimum
11. **TAXI:** taxi allowance for the departure station
12. **RLS FUEL:** release fuel, the figure fuelled to at the gate
13. **TOTAL:** total fuel aboard at the gate

Reading the Summary

RLS FUEL less TAXI is the fuel expected at brake release. Where the actual uplift places the aircraft below MIN T/O after taxi, the flight is not legal to depart. See section [2.4.4 Fuel Below Release Minimum](#).

PLAN ARR FUEL is the fuel expected at the destination after burning ENRT BRN. It is not a reserve. Compare it against the reserve figure to know how much margin exists before a diversion becomes necessary.

2.2.4 On Time Analysis, Weights, and Driftdown

ON TIME ANALYSIS							
	TXO	AIR	TXI	TOTAL		DEP GMT/LCL	ARR GMT/LCL
1	SKDBLK 0020	0605	0010	0635	SKD	0035Z/2035L	0710Z/0810L
2	FLIPLN 0020	0605	0010	0635	EST	0035Z/2035L	0710Z/0810L
3	BIAS 0.0PCT ENDURNC 0729						
4	PAYLOAD 35950 ZFW 145696 RAMP WT 188938 PTOW 188342						
5	AVG WIND DIR/COMP 266/056 AVG TD 007 CI0024						
6	RAMP WT P2000 TIME 06:06 FUEL P316						
	RAMP WT M2000 TIME 06:05 FUEL M404						
	ENGINE OUT DRIFTDOWN						
	M1 INFORMATION						
7	METW - 222666						
8	MOST LIMITING POINT - NONE - NO SIGNIFICANT TERRAIN						

Figure 2.3 Dispatch Release — On Time Analysis, Weights, and Driftdown

1. **SKDBLK**: scheduled taxi-out, air, and taxi-in times, with scheduled departure and arrival in UTC and local
2. **FLIPLN**: what the flight plan computed for the same segments. A material difference is a scheduling assumption that will not hold, and is reported to the dispatcher before departure.
3. **BIAS and ENDURNC**: performance bias applied to the aircraft, and total endurance on the fuel aboard
4. **PAYLOAD, ZFW, RAMP WT, PTOW**: the planned weights
5. **AVG WIND DIR/COMP and AVG TD**: average wind and component, and average temperature deviation from ISA

6. **RAMP WT P2000 / M2000:** how time and fuel change for two thousand pounds above and below the planned ramp weight. Use them to estimate the effect of a late load change without requesting a new plan.
7. **METW:** maximum enroute terrain-limited weight
8. **MOST LIMITING POINT:** the position that constrains the driftdown, or NONE where terrain is not a factor. See section [2.6 Engine-Out Driftdown](#).

2.2.5 Equipment and Remarks

1	MEL ITEMS: - NONE
2	CDL ITEMS: - NONE
3	NEF ITEMS: - NONE
4	SEL ITEMS 320 03 RUNWAY OVERRUN PREVENTION SYSTEM 320 10 RNAV/RNP OR RNP/AR .1 CERT 320 13 AUTOLAND MAX ELEVATION 6500 FT MSL 320 18 OVERWATER: ETOPS 320 23 AP/FD TCAS 320 31 FMS S8 INSTALLED 320 32 AUTO BRAKE OFF CALLOUT INSTALLED 320 34 FLS INSTALLED
5	RMKS/ NONE
6	ACFT RSTR: - NONE
7	FF KZNYZQZX KZBWZQZX CZQMZQZX CZULZQZX CZQXZQZX EGGXZQZX EISNZQZX EC
8	ALTN RTE - FL080 / GRICE3C GRICE DCT STIRA DCT EGPF ALT2 RTE - FL150 / TLA6C TLA Y96 NATEB DCT EGNT

Figure 2.4 Dispatch Release — Equipment Items and Remarks

1. **MEL ITEMS:** minimum equipment list items deferred on this aircraft
2. **CDL ITEMS:** configuration deviation list items
3. **NEF ITEMS:** non-essential equipment and furnishings deferred
4. **SEL ITEMS:** special equipment and authorizations the aircraft carries — ETOPS, RNP AR, autoland with its maximum elevation, runway overrun protection, and the FMS standard. Verify every capability the flight depends on appears here.
5. **RMKS:** dispatcher remarks, including the reason for any fuel added
6. **ACFT RSTR:** any restriction on the aircraft

7. **FF:** the ATC addressee the flight plan was filed to. Informational.

8. **ALTN RTE / ALT2 RTE:** alternate routings, repeated

Each block reads NONE where nothing applies. A blank block is a transmission fault, not an empty list.

2.2.6 Filed Flight Plan

1	(FPL-NAC82-IS
2	-A21N/M-SDE3FGHIRWY/LB1
3	-KJFK0035
4	-N0462F330 DCT GREKI DCT JUDDS DCT MARTN DCT CBIRD DCT TAFY/N0459F350 N643B RIKAL/M078F360 NATV NETKI/N0449F370 DCT REVNU DCT MOLAK DCT ELBOB DCT DUFFY DCT GOTNA DCT BLACA/N0407F250 P600 TUNSO TUNSO1E
5	-EGPH0605 EGPF EGNT
6	-PBN/A1B1C1D1O1S1 DOF/260808 REG/N803NY EET/KZBW0003 CZQM0104 CZUL0117 CZQX0153 53N050W0236 55N040W0319 EGGX0359 56N020W0436
7	EISN0516 EGPX0529 EGT0539 EGPX0542 SEL/KQAJ OPR/NAC PER/C RALT/BG BIKF RMK/TCAS)

Figure 2.5 Dispatch Release — Filed ICAO Flight Plan

1. **Flight identification and flight rules** — I for IFR, S for scheduled
2. **Aircraft type, wake category, and equipment suffixes** — verify against the SEL items
3. **Departure aerodrome and off-block time**
4. **Cruise speed, level, and route as filed**
5. **Destination, elapsed time, and alternates** — verify these are the alternates on the release
6. **Other information** — PBN, date of flight, registration, estimated elapsed times by FIR, SELCAL, operator, and remote alternates
7. **Remaining EET and SELCAL** — verify the SELCAL code against the aircraft

Caution: An incorrect SELCAL code will not be discovered until the first oceanic check. Verify it against the aircraft placard during preflight.

A discrepancy between the filed plan and the release is resolved with the dispatcher before departure.

2.2.7 Navigation Log

	1	2	3	4	5	6	7	8	9		
TO	LAT		LONG	MC	MK	GS	TD	SD	ST	SB	
IDENT	FL	WIND	WCP	MH	TRR	TAS	I	TLDR	TTLT	TTLB	TH
10	11	12	13	14	15	16	17	18	19	20	21
KJFK	N40640		W073779								
13FT	0		2985								
GREKI	N41480		W073314	35	700	457	P14	0078	0015	0031	

Figure 2.6 Dispatch Release — Navigation Log

The navigation log states the full waypoint-by-waypoint plan for the route — the document the time and fuel log in section 2.2.8 condenses into a progress-tracking form.

1. **Waypoint identifier and airway or track** for each leg
2. **Course and distance**, and cumulative distance to the destination
3. **Planned level** for each leg, including step climbs
4. **Wind and temperature deviation** used for the leg
5. **True airspeed and groundspeed**
6. **Leg time and cumulative elapsed time**
7. **Fuel burn for the leg and fuel remaining** at the waypoint

Compare the navigation log against the filed flight plan and the loaded FMS route waypoint by waypoint — not just the endpoints. A route that agrees at the departure and destination can still diverge in the middle.

Note: The navigation log is the planning document. Section 2.2.8 Time and Fuel Log is the one flown against — record actual times and fuel there, not on the navigation log.

2.2.8 Time and Fuel Log

1	-	FLT	NAC82	JFK-EDI	SKD	ARR	0810L
2					ETA		0810L
3					LATE		0000
4	OUT		0035		43.2	TOT /	CTR
5	OFF		0055		42.6		
6	GREKI	0110	226/700		39.5		
	JUDDS	0112	243/730		39.2		
	MARTN	0118	296/780		38.5		
7	TOC	0122	330/780		37.9		
	CBIRD	0137	330/779		36.4		
	TAFFY	0203	330/777		33.8		
	RIKAL	0309	350/783		27.2		
	53N050W	0331	360/780		25.2		
	55N040W	0414	360/780		21.1		
	56N030W	0454	360/780		17.5		
	56N020W	0531	360/780		14.2		

Figure 2.7 Dispatch Release — Time and Fuel Log

1. **Flight and scheduled arrival** with ETA and projected LATE
2. **ETA:** computed arrival, local
3. **LATE:** projected delay against schedule
4. **OUT:** off-blocks time and fuel at that point
5. **OFF:** takeoff time and fuel at that point
6. **Waypoint rows:** estimated time, planned level and wind, and fuel expected to remain, in thousands of pounds
7. **TOC:** top of climb, shown as a computed point

A condensed progress log states, for each waypoint, the estimated time, the planned level and wind, and the fuel expected to remain — in thousands of pounds. OUT, OFF, ON, and IN bracket the flight.

SKD ARR, ETA, and LATE at the head of the block state the scheduled arrival, the computed arrival, and the projected delay.

This is the log used in flight. Record actual time and fuel against the planned figures at each waypoint.

2.2.9 Signature Block and Crew

PAGE 5 RELEASE 1.0 2026-AUG-08 00:04:01Z KJFK-EGPH NAC82				
1	DISP SIGNED BY: SERGIO YANG			
2	DESK: FD01 PHONE: 800-555-0199			
3	CAPT* 		CAT1 YES	
4	F/O BARRY DURHAM		CAT2 YES	
			CAT3 YES	
5	AUZD CAPTAIN SIGNATURE.....			
6	ENRT WX	FL180	FL240	FL300
		TD WIND	TD WIND	TD WIND
			FL340	TD WIND

Figure 2.8 Dispatch Release — Signature Block and Enroute Winds

1. **DISP SIGNED BY:** the dispatcher who exercised operational control
2. **DESK and PHONE:** how to reach that desk
3. **CAPT:** captain
4. **F/O:** first officer
5. **Captain signature line:** records acceptance
6. **ENRT WX:** temperature deviation and wind by waypoint at four flight levels
7. **Waypoint rows:** TD and wind at each level

DISP SIGNED BY names the dispatcher who exercised operational control, with the desk and telephone number.

The crew block lists the captain and first officer. The CAT1, CAT2, and CAT3 entries state the approach capability the crew and aircraft combination is authorized for on this flight. Where an entry reads NO, that capability is not available regardless of the aircraft equipment.

The captain's signature line records acceptance. See section [2.2.19 Acceptance](#).

Note: The approach capability shown accounts for crew qualification as well as aircraft equipment. A high minimums captain will see a reduced capability here. See section [21.8 High Minimums Captain](#).

2.2.10 Enroute Winds and Temperatures

The ENRT WX table states temperature deviation and wind at each waypoint for four flight levels, typically FL180, FL240, FL300, and FL340.

Use it to evaluate a level change — for turbulence, for a better wind, or for a fuel-saving step — before requesting it. The table shows what the alternative level offers without a call to the dispatcher.

2.2.11 Oceanic Track Message

1	NAT-TRACKS
2	EASTBOUND OTS (GANDER/SHANWICK) -----
3	Eastbound North Atlantic Tracks FLS 340/400 INCLUSIVE AUG 08/0100Z TO AUG 08/0800Z U NEEKO 54/50 56/40 57/30 57/20 PIKIL SOVED EAST LVLS 340 350 360 370 380 390 400 WEST LVLS NIL EUR RTS EAST NIL NAR N701A N695A
4	V RIKAL 53/50 55/40 56/30 56/20 RESNO NETKI EAST LVLS 340 350 360 370 380 390 400 WEST LVLS NIL EUR RTS EAST NIL NAR N649B N643B W TINDP 52/50 54/40 55/30 55/20 DOGAT BEYET EAST LVLS 340 350 360 370 380

Figure 2.9 Dispatch Release — Oceanic Track Message

1. **NAT-TRACKS:** the track message block
2. **Direction and control areas** the tracks apply to
3. **Validity period and the levels allocated**
4. **Track definitions** — each lettered track with its waypoints, levels, and associated routings

Where the route uses an organized track system, the current track message is included. It states the tracks, their validity period, the levels allocated, and the entry and exit points.

Verify the cleared track against the message. See section [16c.1 Oceanic Clearance](#).

2.2.12 Station Field Report

PAGE 7 RELEASE 1.0 2026-AUG-08 00:04:01Z KJFK-EGPH NAC82

JFK NAC82
/FC

1 * JFK FIELD REPORT *

2 DATE 07AUG26 TIME 2035 LOCAL

3 WIND 063/03KT ALT 30.08 OAT 23C

4 EXISTING TAA SEE NOTAMS

5 RWY STATUS CONDITIONS REMARKS

6 04L/22R OPEN WET -TAKEOFF RWY 22R-

04R/22L OPEN WET

13L/31R OPEN WET

13R/31L OPEN WET

7 RAMP/TXWY SURFACE WET

8 TAXIWAY ADVISORIES

TWY B SA NC CLSD A6780/26

TWY E Y CLSD A6654/26

TWY X Z CLSD A5977/26

Figure 2.10 Dispatch Release — Station Field Report

1. **Field report header** for the station
2. **Date and time** of issue, local
3. **Conditions:** wind, altimeter, and temperature at issue
4. **TAA:** whether a Temporary Airport Analysis is in effect
5. **Runway table header**
6. **Runway rows:** status and surface condition; the takeoff runway is annotated
7. **RAMP/TXWY:** surface condition of the movement area
8. **Taxiway advisories:** closures, each with its NOTAM number

The field report states the condition of the departure station at the time shown. It is issued by Station Operations, not by ATC, and is not a substitute for the ATIS or NOTAMs.

Block	Contains
Conditions	Wind, altimeter, and temperature at the time of issue

Block	Contains
Runway status	Each runway with its status and surface condition; the takeoff runway is annotated
Ramp and taxiway surface	Surface condition of the movement area
Taxiway advisories	Closed taxiways with the NOTAM number for each
Approach and lighting advisories	Approach aid and lighting outages
Special advisories	Anything else affecting the operation at that station
Frequencies and telephone	Ops, ATIS, maintenance, phone patch, and ACARS, with ground power and air availability
Remarks	Local ATC frequencies and station notes
— End —	

Runway Condition

Where the report states a surface condition other than dry, verify the takeoff performance was computed for that condition. The TPS states the condition it assumed. See section [1p.2.2 Takeoff Performance System \(TPS\)](#).

Warning: The field report is a snapshot at the time of issue. Where conditions have changed since — a shower, a temperature change through freezing — the report no longer describes the runway you are departing from.

2.2.13 Weather

1 JFK - JOHN F KENNEDY INTL DEPART

2 RWYS: 04L/22R 04R/22L
13L/31R 13R/31L

3 TAF

4 2320Z 0800/0906 23008KT P6SM VCSH BKN050 BKN250
300/0801 5SM -SHRA BR
) 25006KT P6SM SCT035 BKN100
) 23009KT P6SM FEW035 SCT150
) 18010KT P6SM SCT050 BKN250

Figure 2.11 Dispatch Release — Weather

1. **Airport block header** with the validity window
2. **Runways** available at that airport
3. **TAF** covering the period of the operation
4. **Forecast groups** — TEMPO, FM, and PROB lines

The weather package satisfies 14 CFR 121.599 and contains, for the departure, destination, and each alternate:

- the current ATIS where available
- the TAF covering the period of the operation
- the current METAR

Enroute advisories follow — SIGMETs, AIRMETs, and convective advisories by FIR.

Review the destination and alternate forecasts against the alternate requirement before accepting. See section [2.5 Alternates](#).

2.2.14 NOTAMs

1 2 3	DEPARTURE	KJFK JFK A5173/26 2026-May-27 03:45
		CREATED:2026-May-27 03:49 FL: SFC
		TWY G CL LGT BTN TWY B AND RWY (
		KJFK JFK A5028/26 2026-May-20 08:30
		CREATED:2026-May-20 08:31 FL: SFC
		TWY K LGT BTN RWY 13R/31L AND TW
		KJFK JFK A4535/26 2026-Apr-29 19:05 -
		CREATED:2026-Apr-29 19:08 FL: SFC
		TWY B CL LGT BTN TWY J AND TWY M
		KJFK JFK A4504/26 2026-Apr-29 09:12 -

Figure 2.12 Dispatch Release — NOTAMs

1. NOTAM number and validity period
2. Creation time and the affected flight level band
3. Text of the NOTAM

NOTAMs are grouped by airport and by FIR, and within each group by category — airspace, obstacle, warning, communication, and general.

Each NOTAM carries its number, validity period, creation time, and affected flight level band, followed by the Q, A, B, C, D, and E fields.

Reading the Set

The set is long and most of it will not apply. Read for:

- runway, taxiway, and approach closures at every airport on the itinerary
- approach minima changes and unusable aids
- cranes and obstacles in the approach and departure areas
- airspace restrictions and temporary danger areas on the route
- oceanic clearance procedure changes

Caution: A NOTAM raising the minima for an approach can invalidate an alternate that was legal when the release was computed. Where an approach at an alternate is affected, confirm the alternate remains suitable.

Where the itinerary changes in flight, the NOTAMs for the new destination are requested from the dispatcher. The release set covers the planned itinerary only.

2.2.15 Charts

The graphical products close the package:

Chart	Contains
Route	The planned track with the alternate routings
Significant weather	Forecast significant weather for the levels flown
Upper air data	Forecast wind and temperature at the levels flown
Vertical profile	The planned vertical path with terrain and step climbs
— End —	

2.2.16 Takeoff Performance

The TPS for the departure station is appended to the release. It is described in section [1p.2.2 Takeoff Performance System \(TPS\)](#).

Where the TPS was computed on assumptions that no longer hold — a different runway, a changed surface condition, a weight beyond tolerance — request a new one rather than adjusting the figures.

2.2.17 Inflight Release Amendment

An inflight release amendment is a **revised abbreviated release**, computed by the dispatcher from the aircraft's present position to the destination, using the fuel on board at that point. It is transmitted by ACARS to the flightdeck printer.

It is issued where the plan has materially changed in flight — most often a reroute, a change of destination or alternate, a level change that alters the burn, or a fuel state that no longer supports the original release.

What It Contains

QUMCIYVFNAC456FTM

1 ATTENTION CAPTAIN
PLEASE REVIEW THIS FLIGHT PLAN/RELEASE
WHICH INCLUDES THE FOLLOWING AMENDMENT
ASDFGHJKL

2 -- INFLIGHT RELEASE AMENDMENT --
RELEASE 3.0

3 - IFR NAC456/28 967/N967NX MIA LIM

4 ALTN CIX MIN T/O FUEL 044347

5 RLS FUEL 046000 TOT BRN 34095
PLAN ARR FUEL 11309 01HR/49MIN
COST INDEX: 040

6 ESTMD ARR FUEL 011309 0149

	ARPT	FUEL	TIME	DIST
7 ENRT BRN	LIM	34095	0511	2339

8 RSV	03700	00041		
MAF	00000			
DISP ADD	00000			
ALTN	CIX	06552	0105	0403

9 MIN FUEL	44347			

10 DISP EXTRA	01057	0008		

11 FOB	046000			

12 PTOW 205824	LDW 171729			

13 DRIFTDOWN - METOW 205824LBS MOST LIMITING POINT - NONE - NO SIGNIFICANT TERRAIN				

14 ROUTE KMIA NNOCE2 FUNDI DCT ALVEK UM335 ARNAL UL465 TBG UL780 TRU UV1 ATATU ATATU4 SPJC				

15 FL300				

16 AMENDED RELEASE FOR A REROUTE				

----- END OF SFP -----				

17 FD70 IGNACIO KENNEDY				

Figure 2.13 Inflight Release Amendment

1. **ATTENTION CAPTAIN** — the amendment banner and the instruction to review
2. **INFLIGHT RELEASE AMENDMENT** with the new release version
3. **IFR line** — flight, registration, origin, and destination

4. **ALTN and MIN T/O FUEL** — the alternate and the recomputed minimum
5. **RLS FUEL and TOT BRN** — fuel and the remaining burn
6. **ESTMD ARR FUEL** — estimated arrival fuel and time on the amended plan
7. **ENRT BRN** — remaining burn, time, and distance to the destination
8. **RSV, MAF, DISP ADD, ALTN** — reserve, additional fuel, and alternate fuel, recomputed from present position
9. **MIN FUEL** — the recomputed minimum required
10. **DISP EXTRA** — discretionary fuel remaining above the minimum
11. **FOB** — fuel on board at the point of computation. Everything above is computed from this figure.
12. **PTOW and LDW** — weights, with landing weight on the amended plan
13. **DRIFTDOWN** — METOW and the most limiting point for the amended routing
14. **ROUTE** — the amended route
15. **Cruise level** for the amended routing
16. **Reason** — a free-text statement of why the amendment was issued
17. **Dispatcher** — desk and name

Crew Action

1. Confirm the amendment is for **this flight** — flight number, registration, and release version.
2. Compare **FOB** on the amendment against actual fuel on board. The amendment is computed from the figure shown; if it is wrong, the whole amendment is.
3. Verify **ESTMD ARR FUEL** remains above reserve on the amended routing.
4. Verify the amended route and level against the clearance, and reload the FMS where the routing has changed.
5. Check the driftdown line where the amended routing crosses terrain that the original did not.
6. **Acknowledge the amendment** to the dispatcher.

Warning: An unacknowledged amendment is not accepted. The flight continues on the previous release until the captain acknowledges, and the dispatcher is entitled to assume the previous plan is being flown until then.

Fuel Below the Amended Minimum

Where the amendment shows arrival fuel below reserve, or minimum fuel above the fuel on board, the routing cannot be flown as amended. Advise the dispatcher and resolve it before the aircraft reaches the point where a diversion becomes the only option. See section [7g.2 Fuel Monitoring](#).

2.2.18 Amendment

An amendment is transmitted by ACARS or delivered by the dispatcher, and carries a new release version. The captain acknowledges each amendment. A flight will not depart against an unacknowledged amendment.

2.2.19 Acceptance

Acceptance signifies that the captain has reviewed the release, the flight plan, the weather, the NOTAMs, and the maintenance status, and concurs that the flight can be conducted as planned.

Do not accept a release that cannot be complied with. Contact the dispatcher and resolve the discrepancy before departure.

Caution: Acceptance of the release is not a formality. A captain who accepts a release and departs assumes joint responsibility for its content.

2.2.20 Re-Release

A flight planned under re-dispatch procedures carries a re-release point. The dispatcher will transmit the re-release before the aircraft reaches that point. If the re-release is not received, proceed to the initial destination shown on the release.

2.3 Computer Flight Plan

2.3.1 Review

Review the computer flight plan for:

- route, including any reroute issued after paperwork generation
- cruise altitudes and step climbs
- forecast winds and temperatures
- fuel burn by waypoint
- ETOPS entry and exit points, where applicable
- terrain and driftdown considerations

2.3.2 Loading the Route

Both pilots will verify FMS entries that alter route, track, or altitude constraints before execution. Compare the loaded route against the computer flight plan and the charted procedure.

Note: Verify the first waypoint sequence against the assigned departure procedure. A route loaded before clearance is received is not a cleared route.

The fuel summary, the on-time analysis, and the ramp weight sensitivity figures are properties of the release itself, not of the computer flight plan — see section [2.2.3 Fuel Summary](#) and section [2.2.4 On Time Analysis](#), Weights, and Driftdown.

2.4 Fuel Planning

Source: 14 CFR 121.639, 121.645

2.4.1 Quantity

Fuel is planned and reported in **pounds**. The release, the TPS, the closeout, and the aircraft all use pounds; no conversion is applied between them.

Where a figure is quoted in thousands — as the TPS and the time and fuel log do — it is stated to one decimal place. 43.2 is 43,200 lb.

Report actual fuel on board to the load agent by ACARS once fuelling is complete. Where the figure entered differs from planned by 4,000 lb or more, an ACARS message is generated for the load agent to verify. See section [1p.5.2 Load Closeout](#).

2.4.2 Minimum Fuel

The minimum fuel shown on the release consists of:

- burn to destination
- alternate fuel, where an alternate is required
- reserve fuel
- Analyzed Contingency Fuel (ACF99), or any holding or ETOPS fuel added by the dispatcher

Analyzed Contingency Fuel

Northern Airways is authorized under **OpSpec B343** to plan contingency fuel using ACF99 — a statistically derived figure computed per city pair from historical fuel burn data — in place of a fixed contingency percentage. It appears on the release as the ACF99 line of the fuel summary. See section [2.2.3 Fuel Summary](#).

Where the ACF99 program is not available for a city pair, or where its output is suppressed, the dispatcher reverts to the fixed contingency fuel required by 14 CFR 121.645.

2.4.3 Additional Fuel

The captain may request additional fuel for any operational reason. The dispatcher will add the requested fuel unless a limitation prevents it, in which case the dispatcher will advise the captain of the limitation.

Additional fuel is appropriate for:

- convective activity on the route or at the destination
- known or forecast holding
- a destination with a single runway or limited approach capability
- an alternate at or near minimums
- enroute turbulence
- departure delays

Enroute Turbulence

This fuel allows the crew to search for a smoother altitude, 4,000 feet above or below the planned altitude. Use the table to determine the ADD fuel where enroute turbulence is known or forecast.

Table 2.2 Enroute Turbulence Fuel (lbs)

Fleet	Lbs/Hour	Fleet	Lbs/Hour
A32F, B737, and E190	200	B757	400
B767	300	A330 and B777	500
— End —			

Plan this fuel only for the affected portion of the flight.

Departure Delays

This fuel accounts for an anticipated taxi-out delay over and above the planned taxi-out time.

Table 2.3 10 Minute Taxi Fuel Burn (lbs)

Fleet	Lbs	Fleet	Lbs
A32F, B737, and E190	300	B767	500
B757	400	A330 and B777	600
— End —			

International

ADD fuel is also used to meet equal time point requirements on an extended overwater operation. See section [16c.1 Oceanic Clearance](#) and section [7g.5 ETOPS](#).

2.4.4 Fuel Below Release Minimum

A flight will not depart with less fuel than the release requires. Where the fuel loaded is less than planned, contact the dispatcher for an amended release before departure.

2.4.5 Enroute Fuel Management

See Enroute General, Fuel Monitoring.

2.5 Alternates

Source: 14 CFR 121.619, 121.621, 121.623, 121.625

2.5.1 Requirement

A destination alternate is required unless the destination forecast, for **1 hour before through 1 hour after** the ETA, shows a ceiling at least **2,000 feet** above airport elevation and visibility at least **3 statute miles**.

2.5.2 Alternate Minimums

Alternate minimums are derived from the approaches available at the alternate and are shown on the release. Where the captain is designated High Minimums, increase the alternate minimums accordingly. See section [21.8 High Minimums Captain](#).

2.5.3 Second Alternate

The dispatcher will file a second alternate where the weather at the first alternate is marginal, where the alternate has limited approach capability, or where required by OpSpecs.

2.5.4 Takeoff Alternate

Source: 14 CFR 121.617

A takeoff alternate is required when the departure airport weather is below the landing minimums for that airport. The takeoff alternate will be within the distance specified in OpSpecs for the aircraft, with one engine inoperative, in still air.

2.6 Engine-Out Driftdown

Where the route crosses terrain that cannot be cleared following an engine failure at cruise altitude, the dispatcher plans a driftdown escape route.

The driftdown plan is annotated on the computer flight plan and identifies:

— the critical point and the escape routing from it

- the driftdown level achievable at the planned weight
- the terrain clearance available along the escape route
- the diversion airport applicable to each segment

Brief the driftdown plan before entering the affected segment. Where the plan cannot be flown at the actual weight, advise the dispatcher before departure.

Warning: Terrain clearance following an engine failure is not assured by the enroute minimum altitude. Fly the driftdown escape route.

2.7 Weight and Balance

Source: 14 CFR 121.665

2.7.1 Final Figures

Final weight and balance figures are transmitted by ACARS or delivered by the agent. Both pilots will verify:

- zero fuel weight and takeoff weight against the aircraft limits
- center of gravity against the certified envelope
- stabilizer trim setting
- that the figures correspond to the correct flight and aircraft

2.7.2 Last-Minute Changes

Apply a last-minute change only within the tolerance published for the fleet. Where the change exceeds the tolerance, request revised figures.

Warning Flight

A flight is designated a **warning flight** where the planned takeoff weight is within **1,500 lb** of a limiting weight — structural, runway, climb, or landing.

On a warning flight there is effectively no margin for a late load change. Any addition after the closeout requires revised figures, and the crew confirms the final weight against the limit itself rather than against the tolerance.

Warning: Do not depart on figures that have not been verified against the aircraft limits. See the applicable AOM for the fleet weight and balance procedure.

2.7.3 TPS Weight and Balance Data

The TPS carries a WEIGHT AND BALANCE DATA block stating the planned weights the takeoff performance was computed against. It is not the load closeout —

the closeout is issued separately once loading is complete and states the final figures. See section [2.7.1 Final Figures](#).

Field	Contains
EOW	Empty operating weight for the registration.
PSGR WT	Passenger weight, computed on standard or actual weights as indicated.
CGO WT	Cargo and mail weight.
BALLAST	Ballast carried, where any.
ZFW / MZFW	Zero fuel weight against its maximum.
FUEL	Planned fuel aboard. A P suffix denotes pounds.
RMP / MRMP	Ramp weight against its maximum.
TXI	Taxi fuel deducted between ramp and takeoff weight.
TOW / MTOW	Takeoff weight against its maximum.
CMPT MAX / AS LDED	Compartment maximum against the weight planned into that compartment.
CNFIG / PSGRS	Cabin configuration by class against passengers planned by class.
— End —	

Verification

These are planning figures. Verify them against the TPS, then verify the load closeout against them once it is issued:

- ZFW, TOW, and ramp weight are each below their stated maximum
- no compartment AS LDED figure exceeds its CMPT MAX
- the PTOW here matches the weight the thrust and V-speeds were computed for
- the figures on the closeout do not exceed any maximum shown here

Standard Weights

Where the block is annotated *** STD ***, passenger weights are standard rather than actual. Standard weights are not valid where the load is non-representative — a charter, a sports team, or a crew move. Request actual weights in those cases.

Caution: This block states structural maximums. It does not state the performance-limited weight for the runway. A weight within every structural limit here may still exceed the runway MTOW on the TPS line.

2.8 Reserved

Chapter 3: Preflight

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3g.1 General

Preflight establishes the condition of the aircraft, the accuracy of the paperwork, and the readiness of the crew. Work performed carelessly at the gate is not recoverable in flight.

For aircraft-specific preflight procedure, see the applicable AOM> 3> Preflight Procedures.

Report Time

Report at the time published for the operation. Report time accounts for paperwork review, exterior inspection, flightdeck setup, and the flight attendant briefing.

Sequence

Accomplish preflight in the sequence established by the AOM flow. Where the sequence is interrupted, resume from the beginning of the interrupted item.

3g.2 Equipment/Documents**Flight Items**

Each crewmember carries, and can produce on request:

- airman certificate
- medical certificate
- passport with any applicable visa
- restricted radiotelephone operator permit, for international operations
- flashlight, serviceable and accessible from the seat
- a timepiece displaying hours, minutes, and seconds
- headset
- Company identification
- high-visibility vest
- spare spectacles, where spectacles are a condition of the medical
- the Company EFB, charged

Documents and Manuals

See section [20.6 Required for Flight for the documents and manuals required to be aboard the aircraft and in personal possession.](#)

Equipment

Verify the emergency equipment required by the equipment list is aboard, accessible, and in date. Record any discrepancy. See section [3n.1 Aircraft Equipment](#).

Dispatch Release Acceptance

See section [2.2.2 Acceptance](#).

3g.3 Flightdeck Preparation

Division of Duties

The AOM flow assigns preflight items by seat. See the applicable AOM> 2> Flow Guide.

Documents

Verify aboard the aircraft:

- aircraft flight manual and required placards
- MEL and CDL
- aircraft registration and airworthiness certificate
- current route manual coverage on the EFB
- emergency equipment per the equipment list

Flightdeck Management

Crew bags, EFBs, and personal items are stowed so that they do not obstruct a control, a circuit breaker panel, an escape path, or either pilot's access to their seat and harness.

Nothing is stowed on the glareshield, on the center pedestal, or behind a seat where it can move during acceleration.

Documents required for the flight are accessible from the seated position with the harness fastened. See section [20.6 Required for Flight](#).

For standing flightdeck management policy, see section [1g.4 Flightdeck Management](#).

Databases

Verify the FMS navigation database is current and covers the period of the flight. An expired database does not permit RNAV or RNP operation.

Note: Where the database expires during the flight, see the applicable AOM for the continued-use provisions.

3g.4 Exterior Inspection

The exterior inspection is accomplished by a pilot before each flight, and after any period during which the aircraft was unattended.

Inspect for:

- evidence of damage, leakage, or contamination
- condition of tires, brakes, and gear
- security of panels, doors, and access covers
- frost, ice, or snow on any surface
- obstruction of probes, ports, and drains

Warning: Any contamination adhering to a critical surface requires deicing before departure. See section [3g.7 Deicing](#).

Report any discrepancy found. See section 3m.2 eAML.

3g.5 Fuel

Fuel Slips

Verify the fuel uplift against the release requirement and against the aircraft indication. Where the three do not agree within the fleet tolerance, resolve the discrepancy before departure.

Fuel Temperature

Verify fuel temperature is above the minimum published for the fleet. Where cold soak is a factor, see Enroute Weather, Cold Fuel.

Reserved

3g.6 Boarding

Flightdeck Access

The flightdeck door is secured in accordance with the AOSSP from the time all external doors are closed. See section [1s.2 Flightdeck Security](#).

Passenger Boarding

The captain is advised when boarding is complete and the cabin is secure. Do not push back until the cabin secure report is received.

Denied Boarding

The captain may refuse carriage of any person consistent with Company policy and applicable law. Advise the dispatcher and complete the required report.

3g.7 Deicing

Source: 14 CFR 121.629

Clean Aircraft Concept

No takeoff will be made with frost, ice, or snow adhering to any critical surface. This is a regulatory prohibition and is not subject to crew judgment as to quantity.

Holdover Time

Holdover time begins at the **start** of the final anti-icing application. Monitor the elapsed time against the holdover table for the fluid, the precipitation type, and the outside air temperature.

Caution: Holdover time is an advisory guideline. The captain determines whether the aircraft is clean immediately before takeoff.

Pre-Takeoff Contamination Check

Where holdover time has been exceeded, or where conditions have changed, a pre-takeoff contamination check is required within **5 minutes** of takeoff. Where the check cannot be accomplished, return for deicing.

Anti-Ice Configuration

See the applicable AOM> 15> Adverse Weather.

3c.1 General

See section [1c.1 General for standing communications policy](#).

3c.2 Flight Attendants**Departure Briefing**

The captain briefs the flight attendants before each departure, or before the first departure of a duty period where the same crew continues together. Include:

- flight time and expected ride
- anticipated turbulence and the plan for cabin service
- weather at the destination
- any MEL item affecting the cabin
- the security and emergency signals in use
- the sterile period and the conditions for contacting the flightdeck

Cabin Secure

The flight attendant in charge reports the cabin secure. Departure will not commence before that report.

3c.3 Ramp

Coordinate with the ground crew before pushback:

- nosewheel steering bypass pin installed, and confirmed removed after pushback
- direction of push and final orientation
- area clear
- all doors closed and equipment clear

See the applicable AOM> 4> Pushback/Start.

3c.4 Passengers**Routine PA**

Make a passenger announcement after boarding is complete, and whenever a delay of **10 minutes** or more occurs or becomes known.

State what is known and when the next update will be made. Do not speculate as to cause and do not commit to a connection.

See section [1c.5 Passengers](#).

3w.1 Resources

The weather package accompanying the release satisfies 14 CFR 121.599. Review in addition:

- current and forecast conditions at the departure airport
- conditions at the takeoff alternate, where required
- conditions at the destination and each alternate
- SIGMETs, AIRMETs, and PIREPs applicable to the route
- field condition reports and braking action reports

See Meteorology for product interpretation.

3w.2 Departure Weather**Takeoff Minimums**

Takeoff minimums are established by OpSpecs and by the airport analysis. Verify the runway and the minimums applicable to the departure.

Adverse Conditions

Where windshear is reported, suspected, or indicated, do not depart. See section [1w.2 Windshear / Microburst](#).

3f.1 Routing

Review the routing on the computer flight plan against the clearance received. Where the clearance differs, reload the route and verify the change with the other pilot before departure.

Verify the special qualification category for every airport on the itinerary, including alternates. See section [21.9 Special Qualification Airports](#).

3f.2 Weight

Verify final weight and balance figures against the aircraft limits. See section [2.6 Weight and Balance](#).

Where a last-minute change exceeds the published tolerance, request revised figures.

3f.3 Alternates

Verify the alternates filed remain suitable against the current forecast. Where an alternate has degraded below the applicable minima, request an amended release. See section [2.5 Alternates](#).

3m.1 General

Airworthiness is determined by Maintenance. The crew observes, records accurately, and does not operate an aircraft known to be unairworthy.

3m.2 eAML

Review the electronic Aircraft Maintenance Log before accepting the aircraft. Verify:

- every open item is recorded, deferred, and placarded
- the deferral matches the MEL or CDL item on the release
- no item conflicts with the planned operation

Record every mechanical discrepancy in objective terms — what was observed, when, in what phase, and the crew action taken. Do not diagnose. Entry types are defined in section 1m.2 eAML.

Note: Recording a discrepancy is not a judgment as to its significance. Engineering determines significance from the record.

Outage

Where the eAML is unavailable, use the paper log procedure published in the Controlled Document Library and advise Maintenance Control.

3m.3 MEL and CDL

The crew complies with the operational (O) procedure for every open item.

Verify at preflight:

- each item is placarded and matches the eAML
- each (O) procedure is understood and any performance penalty is applied
- no item conflicts with ETOPS, RVSM, CAT II/III, or a required system at any airport on the itinerary

3m.4 ETOPS

An ETOPS flight requires a verified ETOPS preflight and an aircraft in ETOPS configuration.

Pre-departure Check

Before any ETOPS operation, verify:

1. the eAML records the aircraft as authorized for ETOPS
2. the aircraft registration matches the eAML
3. an ETOPS pre-departure check has been recorded in the mechanical discrepancy section

4. the ETOPS block on the release is signed
5. no open MEL or CDL item affects an ETOPS significant system

Where any item cannot be verified, the flight will not depart as an ETOPS operation. Contact the dispatcher for a non-ETOPS routing.

System Verification

Where Maintenance requires an in-flight verification of an ETOPS significant system, the requirement is recorded as a mechanical discrepancy entry and identified by a tracking placard.

Verification is accomplished on the first flight capable of making it, whether or not that flight is an ETOPS flight. It may be accomplished on:

1. a functional check flight
2. a ferry flight
3. a non-ETOPS revenue flight
4. an ETOPS revenue flight, before entering the ETOPS portion

See section [7g.5 ETOPS for the in-flight procedure](#).

3s.1 General

Security procedures are governed by 49 CFR Part 1544 and by the Company AOSSP. The AOSSP contains Sensitive Security Information and is issued separately.

Warning: Do not discuss security procedures on an open frequency, in a public area, or on an unsecured system.

3s.2 Aircraft Search

Accomplish the flightdeck security search required by the AOSSP before the first flight of the day and after any period during which the aircraft was unattended.

3s.3 Jumpseat

Source: 14 CFR 121.547

Flightdeck jumpseat occupancy requires authorization and identification in accordance with the AOSSP. The captain has absolute authority over occupancy and may decline any rider without stating a reason.

Brief each rider on:

- seat, harness, and oxygen mask operation
- emergency exit and evacuation
- the sterile period
- the requirement not to interfere, and the expectation that the rider will advise the crew of anything observed

3d.1 Notifications

The captain is provided written notification of every dangerous goods shipment aboard before departure. Verify the notification against the load sheet.

The notification states, for each item:

- proper shipping name and UN number
- hazard class and division
- number of packages and net quantity
- loading position aboard the aircraft
- any emergency response information applicable

Do not depart where dangerous goods are aboard and the notification has not been received. See section [1d.1.3 Notification to Captain](#).

Warning: An undeclared or misdeclared shipment is a safety-of-flight matter. Where a discrepancy exists between the notification and the load sheet, do not depart until it is resolved.

3d.2 Loading

Loading, segregation, and securing are accomplished by ramp personnel in accordance with the loading instruction issued by the load agent. The crew verifies that the loading position shown on the notification matches the load sheet.

Where a shipment requires segregation from another class, confirm with the load agent that the segregation has been applied.

3n.1 Aircraft Equipment

Where an item of required emergency equipment is missing, unserviceable, or out of date, the aircraft will not depart until the item is replaced or the condition is deferred under the MEL.

Verify at preflight:

- portable oxygen, protective breathing equipment, and fire extinguishers
- escape slides armed and in date
- emergency lighting and megaphones
- first aid and emergency medical kits, sealed and in date

Record any discrepancy in the eAML. See section 3m.2 eAML.

3n.2 Dispatch with a Non-Normal Condition

Where a non-normal condition exists before departure, the flight will not depart until the condition is resolved or deferred under the MEL or CDL by Maintenance.

A crew will not apply a QRC/QRH procedure to enable a departure. The QRC/QRH addresses conditions arising in flight.

Warning: A condition that appears during preflight is a maintenance matter, not a crew workaround.

Chapter 4: Taxi

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4g.1 General

Airport surface operations require strict attention and constant situational awareness. Sound flightdeck operating discipline enables the flight crew to plan taxi operations with the same level of attention given to other phases of flight.

4g.1.1 Planning

Anticipate airport surface movements by performing a pre-taxi review based on ATIS and previous experience at the airport. Review the expected taxi route on the airport diagram.

Hot Spots

Identify published hot spots on the airport diagram during the review and brief the mitigation for each.

Bridges and Overpasses

Avoid stopping on a bridge or overpass. If a ground evacuation is required while stopped on a bridge or overpass, passengers may not be able to evacuate safely and emergency vehicles may not be able to access the aircraft.

4g.1.2 Control

The captain will taxi the aircraft. The primary task of the flight crew is to taxi the aircraft safely, and the attention of the flight crew should not be diverted from that task.

Caution: Do not taxi so close behind another aircraft as to ingest foreign objects.

4g.1.3 Communication

See section [1c.2 Radio Communication Guide for clearances requiring a read back](#).

Coordination

Verbally coordinate all taxi instructions with the other pilot to ensure common understanding. If in disagreement, seek clarification from ATC.

Do not taxi until a taxi clearance is received and both pilots verbally coordinate and agree on the runway assigned, any restrictions, and the taxi route.

Note: Ensure the received taxi route is followed, not the expected or briefed route.

4g.1.4 Monitoring

Use a continuous loop process for actively monitoring and updating aircraft position during taxi. This includes knowing the present location and mentally calculating the next location on the route that will require increased attention. Consider writing down the taxi instructions.

While taxiing:

- do not allow other flightdeck duties and non-ATC communications to divert attention from the safe movement of the aircraft, particularly at runway crossings and complex intersections
- maintain the sterile flightdeck whenever the aircraft is moving

Both pilots should:

- have the airport diagram available and reference it as necessary to confirm the taxi clearance is followed
- monitor the tower frequency when number one to cross an active runway
- be heads up to visually monitor progress at critical locations

When approaching an entrance to an active runway, both pilots will confirm compliance with the hold short or crossing clearance by discontinuing non-monitoring tasks.

Note: If uncertain as to position on the airport, taxi clear of any runway, stop the aircraft, and advise the tower.

4g.1.5 Low Visibility

See also sections [4g.2 SMGCS](#) and [4g.5 Airport Markings/Lighting](#).

Exterior Lights

At night or during periods of reduced visibility, use exterior lights except landing lights as appropriate to enhance aircraft visibility.

Accomplishing Checklists

During low visibility taxi operations, conduct checklists only when the aircraft is stopped or taxiing straight ahead without complex intersections.

4g.2 SMGCS

Surface Movement Guidance and Control System procedures apply when visibility is below **1200 RVR**. Operate only on routes published on the SMGCS chart and only with the SMGCS chart available.

Verify the SMGCS plan is in effect with ground control before taxi.

4g.3 Runway Crossing

Read back every runway crossing clearance and every hold short instruction in full, including the runway.

Before crossing:

- both pilots confirm the clearance
- both pilots visually clear the runway in both directions
- the crossing is made without stopping once commenced

Warning: Do not cross a runway hold short line without an explicit clearance naming the runway. A taxi clearance to a point beyond a runway is not a crossing clearance unless the runway is named.

4g.4 Delays

Extended Ground Delay

Where a delay is expected to exceed **30 minutes**, coordinate with the dispatcher and advise the cabin. Consider single-engine taxi, APU operation, and the fuel required to remain at the gate or in the movement area.

Return to Gate

The captain may return to the gate at any time. Advise the dispatcher and Operations.

Tarmac Delay

Company tarmac delay procedures are published in the Controlled Document Library. The captain and the dispatcher jointly manage the delay against the applicable regulatory limit.

4g.5 Airport Markings/Lighting

Source: AIM 2-3; AC 150/5340-18

Markings, signs, and lighting are the airport telling you where you are and what you may not cross. They are the primary defense against a runway incursion.

Where a marking or sign is missing, obscured, contradicts the airport diagram, or contradicts the clearance, **stop**. Request clarification from ATC, then report the condition. See section [23.4 Ground Safety Report](#).

4g.5.1 Markings

Runway markings are **white**. Taxiway markings and any marking on a surface not intended for aircraft are **yellow**.

Marking	Meaning
Runway holding position, taxiway side	Four yellow lines, two solid and two dashed. Solid lines on the side you approach from. Do not cross without a clearance.
Runway holding position, on a runway	Used where a runway is intersected by another runway. Hold here when instructed to hold short of the intersecting runway.
ILS critical area	Yellow ladder. Hold here when ILS critical area protection is in effect.
Taxiway/taxiway holding position	Single dashed yellow line. Hold on ATC instruction.
Enhanced taxiway centerline	Dashed lines flanking the centerline for 150 feet before a runway holding position. A warning that a hold short line follows.
Surface painted holding position sign	White on red, on the pavement left of the centerline. Supplements the elevated sign.
Surface painted location sign	Yellow on black, right of the centerline. Confirms the taxiway you are on.
Demarcation bar	Yellow, 3 feet wide. Separates a displaced threshold from a preceding taxiway, blast pad, or stopway.
Chevrons	Yellow. Pavement unusable for landing, takeoff, or taxi.
Yellow cross	Runway or taxiway closed.
— End —	

Warning: The solid lines of a runway holding position marking are always on the side from which you approach the runway. If the solid lines are behind you, you are on the runway.

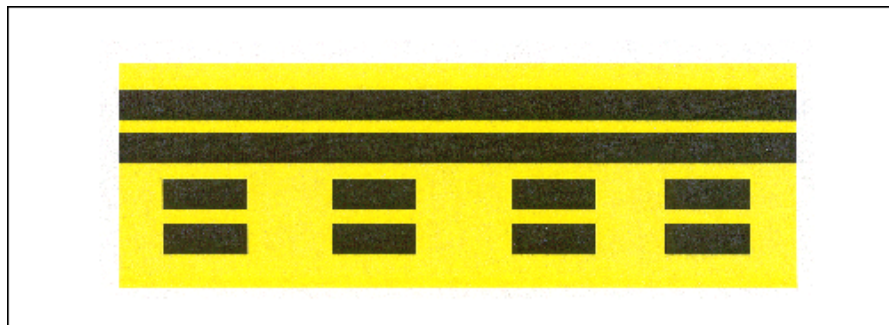


Figure 4.1 Runway Holding Position Markings on a Taxiway

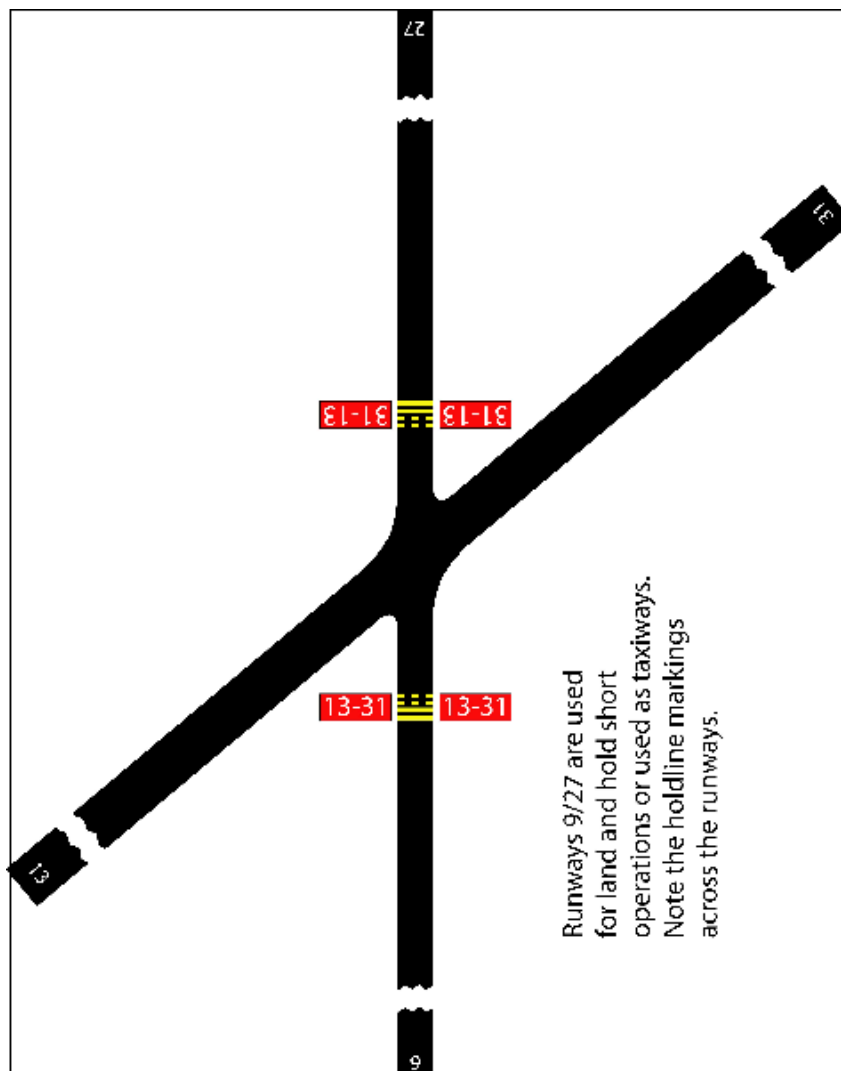


Figure 4.2 Runway Holding Position Markings on a Runway

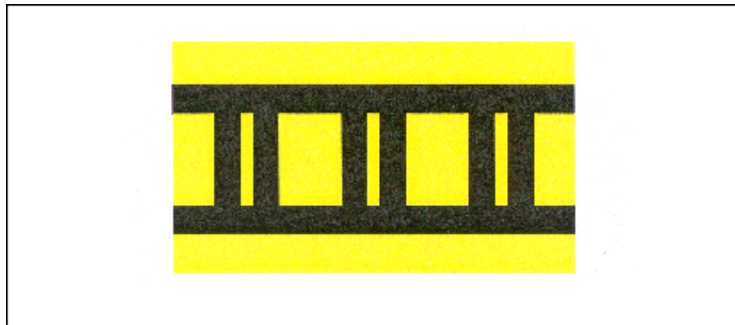


Figure 4.3 Holding Position Marking, ILS Critical Area

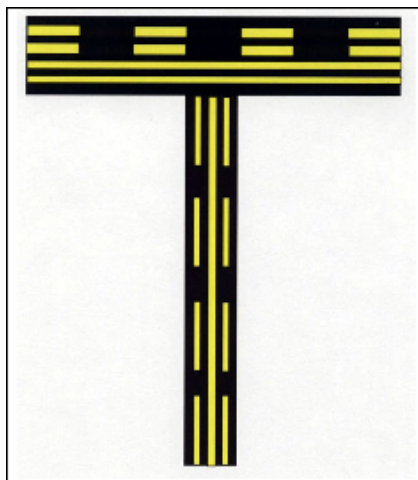


Figure 4.4 Enhanced Taxiway Centerline

4g.5.2 Signs

Six types of sign are installed on an airfield.

Sign	Colours	Meaning
Mandatory instruction	White on red	An entrance to a runway, a critical area, or a prohibited area. Do not pass without a clearance.
Location	Yellow on black, yellow border	The taxiway or runway you are presently on.
Direction	Black on yellow	The designation and direction of intersecting taxiways.
Destination	Black on yellow with arrow	The direction to a destination — a runway, terminal, or cargo area.
Information	Black on yellow	Frequencies, noise procedures, and local conditions.

Sign	Colours	Meaning
Runway distance remaining	White on black	Thousands of feet of runway remaining.
— End —		

Reading a Sign Array

A location sign appears with direction signs to state where you are and what you may turn onto. A runway holding position sign collocated with a location sign means the runway entrance is immediately ahead.

The **ILS critical area boundary sign** faces the taxiway and marks the exit boundary. You are clear of the critical area when you have passed it.



Figure 4.5 Runway Holding Position Sign



Figure 4.6 Holding Position Sign at the Beginning of a Takeoff Runway

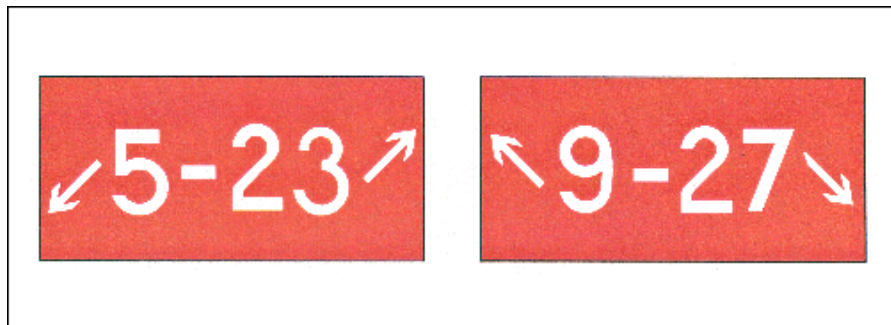


Figure 4.7 Holding Position Sign, Taxiway Intersecting Two Runways



Figure 4.8 Holding Position Sign, ILS Critical Area

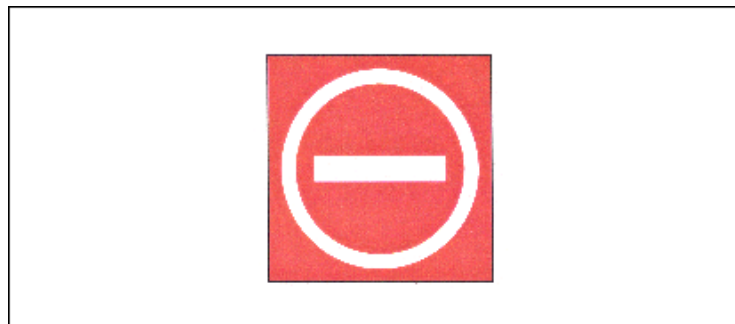


Figure 4.9 No Entry Sign

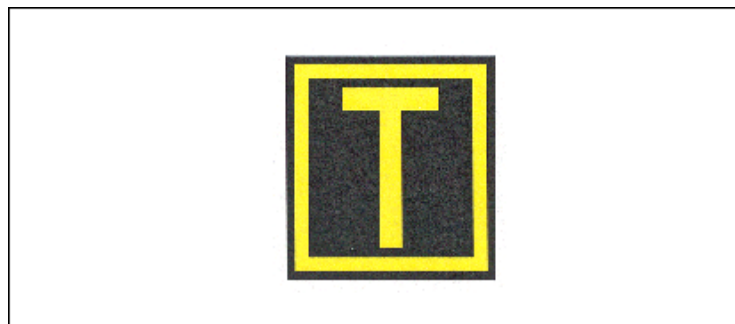


Figure 4.10 Taxiway Location Sign



Figure 4.11 Taxiway Location Sign Collocated with a Runway Holding Position Sign

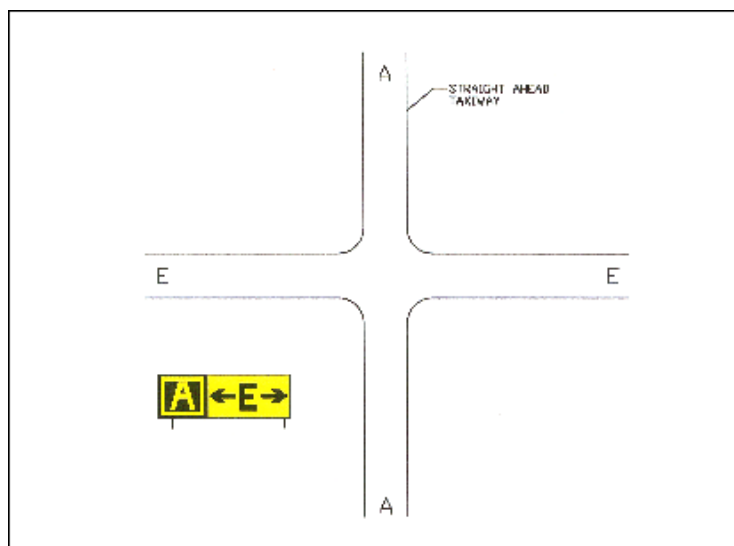


Figure 4.12 Direction Sign Array for a Simple Intersection

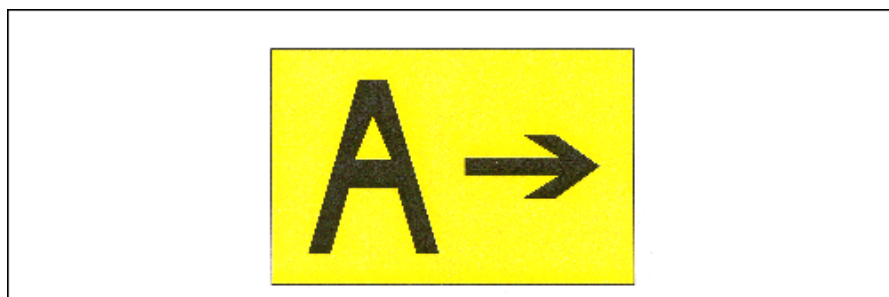


Figure 4.13 Direction Sign for a Runway Exit



Figure 4.14 Destination Sign, Different Taxi Routes to Two Runways



Figure 4.15 Holding Position Sign for Runway Approach and Departure Areas

Light	Meaning
Taxiway edge lights	Blue.
Runway edge lights	White; amber in the last 2,000 feet or half the runway, whichever is less.
Runway centerline lights	White; alternating red and white from 3,000 to 1,000 feet remaining, red in the last 1,000 feet.
— End —	

Caution: A stop bar that fails to extinguish is a system fault, not a clearance. Hold and query ATC.

4c.1 General

See section [1c.1 General](#).

4c.2 Taxi Clearance

Acknowledge a taxi clearance with the full flight number and a verbatim read back of the runway assigned, any hold short instruction, and any runway crossing.

Where the clearance is long or complex, request it in segments rather than accepting a clearance that cannot be read back accurately.

Note: A read back is a verification, not an acknowledgment. Read back what was heard, not what was expected.

4f.1 Weight

4f.1.1 Receipt of the Load Closeout

The load closeout is normally received during taxi. Pushback and taxi may commence without it. **Takeoff may not.** See section [1p.5.2 Load Closeout](#).

When the closeout is available, and as soon as workload permits, the pilot monitoring will:

- check the closeout for accuracy, including the flight number and the aircraft number
- check or enter zero fuel weight and zero fuel weight CG on the FMS fuel prediction page
- check or enter any performance change on the FMS takeoff page — V1, VR, V2, flap setting and stabilizer trim, and the flex temperature where applicable

Caution: Performance data uplink is not available after engine start. Data uplinked before start may still be inserted, but any other change must be entered by hand.

4f.1.2 Checks Against the TPS

Work the TPS Quick Reference Guide in order — weight, then temperature, then wind. See section [1p.4.1 TPS Quick Reference Guide](#).

Table 4.1 Closeout Checks Against the TPS

Check	Flex thrust planned	Maximum thrust planned
Weight	TOW less than ATOW	TOW less than PTOW
Temperature	Current temperature less than AT	Current temperature less than the TPS planned temperature
Wind	Consistent with the wind the TPS assumed	Consistent with the wind the TPS assumed
— End —		

Confirm the gross weight shown by the aircraft, less anticipated taxi burn, does not exceed the limiting weight for takeoff.

Requirements for an unplanned tailwind are in section [1p.4.7 Unplanned Tailwind Takeoff](#).

4f.1.3 Closeout Brief

The pilot monitoring briefs the closeout aloud when it is received. The brief follows a fixed order so that the captain hears the same sequence every time and can tell immediately when something is missing.

Table 4.2 Closeout Brief Sequence

Order	Item	Stated as
1	Receipt and revision	That the closeout is aboard, and its revision number
2	Zero fuel weight	The difference from planned, and its direction
3	Fuel	The difference from planned, and its direction
4	Takeoff weight	The resulting difference from planned takeoff weight
5	Restricted articles	Present or none
6	Center of gravity	Takeoff CG in percent MAC
7	Souls on board	The total from the closeout
8	Security	The security check status
9	Wind, weight, temperature	Whether each is within the TPS assumptions
10	Takeoff configuration	Flap setting, thrust rating, runway, and assumed temperature
— End —		

Weight Build-Up

Items 2 through 4 are briefed as a build-up, not as three separate figures. State the zero fuel weight difference, then the fuel difference, then the takeoff weight difference they produce. The captain can then verify the arithmetic as it is spoken, and a transposed figure is audible rather than silent.

Example

"We've got the closeout, revision zero. We are over the planned zero fuel weight by five hundred pounds, up three hundred on fuel, so eight hundred over the planned takeoff weight. You've got no restricted articles, a CG of twenty-nine point five percent, one hundred eighty-five souls on board, security checks okay. Winds, weights and temps are good for a flap one, twenty-two K takeoff off runway two-niner right, with an assumed temperature of forty-seven degrees."

Stating the Negative

Restricted articles and security are briefed **either way**. "No restricted articles" is a statement that the block was read and was empty. Silence is not.

Where an Item Is Not Good

Items 9 and 10 are a verdict, not a reading. Where wind, weight, or temperature falls outside what the TPS assumed, say which one and by how much, and do not continue to the configuration until it is resolved. See section [1p.4.1 TPS Quick Reference Guide](#).

Where the takeoff configuration differs from the departure briefing — a different flap setting, thrust rating, runway, or a change from flex to maximum thrust — state the change explicitly and brief the difference.

Caution: A closeout brief that ends in the takeoff configuration is confirming the configuration the aircraft is actually set for. Where the FMS has not yet been updated to match, say so.

4f.1.4 Independent Verification

Weight and stabilizer trim are verified by both pilots from **different sources**. The verification is not complete where both pilots read the same document.

Item	Pilot flying	Pilot monitoring
Weight	States the gross weight shown by the aircraft	Reads takeoff weight directly from the load closeout
Stabilizer trim	Reads the gross weight CG from the FMS and confirms trim is set to the corresponding value	Reads takeoff CG directly from the load closeout
— End —		

Where the two weights differ by more than **1,000 lb**, confirm the difference is reasonable for the taxi fuel burned. Where the two CG values differ by more than **5.0 percent MAC**, resolve the discrepancy before takeoff.

Confirm V1 and V2 are displayed on the primary flight display, and that the stabilizer trim is physically set to the value the closeout requires.

Warning: A closeout that does not match the aircraft is not a paperwork error. It is an unverified center of gravity and an unverified takeoff weight.

4f.1.5 Changes After Pushback

Where a weight change occurs after pushback — a passenger removal, a fuel adjustment, or a runway change — verify the revised figures before takeoff.

Where the runway assigned changes during taxi, recompute takeoff performance for the runway and intersection actually assigned. Do not depart on data computed for a different runway.

Warning: A runway change during taxi invalidates the takeoff data. Recompute before entering the runway. See section [5g.1 General](#).

Chapter 5: Takeoff

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5g.1 General

The takeoff briefing establishes the plan for the departure, the rejected takeoff decision, and the engine failure procedure. Brief what is different, threatening, or uncertain about the departure. Do not recite what is standard.

For aircraft-specific procedure, see the applicable AOM> 6> Takeoff.

Runway Verification

Before entering the runway, both pilots will confirm:

- the runway assigned matches the runway entered
- the heading indication agrees with the runway heading
- the runway length available matches the performance data used
- the takeoff data loaded matches the runway and intersection in use

Warning: An intersection departure requires performance data computed for that intersection. Do not depart from an intersection using full-length data.

Takeoff Data

Takeoff data is produced by the Takeoff Performance System. See section [1p.2.2 Takeoff Performance System \(TPS\)](#) and section [1p.3.6 ACARS TPS](#).

Verify the takeoff data against:

- final weight and balance figures
- the runway and intersection in use
- the flap setting and thrust rating selected
- the ambient conditions and any runway contamination

5g.2 Takeoff Minimums

Takeoff minimums are established by OpSpecs and are shown on the airport chart. Lower-than-standard takeoff minimums require the runway lighting and RVR reporting specified in OpSpecs.

The High Minimums designation does not increase takeoff minimums. See section [21.8 High Minimums Captain](#).

Controlling RVR

Where multiple RVR reports are available, the controlling values are those specified in OpSpecs for the minimums in use. All required RVR reports must be at or above the minimum.

Captain's Takeoff

The **captain conducts the takeoff** — ground roll through cleanup and completion of the After Takeoff checklist — where the visibility is at or below:

- **RVR 1600**, or
- **1/4 statute mile** where RVR is not reported

This applies regardless of whose leg it is. Where the first officer is the pilot flying for the sector, control transfers to the captain before the takeoff and back after the After Takeoff checklist is complete. Brief the transfer during the departure briefing.

5g.3 Runway Conditions

Surface Condition

Runway surface definitions and the Runway Condition Assessment Matrix are in section [24.5 Field Condition Reporting](#). Apply the reported Runway Condition Code to the takeoff performance computation.

Contaminated Runway

Apply the performance penalty published for the reported contaminant and depth. Where the contaminant or depth is not reported, or exceeds the tabulated values, do not depart.

Operating Limits

Operations are suspended where accumulation on the runway operating surface exceeds:

Precipitation	Depth
Dry snow	Greater than 2 inches
Wet snow, slush, or standing water	Greater than 1/4 inch
— End —	

Note: Operations may continue in up to 1/2 inch of wet snow, slush, or standing water with the concurrence of both the captain and the dispatcher. Consider wind, runway length, and the proportion of the surface affected.

Braking Action

A braking action report of **NIL** prohibits takeoff. Advise ATC and return to the gate or hold until the condition is corrected.

Minimum Runway Widths

Operate only on runways meeting the minimum width published for the fleet in the airport analysis.

5g.4 Rejected Takeoff

The captain makes the reject decision and executes the reject.

Below 80 Knots

Reject for any condition that would make the aircraft unsafe or unable to fly.

80 Knots to V1

Reject only for:

- fire or fire warning
- engine failure
- predictive or reactive windshear warning
- any condition that would make the aircraft unsafe or unable to fly

At or Above V1

Continue the takeoff. Handle the condition in the air.

Warning: A reject initiated at or above V1 may exceed the runway available. V1 is the decision point, not the initiation point.

After the Reject

Bring the aircraft to a stop on the runway unless conditions require otherwise. Advise ATC. Do not set the parking brake until brake temperature is assessed. Consider the need for an evacuation. See section [1n.1 General and the applicable QRC/QRH](#).

Report every rejected takeoff. See section [22.10 Safety Reporting](#).

5g.5 Line Up and Wait

Read back every line up and wait clearance including the runway.

While holding in position:

- maintain a listening watch on the tower frequency
- maintain external vigilance for traffic on final
- if a takeoff clearance is not received within a reasonable period, query the tower

Note: At night or in reduced visibility, use exterior lighting to maximize conspicuity while holding in position.

5g.6 Foreign Airports

At a foreign airport, values may be published in units other than those the aircraft displays. **Convert before use, and state the units when briefing.**

5g.6.1 Unit Conversions

Table 5.1 Common Unit Conversions

Quantity	Conversion
Metres to feet	Multiply by 3.28
Feet to metres	Multiply by 0.30
Kilometres to nautical miles	Multiply by 0.54
Nautical miles to kilometres	Multiply by 1.85
Metres per second to knots	Multiply by 2
Kilograms to pounds	Multiply by 2.2
Pounds to kilograms	Multiply by 0.45
Litres to US gallons	Multiply by 0.26
Hectopascals to inches of mercury	Divide by 33.86
Inches of mercury to hectopascals	Multiply by 33.86
Degrees Celsius to Fahrenheit	Multiply by 1.8 and add 32
— End —	

Quick Approximations

For a rough check in flight, where an exact figure is not required:

- **metres to feet** — multiply by 3 and add ten percent
- **kilometres to nautical miles** — multiply by 5 and divide by 9
- **metres per second to knots** — double it

Caution: A quick approximation is for situational awareness only. Runway length, obstacle clearance, and fuel are computed on the exact conversion.

5g.6.2 Altimetry

Outside the United States the altimeter setting may be given in **hectopascals**, and altitude may be assigned in **metres**.

Set the units the aircraft expects, and confirm the setting with the other pilot whenever the units change. Where metric altitude assignment is in use, both pilots confirm the assignment and the conversion. See section [7g.6 International](#).

QNH is the setting that makes the altimeter read field elevation on the ground. QFE makes it read zero. **QFE is not used.** Where a QFE setting is issued, request QNH.

5g.6.3 Runway Length and Visibility

Runway length may be published in metres and visibility in metres or kilometres.

Where the takeoff minimums are expressed in metres, convert to the units on the TPS before comparing them. Do not compare a metric visibility against an RVR value without converting.

Warning: A runway published as 2,400 is 7,874 feet if the figure is metres and 2,400 feet if it is not. Confirm the units before computing takeoff performance.

5g.6.4 Phraseology and Readback

Where a clearance is given in units other than those in use, read it back **in the units given** and then state the converted value. This makes a conversion error audible to the controller.

Where the controller's phraseology or the readback is not fully understood, request it again. Do not act on a partially understood clearance at a foreign airport.

5c.1 Takeoff Clearance

Acknowledge a takeoff clearance with the full flight number and a verbatim read back of the clearance including the runway.

Where the clearance includes an initial heading or altitude different from that briefed, both pilots will confirm the change and reset the flight guidance before takeoff.

5w.1 Windshear

Do not depart where windshear is reported, suspected, or indicated. See section [1w.2 Windshear / Microburst](#).

5w.2 Crosswind and Tailwind

Crosswind and tailwind limits are published in the applicable AOM and are reduced for a contaminated runway. Verify the reported wind against the limit for the runway condition in use.

Caution: Gust value is included when assessing the crosswind limit.

Chapter 6: Departure

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6n Non-Normal/Emergency

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6g.1 General

Departure extends from takeoff through the end of the initial climb. The sterile flightdeck applies below **10,000 feet AFE**. See section [1g.5 Sterile Flightdeck Period](#).

For aircraft-specific procedure, see the applicable AOM> 7> Departure.

Departure is a high workload phase flown close to terrain at low energy. The defence against that is that the profile was briefed before the aircraft moved, and is flown as briefed.

6g.2 Departure Procedures

6g.2.1 Assigned Procedure

Fly the departure procedure as cleared. Where the clearance differs from the procedure loaded, resolve the difference with ATC **before takeoff**.

Both pilots verify the loaded procedure against the chart — the transition, the runway, and the first two waypoints in sequence. A procedure loaded for the wrong runway sequences correctly and flies the wrong track.

Warning: A clearance to "fly the departure" is not a clearance to fly what is loaded. Verify what is loaded matches what was cleared.

6g.2.2 Types of Departure

Type	Purpose	Obstacle clearance
SID	ATC routing and traffic flow	Assured on the published track at the published gradient
ODP	Obstacle avoidance where the standard climb gradient is insufficient	Assured on the published track
Diverse departure	No published procedure; obstacle assessment showed none required	Assured on a straight-out climb at the standard gradient
Radar vectors	ATC assigned	ATC is responsible once radar identified, within the vectoring area
— End —		

6g.2.3 Obstacle Departure Procedure

Where an ODP applies, brief the climb gradient required and the altitude at which it is satisfied.

An ODP may not be assigned by ATC. Where one exists for the runway and the conditions require it, fly it and advise ATC.

6g.2.4 Climb Gradient

Verify the aircraft meets the published climb gradient at the planned weight and the ambient conditions. Where it cannot, request an alternate procedure or a weight reduction.

A published gradient is stated in feet per nautical mile. Convert to a rate of climb using groundspeed:

Required rate of climb (fpm) = gradient (ft/NM) × groundspeed (kt) ÷ 60

At 200 knots groundspeed, a 300 ft/NM gradient requires 1,000 fpm. The requirement grows with groundspeed, so a tailwind on the departure track makes a marginal gradient worse.

Caution: The published gradient assumes all engines operating. It is not the engine-out gradient, and meeting it is not evidence that the engine-out case is satisfied. See section 6g.3.

6g.3 Engine-Out Procedures

6g.3.1 The Two Procedures Are Different

The engine-out procedure is published in the airport analysis and **may differ from the assigned departure procedure**. Where the two differ, the engine-out procedure is the one that was analyzed for obstacle clearance with an engine inoperative.

Brief, for every departure:

- the engine-out routing, including the first turn and its direction
- the engine-out acceleration height, above field level and above mean sea level
- the point at which the two procedures diverge
- where the aircraft will level off, and the fuel dump or overweight landing consideration if applicable

Warning: The assigned departure procedure does not guarantee obstacle clearance with an engine inoperative. Where an engine fails, fly the engine-out procedure and tell ATC what you are doing.

6g.3.2 Acceleration Height

Engine-out acceleration height is stated on the TPS for each runway, both above field level and above mean sea level. See section [1p.3.3 Airport Analysis Data](#).

Where terrain requires it, the acceleration height is raised above the standard 1,000 feet AFL. A raised acceleration height lengthens the period at takeoff thrust and delays flap retraction — brief it, because it does not feel normal.

6g.3.3 Obstacle Clearance Corridor

With an engine inoperative after V1, obstacle clearance is assured only:

- along the extended runway centerline, or
- along the published engine-out departure path

It is assured nowhere else. Where it becomes necessary to leave the corridor, the pilot in command assures obstacle clearance visually or by reference to terrain displays.

See section [1p.2.6 Takeoff Obstructions](#).

6g.3.4 Special Engine-Out Procedures

At a terrain-constrained airport the TPS airport notes will state that special engine-failure procedures exist for one or more runways.

Where that note appears, the procedure is obtained from the airport pages and briefed in full. Do not depart from a runway with a special engine-out procedure without having briefed it.

6g.4 Climb

6g.4.1 Altitude Awareness

See section [1g.9 Altitude Awareness](#).

Both pilots confirm each altitude clearance, and the altitude selected is verified against the clearance before it is flown.

6g.4.2 Speed Restrictions

Comply with the **250 knot** restriction below 10,000 feet MSL unless a higher speed is required for safe operation, in which case advise ATC.

Comply with any speed published on the departure procedure. A speed restriction on a SID remains in force through a level change unless ATC cancels it.

Outside the United States the low-altitude speed limit and the altitude it applies to vary. Verify both before departure.

6g.4.3 Acceleration and Cleanup

Accelerate and retract at the altitude briefed. Where the briefed acceleration altitude is raised for terrain or noise, it is stated in the departure briefing and not changed in the climb without both pilots agreeing.

6g.4.4 Transition Altitude

Set altimeters to standard **climbing through** the transition altitude. Both pilots confirm the setting. See section [1g.11 Altimetry](#).

6g.4.5 Climb Profile

Climb at the speed and thrust published for the fleet. Where a level-off is issued below the planned cruise altitude, advise the dispatcher if the fuel penalty is material — a long level-off at low altitude is a fuel event, not just a delay.

6g.5 Noise Abatement

Where a noise abatement departure procedure applies, comply with the published profile.

Noise abatement is complied with **only where it does not conflict with safety**. The captain may deviate from a noise abatement procedure for terrain, weather, windshear, traffic, or an aircraft malfunction, and will report the deviation.

Note: A noise abatement procedure is never a reason to accept a degraded climb profile, a late configuration change, or an unstable departure.

6g.6 International

Transition altitudes and levels vary by state and are published on the chart. Verify the transition altitude for the departure airport **before takeoff**, not in the climb.

Where the departure state applies metric altitude assignment, see section [7g.6 International](#).

Where the departure state issues altimeter settings in hectopascals, see section [5g.6.2 Altimetry](#).

6c.1 General

See section [1c.1 General](#).

On initial contact with departure control, state the assigned altitude and, where applicable, the assigned heading and airspeed.

Where a clearance received in the climb differs from what was briefed, both pilots confirm the change and reset the flight guidance before it is flown.

6c.2 Flight Attendants

Notify the cabin at the end of the sterile period using the signal published for the fleet.

Where turbulence is anticipated, advise the expected onset and duration and direct the flight attendants to remain seated. See section [1c.4 Flight Attendants](#).

6c.3 Passengers

Make the after-takeoff announcement when workload permits above 10,000 feet. Include flight time, expected ride, and destination weather.

Where the departure was delayed, or the ride on climb-out was poor, say so. See section [1c.5 Passengers](#).

6n.1 Return to the Departure Airport

A return to the departure airport is flown at the captain's discretion. It is frequently the right answer and rarely the fast one.

Consider before committing:

- **Weight.** The aircraft is at or near maximum takeoff weight. An immediate return is an overweight landing unless fuel is burned or dumped. See section [10n.1 Overweight Landing](#).
- **Time.** Burning down to maximum landing weight takes time that a fire or a structural concern does not allow.
- **The runway.** Landing distance at takeoff weight on the departure runway may not be available, particularly where the departure was performance limited.
- **The weather.** The departure airport was legal for takeoff, which is a lower standard than legal for approach and landing.

Where the condition is time-critical, land at the nearest suitable airport at whatever weight the aircraft is at, and accept the overweight landing. Advise ATC first, then the dispatcher, then the cabin.

Accomplish the applicable QRC/QRH procedure. Where the return follows an engine failure, fly the engine-out procedure until obstacle clearance is assured before manoeuvring for the return.

Chapter 7: Enroute

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7g.1 General

Cruise is the phase in which planning for the arrival is accomplished. Workload is lowest and information is most complete.

For aircraft-specific procedure, see the applicable AOM> 8> Enroute.

Monitoring

Both pilots maintain awareness of position, fuel state, weather, and aircraft configuration. One pilot monitors at all times.

RVSM

RVSM operation requires the equipment and procedures published in OpSpecs. Where an RVSM significant system becomes inoperative, advise ATC and request an altitude outside RVSM airspace.

7g.2 Fuel Monitoring

Compare actual fuel remaining against the computer flight plan at each waypoint, or at intervals not exceeding **30 minutes** where waypoints are widely spaced.

Advise the dispatcher where the deviation from plan exceeds the fleet tolerance or where arrival fuel is projected below the release requirement.

Minimum Fuel

Declare **minimum fuel** to ATC when arrival at the destination with less than planned reserve is expected. Minimum fuel is an advisory. It is not a declaration of emergency.

Emergency Fuel

Declare an emergency when the calculated usable fuel on landing at the nearest suitable airport would be less than final reserve. See section [1n.2 Declaring an Emergency](#).

7g.3 In-Seat Rest

Source: 14 CFR 117; 14 CFR 121.542

Controlled rest in the seat — a planned nap by one pilot at the controls — is **not authorized** at Northern Airways. The FAA does not permit it in 14 CFR Part 121

operations. It is an ICAO practice, permitted by a number of foreign authorities, and it is not available to a US certificate holder.

Both pilots remain awake and on duty for the whole of the flight, except during a rest period taken in an approved crew rest facility on an augmented operation. See section [7g.4 Crew Rest Facilities](#).

Warning: Sleeping at the controls is not a fatigue mitigation. It is a loss of one crewmember and, on a two-pilot flight deck, of the monitoring function entirely.

If You Are Too Fatigued to Continue

Fatigue that develops in flight is managed by the mitigations in section 7g.3.1, and reported. It is not managed by sleeping.

7g.3.1 In-Flight Fatigue Mitigation

Where a crewmember becomes fatigued in flight:

- tell the other pilot; fatigue is stated plainly, not concealed
- transfer the flying duty where that reduces workload
- raise the cabin temperature setting toward the cool end, and increase airflow
- take a meal or a caffeinated drink
- leave the flight deck briefly where relief and security procedure allow
- brief the arrival early, while workload is low
- request a lower workload routing, a delay, or holding rather than accept a rushed arrival

Where fatigue reaches the point that safe completion is in doubt, advise the dispatcher and consider a diversion. A diversion for fatigue is supported.

Report the occurrence. See section [21.11 Fatigue Risk Management](#).

7g.4 Crew Rest Facilities

On an augmented operation, rest is taken in the facility approved for the fleet. The captain establishes the rest rotation before departure and designates the pilot in command during each rest period.

7g.5 ETOPS

Entry

Before the ETOPS entry point, verify:

- fuel at entry meets or exceeds the release requirement
- all ETOPS significant systems are operative
- the weather at each ETOPS alternate is at or above the applicable minima

Where any condition is not met, do not enter ETOPS airspace. Coordinate an alternate routing with the dispatcher.

In-Flight Verification

Where an ETOPS system verification is required, accomplish it before entering ETOPS airspace. Maintenance procedures require a minimum of **1 hour** of flight time before entry for any system verification.

Where the system operates normally with no discrepancy observed, make an in-flight verification entry in the eAML. The recommended format is:

PRIOR TO ENTERING ETOPS; IN-FLIGHT VERIFICATION CHECK COMPLETED,
NO DISCREPANCIES NOTED

Where the system cannot be verified as satisfactory before entry, the aircraft is **not** authorized to enter ETOPS airspace. Contact the dispatcher to coordinate a non-ETOPS reroute or a diversion to a maintenance base, and make a mechanical discrepancy entry in either case.

In ETOPS Airspace

Monitor the ETOPS alternates continuously. Advise the dispatcher of any change in aircraft status.

Diversion

An ETOPS diversion is flown to the nearest suitable alternate consistent with the conditions. See section [7n.1 Diversion](#).

7g.6 International

Metric Altitude

Where metric altitude assignment is in use, set and cross-check the metric display in accordance with the applicable AOM. Both pilots confirm the assignment and the conversion.

Oceanic Clearance

Obtain the oceanic clearance before the oceanic entry point. Read back the clearance in full. Confirm the cleared track, altitude, and Mach against the loaded route.

Warning: Do not enter oceanic airspace without a clearance. Advise ATC and hold or divert as required.

7c.1 General

Maintain a continuous listening watch on ACARS, SELCAL, or the appropriate VHF/HF frequency. See section [1c.3 Dispatch](#).

7c.2 Position Reporting

Make position reports as required by the airspace. Where CPDLC and ADS-C are in use, verify the connection is active and report by voice if it is not.

7c.3 Passengers

Make the enroute passenger announcement when workload permits. Where the ride is expected to deteriorate, advise the cabin before it does. See section [1c.5 Passengers](#).

7c.4 Lost Communications

Where two-way communication is lost, comply with 14 CFR 91.185 in domestic airspace, and with the procedure published for the airspace elsewhere.

Attempt to re-establish communication on:

- the previously assigned frequency
- the emergency frequency
- company radio, SATCOM, or ACARS
- relay through another aircraft

7w.1 Turbulence

7w.1.1 Operations

Turbulence penetration speed and autoflight configuration are published in the applicable AOM. Do not chase altitude in severe turbulence.

7w.1.2 Procedures

Intensity	Action
Light	Seat belt sign on. Advise the cabin.
Moderate	Seat belt sign on. Flight attendants seated. Advise ATC. PIREP.
Severe	Seat belt sign on. All occupants seated. Turbulence penetration speed. Advise ATC. PIREP. Advise the dispatcher. File a safety report.
— End —	

7w.1.3 Communications

Advise the cabin of expected onset and duration. Where moderate or greater turbulence is anticipated, direct the flight attendants to be seated and secured. See section [1c.4 Flight Attendants](#).

7w.1.4 Terminology

Turbulence intensity is reported in accordance with the definitions in the Aeronautical Information Manual. Report the aircraft type with the PIREP.

7w.2 Thunderstorms

7w.2.1 Airborne Weather Radar

The pilot monitoring operates, tilts, and interprets the radar actively. A radar left at a fixed tilt is not in use.

Caution: Attenuation makes the far side of a strong cell appear clear. A radar shadow is not a gap.

7w.2.2 Circumnavigation

Circumnavigate laterally. Do not overfly a cell and do not pass beneath one. Avoidance distances increase with altitude and are published in the applicable AOM> 15> Adverse Weather. See section [1w.3 Thunderstorms](#).

Request a deviation early. A deviation requested well ahead is routinely approved; the same request close in may not be. Where a required deviation is not approved, advise ATC that the deviation is required and turn.

7w.3 Cold Fuel

Monitor fuel temperature against the minimum published for the fuel type in use. Where the temperature approaches the limit, request a lower altitude, a warmer routing, or an increase in Mach in accordance with the applicable AOM.

7w.4 Volcanic Ash

Avoid forecast and reported volcanic ash by the distance published in the advisory. Where ash is encountered, accomplish the applicable QRH> Volcanic Ash procedure and exit by the shortest route consistent with terrain.

7f.1 Fuel**Fuel Log**

Record fuel remaining against the computer flight plan at each waypoint, or at intervals not exceeding 30 minutes where waypoints are widely spaced. The fuel log is the record used to identify a developing shortfall before it becomes a diversion. See section [7g.2 Fuel Monitoring](#).

Reclearance

Where the flight is planned under re-dispatch, verify fuel at the re-release point against the requirement before accepting the reclearance. See section [2.2.3 Re-Release](#).

7m.1 Inflight Discrepancies

Advise Maintenance Control through the dispatcher of any mechanical irregularity affecting the operation. Record the discrepancy in the eAML after landing. See section 3m.2 eAML.

7s.1 Flightdeck

The flightdeck door remains closed and locked. Opening in flight is accomplished only in accordance with the AOSSP procedure. See section [1s.2 Flightdeck Security](#).

7n.1 Diversion

The decision to divert is made jointly by the captain and the dispatcher. Either may initiate it.

Consider:

- the nature of the condition and the time available
- runway length, width, and condition
- approach capability and weather
- maintenance, medical, and handling capability on the ground
- terrain and fuel required

Advise ATC, then the dispatcher, then the cabin. Where the condition is time-critical, advise ATC and divert. Coordinate afterward.

7n.2 Emergency Equipment

Emergency equipment carried aboard is listed in the applicable AOM and is verified at preflight. See section [3n.1 Aircraft Equipment](#).

In Flight

Where an item of emergency equipment is used or found unserviceable in flight:

- continue the flight; an item used in flight does not by itself require a diversion
- record what was used, where it was stowed, and its condition, in the eAML after landing
- advise the dispatcher where the item affects the capability of the aircraft for the next sector — an expended extinguisher, a used protective breathing unit, a depleted oxygen bottle

Locations

Both pilots know the location of the flightdeck extinguisher, the protective breathing equipment, the crash axe, the escape rope or slide, and the emergency oxygen.

Confirm the locations during the preflight, not during the event.

Warning: An item of required emergency equipment that is missing or unserviceable at the next departure is a dispatch matter. The aircraft will not depart until it is replaced or deferred under the MEL. See section [3n.1 Aircraft Equipment](#).

7n.3 Navigation and Altitude

Where a navigation system failure affects the ability to comply with the clearance, advise ATC and request routing consistent with the remaining capability.

Where an altitude cannot be maintained, advise ATC, state the altitude that can be maintained, and comply with the offset procedure published for the airspace.

7m.2 Early Notification

Advise Maintenance Control through the dispatcher as early as practicable of any condition that will require maintenance action on arrival. Early notification allows parts, personnel, and a gate to be arranged before the aircraft lands.

Include the condition observed, the phase in which it occurred, the crew action taken, and the current status.

Chapter 8: Arrival

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8g.1 General

Arrival extends from top of descent to the beginning of the approach. Planning accomplished in cruise reduces workload during descent.

For aircraft-specific procedure, see the applicable AOM> 9> Arrival.

Almost everything that makes an approach unstable was decided before top of descent. The arrival is where the approach is set up to succeed or made difficult.

8g.2 Descent Planning

Complete descent planning **before top of descent**, while workload is low and the information is complete.

Include:

- destination weather, field condition, and braking action
- runway in use and the approach expected
- landing performance for the conditions
- fuel at destination, and the plan if holding is required
- terrain and minimum safe altitudes
- the go-around procedure and the missed approach routing

8g.2.1 Information Sources

Obtain the destination ATIS, and the current METAR where the ATIS is not available. Where the arrival is more than a short sector, obtain an update approaching top of descent — the package issued with the release is old by then.

Check for a TWIP or a terminal forecast amendment. See section [24.3 Significant Terminal Weather Information](#).

8g.2.2 Landing Performance

Compute landing performance for the runway, the reported wind, and the reported surface condition. Where the reported condition changes, **recompute**. See section [1p.6 Landing Performance](#).

Confirm the computed landing distance against the landing distance available with the intended exit, not the runway end.

Warning: A braking action report of NIL prohibits landing.

8g.2.3 Terrain

Establish the minimum safe altitude for the arrival area and the sector minimum altitudes on the approach chart before descending.

Where the arrival is at a special qualification or terrain-constrained airport, review the airport qualification page during the arrival planning, not on the approach. See section [21.9.4 Airport Qualification Pages](#).

8g.3 Descent

8g.3.1 Descent Planning Arithmetic

For a rough top of descent point, use **three times the altitude to lose, in thousands of feet**, in nautical miles. To lose 30,000 feet takes about 90 NM.

Add for:

- a tailwind — roughly 1 NM per 10 knots of tailwind component
- speed to lose at level-off — roughly 1 NM per 10 knots of speed reduction
- an expected level-off or a crossing restriction well before the field

Note: The rule of thumb is for the cross-check, not the plan. The FMS computes the descent; the arithmetic tells you whether to believe it.

8g.3.2 Managing the Descent

Monitor progress against the planned profile at each constraint. A descent that is behind profile does not recover itself — it is recovered with speedbrake, an early speed reduction, or by asking ATC for track miles.

Ask early. A request for track miles 40 NM out is routinely approved. The same request at 15 NM usually is not, and by then the only remaining options are degraded.

Caution: Arriving high and fast at the final approach fix is the origin of most unstable approaches. The decision that prevents it is made in the descent, not on final.

8g.3.3 Transition Level

Set altimeters to local **descending through** the transition level. Both pilots confirm the setting. See section [1g.11 Altimetry](#).

Where the arrival is outside the United States, verify the transition level for the arrival airport during the descent planning.

8g.3.4 Cabin Preparation

The cabin needs time. Advise the flight attendants early enough that the cabin is secure without a rushed final call — see section [8c.4 Flight Attendants](#).

At a high elevation destination, the cabin is secured earlier because the descent from cruise is shorter in altitude terms and the ride is often worse. See section [1i.2 South \(Latin\) America](#).

8g.4 Arrival Procedures

Fly the arrival as cleared. Where a descend-via clearance is issued, verify every published constraint is loaded and armed, and cross-check against raw data.

Note: A crossing restriction met by the automation is verified, not assumed. See section [1g.9 Altitude Awareness](#).

8g.4.1 Descend Via

A **descend via** clearance authorizes descent at pilot discretion to comply with all published altitude and speed constraints on the procedure.

Before accepting it, confirm:

- the correct arrival and transition are loaded
- every published constraint is present in the FMS
- the constraints in the FMS match the chart, constraint by constraint
- the bottom altitude is set

Where a constraint is missing from the FMS, it is still a constraint. Fly it.

8g.4.2 Speed Restrictions

Comply with published and assigned speed restrictions. Advise ATC where a restriction cannot be met.

A speed published on the arrival remains in force through an altitude change unless ATC cancels it. Where ATC cancels altitude restrictions, **confirm whether the speed restrictions remain**.

8g.4.3 Runway or Approach Change

A runway or approach change during the arrival invalidates the briefing, the landing performance, and frequently the descent profile.

Where a change is issued:

1. Confirm the aircraft can accept it before agreeing. Where it cannot be flown safely, decline it.
2. Reload and re-verify the procedure.
3. Recompute landing performance for the new runway.
4. Re-brief the changed items — not the whole briefing, the changes.

Warning: Accepting a late runway change is a decision, not an instruction to comply with. Where accepting it would produce a rushed or unstable approach, decline and request the original runway or delay vectors.

8g.5 Holding

Where holding is expected, determine and advise ATC of the Expect Further Clearance time and the fuel available for holding.

8g.5.1 Entering the Hold

Confirm the holding fix, the inbound course, the turn direction, the leg length or time, and the maximum holding speed for the altitude.

Where the hold is not published, read back the full clearance including all five elements.

8g.5.2 Fuel in the Hold

Establish, before entering the hold, the **fuel figure at which the flight will leave for the alternate**. State it out loud and note it.

Do not hold beyond the fuel required to reach the alternate with final reserve intact. See section [7g.2 Fuel Monitoring](#).

Advise the dispatcher of the holding, the EFC, and the divert fuel figure. The dispatcher may have information on the delay that ATC does not pass.

8g.5.3 Declaring

Declare **minimum fuel** to ATC when arrival at the destination with less than planned reserve is expected. It is an advisory, not an emergency.

Declare an **emergency** when the calculated usable fuel on landing at the nearest suitable airport would be less than final reserve. See section [1n.2 Declaring an Emergency](#).

8g.6 Diversion Decision

The decision to divert is made jointly by the captain and the dispatcher. Either may initiate it. See section [7n.1 Diversion](#).

A diversion decided early is an ordinary flight to another airport. A diversion decided late is a fuel emergency. The difference is when the divert fuel figure was established and whether it was honoured.

Where the destination weather is deteriorating toward minimums, or holding is open-ended with no EFC, brief the alternate as though you are going there.

8c.1 General

See section [1c.1 General](#).

8c.2 Briefings

Arrival Briefing

The arrival briefing is made before top of descent and covers the arrival, the approach, the landing, and the go-around.

Brief what is **different, threatening, or uncertain** about this arrival. Do not recite the chart — both pilots have it.

Include:

- the arrival and any descend-via constraint
- the approach expected and the runway in use
- landing performance and the planned exit
- the go-around procedure and the missed approach routing
- the fuel available and the point at which the flight will divert
- any threat specific to this arrival — terrain, weather, a special qualification airport, an unfamiliar procedure, a known ATC practice

Re-Briefing

Where the runway, the approach, or the landing performance changes, re-brief the changes. A re-brief is short and covers only what moved.

8c.3 Standard Terminal Arrival Route (STAR)

Where a descend-via clearance is issued, both pilots verify every published constraint is loaded and armed, and cross-check against raw data.

Where ATC issues an altitude that cancels the published constraints, confirm whether the speed constraints remain in effect.

Note: "Descend via" and "descend and maintain" are different clearances.

Confirm which was issued.

8c.4 Flight Attendants

Advise the cabin **10 minutes** before landing and at the start of the sterile period on descent through 10,000 feet AFE.

The flight attendant in charge reports the cabin secure for landing. Landing will not be made without that report — where it has not been received approaching the final approach fix, request it.

Where turbulence is expected on descent, advise the cabin before it starts, and secure the cabin earlier.

8c.5 Passengers

Make the arrival announcement when workload permits.

Where the arrival is off schedule, limit the announcement to advising that passenger service personnel will assist in the terminal. Do not commit to connections or accommodations.

See section [1c.5 Passengers](#).

Chapter 9: Approach

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9g.1 General

For aircraft-specific procedure, see the applicable AOM> 10> Approach.

Approach Selection

Fly the most capable approach available for the runway in use. Where a precision approach is available, use it.

Autoland/HUD

Conduct an autoland only where the aircraft, the runway, and the ground equipment support it, and where the crew is qualified. Where an autoland is conducted in other than low visibility conditions, advise ATC.

Head-Up Display

Where the fleet is HUD equipped and the crew is qualified, the HUD may be used in place of an autoland for a lower-visibility approach, to the capability published in OpSpecs for that aircraft and runway.

The HUD is flown by the pilot in the HUD-equipped seat. Where only one seat is equipped, that pilot flies the approach regardless of whose leg it is, and the transfer is briefed during the approach briefing.

The pilot monitoring uses head-down instruments throughout and makes the standard callouts. The HUD does not change the monitoring duty.

9g.2 Stabilized Approach

An approach is stabilized when all of the following are met:

- the aircraft is on the correct lateral and vertical path
- the aircraft is in the landing configuration
- thrust is stabilized and the speed is within the tolerance published for the fleet
- the rate of descent does not exceed **1,000 feet per minute**, or the value briefed for a steeper procedure
- all briefings and checklists are complete

Stabilization Gate

Condition	Gate
Instrument meteorological conditions	1,000 feet AFE
Visual meteorological conditions	500 feet AFE
— End —	

Go-Around Required

An approach not stabilized at the gate requires a go-around. An approach that becomes unstabilized after the gate requires a go-around.

Warning: Do not troubleshoot an unstabilized approach. Go around, then discuss it. See section [22.4 Undesired Aircraft State](#).

Either pilot may call for a go-around. The call is executed without discussion.

Report every go-around. See section [22.10 Safety Reporting](#).

9g.3 Approach Minimums

Approach minimums are published on the chart and are increased by:

- the High Minimums designation, where applicable — see section 21.8
- any MEL or CDL item affecting approach capability
- any inoperative ground component

Descent Below Minimums

Source: 14 CFR 121.651

Do not descend below DA/DH or MDA unless the required visual reference is distinctly visible and identifiable, the aircraft is continuously in a position to land using normal maneuvers, and the reported visibility is at or above the minimum.

Reported Visibility

Where the reported visibility falls below minimums before the aircraft passes the final approach fix, the approach will not be commenced. Where it falls below after the final approach fix, the approach may be continued to DA/DH or MDA.

9g.4 ILS Approaches

Simultaneous Operations

Where simultaneous parallel or converging approaches are in use, monitor the appropriate frequency continuously and comply immediately with any breakout instruction. A breakout is flown manually where the required rate of turn or climb exceeds the autoflight capability.

Autoland/HUD

See section [9g.1 General](#).

ILS Approaches with Unreliable Rollout Guidance

Some ILS installations do not provide usable localizer guidance along the rollout. The chart states it, commonly as **unreliable rollout guidance, localizer unusable beyond the threshold**, or a note limiting autoland.

Where the chart carries such a note:

- **Autoland is not authorized.** The rollout mode has no reliable signal to track, and the aircraft may steer off the centerline after touchdown.
- A CAT II or CAT III approach is not conducted to that runway.
- The approach is flown to CAT I minimums, or to the minimums published, and landed manually.

Briefing

Where the note applies, brief it explicitly with the approach — state that autoland is not available and that directional control after touchdown is manual and visual.

After Touchdown

Track the centerline visually. Do not follow flight director or autoflight rollout guidance. Disconnect the autopilot at or before touchdown in accordance with the applicable AOM.

Warning: Unreliable rollout guidance is a limitation on what the automation can do, not on what the approach looks like. The approach may fly normally to minimums and then steer the aircraft off the runway after touchdown. Confirm the note during the approach briefing, not on rollout.

9g.5 RNAV Approaches

Verify the required navigation performance is available and that the FMS indicates the required accuracy. Where an RNP alert occurs, execute a go-around unless visual reference is established.

RNAV Substitute

Use of RNAV to substitute for an out-of-service ground navaid is authorized only where published for the procedure and permitted by OpSpecs.

9g.6 Non-ILS Approaches**Visual Approach**

Accept a visual approach only when the airport or the preceding aircraft is in sight and the flight can be conducted clear of cloud with the terrain in view.

Back up every visual approach with an available instrument procedure.

Charted Visual Flight Procedure

Fly the CVFP as charted. The charted track and altitudes are mandatory.

Circle-to-Land Approach

Circling is authorized only where published, and only at or above the circling minimums for the aircraft category. Maintain visual reference with the runway environment throughout.

9g.7 Low Visibility Operations

CAT II and CAT III operations require the aircraft, the crew, the runway, and the ground equipment to be qualified. Verify each before commencing the approach.

Requirements

- autoflight configuration per the applicable AOM
- required RVR reports at or above minimums
- the crew qualified and current for the category
- no MEL item affecting a required system

Alert Height and Decision Height

Alert height and decision height are published for the fleet and the category. Comply with the failure recognition and response criteria in the applicable AOM.

9g.8 Cold Temperature Corrections

Where the reported temperature is at or below the cold temperature restricted airport threshold, apply the published altitude correction to the affected segments and advise ATC of the correction.

Corrections are applied to the published segment altitudes, not to the assigned altitude, unless ATC concurs.

9g.9 Go-Around

Fly the published missed approach unless ATC issues alternate instructions.

After the go-around:

- advise ATC and state intentions
- advise the cabin
- reassess fuel, weather, and the plan before commencing a second approach
- advise the dispatcher

Note: A second approach is flown only where the condition that caused the go-around has been identified and resolved.

9g.10 PRM and Simultaneous Parallel Approaches

Source: AIM 5-4-16

9g.10.1 Attention All Users Page

A PRM approach chart carries an **Attention All Users Page (AAUP)**. Both pilots review it before accepting a PRM approach. The chart note directs the review; it is not optional reading.

The AAUP states the runway pairing, the monitor frequency, and the breakout procedure specific to that airport.

9g.10.2 No Transgression Zone

A **no transgression zone** is established between the two approach courses. It is **2,000 feet wide**, and it exists so that a deviation by one aircraft can be detected before it becomes a conflict.

An aircraft entering the NTZ is a blunder, and the monitor controller will break out the threatened aircraft.

9g.10.3 Monitor Frequency

Both communication radios are used. The primary is tuned to the tower; the second is tuned to the PRM monitor frequency with the volume up. The monitor controller can override the tower transmission.

Do not turn the monitor frequency down. A breakout instruction may arrive on it.

9g.10.4 Breakout

A breakout is **hand flown**. Disconnect the autopilot and autothrottle and maneuver immediately and positively — a breakout is not a normal go-around and is not delayed to reconfigure.

Breakout Conflicting with a TCAS RA

Where a TCAS resolution advisory conflicts with the vertical guidance in the breakout instruction:

- **Follow the TCAS RA vertically.**
- **Follow the controller's instruction laterally** — comply with the turn.

Advise ATC as soon as practical that an RA is being followed.

Warning: The turn is complied with in both cases. It is the vertical guidance that the RA overrides, not the breakout itself.

9g.10.5 Weather Below CAT I Minimums

Where the weather is below CAT I minimums, the **pilot in command** flies the approach. See section [9g.7 Low Visibility Operations](#).

9c.1 Approach Clearance

Read back the approach clearance including the runway and any altitude restriction.
Read back the landing clearance including the runway.

Where a landing clearance has not been received by the stabilization gate, query the tower. Where a landing clearance has not been received passing **300 feet AFE**, execute a go-around.

9w.1 Windshear

Do not continue an approach where windshear is reported, suspected, or indicated.
See section [1w.2 Windshear / Microburst](#).

9w.2 Runway Conditions

Verify the reported runway surface condition and braking action against the landing performance computed. Where the condition degrades, recompute before continuing.
A braking action report of **NIL** prohibits landing.

9w.3 Cold Weather Operations

9w.3.1 General

Cold temperature affects the altimeter, the approach path, and the aircraft. The colder it is relative to ISA, the more the true altitude sits **below** the indicated altitude.

Barometric altimetry assumes a standard temperature lapse. In cold air the altimeter over-reads, so an aircraft flying an indicated altitude is lower than it believes. The error grows with height above the altimeter source and with the departure from standard.

Warning: The error is always in the unsafe direction. Cold makes the aircraft lower than indicated, never higher.

9w.3.2 Altitude Corrections

Apply a cold temperature correction to published altitudes where the reported aerodrome temperature is at or below the correction threshold. See section [9g.8 Cold Temperature Corrections](#).

Corrections are applied to:

- the final approach fix altitude and any step-down fix
- the decision altitude or minimum descent altitude, where the procedure requires it
- the missed approach altitude
- minimum sector and holding altitudes where they are terrain critical

Advise ATC whenever a corrected altitude is flown, so the controller knows the aircraft is not at the altitude assigned.

Where the procedure is flown with vertical guidance from a satellite-based source, no correction is applied to the glidepath — the guidance is geometric. The published altitudes for cross-checking are still corrected.

9w.3.3 Extreme Cold Temperature

At extreme cold the correction becomes large enough to change the shape of the approach, and some procedures become unavailable.

Where the temperature is below the lowest temperature published for the procedure, **the procedure is not authorized**. This is a charted limitation, not a correction to be extrapolated.

Aircraft Considerations

Extreme cold also affects the aircraft. Confirm before the approach:

- fuel temperature is above the minimum for the fuel in use — see section [7w.3 Cold Fuel](#)
- anti-ice is configured per the applicable AOM
- the landing performance accounts for the surface condition, which at extreme cold is unlikely to be merely wet

9w.3.4 Cold Temperature Airports

The FAA publishes a **Cold Temperature Restricted Airport** list. At a listed airport, an altitude correction is **required** on the affected segment when the reported temperature is at or below the published threshold for that airport.

The list identifies which segment is restricted — some airports restrict only the final segment, others the intermediate or the missed approach.

Crew Action

1. Check whether the destination and each alternate is on the list during preflight planning.
2. Note the threshold temperature and the restricted segment for each.
3. Where the reported temperature is at or below the threshold, compute the correction, apply it to the restricted segment, and advise ATC.

The current list, and the threshold for each airport, is published in the airport pages on the EFB and in the Controlled Document Library.

Caution: A cold temperature airport may be legal at dispatch and restricted by the time you arrive. The threshold is checked against the reported temperature at the time of the approach, not the forecast at dispatch.

Chapter 10: Landing

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10n Non-Normal/Emergency

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10g.1 General

For aircraft-specific procedure, see the applicable AOM> 12> Landing.

10g.1.1 Crosswind Limits

Crosswind limits are published in the applicable AOM and are **reduced for a contaminated runway**. The gust value is included when assessing the limit — a wind of 20 gusting 35 is assessed against 35.

The demonstrated crosswind is a demonstrated value, not a target. Where the crosswind is at or near the limit and the runway is other than dry, the limiting figure is the contaminated one.

10g.1.2 Landing Performance

See section [1p.6 Landing Performance](#).

Landing performance is recomputed where any of the following change after the original computation: the runway, the surface condition, the braking action report, the wind, or the aircraft configuration.

10g.1.3 Autobrakes

Select the autobrake setting appropriate to the runway length, the surface condition, and the planned exit.

Where the runway is short, contaminated, or the landing distance margin is small, select a higher setting than the one that would produce a comfortable rollout.

Deceleration can be reduced after the landing is assured; it cannot be recovered after it is not.

10g.2 Contaminated Runway

10g.2.1 Runway Condition Assessment

Runway condition is reported as a **runway condition code** by third of runway, with a description of the contaminant and its depth.

Table 10.1 Runway Condition Codes

Code	Braking action	Description
6	—	Dry
5	Good	Wet, or frost, or up to 3 mm of water, slush, or dry snow
4	Good to medium	Compacted snow at low temperature

Code	Braking action	Description
3	Medium	Wet snow, dry snow, or compacted snow at higher temperature
2	Medium to poor	More than 3 mm of water or slush
1	Poor	Ice
0	Nil	Wet ice, or water or slush over ice
— End —		

Nil

A reported braking action of **NIL**, or a condition code of **0**, prohibits landing. The runway is closed to the operation until a better report is issued.

Warning: A NIL report is not a judgement call and is not improved by a light aircraft landing successfully ahead of you. Do not land on a runway reported NIL.

10g.2.2 Assessing the Report

The condition code applies to a **third of the runway**. Where the thirds differ, compute performance on the worst.

A pilot braking action report is a report of what one aircraft experienced, at one weight, on one part of the runway, at one moment. Treat it as information, not as a measurement.

Where the deceleration achieved is worse than the report indicated, **report it** — the next aircraft is planning on the figure you are about to correct. See section [23.4 Ground Safety Report](#).

10g.2.3 Crosswind on a Contaminated Runway

The crosswind limit for the reported condition governs, not the dry limit.

Directional control on a contaminated runway is lost before braking is lost. Where the crosswind approaches the limit for the condition, consider another runway or another airport.

10g.3 Touchdown

Land within the touchdown zone. Where touchdown will occur beyond the touchdown zone, **execute a go-around**.

Warning: Do not attempt to salvage a long landing. Landing performance is computed from a touchdown within the zone, and every foot beyond it is a foot removed from the stopping distance.

Do not extend the flare to achieve a smooth touchdown on a wet or contaminated runway. A firm touchdown breaks the water film, spins the wheels up, and starts the deceleration devices working.

Where the aircraft floats, is not tracking the centerline, or the touchdown will be long, go around. A go-around from the flare is a normal manoeuvre and is always available until reverse thrust is selected.

10g.4 Rollout and Braking

Apply reverse thrust and braking in accordance with the applicable AOM. Reduce reverse thrust by the speed published for the fleet.

Where the deceleration achieved does not match the deceleration expected, apply **maximum manual braking** and advise ATC. Do not wait to see whether it improves.

10g.4.1 Directional Control

Directional control is maintained with rudder, and with differential braking only where rudder is insufficient. Do not use nosewheel steering at high speed.

Where the aircraft begins to drift on a contaminated runway, reduce reverse thrust to idle before it becomes a factor — asymmetric reverse aggravates a directional problem.

10g.4.2 Brake Temperature

Assess brake temperature after landing.

Where the temperature exceeds the limit published for the fleet:

- do not set the parking brake
- advise the ground crew before anyone approaches the gear
- observe the cooling schedule in the applicable AOM
- consider a remote parking position rather than a gate

Record the event where the fleet requires it. See section 1m.2 eAML.

Caution: Hot brakes are a hazard to the ground crew before they are a hazard to the aircraft. The ground crew is told before they arrive at the aircraft, not after.

10g.5 Clearing the Runway

Clear the runway completely before stopping. The aircraft is clear when the **entire aircraft** is beyond the runway hold short line, including the tail.

Do not accomplish the after-landing flow until the aircraft is clear of the runway and either stopped or taxiing straight ahead. See section [11g.1 General](#).

Advise ATC when clear where required by the airport.

Warning: A runway is not vacated when the nose passes the hold short line. Stopping short of fully clear leaves the tail in the protected area with an aircraft on short final behind you.

10g.6 Land and Hold Short Operations

LAHSO is accepted only where:

- the available landing distance meets or exceeds the requirement for the aircraft, the runway condition, and the conditions
- the runway is **dry**
- the ceiling is at least **1,000 feet** and the visibility at least **3 statute miles**
- the crew has briefed the hold short point

The captain may decline LAHSO at any time without stating a reason. Decline before the approach where practicable, so that ATC can plan.

Note: A LAHSO clearance is not accepted by silence. State acceptance, or decline it.

Where LAHSO has been accepted and the landing cannot be made to stop short, go around. Do not continue and attempt to stop.

10c.1 General

Read back the landing clearance including the runway. See section [9c.1 Approach Clearance for the missing clearance procedure](#).

Where the runway condition or braking action reported differs from what was briefed, state it aloud so that both pilots are working from the same figure.

10n.1 Overweight Landing

An overweight landing is authorized when required by the condition. It is not a reason to delay a landing that the condition requires.

Where an overweight landing is made:

- fly the approach and landing in accordance with the applicable AOM overweight landing procedure
- plan for a longer landing distance and a higher approach speed
- advise the dispatcher and Maintenance Control
- record the landing in the eAML for the overweight landing inspection — see section 1m.2 eAML

Note: An overweight landing on a long dry runway in good conditions is a maintenance inspection. Holding to burn fuel with a fire, a structural concern, or a medical emergency aboard is a different and worse risk.

10n.2 Rejected Landing

A rejected landing is a go-around initiated after the flare has begun, or after touchdown before reverse thrust is selected.

It is a normal manoeuvre. Apply go-around thrust, rotate to the go-around attitude, and fly the missed approach. Expect the aircraft to touch down during the manoeuvre and accept it.

Warning: Once reverse thrust has been selected, the landing is committed. Do not attempt to go around after reverse has been deployed.

10n.3 Evacuation

The decision to evacuate is the captain's. Where the captain is incapacitated, any crewmember may initiate an evacuation where an immediate hazard to life exists.

Before ordering an evacuation:

- stop the aircraft and set the parking brake
- shut down the engines
- assess the hazard and the wind direction
- advise ATC and the cabin

See the applicable QRC/QRH and AOM> 16> Non-Normals.

Warning: An unnecessary evacuation causes injuries. Assess the condition before ordering one, unless fire, smoke, or structural damage requires immediate action.

Where an Evacuation Is Not Required

Where the aircraft is stopped and there is no immediate hazard, tell the cabin so explicitly — **"Cabin crew, remain seated"** or the phrase published for the fleet. Silence from the flightdeck after an abnormal stop is read as an evacuation order by a cabin crew who cannot see what you can see.

Chapter 11: After Landing

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11c Communications

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11g.1 General

After landing extends from the runway exit to arrival at the gate. The sterile flightdeck remains in effect until the aircraft is parked with the brake set. See section [1g.5 Sterile Flightdeck Period](#).

For aircraft-specific procedure, see the applicable AOM> 13> After Landing.

The after-landing flow is not accomplished while the aircraft is still moving through the high-workload part of the exit. Complete it only once the taxi is stabilized — see section 11g.2.

11g.2 Taxi In

Accomplish the after-landing flow only when the aircraft is clear of the runway and stopped, or taxiing straight ahead without complex intersections.

The taxi requirements of section [4g.1 General apply to taxi in](#). Read back every crossing clearance and hold short instruction.

11g.2.1 Taxi Speed

Taxi at a speed that can be stopped within the distance seen to be clear. Reduce speed further:

- on a wet, icy, or contaminated taxiway
- approaching a turn, an intersection, or another aircraft
- where a wing walker or ground crewmember is alongside the aircraft

Caution: Brake temperature from the landing has not yet been assessed. Avoid additional braking beyond what the taxi requires until the assessment in section 11g.4 is complete.

11g.2.2 Ramp Entry

Confirm the gate assignment before entering the ramp. Where the gate is occupied or the marshaller is not in position, stop clear of the ramp and hold.

Caution: Do not enter the ramp without a marshaller or a docking guidance system in operation. See section [12g.3.1 Guidance Systems](#).

Where the ramp is closed, see section [12g.3.2 Parking Aircraft During Ramp Closure \(PARC\)](#).

11g.2.3 Taxi-In Checklist Items

Accomplish the after-landing flow when the taxi is stabilized. Items typically include:

- flaps and spoilers
- transponder and weather radar
- exterior lighting appropriate to the ramp environment
- APU start, where required before engine shutdown

See the applicable AOM for the fleet-specific flow and sequencing.

11g.3 Single-Engine Taxi

Single-engine taxi is authorized after landing when:

- the engine cool-down period published for the fleet has elapsed
- the taxi route does not require a runway crossing in a high-traffic environment
- the ramp surface is dry and free of contamination
- the aircraft is not required to taxi up an incline

Advise the ground crew when arriving with an engine shut down.

See the applicable AOM> 13> After Landing.

11g.4 Brake and Tire Status

Assess brake temperature immediately after clearing the runway, before the taxi to the gate. See section [10g.4.2 Brake Temperature](#).

Where a tire failure, an overweight landing, a hard landing, or a suspected foreign object strike occurred:

- taxi at reduced speed and avoid unnecessary braking or turning
- advise the ground crew before the marshaller approaches the aircraft
- request a visual inspection before continuing to taxi where damage is suspected

Warning: A hot brake or a damaged tire is a hazard to the ramp before it is a hazard to the flight. Tell the ground crew before they walk up to the aircraft, not after.

Record the condition in the eAML. See section [13m.1 Aircraft Maintenance Log](#).

11c.1 General

See section [1c.1 General](#).

Report clear of the runway to ATC where required by the airport, and change to ground control frequency as instructed.

11c.2 Ground Control

Read back every taxi instruction, crossing clearance, and hold short instruction given by ground control. Where an instruction is not understood or the taxi route is unclear, stop and ask rather than taxi on an assumption.

Advise ground control of any condition affecting the taxi — a disabled aircraft ahead, a ramp closure, or a need for a tow or a marshaller not yet in position.

11c.3 Flight Attendants

The flight attendants remain seated with belts fastened until the aircraft is parked at the gate with the parking brake set and the seatbelt sign is turned off. See section [1c.4 Flight Attendants](#).

Where the taxi in is extended, or a hold on the ramp is expected, advise the cabin so that the delay does not appear unexplained.

11c.4 Passengers

Do not make an extended announcement during taxi in. A brief acknowledgment of arrival is appropriate; defer connection and baggage information to the arrival announcement made after parking. See section [1c.5 Passengers](#).

Where the taxi in is unusually long, or the gate is not immediately available, advise the passengers of the reason.

Chapter 12: Parking

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12g.1 General

For aircraft-specific procedure, see the applicable AOM> 14> Shutdown.

The sterile flightdeck ends when the aircraft is stopped with the parking brake set.

12g.2 Shutdown

Set the parking brake only after brake temperature has been assessed. See section [10g.3 Rollout and Braking](#).

Before shutting down:

- confirm chocks in place and ground power or APU established
- confirm the ground crew is clear of the engines
- confirm the beacon remains on until the engines have spooled down

Turn off unnecessary flightdeck lighting and reduce display brightness before leaving the flightdeck.

Flightdeck Door

The flightdeck door remains secured until the aircraft is parked at the gate. See section [1s.2 Flightdeck Security](#).

12g.3 Aircraft Parking Systems**12g.3.1 Guidance Systems**

A gate may be equipped with a visual docking guidance system, marshalled by hand, or both. The system in use at each gate is published in the airport pages.

Where a docking guidance system is in use:

- confirm the display shows the **correct aircraft type** before following it
- stop immediately on the stop indication, and confirm the aircraft is at the stop bar before setting the parking brake
- where the display is blank, frozen, or shows another type, **stop and request a marshaller**

Warning: A docking guidance system set for the wrong type will stop the aircraft in the wrong place. Verify the type indication before you commit to the lead-in line.

Where the gate is marshalled by hand, the marshaller has control of the movement. See section [1c.8 Flightdeck to Ground Hand Signals](#).

12g.3.2 Parking Aircraft During Ramp Closure (PARC)

A ramp may be closed to personnel for lightning, high wind, or another hazard. When a ramp is closed there is no marshaller, no wing walker, and no one to receive the aircraft.

Arriving During a Ramp Closure

Where the ramp is closed on arrival, the aircraft holds clear of the ramp until the closure is lifted, unless a PARC position has been assigned.

A **PARC position** is a pre-designated hold position, published for the station, where an aircraft may be parked and shut down without ground personnel present.

Where a PARC position is assigned:

1. Confirm the position and the routing to it with Ramp Control.
2. Taxi to the position and stop clear of all other aircraft and equipment.
3. Set the parking brake. **Do not shut down until the aircraft is stopped and the brake is set.**
4. Leave the beacon on.
5. Advise Operations that the aircraft is at the PARC position.

Restrictions

- Passengers remain aboard. Do not deplane at a PARC position.
- The parking brake remains set. Chocks will not be available.
- The APU provides electrical power and air conditioning. Where the APU is unserviceable, advise Operations before accepting the position.
- Do not shut down the APU while passengers are aboard.

Caution: Brake temperature is a factor at a PARC position because chocks are unavailable and the brake stays set. Assess brake temperature before setting the brake, and monitor it while held. See section [10g.3 Rollout and Braking](#).

Extended Hold

Where the hold is expected to be extended, the tarmac delay clock applies. See section [15g.5 Tarmac Delay](#).

Advise the dispatcher and Operations of the fuel state and the passenger situation. The captain decides whether to return to a gate at another concourse, where one is available.

12g.4 Clean Ramp Policy

Do not discard any item on the ramp. Foreign object debris is a hazard to engines, tires, and personnel.

Nothing is thrown, dropped, or emptied from the flightdeck window or the aircraft doors — including liquids, paper, packaging, and galley waste.

Report any debris observed on the ramp or the taxiway to Operations, stating the location precisely enough for it to be found.

Note: A single item of debris ingested on the next departure is an engine change. The cost of reporting it is a radio call.

Chapter 13: Post Flight/RON

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13m Maintenance

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13g.1 General

Post flight duties are complete when the aircraft has been secured, the log has been made, and any required report has been filed.

For aircraft-specific procedure, see the applicable AOM> 13> After Landing and shutdown flow.

13g.2 Post Flight Inspection

A pilot accomplishes a post flight walkaround at the terminating station and after any flight in which a condition was observed that warrants inspection.

Inspect for:

- damage, leakage, or contamination
- condition of tires and brakes
- evidence of a lightning strike, bird strike, or ground contact
- engine inlet and exhaust for foreign object damage
- panels and access doors secure

Record any discrepancy found. See section [13m.1 Aircraft Maintenance Log](#).

Note: The walkaround is not optional because the aircraft "flew fine." Damage from a hard landing, a bird strike, or a ramp equipment contact is frequently not felt on the flightdeck.

13g.2.1 Overweight or Hard Landing

Where an overweight landing or a hard landing occurred, accomplish the inspection specified in the applicable AOM before the aircraft is released for its next flight.

Advise Maintenance Control before the walkaround so that a mechanic can be present if required by the fleet procedure.

13g.3 Crew Debriefing

Debrief after every flight in which:

- a go-around, diversion, or rejected takeoff occurred
- the ABCs were escalated beyond Advise — see section [22.7 Applying the ABCs](#)
- a threat or an error affected the operation
- any crewmember requests it

The debrief addresses the event, not the individual. File a safety report where required. See section [22.10 Safety Reporting](#).

Note: A debrief is most useful within minutes of the event, while the details are still fresh and before the crew disperses.

13g.4 Remain Overnight

At an RON station:

- secure the flightdeck and confirm the aircraft is secured per the AOSSP
- confirm ground power or APU status with the ground crew
- confirm the aircraft is chocked and the parking brake set per the fleet procedure for the expected temperature range
- report the aircraft status to Maintenance Control where a discrepancy exists
- confirm the departure report time and any known change to the next day's schedule before leaving the airport

13g.4.1 Cold Weather

Where freezing precipitation is forecast overnight, coordinate with Maintenance Control on engine covers, pitot covers, and the deicing plan for departure.

Where the aircraft will sit at a temperature below the fleet's cold-soak limit, confirm with Maintenance Control whether a preconditioned start or an engine cover is required before the next departure.

13g.4.2 Security

The aircraft is not left unattended and unsecured. Where ground security is not provided by the station, confirm with Operations that the aircraft is on a monitored or restricted ramp before leaving it for the night.

Caution: An unsecured, unattended aircraft overnight is a security exposure, not only a housekeeping one. See section [1s.2 Flightdeck Security](#).

13g.5 Quick Turn

Where the aircraft continues on the same crew and ship without an overnight, the post flight inspection and the eAML review still apply before the next departure. A quick turn does not reduce the inspection — it compresses the time available to do it.

Brief the incoming crew, where a crew change occurs, on any discrepancy noted and any item still open with Maintenance Control.

13c.1 General

See section [1c.1 General](#).

Advise the dispatcher and Operations of the arrival time, the aircraft status, and any discrepancy affecting the next flight before leaving the aircraft.

13c.2 Handoff to Ground Personnel

Brief the arriving ground crew on:

- brake temperature and any restriction on chocking or towing
- any open discrepancy affecting ground handling
- fueling requirements for the next departure, where known

Where a crew change occurs, brief the following crew directly where practicable, rather than relying on the eAML alone to carry the information.

13m.1 Aircraft Maintenance Log

Complete the eAML entry before leaving the aircraft. Record:

- flight times and cycles
- every discrepancy observed in flight or during the post flight inspection
- the crew action taken

Record the discrepancy in objective terms. Do not diagnose. See section 3m.2 eAML.

Note: A discrepancy that resolved itself in flight is still recorded.

13m.2 Deferred Maintenance Carryover

Where an item is deferred under the MEL or CDL, confirm the deferral is correctly reflected in the eAML and carries forward to the next flight's release. See section [1p.4 Minimum Equipment List](#).

Do not assume a deferred item carries forward automatically. Verify it against the release for the next flight before accepting the aircraft.

Chapter 14: Non-Scheduled Flights

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14d Crew and Documentation

- [14d.1 Crew Change](#)
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14g.1 General

A non-scheduled flight is any Company flight not operated under the published schedule. Every non-scheduled flight requires a dispatch release and is subject to this manual in full, except where this chapter states otherwise.

Emergency

Nothing in this chapter limits the authority of the captain in an emergency. See section [20.2.3 Authority of the Pilot in Command](#).

14g.2 Positioning Flights

A positioning flight carries no revenue passengers and repositions the aircraft for schedule recovery, maintenance, or crew coverage.

Positioning flights are operated:

- under a dispatch release, in the same manner as a revenue flight
- with the cabin crew complement required for the persons carried
- with all persons aboard manifested

Persons Carried

Company employees and authorized contractors may be carried on a positioning flight. Passengers holding revenue tickets will not be carried.

14g.3 Maintenance Ferry Flights (MFF)

A maintenance ferry flight (MFF) moves an aircraft that does not meet the standard airworthiness requirements for revenue service.

Authorization

An MFF requires:

- a special flight permit or an MEL-authorized ferry provision
- a written ferry procedure issued by Maintenance
- a dispatch release annotated as an MFF
- concurrence of the captain

Restrictions

Unless the ferry procedure states otherwise:

- no passengers are carried

- the minimum flight crew is carried, plus any Maintenance personnel required by the ferry procedure
- the flight is conducted in day VMC
- the flight is conducted with the minimum fuel consistent with the route and a suitable alternate

Warning: The ferry procedure may impose limitations that differ from the AOM and the QRC/QRH. Where they differ, the ferry procedure governs for the ferry flight only.

Crew Qualification

An MFF is flown by a captain designated by the chief pilot. The captain reviews the ferry procedure with Maintenance before departure and briefs the first officer on every limitation and non-normal contingency it contains.

14g.4 Charter and Supplemental

A charter flight is operated under the terms of the charter contract and this manual. Where the contract and this manual conflict, this manual governs.

Sports and Entertainment Charters

Coordinate with Operations on baggage handling, security screening status, and the ground handling plan before departure. Verify screening status is documented on the release.

Government and Military Charter

Operate in accordance with the applicable contract annex and OpSpecs.

14g.5 Proving and Demonstration Flights

Proving flights conducted for the addition of an aircraft, a route, or an authorization are flown in accordance with the test plan approved by the Administrator and the Director of Operations.

Demonstration and acceptance flights are flown by crews designated by the chief pilot.

14d.1 Crew Change

Where a crew change occurs away from a domicile:

- the departing captain briefs the arriving captain on the aircraft status, open discrepancies, and any item of the release affecting the next leg
- the departing crew completes the eAML entry before leaving the aircraft
- the arriving crew accomplishes the preflight applicable to a crew change in the applicable AOM

Where a crew change occurs with an engine running, both crews coordinate through the ground crew and one qualified pilot remains at the controls throughout.

Caution: Do not commence a crew change until the aircraft is stopped, the parking brake is set, and the ground crew confirms chocks in place.

14d.2 Documentation

Every non-scheduled flight is documented in the same manner as a revenue flight:

- dispatch release retained per the Company record retention schedule
- eAML entry completed
- manifest of all persons aboard
- any special authorization, ferry permit, or contract annex attached to the release

Report the completion of every MFF, positioning flight, and proving flight to the dispatcher.

Chapter 15: Irregular Operations

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15c Communications

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15g.1 General

An irregular operation is any operation that departs from the published schedule. IROPS increases workload, compresses decision time, and introduces schedule pressure. Each of these is a threat. See section [22.2 Threats](#).

The operational priorities do not change during IROPS. See section [1g.1 Operational Priorities](#).

Note: Schedule pressure is the threat most often present in an irregular operation and the one least often briefed. Brief it.

Authority

The captain and the dispatcher retain joint operational control throughout. Neither Operations, nor Crew Scheduling, nor any Company department may direct either party to accept or continue a flight. See section [20.4.2 Limitations](#).

15g.2 Delay**15g.2.1 General**

On becoming aware of a delay of **10 minutes** or more, advise the passengers and the flight attendants, and confirm the revised plan with the dispatcher.

15g.2.2 Crew Legality

Verify against the revised departure time:

- flight duty period limit for the scheduled report time — see section 21.10
- flight time limit for the pairing
- rest available before the next duty period

Warning: The FDP limit is fixed by the **scheduled** report time. A delay does not extend it. See section [21.10.3 Flight Duty Period](#).

Where the delay will exceed the FDP limit, advise Crew Scheduling and the dispatcher immediately. Do not wait for the aircraft to be ready.

15g.2.3 Extension

An FDP may be extended up to **2 hours** for unforeseen operational circumstances arising after departure, with the concurrence of the captain and the dispatcher. See section [21.10.4 Extension](#).

The captain may decline an extension without stating a reason.

15g.2.4 Fuel and Aircraft

Where a delay affects the fuel plan, request an amended release. Where the aircraft is at the gate with passengers aboard, see section [15g.5 Tarmac Delay](#).

15g.3 Cancellation

A flight is cancelled by the dispatcher, or by the captain where the flight cannot be conducted safely or legally.

On cancellation:

- advise the passengers and direct them to passenger service
- complete the eAML entry where a mechanical condition caused the cancellation
- advise Crew Scheduling of crew location and status
- secure the aircraft in accordance with section [12g.2 Shutdown](#)

15g.4 Diversion

The decision to divert is made jointly by the captain and the dispatcher. Either may initiate it. See section [7n.1 Diversion](#).

15g.4.1 Selecting the Airport

Consider:

- the nature and urgency of the condition
- runway length, width, and surface condition
- approach capability and current weather
- Company or contract handling available on the ground
- customs and immigration facilities for an international diversion
- fuel required and terrain enroute

Caution: An airport that is legal is not necessarily suitable. A diversion airport with no handling, no stairs, and no fuel creates a second problem.

15g.4.2 On the Ground

After a diversion:

- advise the dispatcher of aircraft status, fuel state, and crew legality
- do not deplane passengers without coordination with the dispatcher and Operations
- establish the earliest realistic departure time and hold to it

— update the passengers at intervals not exceeding **30 minutes**

15g.4.3 Fuel

Order fuel through the dispatcher. Where the diversion airport is not a Company station, confirm the fuel is the correct grade and that the uplift is authorized before it is loaded.

15g.5 Tarmac Delay

A tarmac delay begins when passengers are aboard and the aircraft is not at a gate with the door open.

The captain and the dispatcher jointly manage the delay against the applicable regulatory limit. Company tarmac delay procedures are published in the Controlled Document Library.

The captain will:

- advise the passengers of the delay and the reason at intervals not exceeding **30 minutes**
- confirm lavatories are serviceable and that water and food service can be provided as required
- advise the dispatcher of the time at which a return to the gate becomes necessary
- return to the gate where the limit would otherwise be exceeded, unless prevented by safety or by ATC

Warning: The captain may return to the gate at any time. Do not allow a tarmac delay to run to the limit while waiting for a departure that is not going to happen.

15g.6 Recovery

15g.6.1 Crew Reassignment

A reassignment is issued by Crew Scheduling and confirmed by the crew. Verify legality for the new pairing before accepting it.

Where a reassignment cannot be flown legally, or where the crewmember is unfit or fatigued, decline it and state the reason. See sections [21.4 Fitness for Duty](#) and [21.11 Fatigue Risk Management](#).

15g.6.2 Aircraft Swap

On an aircraft swap, accomplish a full preflight for the new aircraft. Verify the release, the eAML, the MEL items, and the weight and balance correspond to the aircraft actually being flown.

Caution: An aircraft swap is a common source of paperwork mismatch. Verify the tail number on the release against the tail number on the aircraft.

15g.6.3 Ferry and Positioning

Recovery may require a positioning or ferry flight. See Chapter 14 Non-Scheduled Flights.

15c.1 General

An informed cabin is a manageable cabin. During an irregular operation, frequent short announcements are more effective than one comprehensive announcement delivered late.

See section [1c.5 Passengers](#).

15c.2 Passengers

State what is known, what is not known, and when the next update will be made. Give a time for the next update and meet it.

Do not:

- speculate as to the cause of a mechanical delay
- attribute the delay to another Company department
- commit to a connection, a gate, or an accommodation

15c.3 Flight Attendants

Brief the flight attendants on the revised plan before briefing the passengers. The cabin crew will be asked the same questions and should have the same information.

Advise the flight attendants of any change to crew legality that affects the cabin crew.

Chapter 16: International Operations

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16c Communications

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16n Non-Normal/Emergency

- [16n.1 Diversion Outside the United States](#)
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16g.1 General

This chapter consolidates the requirements applicable to operations outside the United States. Phase-specific international items remain in the phase chapters and point here.

Where a foreign requirement is more restrictive than this manual, the foreign requirement governs. Where this manual is more restrictive, this manual governs.

ETOPS

Extended operations requirements are in section [7g.5 ETOPS](#). Northern Airways does not conduct polar operations.

16g.2 Documentation

Carry in personal possession:

- passport valid for at least **6 months** beyond the trip
- Crewmember Certificate (FAA Form 8060-42)
- restricted radiotelephone operator permit
- visas and crew documentation for each country on the itinerary

Verify aboard the aircraft:

- aircraft registration and airworthiness certificate
- radio station license
- noise certification, where required by the state
- general declaration and any required health documentation
- insurance certificate, where required by the state

Note: A missing document is a ground stop at a foreign station, not a paperwork issue. Verify before leaving for the airport. See section [20.10 Identification and Credentials](#).

16g.3 Differences from Domestic Operation

16g.3.1 Altimetry

Transition altitude and transition level vary by state and are published on the chart. Verify both for every departure and arrival outside the United States. See section [1g.11 Altimetry](#).

Metric Altitude

Where metric altitude assignment is in use, both pilots confirm the assignment and the conversion, and set the metric display in accordance with the applicable AOM.

16g.3.2 Units

Verify the units in use for altimeter setting, visibility, runway condition, and fuel. Where an altimeter setting is issued in hectopascals, confirm the setting in the unit displayed.

16g.3.3 Phraseology

ICAO phraseology differs from FAA phraseology. Where a clearance is unclear, request it again. Do not accept a clearance that cannot be read back accurately. See section [1c.2 Radio Communication Guide](#).

16g.3.4 Airspace and Separation

Verify the RVSM, RNP, and PBN requirements applicable to the airspace before departure. Where a required capability is unavailable, advise the dispatcher and request routing consistent with the remaining capability.

16g.3.5 Age 60 Limitation

A pilot age 60 or older may serve as pilot-in-command in international commercial air transport only where the other pilot is under 60. See section 21.2.4.

16g.4 Customs and Immigration

16g.4.1 Arrival in the United States

Comply with the arrival procedure published for the station. Crewmembers present themselves for inspection with the general declaration and crew list.

Report any discrepancy in the manifest to Operations before deplaning.

16g.4.2 Departure from a Foreign Station

Confirm outbound clearance is complete before pushback. Where a state requires crew documentation to be presented at departure, allow time in the report time for it.

16g.4.3 Prohibited and Restricted Items

Do not carry any item across a border on behalf of any person other than yourself. This includes items offered by Company employees, contractors, or passengers.

Warning: Carriage of an item on behalf of another person is the mechanism by which crewmembers are used as couriers. Decline every such request and report it.

16g.5 Foreign Station Operations**16g.5.1 Ground Handling**

At a station handled by a contract agent, the agent provides services only. The agent exercises no operational control. See section [20.4.3 Delegation](#).

Confirm directly with the handler:

- fuel grade, quantity, and the uplift authorization
- deicing capability and fluid type where required
- pushback and marshalling arrangement
- ground power and air availability

16g.5.2 Fuel

Verify the fuel grade before uplift. Where the grade cannot be confirmed, do not accept the fuel and advise the dispatcher.

16g.5.3 Security

Security requirements at a foreign station are governed by the AOSSP and by the requirements of the state. Where the two differ, comply with the more restrictive. See section [1s.1 General](#).

Northern Airways requires a second person in the flightdeck whenever one pilot leaves, at every station, foreign or domestic — see section [1s.2 Two-Person Flightdeck](#). Where a foreign authority imposes an additional requirement beyond this policy, comply with the more restrictive of the two.

16g.5.4 Language

Where the working language at the station is not English, confirm every operationally significant instruction in writing or through the handler's English-speaking representative.

16c.1 Oceanic Clearance

Obtain the oceanic clearance before the oceanic entry point. Read back the clearance in full and confirm the cleared track, altitude, and Mach against the loaded route.

Warning: Do not enter oceanic airspace without a clearance. Advise ATC and hold or divert as required.

Mach Number Technique

Maintain the assigned Mach. Advise ATC of any change of **0.01 Mach** or greater.

16c.2 HF and SATCOM

Establish and verify the primary and secondary communication means before the entry point. Where SELCAL is used, confirm the SELCAL check with the controlling agency.

Where communication cannot be established before the entry point, do not enter. Advise ATC and request an alternate routing.

16c.3 Position Reporting

Report position as required by the airspace. Where CPDLC and ADS-C are in use, verify the connection is active. Where either fails, revert to voice reporting and advise ATC.

See section [7c.2 Position Reporting](#).

16n.1 Diversion Outside the United States

Where a diversion outside the United States is required, consider in addition to the criteria in section 15g.4.1:

- customs, immigration, and quarantine facilities
- the Company or contract handling available
- passenger documentation and visa status for the diversion country
- the ability to obtain fuel and maintenance support

Advise the dispatcher as early as practicable so that handling and clearance can be arranged before arrival.

16n.2 Contingency Procedures

Where a contingency requires deviation from the cleared oceanic track and a clearance cannot be obtained, comply with the contingency procedure published for the airspace.

In general terms, and subject to the published procedure:

- leave the assigned route by the published offset
- climb or descend to a level differing from those in use
- broadcast position and intentions on the air-to-air frequency and on 121.5
- display exterior lights
- advise ATC as soon as communication is established

Warning: The published contingency procedure for the airspace governs. The summary above does not replace it.

Chapter 17: Training and Checking

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- [17.3 Training Events](#)
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- [17.5 Check Airmen](#)
- [17.6 Operating Experience](#)
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- [17.8 Reserved](#)

17.1 General

This chapter describes how training and checking are conducted. The qualification and currency **requirements** are in Chapter 21. Where the two appear to differ, Chapter 21 states the requirement and this chapter states the method.

The Flight Training Program Manual is the controlling document for curriculum content.

17.1.1 Responsibility

Flight Training conducts training. Flight Standards owns the procedures being trained. The Chief Pilot — Flight Training holds the regulatory position under 14 CFR 119.65.

17.1.2 Training Environment

Training and checking are conducted in an environment in which a crewmember can make an error, identify it, and correct it. An event in which no error occurs has demonstrated little.

Note: A crewmember who conceals an error in training carries it to the line.

17.2 Advanced Qualification Program

Northern Airways operates under an Advanced Qualification Program approved under 14 CFR Part 121 Subpart Y.

Under AQP:

- curriculum content and intervals are established by the approved program, not by the Subpart N and O tables
- proficiency is assessed against published, observable standards
- data from every event feeds the curriculum revision cycle

Where the AQP curriculum specifies an interval different from the regulation cited in Chapter 21, **the AQP curriculum governs.**

17.3 Training Events

Event	Purpose
Initial	First qualification with the Company
Transition	Qualification in a new fleet
Upgrade	Qualification as captain
Requalification	Return after a lapse in qualification
Recurrent	Continuing qualification
Differences	Qualification for a variant within a fleet
Special purpose	ETOPS, CAT II/III, special qualification airports
— End —	

17.3.1 Ground Training

Ground training is delivered by distance learning, in the classroom, or in a procedures trainer, in accordance with the curriculum.

17.3.2 Simulator Training

Simulator sessions are conducted to the published lesson plan. The instructor may repeat a maneuver within the session to achieve the standard.

17.3.3 Line Oriented Flight Training

LOFT is conducted in real time without instructor intervention. The scenario is flown to completion. Debriefing is conducted against the CRM/TEM skills in section 22.6.

17.4 Checking Events

17.4.1 Proficiency Check

Conducted in accordance with 14 CFR 121.441 and the AQP curriculum. Intervals are in section 21.5.3.

17.4.2 Line Check

Conducted on a scheduled revenue flight by a line check pilot. The line check assesses the conduct of a normal operation, not maneuvers.

17.4.3 Conduct

During a checking event the check airman:

- briefs the scope of the event before it begins
- does not act as a required crewmember unless the event requires it
- intervenes only where safety requires it
- debriefs the event against the published standard

17.4.4 Crewmember Rights

A crewmember may:

- request a repeat of any maneuver where an equipment malfunction affected the result
- decline to continue an event where unfit or fatigued, without prejudice
- request a different check airman where a conflict exists, through the chief pilot

17.5 Check Airmen**17.5.1 Designation**

Check airmen are designated by the Chief Pilot — Flight Training and are qualified in accordance with 14 CFR 121.411.

Designation	Conducts
Check airman (simulator)	Proficiency checks and training in the simulator
Check airman (aircraft)	Operating experience and checks in the aircraft
Line check pilot	Line checks on revenue flights
Aircrew program designee	Checks on behalf of the Administrator
— End —	

17.5.2 Currency

A check airman maintains line currency in the seat and fleet in which checks are conducted.

17.5.3 Conduct on the Line

A check airman occupying a required seat is a required crewmember and is subject to this manual in full. The presence of a check airman does not alter the authority of the captain. See section 20.2.3.

17.6 Operating Experience

Operating experience requirements are in section 21.7. This section describes its conduct.

17.6.1 Conduct

The check airman occupies a pilot station and acts as pilot in command unless the conditions of 14 CFR 121.434(c)(1) permit otherwise. The trainee performs the duties of the seat.

17.6.2 Progression

The check airman progressively reduces intervention as the trainee demonstrates the standard. Operating experience is complete when the check airman determines the trainee can perform the duties unsupervised, and not before, regardless of hours flown.

17.6.3 Limitations

Limitations applicable during operating experience are in section 21.7.4.

17.7 Unsatisfactory Performance**17.7.1 During Training**

Additional training is provided where a crewmember does not achieve the standard in a training event. Additional training is a normal part of the program and is not a disciplinary matter.

17.7.2 During Checking

Where a checking event is unsatisfactory, the check airman:

- discontinues the event
- documents the specific standard not met
- refers the crewmember to Flight Training for additional training
- notifies the domicile chief pilot

The crewmember is removed from line flying until the standard is met.

17.7.3 Return to Line

A crewmember returns to line flying on satisfactory completion of the additional training and the associated recheck.

17.8 Reserved

Chapter 18: Passengers and Cargo

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18d Cargo

- [18d.1 General](#)
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- [18d.3 Weight and Balance](#)
- [18d.4 Dangerous Goods](#)

18g.1 General

The captain is responsible for the safety of every person aboard the aircraft. Passenger handling is performed by Customer Service and by the cabin crew. The captain decides whether a person is carried.

Passenger service policy is published by Customer Service. This chapter states the policy applicable to the flight crew.

18g.2 Boarding and Deplaning**Flight Crew Present**

A flight crewmember is aboard the aircraft whenever passengers are aboard.

Boarding Complete

The flight attendant in charge reports boarding complete and the cabin secure. Departure will not commence before that report. See section [3c.2 Flight Attendants](#).

Boarding with an Engine Running

Boarding or deplaning with an engine running is not authorized.

Deplaning Away from a Gate

Deplaning at a remote stand or by airstair requires coordination with Operations and a cabin crew briefing before the doors are opened. The captain confirms the ramp is secure and the engines and beacon are off.

18g.3 Seating

Exit row seating is assigned by Customer Service in accordance with 14 CFR 121.585. The cabin crew confirms each exit row occupant meets the criteria and rebriefs or reseats where required.

The captain is advised of any seating irregularity that affects the emergency evacuation plan.

18g.4 Special Categories

Category	Crew action
Unaccompanied minor	Cabin crew confirms count and destination with the agent before departure
Passenger requiring mobility assistance	Confirm the boarding and deplaning plan and any equipment stowage
Passenger requiring medical oxygen	Confirm the approval and the equipment against the Company policy
Passenger in custody	Captain is notified before boarding, with the escort arrangement
Deportee	Captain is notified before boarding
Human remains	Captain is notified of the loading position
— End —	

Where the captain is not notified in advance of a category requiring notification, do not depart until the arrangement is confirmed.

18g.5 Denied Boarding and Removal

The captain may refuse to transport any person, and may remove any person, consistent with Company policy and applicable law.

Grounds include intoxication, threatening or abusive behavior, refusal to comply with a crewmember instruction, and any condition that would compromise the safety of the flight.

Where a person is removed:

- request law enforcement to meet the aircraft where the conduct warrants it
- obtain written statements from the crewmembers involved before the crew separates
- advise the dispatcher
- file a report before the end of the duty period — see section [23.6.3 Removal of a Passenger](#)

18g.6 Live Animals**In the Cabin**

Animals carried in the cabin are accepted by Customer Service under Company policy. The cabin crew confirms the container is stowed and remains stowed for taxi, takeoff, and landing.

In the Hold

Live animals carried in the hold are loaded in a compartment with temperature control and ventilation appropriate to the species and the sector.

The captain is advised before departure of:

- the species and number carried
- the loading position
- any temperature or ventilation requirement
- any planned ground delay that would affect the compartment environment

Caution: An extended ground delay with live animals in the hold requires coordination with Operations. Advise the dispatcher where a delay is expected to exceed the period stated in the loading instruction.

18d.1 General

The load agent builds the load plan and issues the loading instruction. Ramp personnel load, secure, and document the cargo against that instruction. The crew verifies the documentation, not the loading.

18d.2 Load Sheet

The load sheet states the distribution of passengers, baggage, cargo, and mail by compartment, and is the basis for the weight and balance computation.

Verify before departure:

- the load sheet corresponds to the correct flight and aircraft registration
- compartment loads do not exceed the published compartment limits
- the dangerous goods notification, where applicable, matches the load sheet
- any last-minute change is within the published tolerance

See section [2.7 Weight and Balance](#).

18d.3 Weight and Balance

Final figures are transmitted by ACARS or delivered by the agent. Both pilots verify zero fuel weight, takeoff weight, center of gravity, and the stabilizer trim setting against the aircraft limits.

Warning: Do not depart on figures that have not been verified against the aircraft limits. A load sheet that does not match the aircraft is not a paperwork error — it is an unverified center of gravity.

18d.4 Dangerous Goods

Dangerous goods policy, classification, and responsibilities are in section [1d.1 Dangerous Goods/Hazardous Materials](#). Preflight notification and loading verification are in section [3d.1 Notifications](#).

Chapter 20: Administration

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20.1 General

20.1.1 General

This chapter establishes the organization of Flight Operations, the regulatory management positions, and the standards of conduct applicable to crewmembers.

20.1.2 Operational Priorities

Every decision made in the conduct of a Northern Airways flight is measured against four priorities, **in this order**:

1. **Safety**
2. **Regulatory compliance**
3. **Passenger comfort and service**
4. **Schedule and efficiency**

A lower priority is never traded against a higher one. When a decision appears to require such a trade, it does not — the higher priority governs and the decision is already made.

Note: These priorities are stated again in section [1g.1 Operational Priorities](#) because they are operative in flight, not only in the abstract. They are identical in both places.

20.1.3 Assumptions

The policies in this manual assume:

- Two qualified pilots, current and legal for the operation being conducted.
- An airworthy aircraft dispatched under a valid release.
- Automation used at the level appropriate to the situation.
- Crewmembers who have read the material applicable to their duty.

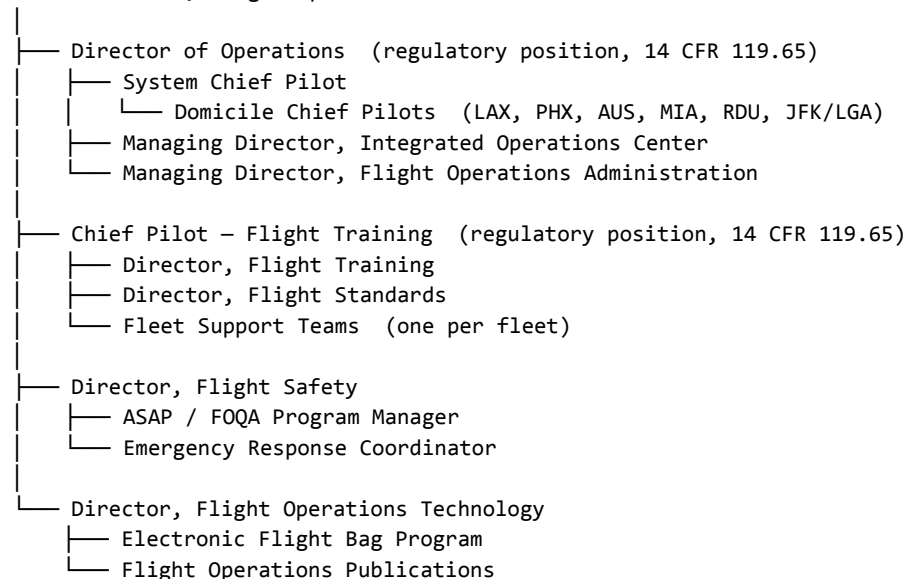
Where an assumption does not hold, the operation is not normal. Refer to the applicable non-normal section.

20.2 Flight Department Organization

20.2.1 Structure

Northern Airways Flight Operations reports to the Senior Vice President, Operations. Within Flight Operations:

Vice President, Flight Operations



20.2.2 Regulatory Management Positions

The following positions are named in the Northern Airways Operations Specifications and are required by 14 CFR 119.65. The current holders are listed in

the Controlled Document Library, which is the authoritative source. Names are not printed in this manual so that a personnel change does not require a revision.

Position	Regulatory Basis
Director of Operations	14 CFR 119.65(a)(1)
Chief Pilot	14 CFR 119.65(a)(3)
Director of Maintenance	14 CFR 119.65(a)(2)
Chief Inspector	14 CFR 119.65(a)(4)
— End —	

20.2.3 Authority of the Pilot in Command

The captain is in command of the aircraft and of all persons aboard it. The captain's authority is established by 14 CFR 91.3 and is not delegated, limited, or shared by any provision of this manual.

The captain:

- Is directly responsible for, and is the final authority as to, the operation of the aircraft.
- Shares responsibility for operational control with the dispatcher. Neither may originate or continue a flight the other refuses. See section [20.4 Operational Control](#).
- May deviate from any rule in this manual, and from 14 CFR, to the extent required to meet an emergency.
- May remove any person from the aircraft, or refuse carriage, consistent with company policy and applicable law.

Note: The final authority of the captain does not authorize an operation prohibited by dispatch, by maintenance, or by this manual.

20.2.4 Authority of the First Officer

The first officer is a required crewmember and shares responsibility for the safe conduct of the flight. The first officer:

- Will bring to the captain's attention any condition, action, or omission the first officer believes compromises safety, and will persist until it is resolved. See section [22.7 Applying the ABCs](#).
- Assumes command if the captain becomes incapacitated. See section [1n.3 Crew Incapacitation](#).
- Is not relieved of responsibility by the captain's seniority, position, or instruction.

20.2.5 Succession of Command

If the captain becomes incapacitated or is otherwise unable to act:

1. The first officer assumes command.
2. If a check airman or line check pilot is aboard in a required seat, that individual assumes command only if senior in qualification for the operation being conducted.
3. Relief pilots assume command in order of seat qualification, then seniority.

Command, once assumed, is not returned in flight.

20.3 Contact Information

Contact numbers, ACARS addresses, and radio frequencies are maintained in the Controlled Document Library and on the EFB Quick Reference page. They are not printed here.

The following are the standing contacts every crewmember must be able to reach without reference to a document:

Need	Contact
Anything in flight	Dispatch, via ACARS or company radio
Anything on the ground, immediate	Integrated Operations Center (IOC)
Emergency, in flight	Declare with ATC first. Then IOC.
Crew scheduling / legality	Crew Scheduling
Maintenance	Maintenance Control, via IOC
Chief pilot, urgent, after hours	Duty Chief Pilot line
Report a safety concern	ASAP, or Flight Safety directly
Report a crewmember conduct concern	Professional Standards
— End —	

Warning: Declare an emergency with ATC before contacting the Company. Do not delay a declaration to consult dispatch.

20.4 Operational Control

20.4.1 Joint Responsibility

Operational control of a Northern Airways flight is exercised jointly by the **captain** and the **aircraft dispatcher**. This is a requirement of 14 CFR 121.533(a) through (e), not a company convention.

- The dispatcher is responsible for the initiation, continuation, diversion, and termination of the flight in the manner authorized by 14 CFR 121.533(c).
- The captain is responsible for the safe conduct of the flight from the moment the aircraft moves under its own power until it comes to rest at the end of the flight, per 14 CFR 121.533(d).
- **Both must agree** to originate a flight. Both must agree to continue one. Either may terminate or divert.

20.4.2 Limitations

A dispatcher may not be directed to release a flight by anyone. A captain may not be directed to accept one. Any attempt by any person — including management — to override either party's refusal will be reported immediately to the Director of Operations and to Flight Safety.

20.4.3 Delegation

Operational control may not be delegated to any person or entity outside Northern Airways. Where a flight is handled by a contract ground agent or a foreign handler, that agent acts only as a service provider and exercises no operational control.

20.4.4 Integrated Operations Center

Integrated Operations Center (IOC) is the physical facility and the organization through which Northern Airways exercises operational control. IOC houses Dispatch, Meteorology, Maintenance Control, Crew Scheduling, ATC Coordination, and the Emergency Response Center.

See Flight Plan/Dispatch Release for the dispatch process, and the **Dispatch Operations Manual** for dispatcher policy.

20.5 Chief Pilot and Domiciles

20.5.1 Domiciles

Northern Airways maintains pilot domiciles at:

Domicile	Airport(s)
Los Angeles	LAX
Phoenix	PHX
Austin	AUS
Miami	MIA
Raleigh–Durham	RDU
New York	JFK, LGA
— End —	

Fleet assignment by domicile is published in the bid documents issued by Crew Resources. It is not stated in this manual.

Note: Northern Airways operates at EWR but does not domicile crews there. EWR assignments originate from the New York domicile.

Group bases at PDX, LAS, DEN, and BOS are held by subsidiary and regional operators. They are not Northern Airways mainline domiciles. See Scope, Applicability to Subsidiary and Regional Operators.

20.5.2 Chief Pilot

The domicile chief pilot is the crewmember's primary point of contact for:

- Interpretation of policy in this manual
- Qualification, currency, and training issues
- Attendance, availability, and fitness for duty
- Discipline and performance
- Authorization for anything this manual describes as requiring chief pilot approval

A crewmember uncertain whether an operation is permitted will contact the chief pilot **before** conducting it. A chief pilot's verbal authorization is valid; the crewmember will document it in the flight record and submit a written report.

20.5.3 Duty Chief Pilot

A duty chief pilot is available at all times. The duty chief pilot has full authority to act for the domicile chief pilot.

20.6 Required for Flight

20.6.1 Required Checklists, Manuals, and Documents

The following are required to be aboard the aircraft:

Item	Held
Aircraft Flight Manual and placards	Aircraft
Aircraft Operating Manual	EFB
Quick Reference Handbook	Flightdeck, and EFB
Normal Procedures Checklist	Flightdeck
Minimum Equipment List and CDL	EFB
Flight Operations Manual	EFB
Route manual coverage for the itinerary	EFB
Aircraft registration and airworthiness certificate	Aircraft
Radio station license, international	Aircraft
Dispatch release and computer flight plan	EFB or paper
— End —	

20.6.2 Personal Possession

See section [20.10 Identification and Credentials](#).

20.6.3 Verification

Verify at preflight. See section [3g.2 Required for Flight](#).

20.7 Fleet Support Teams

20.7.1 Role

Each fleet has a Fleet Support Team responsible for the aircraft-specific manuals — AOM, QRC/QRH, and the fleet's portion of the training curriculum. The Fleet Support Team is the correct destination for a question about **how** an aircraft is operated. Questions about **whether** an operation is permitted go to the chief pilot.

Fleet Support Team	Aircraft	AOM
Airbus Narrowbody	A319, A320, A321, A321neo	A32F
Airbus Widebody	A330-900neo	A33F
Boeing Narrowbody	737-800, 737 MAX 8	73F
757	757-200	757

Fleet Support Team	Aircraft	AOM
767	767-300	767
777	777-200ER	777
Embraer	E190	E90
— End —		

20.7.2 Submitting a Question

Submit through the Fleet Support form in the Controlled Document Library. Responses of general applicability are published in the fleet bulletin and, where they change policy or procedure, incorporated at the next revision.

Note: A Fleet Support Team response is not authorization to deviate from this manual. Only the chief pilot's office can authorize a deviation from FOM policy.

20.8 Professional Standards

20.8.1 Purpose

Professional Standards is a joint Company and NAPA program for resolving crewmember conduct and performance concerns between peers, without discipline, where the concern does not involve safety of flight, intentional non-compliance, or a violation of law.

20.8.2 Routing

Concern	Route To
Interpersonal conflict, professionalism, demeanor	Professional Standards
Unintentional error, procedural deviation, safety event	ASAP
Intentional non-compliance	Chief Pilot
Suspected substance use or impairment	Chief Pilot, immediately
Suspected medical unfitness	Chief Pilot, immediately
Threat, violence, or criminal act	IOC and law enforcement
— End —	

Warning: Do not route a safety-of-flight concern through Professional Standards. It is not a safety reporting system and does not feed the safety data program.

20.8.3 Confidentiality

Professional Standards proceedings are confidential and are not reported to management, are not placed in a personnel file, and are not used in a disciplinary proceeding.

20.9 Uniform and Appearance

20.9.1 Standard

Crewmembers in uniform represent Northern Airways continuously — in the terminal, in transit, in the hotel lobby, and on the jumpseat of another carrier.

The uniform will be complete, clean, and correctly worn at all times while on duty and while in public view in the course of duty. The full specification is published in the Uniform Standards document in the Controlled Document Library.

20.9.2 Standing Requirements

- The uniform jacket or an approved sweater will be worn through the terminal.
- The uniform hat, where issued, is optional in the terminal and is not worn in the flightdeck.
- Company identification is worn visibly on the outermost garment at all times in the sterile area. See section [20.10 Identification and Credentials](#).
- Only company-issued or company-approved pins, wings, and insignia are worn. Refer to the Uniform Standards document for the approved list.
- The uniform is not worn in an establishment whose principal business is the service of alcohol, or in any establishment or activity that would reflect poorly on Northern Airways.

20.9.3 Alcohol in Uniform

A crewmember will not consume alcohol while in uniform, whether or not on duty. This is a company standard more restrictive than 14 CFR 121.458 and is not waivable.

20.10 Identification and Credentials

20.10.1 Required Documents

A pilot will have the following in personal possession when performing duty:

Document	Requirement
Airline transport pilot certificate	14 CFR 61.3(a)
Current medical certificate, first class	14 CFR 61.23, 61.3(c)
Radiotelephone operator permit (restricted)	International operations
Passport, valid $\geq$ 6 months beyond the trip	International operations
Company identification	Company
Crewmember Certificate (FAA Form 8060-42)	49 CFR 1544, international
Applicable visas and crew documentation	Destination requirement
— End —	

20.10.2 Verification

Verify possession of all required documents **before leaving for the airport**. A pilot who discovers a missing or expired document will notify Crew Scheduling and the chief pilot immediately and will not operate.

Note: Possession is the requirement. A pilot who arrives at the aircraft without a required certificate is not legal to operate, regardless of the validity of the certificate.

20.10.3 Expired or Lost Credentials

A pilot whose medical certificate expires, is suspended, or whose condition changes such that the medical is no longer valid, will immediately cease operating and notify the chief pilot. See section [21.3 Medical](#).

20.11 Personal Conduct

20.11.1 Fitness for Duty

A crewmember will report for duty rested, sober, and physically and mentally fit. A crewmember who is not fit will remove themselves from duty and notify Crew Scheduling. Removal from duty for fitness is not, by itself, a disciplinary matter.

See section [21.4 Fitness for Duty for the IM SAFE assessment and for fatigue policy](#).

20.11.2 Alcohol and Drugs

Company policy is more restrictive than 14 CFR 91.17 and 121.458:

Restriction	Northern Airways	14 CFR
Alcohol before duty	12 hours bottle to throttle	8 hours
Blood alcohol concentration	0.00% while on duty or on call	0.04%
Alcohol while in uniform	Prohibited	Not addressed
Alcohol while on reserve/on call	Prohibited	Not addressed
— End —		

A crewmember will not report for duty within 12 hours of consuming alcohol, and will not report for duty if any effect of alcohol remains, regardless of elapsed time.

Any medication, prescription or over-the-counter, will be evaluated for compatibility with flight duty before use. See section [21.3.3 Medication](#).

20.11.3 Self-Reporting

A crewmember who recognizes a substance dependency issue and self-reports before being identified through testing or observation will be referred to the company HIMS program. Self-reporting is not a disciplinary event. Refer to the HIMS program description in the Controlled Document Library.

20.11.4 Social Media and Public Comment

A crewmember will not:

- Post images taken in the flightdeck during any phase of flight, or images that identify a passenger.
- Discuss an incident, accident, or investigation involving Northern Airways.
- Represent a personal opinion as the position of Northern Airways.
- Disclose the contents of this manual or any other controlled document.

Media inquiries are directed to Corporate Communications without comment. This applies at an accident scene and applies to a crewmember not involved in the event.

20.11.5 Personal Electronic Devices

See section [1g.4 Flightdeck Management for the use of personal electronic devices in the flightdeck](#), and to see Electronic Flight Bag (EFB). for company devices.

20.12 Reserved

Chapter 21: Qualifications/Limitations

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21.1 General

21.1.1 Introduction

This chapter establishes the qualification requirements, currency standards, bid status designations, and operating limitations applicable to Northern Airways pilots. It states the **company** requirement. Where the Company requirement is more restrictive than the regulation, the Company requirement governs and the regulation is cited for reference.

21.1.2 Personal Responsibility

Qualification and currency are the individual pilot's responsibility. Crew Scheduling maintains records and will not knowingly assign an unqualified pilot, but a scheduling system error does not make an unqualified pilot legal.

Before each trip, verify:

- Medical certificate valid for the duration of the trip
- Recency of experience (takeoffs and landings) satisfied

- Recurrent training and proficiency events current
- Passport and visas valid, for international
- Special qualification satisfied for any airport on the itinerary

A pilot who discovers a qualification deficiency will notify Crew Scheduling and the chief pilot immediately and will not operate.

21.2 Pilot Certification Requirements

21.2.1 Captain

Requirement	Standard
Certificate	Airline Transport Pilot
Type rating	For the aircraft flown, with no limitation prohibiting the operation
Medical	First class, current per 14 CFR 61.23
Age	Under 65 (14 CFR 121.383(d))
English proficiency	ICAO Level 4 minimum, noted on certificate
Line check	Current per 14 CFR 121.440
Proficiency check	Current per 14 CFR 121.441
— End —	

21.2.2 First Officer

Requirement	Standard
Certificate	Airline Transport Pilot, or Restricted ATP per 14 CFR 61.160
Type rating	For the aircraft flown
Medical	First class, current per 14 CFR 61.23
Age	Under 65
English proficiency	ICAO Level 4 minimum
Proficiency check	Current per 14 CFR 121.441
— End —	

Note: A pilot holding a Restricted ATP may serve only as second in command and may not upgrade until the full ATP aeronautical experience requirements of 14 CFR 61.159 are met.

21.2.3 Age 65

A pilot may not serve as a required crewmember in Part 121 operations after reaching age 65 (14 CFR 121.383(d)). The pilot's final day of eligibility is the day before the 65th birthday. Crew Scheduling will remove the pilot from all pairings extending beyond that date.

21.2.4 Age 60 in International Operations

Under ICAO Annex 1, a pilot age 60 or older may serve as pilot-in-command in international commercial air transport **only** if the other pilot is under 60. Both pilots may not be 60 or older on an international flight. Crew Scheduling builds and adjusts pairings to satisfy this; the crew will verify it at check-in for international operations.

21.3 Medical**21.3.1 Certificate Validity**

A first class medical certificate is required. Duration per 14 CFR 61.23:

Age at Examination	First Class Valid For
Under 40	12 calendar months
40 and over	6 calendar months
— End —	

21.3.2 Change in Medical Condition

A pilot will not act as a required crewmember while having a known medical condition, or taking medication, that would make the pilot unable to meet the requirements of the medical certificate (14 CFR 61.53).

This is self-executing. It does not require a physician's grounding, an FAA action, or a company determination. If the condition exists, the pilot is not legal.

Notify the chief pilot of:

- Any hospitalization, surgery, or emergency department visit
- Any new diagnosis or new prescription
- Any loss of consciousness, however brief
- Any condition affecting vision, hearing, balance, or cognition
- Blood donation — **72 hours** duty free after donating
- Plasma donation — **4 hours** duty free after donating

- Platelet donation — return to duty only once the platelet count is restored to **100,000 or higher**
- Scuba diving — no diving within **24 hours of sign-in**

Note: The periods above are measured from the end of the donation to the start of duty, not to report time. Where a donation is made on a layover, the period runs against the next sign-in.

A crewmember who has donated and is unsure whether the period has elapsed is unfit for duty. See section [21.4 Fitness for Duty](#).

21.3.3 Medication

Before using any medication, prescription or over-the-counter, verify compatibility with flight duty. Where a medication is newly prescribed, a pilot will observe a **ground trial period** appropriate to the medication — no fewer than 48 hours, and longer where the medication class requires it — before returning to duty.

Warning: Many common over-the-counter antihistamines, sleep aids, and cold preparations are disqualifying. The absence of a warning label is not clearance to fly.

21.3.4 Corrective Lenses

A pilot whose medical certificate bears a corrective lens limitation will have a **second** pair of corrective lenses in personal possession when performing duty.

21.3.5 Pregnancy

A pregnant crewmember may continue to operate with the concurrence of her physician and an aviation medical examiner. Notify the chief pilot upon confirmation of pregnancy so that scheduling accommodations, including radiation exposure on ultra-long-haul and high-latitude routing, can be evaluated.

21.4 Fitness for Duty

21.4.1 General

A pilot will not operate unless fit. Fitness is self-assessed before every duty period and continuously reassessed during it.

21.4.2 IM SAFE

Letter	Assess
I — Illness	Am I sick? Do I have any symptom, however minor, that would distract me?
M — Medication	Am I taking anything that could impair performance?
S — Stress	Am I preoccupied by a personal, financial, or work matter?
A — Alcohol	Has it been at least 12 hours? Is any effect remaining?
F — Fatigue	Did I sleep adequately? Am I well-rested for the duty ahead?
E — Eating / Emotion	Have I eaten and hydrated? Am I emotionally able to perform?
— End —	

21.4.3 Removal from Duty

A pilot who determines they are unfit will contact Crew Scheduling and state that they are unfit for duty. The pilot is **not** required to disclose the reason to Crew Scheduling.

Calling out unfit is not a disciplinary event and is not counted as an attendance occurrence. A pattern may prompt a conversation with the chief pilot; that conversation is supportive, not punitive.

Note: Removal from duty for fitness takes precedence over schedule. See section [1g.1 Operational Priorities](#).

21.4.4 Other Crewmembers

A pilot who believes the other pilot is unfit will raise the concern directly, and if unresolved, will contact the chief pilot before departure. See section [22.7 Applying the ABCs](#).

21.5 Currency and Recency**21.5.1 Takeoff and Landing Recency**

Per 14 CFR 121.439, a pilot may not serve as a required crewmember unless, within the preceding **90 days**, that pilot has made at least **three takeoffs and three landings** in the same type of aircraft, or in an approved simulator.

Northern Airways applies this by AOM fleet grouping. A landing in an A319 counts toward A32F recency; it does not count toward 73F recency.

Note: For the 737-800 and 737 MAX 8, and for the A319/A320/A321/A321neo, recency is common within the fleet grouping. Differences training does not create a separate recency requirement.

21.5.2 Loss of Recency

A pilot who lapses beyond 90 days will complete reestablishment of recency in an approved simulator or under the supervision of a check airman before serving as a required crewmember (14 CFR 121.439(b)).

21.5.3 Recurrent Training and Checking

Event	Interval	Reference
Recurrent ground training	12 calendar months	14 CFR 121.427
Proficiency check / AQP equivalent	12 calendar months (captain), 24 (first officer, with a proficiency check or training event at 12)	14 CFR 121.441
Line check (captain)	12 calendar months	14 CFR 121.440
Emergency equipment / drills	Per AQP curriculum	14 CFR 121.417
CRM	Per AQP curriculum	14 CFR 121.404
— End —		

Northern Airways operates under an Advanced Qualification Program. Where the AQP curriculum specifies an interval different from the table above, the **AQP curriculum governs**. Refer to the Flight Training Program Manual.

21.5.4 Grace Month

A training or checking event completed in the calendar month before, during, or after the month in which it is due is considered completed in the month it was due. This does not extend the underlying medical certificate or the 90-day recency requirement, neither of which has a grace period.

21.6 Bid Status and Seat Qualification

21.6.1 Status Designations

Status	Meaning
CA	Captain, qualified and current in seat and fleet
FO	First officer, qualified and current in seat and fleet
CA-HM	Captain, high minimums. Refer to 21.8.
CA-OE / FO-OE	In operating experience. Refer to 21.7.
CA-TR / FO-TR	In training, not released to the line
RSV	On reserve availability
INAC	Inactive — medical, leave, or qualification lapse
— End —	

21.6.2 Fleet and Scope Designations

Bid status also states the fleet and the operational scope a pilot is assigned to. It is awarded on completion of the applicable training and operating experience.

Bid Status	Fleet	Scope
E190	Embraer E190	Domestic only
A32D	A319 / A320 / A321 / A321neo / A321XLR	Domestic only
A32I	A319 / A320 / A321 / A321neo / A321XLR	Domestic and international
B73D	Boeing 737	Domestic only
B73I	Boeing 737	Domestic and international
B756	Boeing 757 / 767	Domestic and international
756C	Boeing 757 (cargo)	Cargo only — same type rating as B756
A330	Airbus A330	Domestic and international
— End —		

Domestic and International Positions

On the 737 and A320 family, domestic (D) and international (I) are separate bid positions.

- An I-qualified pilot may fly domestic operations.
- A D-qualified pilot may **not** fly international operations.

— Some bases are domestic only and some international only. Base designations are published in the bid package.

The 756C status designates 757 cargo flying. It requires the same type rating and training as B756. A pilot holding 756C alone is not authorized to operate a passenger-configured aircraft.

What Counts as International

An international operation is any flight that crosses a body of water larger than the Great Lakes, operates outside United States sovereign airspace with an overwater segment, or flies an instrument approach procedure that swings out over open water.

Note: Canada and Mexico are domestic for qualification purposes unless the routing crosses the Gulf of Mexico or another qualifying overwater segment.

21.6.3 International Qualification Tiers

International qualification is a standard component of the bid position at a given base, not a separately earned achievement. A pilot who bids an international position at a base holds the qualifications that go with it. Regional items are assessed and signed off during operating experience.

Basic International Qualification

Required for every pilot holding an international bid status. Covers RVSM, international altimetry and transponder procedure, position reporting, customs and overflight documentation, HF communication, and international weather and NOTAM interpretation.

Extended International Qualification (EIQ)

Required for Latin America, South America, and Hawaii. Approximately 1.5 days of additional training covering extended overwater operation, oceanic contingency procedure, regional ATC and airspace procedure, and the Hawaii overwater requirements.

Long-Range International Qualification (LIQ)

Required for Europe, Asia, and the Pacific. Three to four days depending on equipment, covering ETOPS authorization and procedure, the NAT and PACOTS track systems, PBCS requirements in Asia and the Pacific, long-range fuel planning and equal time points, and oceanic position reporting and HF requirements.

21.6.4 Division Qualification

In addition to EIQ or LIQ, a pilot holds a division qualification for each region they operate in. Division qualification is conducted with a check airman and is base specific — a pilot's base determines the division held.

Division	Tier Required	Terrain Qualification Included
Latin / South America	EIQ + division qualification	Yes — see 21.6.5
Hawaii	EIQ + division qualification	No
Atlantic / Europe	LIQ + division qualification	Yes, select routes
Asia / Far East	LIQ + division qualification	Yes
Pacific	LIQ + division qualification	Yes
— End —		

21.6.5 Latin Terrain Qualification

A separate **Latin Terrain Qualification** is required to operate into the terrain-challenged cities of Latin and South America. It is held in addition to EIQ and the Latin/South America division qualification, and is a standard component of a Latin/South America base assignment.

Terrain awareness and terrain procedure for every other region are incorporated into the applicable division qualification rather than issued separately.

What It Covers

- Escape and engine-out procedure at high-elevation, terrain-bounded airports
- RNP AR authorization and the approaches authorized by fleet
- Circling and night visual restrictions at designated airports
- Drift-down and terrain clearance planning on Andean and Central American routes
- High-elevation performance, including the effect of temperature on climb limit weight

Warning: At several Latin division airports only the captain may conduct the takeoff or the landing, and transfer of control is completed before top of descent and after top of climb. Verify the restriction on the airport qualification page before every operation.

Airport Qualification Is Separate

Airport qualification for a specific special airport is the only qualification at Northern Airways earned individually by a pilot on top of the bid position. See section [21.9 Special Qualification Airports](#).

21.6.6 One Seat, One Fleet

A pilot holds a bid status in **one seat of one fleet** at a time. A pilot may not operate in a seat or fleet other than the one held.

Exceptions, each requiring chief pilot authorization and a valid qualification record:

- Check airmen and line check pilots qualified in multiple fleets
- Management pilots holding a currency qualification in a second fleet
- A pilot in transition, during the period both qualifications are valid

21.6.7 Transition

A pilot awarded a new fleet or seat is removed from line flying in the previous status at the start of training. The previous qualification lapses on completion of operating experience in the new status.

21.7 Operating Experience

21.7.1 Requirement

Following initial, transition, or upgrade training, a pilot will complete operating experience in line operations under the supervision of a check pilot before serving unsupervised (14 CFR 121.434).

All Northern Airways aircraft are **Group II** airplanes (turbojet).

Training	Seat	Minimum
Initial	Captain and first officer	25 hours
Transition	Captain	25 hours
Transition	First officer	15 hours
— End —		

Operating experience will include at least **4 operating cycles**, at least 2 of them as the pilot flying (14 CFR 121.434(c)(3)).

Northern Airways may extend operating experience beyond the minimum at the discretion of the check pilot.

Note: The hourly reduction for additional takeoffs and landings permitted by 14 CFR 121.434(f) is **not available** for Group II initial training, or for second in command pilots in Group II transition training. Northern Airways does not apply it in any case.

21.7.2 FAA Observation — Qualifying Captains

A pilot completing initial or upgrade training to captain will be observed by an FAA inspector during at least one flight leg including a takeoff and landing (14 CFR 121.434(c)(1)(ii)).

21.7.3 Consolidation of Knowledge and Skills

Following a proficiency check or a type rating practical test, a pilot will acquire at least **100 hours of line operating flight time** — inclusive of operating experience — within **120 days** (14 CFR 121.434(g)).

If the 100 hours are not completed within 120 days, the period may be extended to no more than **150 days** provided the pilot completes refresher training on or before the 120th day, or a check pilot determines the pilot has retained an adequate level of proficiency after observing a supervised line operating flight (14 CFR 121.434(h)(4)).

A pilot who serves in another airplane type before completing the 100 hours will complete refresher training before returning to the newly qualified airplane (14 CFR 121.434(h)(3)).

Crew Scheduling tracks consolidation. The pilot is responsible for verifying it.

21.7.4 Limitations During Operating Experience

While in OE status, a pilot will not:

- Operate into a Special Qualification airport (Category B, C, or D) unless the check airman is qualified for that airport
- Operate with another pilot who is in OE status, or who is a High Minimums Captain
- Serve as pilot flying below Category I minimums

21.8 High Minimums Captain

21.8.1 Applicability

A captain is designated **High Minimums (CA-HM)** until having served 100 hours of line operating time as pilot in command in the aircraft type (14 CFR 121.652).

21.8.2 Increased Minimums

A High Minimums Captain will add:

- **100 feet** to the published DA/DH or MDA, and
- **½ statute mile** (or 800 meters RVR) to the published visibility

for every instrument approach flown, at every airport.

The increased minimums apply to the **approach minimums** and to the **alternate minimums** for any alternate filed. They do not increase takeoff minimums.

21.8.3 Crew Pairing

A High Minimums Captain will not be paired with a first officer in OE status. Crew Scheduling enforces this. If the pairing occurs through irregular operations, contact the chief pilot before departure.

21.8.4 Removal of the Designation

The designation is removed automatically on reaching 100 hours PIC in type. The pilot will verify removal in the crew record; until removed, the increased minimums apply.

21.9 Special Qualification Airports

21.9.1 Categories

Certain airports require qualification beyond ordinary line qualification because of terrain, obstacles, approach design, runway characteristics, or airspace complexity (14 CFR 121.445). Northern Airways categorizes them:

Category	Requirement
A	No special qualification. The majority of airports.
B	Self-briefing. Captain reviews the airport qualification package before the first operation and annually thereafter.
C	Formal briefing. Captain completes a training module and a check airman or simulator qualification before the first operation.
D	Captain must have operated into the airport within the preceding 12 months, or be accompanied by a qualified check airman.
— End —	

21.9.2 Currency

The **captain** holds the special airport qualification. The first officer is not required to hold it, but will review the airport qualification package before operating there.

21.9.3 Designated Special Airports

Northern Airways designates the following as special airports requiring individual captain airport qualification. They are referred to in scheduling and in crew tracking as **HIT cities**.

ICAO	Airport	Location	Reason
KDCA	Ronald Reagan Washington National	Washington, D.C.	River visual, restricted airspace, short runways
KONT	Ontario International	Ontario, CA	Terrain, high summer temperatures
KRNO	Reno-Tahoe International	Reno, NV	Mountainous terrain, elevation, escape procedure
KSAN	San Diego International	San Diego, CA	Obstacle-limited approach, single runway
KSFO	San Francisco International	San Francisco, CA	Parallel operations, terrain, low visibility
MHTG	Toncontín International	Tegucigalpa, Honduras	Terrain, short runway, steep visual approach
MMMX	Benito Juárez International	Mexico City, Mexico	High elevation, terrain all quadrants
PHLI	Lihue	Kauai, HI	Terrain, weather, single runway
PHOG	Kahului	Maui, HI	Terrain, wind, downdraft
SEQM	Mariscal Sucre International	Quito, Ecuador	High elevation, terrain all quadrants, RNP AR
SKBO	El Dorado International	Bogotá, Colombia	High elevation, terrain, weather
TIST	Cyril E. King	St. Thomas, USVI	Short runway, terrain, obstacle
TNCM	Princess Juliana International	St. Maarten	Short runway, obstacle, low approach
VHHH	Hong Kong International	Hong Kong	Terrain, airspace complexity
KASE	Aspen-Pitkin County	Aspen, CO	Elevation, one-way operation, steep visual, terrain

ICAO	Airport	Location	Reason
KEGE	Eagle County Regional	Eagle, CO	Elevation, one-way operation, escape procedure
KJAC	Jackson Hole	Jackson, WY	Terrain, escape procedure, elevation
MGGT	La Aurora International	Guatemala City, Guatemala	Elevation, terrain, single runway
MROC	Juan Santamaría International	San José, Costa Rica	Terrain, weather, escape procedure
SKRG	José María Córdova International	Medellín, Colombia	High elevation, terrain, weather
— End —			

Note: Additional airports may be designated at the discretion of the Director of Operations or the Chief Pilot. Crews are notified of an addition by Flight Operations Bulletin.

Latin Division Airports

MGGT, MHTG, MMMX, MROC, SEQM, SKBO, and SKRG also require Latin Terrain Qualification. See section [21.6.5 Latin Terrain Qualification](#). Holding one does not substitute for the other — a captain flying into Quito holds the Latin Terrain Qualification **and** the SEQM airport qualification.

21.9.4 Airport Qualification Pages

Each special airport has an airport qualification page in the Route Manual on the EFB. The page states the reason for designation, terrain and obstacle information, authorized and prohibited procedures, the approaches authorized by fleet, and any restriction on who may fly the takeoff or the landing.

Both pilots review the applicable page before departing to or from a special airport, whether or not either has operated there before.

Common restrictions found on those pages:

- Captain-only takeoff or landing, with transfer of control before top of descent and after top of climb
- Night visual and circling operations prohibited
- Specific RNAV departures prohibited on climb-gradient grounds
- Cabin secured earlier than normal — flight attendants seated by a stated flight level on descent

21.9.5 Diversion to a Special Qualification Airport

Special qualification requirements do not apply when diverting for an emergency or when the captain determines that landing at the airport is the safest available course of action. Report the diversion. See section [23.1 General](#).

21.9.6 Extended Operations (ETOPS)

ETOPS requires separate qualification. See Enroute ETOPS.

21.10 Scheduling**21.10.1 Governing Regulation**

Northern Airways flightcrew members operate under **14 CFR Part 117**. Part 117 governs flight time, duty period, and rest for all passenger operations conducted under Part 121.

This section summarizes Part 117 for planning. **It is not a substitute for the regulation.** Where this section and Part 117 differ, Part 117 governs, and the discrepancy will be reported to Flight Operations Publications.

21.10.2 Flight Time Limits

Limit	Value	Reference
Flight time, unaugmented, 2 pilots	8 or 9 hours, by scheduled report time	§117.11(a), Table A
Flight time, augmented	13 hours (3 pilots), 17 hours (4 pilots)	§117.11(b)
Flight time	100 hours in any 672 consecutive hours	§117.23(c)(1)
Flight time	1,000 hours in any 365 consecutive calendar days	§117.23(c)(2)
Flight duty period	60 hours in any 168 consecutive hours	§117.23(b)(1)
Flight duty period	190 hours in any 672 consecutive hours	§117.23(b)(2)
— End —		

21.10.3 Flight Duty Period

Maximum FDP is determined by **scheduled report time** and **number of flight segments**, per **§117.13 and Table B** for unaugmented operations and **§117.17 and**

Table C for augmented operations. The FDP begins when the crewmember reports and ends when the aircraft parks after the final segment.

Note: A flightcrew member who is **not acclimated** has the Table B maximum reduced by **30 minutes**, and the applicable FDP is based on local time in the theater in which the crewmember was last acclimated (§117.13(b)).

Note: The applicable FDP limit is fixed by the **scheduled** report time. A delayed report does not shift the crewmember into a more favorable row of the table.

21.10.4 Extension

An FDP may be extended up to **2 hours** due to unforeseen operational circumstances arising after departure per §117.19, with the concurrence of the pilot in command and the dispatcher. An extension of more than 30 minutes may occur no more than twice in any 168 consecutive hours.

21.10.5 Rest

Requirement	Standard
Minimum rest before an FDP	10 hours, with opportunity for 8 uninterrupted hours of sleep (§117.25(e))
Minimum rest, reduced	Not permitted under Part 117
Weekly rest	30 consecutive hours free from all duty in any 168 consecutive hours (§117.25(b))
— End —	

Warning: Rest is a **period free from all duty and from any present responsibility for work**. A crewmember contacted by the Company during a rest period should note the interruption and report it. Contact during rest may invalidate the rest period.

21.10.6 Reserve

Short call and long call reserve are governed by §117.21. The reserve availability period and its interaction with FDP limits are described in the collective bargaining agreement and in the Crew Scheduling Handbook.

21.10.7 Responsibility

Per §117.5, **both** the Company and the crewmember are responsible for ensuring the crewmember is fit and legal. A crewmember will not accept an assignment, and the Company will not issue one, if the crewmember cannot legally complete it.

21.11 Fatigue Risk Management

21.11.1 Policy

Fatigue is a threat, and it is managed like any other threat — identified, assessed, and mitigated. See Threat & Error Management.

Compliance with Part 117 does not establish fitness. A crewmember who is legal under Part 117 and too fatigued to operate safely will not operate.

21.11.2 Fatigue Call

A crewmember who is too fatigued to operate safely will notify Crew Scheduling and state that they are fatigued. The crewmember will then submit a fatigue report.

A fatigue call is not a disciplinary event and is not an attendance occurrence. Crew Scheduling will not require a fatigued crewmember to operate.

21.11.3 Fatigue Reporting

Submit a fatigue report through the ASAP portal for every fatigue call, and for any duty period in which fatigue was a factor even if the flight was completed. Reports feed the Fatigue Risk Management Program and are used to identify pairing constructions that reliably produce fatigue.

21.11.4 In-Flight Mitigation

In-seat rest is not authorized — see section [7g.3 In-Seat Rest](#). For rest on an augmented operation see section [7g.4 Crew Rest Facilities](#).

21.12 Operating Limitations by Aircraft Type

The table summarizes key operating limitations by fleet. Detailed performance limitations are in the applicable AFM and AOM. **The AFM is the controlling document in every case.**

Aircraft	Max Crosswind (Dry)	Max Crosswind (Wet)	Max Tailwind	Max Altitude
E190	33 kt	23 kt	10 kt	FL410
A319 / A320 / A321	38 kt	29 kt	10 kt	FL390
A321neo / XLR	38 kt	29 kt	10 kt	FL410

Aircraft	Max Crosswind (Dry)	Max Crosswind (Wet)	Max Tailwind	Max Altitude
B737	33 kt	25 kt	10 kt	FL410
B757	33 kt	25 kt	10 kt	FL420
B767	33 kt	25 kt	10 kt	FL430
B777	38 kt	31 kt	10 kt	FL430
A330	38 kt	29 kt	10 kt	FL410
— End —				

Caution: These are demonstrated or certificated values. They are not a target. Contaminated-runway crosswind limits are lower and are stated with the runway condition guidance — see section [10g.2 Contaminated Runway](#).

21.13 **Reserved**

Chapter 22: Threat & Error Management

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22.1 General

22.1.1 Purpose

Threat and Error Management is how Northern Airways expects crews to think. It is not a training subject that ends at the simulator door. Every policy in this manual exists because someone, somewhere, encountered a threat, made an error, or arrived in an undesired aircraft state.

TEM assumes three things:

1. **Threats are always present.** The operational environment is never clean.
2. **Errors will occur.** Competent, well-trained, well-rested pilots make errors.
3. **The outcome depends on management, not on avoidance.** Safety is produced by detecting and correcting, not by being error-free.

22.1.2 Components

Component	Definition
Threat	An event or condition that occurs outside the influence of the crew, increases operational complexity, and requires crew attention to maintain safety margins.

Component	Definition
Error	A crew action or inaction that leads to a deviation from crew or organizational intentions or expectations.
Undesired Aircraft State	A position, attitude, condition, or configuration of the aircraft that reduces safety margins.
— End —	

These are sequential. An unmanaged threat induces an error. An unmanaged error produces an undesired aircraft state. An unmanaged undesired aircraft state produces an incident or an accident.

The crew's job is to break the sequence at the earliest point available.

22.2 Threats

22.2.1 Categories

Category	Examples
Environmental	Weather, terrain, turbulence, icing, night, contaminated runway, high or hot conditions
Airport	Complex taxi layout, construction, hot spots, short runway, LAHSO, non-standard signage
ATC	Late runway change, slam-dunk descent, similar call signs, congested frequency, non-standard phraseology, language
Aircraft	MEL/CDL item, unfamiliar configuration, automation quirk, recent maintenance
Operational	Schedule pressure, minimum turn, late aircraft, close-in fuel change, last-minute weight and balance
Crew	Unfamiliar crew pairing, fatigue, first flight of a trip, last flight of a trip, distraction
Organizational	Recent procedure change, new manual revision, new station, new fleet
— End —	

22.2.2 Anticipated and Unanticipated

An **anticipated** threat is one the crew identified in planning or briefing. It is managed by a plan made when workload was low.

An **unanticipated** threat arrives without warning and is managed in real time at whatever workload exists. Unanticipated threats are more dangerous, which is why briefing quality matters — a good brief converts unanticipated threats into anticipated ones.

22.2.3 Identifying Threats

Threat identification is a continuous, out-loud activity. During preflight and descent planning, the crew will state the threats applicable to the operation and the specific mitigation for each. A threat named without a mitigation has not been managed.

"The runway is wet, we're heavy, and the crosswind is near the limit. If we're not stable by 1,000 feet, or if we touch down beyond the aiming point, we go around. I'll use max reverse and I'm planning to exit at the end."

22.3 Errors

22.3.1 Types

Type	Description
Aircraft handling	Manual flying, automation input, systems operation, ground handling
Procedural	Correct intention, wrong execution — checklist, callout, briefing, documentation
Communication	Missing or misinterpreted information, within the crew or with ATC or ground personnel
— End —	

22.3.2 Error Management

Effective error management mitigates the negative consequences of human error. The first step is to **identify** the error. The earlier an error is identified, the more quickly it can be repaired, returning the flight to a safe operation.

Strategies are the applicable CRM/TEM skills:

- **Monitor / cross-check** by maintaining awareness of aircraft status and crewmember actions
- **Workload management** by ensuring tasks are properly prioritized
- **Automation management** by ensuring the proper automation level is selected

*Stated simply: **Identify and repair.***

22.3.3 Response to an Error

There are three responses to a detected error, and only one is acceptable:

Response	Result
Trap — detect and correct before consequence	Acceptable. This is the objective.
Exacerbate — detect and respond in a way that worsens the situation	Unacceptable
Fail to respond — detect and take no action, or fail to detect	Unacceptable
— End —	

22.3.4 Reporting

Report every error, including an error that was trapped and corrected. See section [22.10 Safety Reporting](#).

22.4 Undesired Aircraft State

22.4.1 Definition

An Undesired Aircraft State (UAS) is a position, attitude, condition, or configuration of an aircraft that reduces safety margins. It is a safety-compromising state resulting from ineffective error management.

22.4.2 Examples

- Unstable approach continued below the stabilization gate
- Aircraft not configured for the phase of flight
- Vertical or lateral deviation from clearance
- Speed outside the target band
- Unintended runway or taxiway occupancy
- Incorrect systems configuration — fuel, pressurization, anti-ice
- Approaching a limitation — flap, gear, MMO/VMO

22.4.3 Recovery

Identifying a UAS is the first step to returning the flight to safe operations. The earlier a UAS is identified, the earlier recovery can occur. Pilots will take immediate action, which may include:

- **Monitor / cross-check** to actively verify aircraft position and configuration

- **Automation management** to ensure the proper automation level is selected — including disconnecting and hand-flying
- **Technical skill** — fly the aircraft

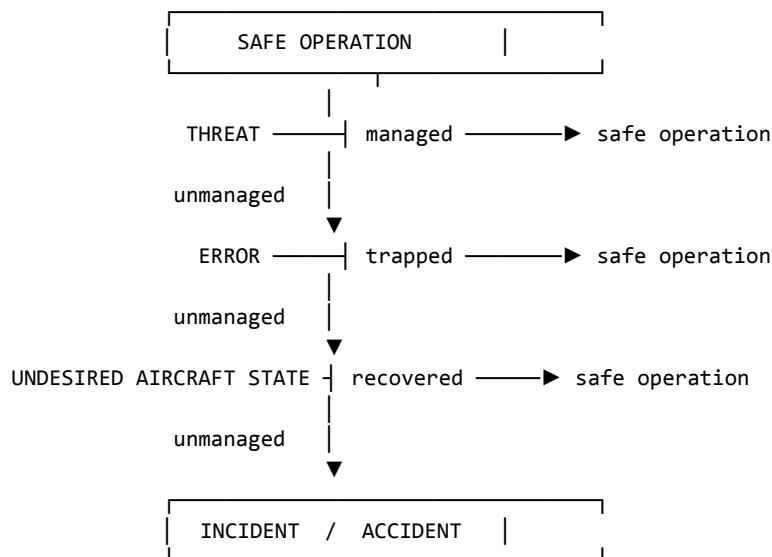
*Stated simply: **Identify and recover.***

22.4.4 Priority

Warning: A crew in an undesired aircraft state will recover the aircraft before diagnosing how it got there. Do not troubleshoot an unstable approach. Go around, then discuss it.

An aircraft operating in an undesired state and not promptly recovered may lead to an incident or accident.

22.5 TEM Model



Safe operation is at the top and represents the desired operating environment. As a crew encounters threats or errors there is potential to move away from it. If the trend continues, the result may be an incident or accident. Effective application of CRM/TEM skills creates a path away from that outcome and turns a divergent trend back toward safe operation.

The model is read in both directions. Every downward arrow has a corresponding recovery. There is no point in the sequence at which the crew has run out of options.

22.6 CRM/TEM Skills

A safe operation is maintained by combining technical skill with the management of threats and errors. The following observable, assessable behaviors are trained, evaluated, and expected on every flight:

Skill	Definition
Situational Awareness	Correctly assess the current and anticipated environment; identify and anticipate threats and errors.
Leadership Effectiveness	Exercise responsibilities in a manner that promotes teamwork, professionalism, and mentoring.
Planning and Decision Making	Incorporate relevant information, develop operational strategies, and choose the best course of action consistent with safe, reliable, and efficient operations.
Communication	Exchange information, ideas, and instructions in a timely manner.
Monitor / Cross-Check	Actively verify aircraft systems, aircraft position and configuration, and crewmember actions; resolve inconsistency and uncertainty.
Workload Management	Effectively manage tasks to optimize overall performance.
Automation Management	Select the appropriate level of automation for the situation and verify automation status.
— End —	

22.6.1 Monitoring

Monitoring is an assigned duty. The pilot monitoring actively verifies aircraft state, flight path, and the actions of the other pilot, and calls any deviation.

See section [1g.3 PF and PM Duties](#).

22.6.2 Automation Management

Select the level of automation appropriate to the situation, and state the change out loud. When the aircraft is not doing what is expected, reduce the level of automation rather than attempting to reprogram it.

Reduce the level of automation rather than reprogramming it in flight.

See section [1g.8 Automation Policy and the applicable AOM> 1> Automation](#).

22.7 Applying the ABCs

22.7.1 Model

Any crewmember who observes a threat, an error, or a developing undesired aircraft state will speak, using an escalating structure:

Step	Action	Example
A — Advise	State the observation factually.	"We're 200 feet high on the profile."
B — Be direct	State the required correction.	"You need to descend now to make the crossing restriction."
C — Command / Correct	Take control or take action.	"My aircraft."
— End —		

Escalate when the previous step does not produce a response. Do not stop at Advise because the other pilot is senior.

22.7.2 Two-Communication Rule

If a crewmember makes **two** communications regarding a safety-critical deviation and receives no adequate response, that crewmember will assume the other pilot is incapacitated or unresponsive and will **take control**.

State **"My aircraft."** verbatim. Do not phrase it as a question.

22.7.3 Applicability

The ABCs apply to every crewmember including flight attendants, relief pilots, jumpseat riders, and check airmen. A first officer is expected to take control from a captain when the criteria are met, and will be supported for doing so.

Note: A crewmember who takes control under this policy will be supported by the Company.

22.7.4 Debriefing

Any use of Be Direct or Command will be debriefed by the crew after the flight, and will be reported through ASAP. The debrief is about the event, not about the individual.

22.8 Briefings

22.8.1 Purpose

A briefing converts unanticipated threats into anticipated ones and establishes a shared mental model. A briefing that recites what both pilots already know accomplishes nothing.

22.8.2 Content

Brief what is **different, threatening, or uncertain** about this operation. Omit what is standard.

Brief this	Not this
"Runway 25L, we're landing beyond the intersection, brakes will be hot for the turn"	A recitation of the approach plate both pilots are looking at
"If we lose an engine after V1 the terrain turn is at 2.5 DME, I'll call for it"	"Standard engine failure procedure"
"This crew is new to each other and it's the first leg — let's be deliberate"	Nothing
— End —	

22.8.3 Required Briefings

Briefing	When	Reference
Departure	Before pushback	Preflight Communications, Departure Briefing
Taxi	Before taxi, and again on any clearance change	Taxi Communications
Takeoff	Included in departure briefing	Takeoff General
Approach	Before top of descent	Approach General, Approach Briefing
Flight attendant	Before each departure	Preflight Communications, Flight Attendants
— End —		

22.8.4 Interruption

A briefing interrupted will be resumed **from the beginning of the interrupted item**, not from the point of interruption. Interruption is a leading cause of omitted items.

22.9 Intentional Non-Compliance

22.9.1 Definition

Intentional non-compliance (INC) is a deliberate deviation from an established procedure or policy. It is categorically different from an error. An error is unintended; INC is chosen.

Line Operations Safety Audit data identifies errors caused by INC as "**intentional non-compliance errors**," and establishes that non-compliant flight crews are far more likely to commit errors of every type.

22.9.2 Normalization of Deviance

When INC becomes habitual in the absence of consequences, **Normalization of Deviation** is established as an alternative to compliance. This creates opportunity for exponential errors and a growing systemic threat to safe operations.

The mechanism is simple and it is not obvious from inside it: a deviation that produces no bad outcome is experienced as evidence that the procedure was unnecessary. Repeated, it becomes the crew's actual procedure. The margin it consumed is invisible until the day it is needed.

22.9.3 Compliance

The best defense against intentional non-compliance is holding one's self and each other accountable to the established standard operating policies and procedures.

This is a two-part obligation. Comply. And say something when the other pilot does not.

22.9.4 Reporting a Procedure

A procedure that cannot be complied with, or that is unsafe as written, will be **reported and changed** — not worked around. Submit through the Manual Feedback form and, if it affected a flight, through ASAP.

A workaround applied by one crew and not reported leaves the procedure in place for every other crew.

22.10 Safety Reporting

22.10.1 Aviation Safety Action Program (ASAP)

ASAP is a voluntary, non-punitive reporting program operated jointly by Northern Airways, the Northern Airways Pilots Association (NAPA), and the FAA under an Memorandum of Understanding.

Report through ASAP:

- Any deviation from a clearance, procedure, or regulation
- Any error, whether or not it had a consequence
- Any threat encountered that others should know about
- Any event in which the ABCs were escalated beyond Advise
- Any go-around, diversion, or rejected takeoff
- Anything that made you uncomfortable and that you are still thinking about

Timeliness: file within **24 hours** of the event, and before the end of the trip where practicable. Late reports may be excluded from ASAP protection.

22.10.2 Exclusions

ASAP protection does not extend to:

- Intentional disregard for safety
- Criminal activity, substance abuse, controlled substances, or intentional falsification
- Events already known to the FAA through another source, where the report is not timely

22.10.3 Flight Operations Quality Assurance (FOQA)

FOQA collects de-identified flight data to identify trends. Data is not used for discipline and is not attributable to an individual crew, per the FOQA Memorandum of Understanding.

22.10.4 Line Operations Safety Audit (LOSA)

LOSA observations are conducted periodically by trained observers on normal line flights. Observations are de-identified, are never used for discipline, and are declined by the crew at any time without consequence.

22.10.5 Just Culture

Northern Airways operates a just culture. The distinction that matters is not between a good outcome and a bad one, but between:

Behavior	Response
Human error	Console and learn. Examine the system that permitted it.
At-risk behavior — a drift the individual did not recognize as risky	Coach. Remove the incentive to drift.
Reckless behavior — conscious disregard of substantial and unjustifiable risk	Discipline.

Behavior	Response
— End —	

The same outcome may follow from any of the three. The response depends on the behavior, not the damage.

22.11 **Reserved**

Chapter 23: Reports and Forms

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23.1 General

This chapter identifies what is reported, to whom, and within what period. Where a report is required by more than one section, one submission to each destination satisfies all of them.

Reports are submitted through the Flight Operations portal unless this chapter specifies otherwise.

Note: A report filed late is worth more than a report not filed. File it.

23.2 Reports to Dispatch

23.2.1 General

Advise the dispatcher of any condition affecting the conduct of the flight as released. The dispatcher shares operational control and cannot exercise it without information. See section [20.4 Operational Control](#).

23.2.2 Required Reports

Advise the dispatcher of:

- any deviation from the release — route, altitude, destination, or alternate
- any diversion, and the intent to divert where time permits
- any mechanical irregularity affecting the operation
- any emergency declaration, once the aircraft is under control
- any medical event, security event, or passenger disturbance
- any encounter with severe turbulence, windshear, or volcanic ash

- any fuel state projected below the release requirement
- any use of the captain's emergency authority to deviate from this manual or from 14 CFR

23.2.3 Follow-Up Reports

Where a verbal or ACARS report was made in flight, submit the written follow-up before the end of the duty period.

23.3 Safety Reports

23.3.1 Aviation Safety Action Program

ASAP is a voluntary, non-punitive reporting program operated jointly by Northern Airways, the Northern Airways Pilots Association (NAPA), and the FAA under a Memorandum of Understanding.

File an ASAP report for:

- any deviation from a clearance, procedure, or regulation
- any error, whether or not it had a consequence
- any go-around, diversion, or rejected takeoff
- any escalation of the ABCs beyond Advise — see section [22.7 Applying the ABCs](#)
- any in-flight fatigue event and the mitigation applied — see section [7g.3 In-Seat Rest](#)
- any threat encountered that other crews should know about

File within **24 hours** of the event, and before the end of the trip where practicable. A late report may be excluded from ASAP acceptance.

Exclusions

ASAP does not cover intentional disregard for safety, criminal activity, substance abuse, or intentional falsification. See section [22.10 Safety Reporting](#).

23.3.2 Fatigue Report

File a fatigue report for every fatigue call, and for any duty period in which fatigue was a factor even where the flight was completed. See section [21.11 Fatigue Risk Management](#).

23.3.3 Flight Operations Quality Assurance

FOQA data is collected automatically and de-identified. No crew action is required. Data is not used for discipline and is not attributable to an individual crew.

23.4 Flight Crew Irregularity Report

File a Flight Crew Irregularity Report (FCIR) for an event that is operationally significant but is not a safety event, including:

- a ground handling irregularity or damage discovered post flight
- a passenger or crewmember injury
- a service failure affecting the conduct of the flight
- an interaction with a Company department requiring resolution

The FCIR is routed to the domicile chief pilot.

23.5 Regulatory Reports

23.5.1 NTSB Notification

Source: 49 CFR Part 830

Notify the dispatcher immediately of any event that may require NTSB notification. The dispatcher notifies the Integrated Operations Center, which notifies Flight Safety. Flight Safety makes the NTSB notification.

Events requiring immediate notification include flight control system malfunction, in-flight fire, aircraft collision in flight, serious injury, and damage meeting the substantial damage definition.

23.5.2 Cockpit Voice Recorder

Source: 14 CFR 121.359 and 91.609

The CVR is limited to use in an accident or incident investigation. The recording will not be erased before completion of the parking checklist.

Where an incident requiring immediate NTSB notification has occurred within the preceding 30 minutes, the flight crew will:

1. preserve the recording after landing or after parking by removing power from the CVR at the circuit breaker
2. contact the Integrated Operations Center so that Flight Safety can be notified

See also section [23.5.1 NTSB Notification](#).

23.5.3 Near Midair and Airspace Deviation

File an ASAP report and advise the dispatcher. Where ATC requests a phone number, provide it and advise the chief pilot before speaking with the FAA.

23.5.4 Bird and Wildlife Strike

Record the strike in the eAML, file an ASAP report, and submit an FAA wildlife strike report through the Flight Operations portal.

23.5.5 Lightning Strike

Record in the eAML. The aircraft requires inspection before further flight where required by the applicable maintenance procedure.

23.6 Cabin and Passenger Reports

23.6.1 Passenger Disturbance

Where a passenger interferes with a crewmember, the captain will:

- advise ATC and request law enforcement to meet the aircraft
- advise the dispatcher
- obtain written statements from the crewmembers involved before the crew separates
- file a report through the Flight Operations portal

23.6.2 Medical Event

Advise the dispatcher and request the ground-based medical advisory service. Record the advisory recommendation and the action taken.

Where a diversion is made for a medical event, see section [7n.1 Diversion](#).

23.6.3 Removal of a Passenger

Document the reason for removal, the persons involved, and the time. File before the end of the duty period.

23.7 Delay Coding

23.7.1 General

Every delay is assigned a delay code so that the cause is captured consistently across the system. The captain is responsible for the flight crew's portion of the delay report; the station is responsible for assigning the code. See section [1c.3.3 Delay Report](#).

Where the station has not assigned a code by departure, the delay is automatically coded **SD — Station Default**. Assigning the correct code afterward is the station's responsibility, done as soon as possible and no later than the end of the day.

23.7.2 Reference

The full delay code list is published in **Appendix A — Delay Codes**. A delay code is not selected by guess or by habit — the code that most specifically describes the cause is used, and the remarks state anything the code alone does not capture.

23.7.3 Split Delays

Where more than one cause contributed, the delay is split between the applicable codes rather than forced into a single one. Where a workgroup's late response contributed to a delay that was otherwise caused elsewhere — for example, a precautionary maintenance inspection called later than it could have been — the delay may be split between the originating code and the responding workgroup's code.

23.7.4 Crew Responsibility

The captain judges the minutes attributable to a flight crew cause and ensures they are reported accurately. See section [1c.3.3 Delay Report](#). A delay is not assigned to a crew-side code to avoid attributing it elsewhere, and it is not absorbed silently to protect an on-time statistic.

Chapter 24: Meteorology

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24.1 TAF/METAR

24.1.1 General

The weather package issued with the release satisfies 14 CFR 121.599. This chapter describes the products and their interpretation. Operating policy for each weather phenomenon is in the applicable phase chapter and in Gen Ops Weather.

Weather used for dispatch is obtained from the Company meteorology provider through SimBrief and the FOS overlay. Weather obtained from any other source is advisory only and will not be used to satisfy a regulatory requirement.

24.1.2 US Aerodrome Forecast (TAF)

A TAF is valid for the period stated and applies within **5 statute miles** of the center of the runway complex.

Group	Meaning
FM	From. A rapid change. The new conditions replace all preceding conditions.
BECMG	Becoming. A gradual change over the stated period.
TEMPO	Temporary. Fluctuations lasting less than 1 hour, in total less than half the period.
PROB30 / PROB40	Probability of the stated conditions during the period.
— End —	

Conditional Groups and Alternate Requirements

TEMPO and PROB groups are considered when determining whether an alternate is required and when assessing alternate suitability. Where a conditional group would

place the airport below the applicable minima, treat the airport as being at those conditions.

24.1.3 METAR and SPECI

A METAR is the routine observation. A SPECI is issued when conditions change significantly between routine observations.

Verify the observation time before use. An observation more than **1 hour** old is not current for the purpose of commencing an approach.

24.1.4 Ceiling and Visibility

Ceiling is the lowest broken or overcast layer, or vertical visibility. Scattered layers do not constitute a ceiling.

Where both prevailing visibility and RVR are reported, RVR controls for the runway to which it applies.

24.2 FOS/DECS Commands

24.2.1 General

Weather products are retrieved through the FOS overlay. The DECS command set is retained for compatibility and is the fallback where the overlay is unavailable.

Note: Where the FOS overlay is unavailable, see section [26.2 FOS Outage Backup Plan](#).

24.2.2 SIGMET and AIRMET

A SIGMET is issued for severe icing, severe or extreme turbulence, widespread duststorm or sandstorm, and volcanic ash. A Convective SIGMET is issued for thunderstorm activity meeting the published criteria.

An AIRMET is issued for conditions of lesser intensity affecting aircraft operations.

Review every SIGMET applicable to the route before departure and monitor for new issuances in flight.

24.2.3 Weather Radar Reports

Ground-based radar summaries indicate echo intensity, tops, and movement. They are a planning product. They do not replace airborne radar interpretation. See section [7w.2.1 Airborne Weather Radar](#).

24.2.4 Wind and Temperature Aloft

Forecast winds and temperatures are used in flight planning and are shown by waypoint on the computer flight plan. Compare actual against forecast and advise the dispatcher where the difference affects the fuel plan. See section [7g.2 Fuel Monitoring](#).

24.2.5 Volcanic Ash Messages

Volcanic Ash Advisories are issued by the responsible VAAC and identify the forecast ash cloud position by altitude and time. See section [7w.4 Volcanic Ash](#).

24.2.6 Pilot Reports

A PIREP is the most timely report of conditions actually encountered. File a PIREP for:

- moderate or greater turbulence
- any icing encountered
- any condition differing materially from the forecast
- braking action on landing where it differs from the report received

24.2.7 Severe Weather Watches

Convective outlooks, watches, and warnings identify areas of expected severe weather. They are a planning product and do not by themselves restrict the operation.

24.3 Significant Terminal Weather Information

24.3.1 Terminal Aerodrome Products

Terminal products include the TAF, the METAR, and any Company-issued terminal forecast for a hub or a station with limited reporting.

24.3.2 TWIP-Equipped Airports

Terminal Weather Information for Pilots is available by ACARS at equipped airports and provides microburst, windshear, and precipitation information for the terminal area.

Request TWIP before descent at an equipped airport when convective activity is present.

24.3.3 Low-Level Windshear Alert System

An LLWAS or TDWR alert reported by ATC is treated as a windshear report. See section [1w.2 Windshear / Microburst](#).

24.4 Enroute Weather Information

Enroute products include SIGMETs, AIRMETs, PIREPs, significant weather charts, and the turbulence forecast index shown on the computer flight plan.

Turbulence Index

The turbulence index on the flight plan is a forecast severity value by segment. It is advisory. Reported conditions and observed conditions take precedence.

24.5 Field Condition Reporting

24.5.1 Runway Surface Definitions

Condition	Definition
Dry	Less than 25% of the surface area within the reported length and width in use is covered by visible moisture.
Wet	25% or more of the surface area is covered by visible moisture not exceeding 1/8 inch.
Slippery	A wet runway with braking action equivalent to medium or worse.
Contaminated	More than 25% of the surface area is covered by water, frost, slush, or snow greater than 1/8 inch, or by any depth of compacted snow or ice.
— End —	

24.5.2 Runway Condition Assessment Matrix

Field condition reports use the Runway Condition Assessment Matrix (RCAM) and are issued as a Runway Condition Code (RwyCC) for each third of the runway.

Code	Runway Condition	Deceleration and Directional Control	PIREP
6	Dry	—	—
5	Frost. Wet, including damp and 1/8 inch or less of water. 1/8 inch or less of slush, dry snow, or wet snow.	Normal for the wheel braking effort applied; directional control normal.	Good
4	Compacted snow, OAT -15°C or colder.	Between good and medium.	Good to medium

Code	Runway Condition	Deceleration and Directional Control	PIREP
3	Wet where reported "slippery when wet". Dry or wet snow of any depth over compacted snow. More than 1/8 inch of dry snow, wet snow, or compacted snow at OAT warmer than -15°C.	Noticeably reduced for the wheel braking effort applied, or directional control noticeably reduced.	Medium
2	More than 1/8 inch of water or slush.	Between medium and poor.	Medium to poor
1	Ice.	Significantly reduced for the wheel braking effort applied, or directional control significantly reduced.	Poor
0	Wet ice. Water on top of compacted snow. Dry or wet snow over ice.	Minimal to non-existent for the wheel braking effort applied, or directional control uncertain.	NIL
— End —			

Apply the code to the takeoff and landing performance computation. See sections [5g.3 Runway Conditions](#) and [9w.2 Runway Conditions](#).

24.5.3 FICON NOTAM

Runway condition codes are published in a field condition NOTAM following the word **FICON**, given by runway third from the approach end.

ORD RWY 04L FICON 5/3/5 50PRCT WET, 50PRCT 1/8IN WET SN OVR
COMPACTED SN, 50PRCT 1/8IN SL. 1602251625-1602261625

Note: Where the three codes differ, apply the lowest code to the performance computation unless the applicable AOM states otherwise.

24.5.4 Pilot Braking Action Reports

Report braking action after landing on a contaminated runway using the terms **good**, **good to medium**, **medium**, **medium to poor**, **poor**, and **nil**.

Warning: A report of NIL braking action prohibits takeoff and landing. See sections [5g.3 Runway Conditions](#) and [9w.2 Runway Conditions](#).

24.6 Reserved

Chapter 25: Electronic Flight Bag (EFB)

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25.1 General

The EFB is a Company-issued device authorized under OpSpecs A061. It is required equipment. A flight will not depart without at least one serviceable EFB per pilot unless the reversion provisions of section 25.4 are satisfied.

25.1.1 Authorized Use

The EFB is used only for purposes related to the operation of the flight. Personal use of the Company EFB is not authorized.

For personal devices, see section [1g.4 Flightdeck Management](#).

25.1.2 Currency

Verify before each flight:

- the chart database is current for the period of the flight
- the performance application is at the released version
- the manual library has synchronized within the period published by Flight Operations Technology

Note: A device that has not synchronized within the published period is not current, regardless of the revision it displays.

25.2 EFB Operations

25.2.1 Mounting and Stowage

Table 25.1 EFB Mount by Fleet

Fleet	Mount
E190	Suction mount, side window
A319 / A320 / A321 / A321neo	Certified clip mount, side stick table
B737	Certified yoke mount
B757 / B767	Certified yoke mount
A330	Certified clip mount, side stick table
B777	Certified yoke mount
— End —	

The EFB is mounted only on the mount certified for that fleet. It is not rested on the yoke, the glareshield, the side console, or a knee.

Where a mount is unserviceable, the EFB is stowed for taxi, takeoff, and landing and used only in the cruise, held or braced clear of the controls. Record the unserviceable mount in the eAML.

The EFB is mounted in the approved mount during all phases of flight. It is not placed on the control column, the glareshield, or any surface from which it could become a projectile or obstruct a control.

Caution: A loose device in the flightdeck is a hazard in turbulence and can jam a control. Mount it or stow it.

25.2.2 Use During Critical Phases

EFB use during a critical phase of flight is limited to the task at hand. Complete chart selection and performance computation before taxi where practicable.

Taxi

During taxi, the pilot monitoring may reference the airport diagram. Extended interaction with the EFB during taxi is not authorized. See section [4g.1.4 Monitoring](#).

25.2.3 Custody

The EFB is issued to the pilot and remains in the pilot's possession. It is not left in the aircraft at the end of a sector, on a layover, or at the end of a duty period.

An EFB left aboard is a security exposure and a dispatch exposure — the next crew has no way of knowing whether it is current, charged, or has been tampered with.

Where an EFB is left aboard in error, report it to the Chief Pilot's office and to the EFB Help Desk. It is not returned to service until it has been checked.

25.2.4 Power and Charging

Both EFBs will not be operated on battery simultaneously. Where aircraft power is unavailable, manage battery state so that at least one device remains available through arrival.

25.2.5 Cross-Check

Performance figures computed on the EFB are cross-checked by the other pilot against an independent entry. Both pilots verify the inputs, not only the output.

Warning: Two pilots reading the same wrong output is not a cross-check. Enter the data independently.

25.3 Applications

25.3.1 Authorized Applications

Only the applications below are authorized on the Company EFB. No other application is installed, and none is used for an operational purpose.

Table 25.2 Authorized EFB Applications

Application	Function
Jeppesen FliteDeck Pro	Terminal and enroute charts, airport diagrams, and airport qualification pages
Comply365	Manual library and the Controlled Document Library — FOM, AOM, QRC/QRH, MEL, CDL, bulletins, forms, and regulatory documents
Land App	Landing performance
FMS+	Flight profile optimizer — altitude and cost index recommendations
WSI Pilotbrief	Weather — TAF, METAR, SIGMET, radar, satellite, turbulence, and TWIP
Flight Plan Viewer	The dispatch release and computer flight plan, delivered from FlightKeys
eHOT	Holdover time computation
CCI	Flight details view — crew, aircraft, and flight status

Application	Function
eAML	Electronic aircraft maintenance log
— End —	

Note: Takeoff performance is **not** computed on the EFB. It is issued as the TPS, requested by ACARS, or computed on the TPS manual calculation worksheet. See section [1p.2.2 Takeoff Performance System \(TPS\)](#).

25.3.2 Authoritative Source

Where two sources disagree, the following governs.

Table 25.3 Authoritative Source by Function

Function	Governs	Backup
Takeoff performance	The TPS	ACARS TPS, then the TPS manual calculation worksheet
Landing performance	Land App	ACARS Landing Report
Cruise optimization	FMS+ is advisory only	The filed flight plan governs
Release and flight plan	The electronic release on the EFB	The dispatcher, by voice
Charts	The current chart cycle in FliteDeck Pro	A chart obtained from Dispatch
Manuals	The revision in Comply365	None — an uncontrolled copy is not used
Weather	The package issued with the release	WSI Pilotbrief, and the dispatcher
Holdover time	eHOT	The printed holdover tables
— End —		

Caution: An application is a calculator, not an authority. Where the result it produces is inconsistent with the TPS, the release, or the aircraft, the result is wrong until it is reconciled.

25.3.3 Currency

Verify before each flight that the chart cycle, the performance database, and the manual revisions are current for the period of the flight.

An expired chart cycle or performance database is not used. Where either expires during the flight, see section [3g.3 Databases](#).

25.3.4 Performance Data Entry

Performance inputs are entered by one pilot and **independently verified by the other against the source document** — the TPS, the closeout, or the ATIS.

Verification is of the **inputs**, not only the result. Both pilots confirm:

- runway and intersection
- aircraft weight and CG
- temperature, altimeter setting, and wind
- runway condition and any contamination
- flap setting, thrust rating, and anti-ice state

Warning: A performance application will return a confident answer to a wrong input. A transposed weight or a wrong runway produces a plausible V-speed. The verification is the only defense, and it is verification of what was typed in.

25.3.5 Training and Help

This manual states **when** an application is used and **which** one governs. It does not describe how to operate one.

For operation, use the in-application tutorial and the EFB training module in Comply365. Both are maintained against the current release of each application; this manual is not.

Report an application defect through the EFB Help Desk. Do not develop a workaround and continue using it without reporting it.

25.3.6 Task Index

Table 25.4 Which Application for Which Task

Task	Application
Pull a terminal chart or airport diagram	FliteDeck Pro
Review an airport qualification page	FliteDeck Pro
Read the release or the navigation log	Flight Plan Viewer
Request an ACARS TPS	Aircraft ACARS, not the EFB
Compute landing performance	Land App
Compute holdover time	eHOT
Optimize cruise altitude or cost index	FMS+

Task	Application
Check crew, aircraft, or flight status	CCI
Check a TAF, METAR, or SIGMET	WSI Pilotbrief
Look up an MEL or CDL item	Comply365
Read a Flight Operations Bulletin	Comply365
File a safety or irregularity report	Comply365
Make an eAML entry	eAML
— End —	

25.4 Failure and Reversion

25.4.1 Single EFB Failure

Where one EFB fails, the flight may continue and may depart, provided:

- the remaining device is serviceable and current
- the remaining device is powered from aircraft power
- the crew briefs the sharing arrangement for charts and performance

Record the failure in the eAML and advise Flight Operations Technology.

25.4.2 Dual EFB Failure

Where both EFBs fail:

- advise ATC and the dispatcher
- continue to a suitable airport using the aircraft navigation systems and ATC assistance
- do not commence an instrument approach without a current chart for that approach

A flight will not depart with both EFBs inoperative.

25.4.3 Application Failure

Where a single application fails but the device is otherwise serviceable, use the backup source in the table below.

Failed application	Backup
Charts	Other pilot's device
Performance	Airport analysis tables
Manual library	Other pilot's device
Flight plan viewer	Printed release from Operations
— End —	

25.5 Securing

Secure the EFB in the approved stowage before leaving the aircraft. Do not leave a device in the flightdeck at an RON station unless the aircraft is secured in accordance with the AOSSP.

Report a lost or stolen device immediately to the chief pilot and to Flight Operations Technology.

25.6 Reserved

Chapter 26: Sabre/DECS

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- [26.1 General](#)
- [26.2 FOS Outage Backup Plan](#)
- [26.3 Common Entries](#)
- [26.4 Access and Security](#)
- [26.5 Message Traffic and Printer Routing](#)
- [26.6 Reserved](#)

26.1 General

Northern Airways flight planning is produced in **SimBrief** and delivered through the Company **Flight Operating System (FOS)** overlay. Host reservations and crew records reside in **Sabre**, and the legacy **DECS** command set remains available for direct entry.

System	Holds
SimBrief	Computer flight plan, fuel plan, weather package
FOS	Dispatch release, crew assignment, aircraft routing, eAML
Sabre	Reservations, passenger records, jumpseat listing
DECS	Direct command access to FOS and Sabre data
— End —	

Crewmembers are not expected to work in DECS routinely. The command set is published so that a crew can retrieve its own release, weather, and aircraft status when the overlay is unavailable.

Note: DECS is a retrieval and reference tool for the flight crew. It is not a substitute for dispatcher-issued instructions, and an entry made in DECS does not itself amend a release. See section 26.2.1.

26.2 FOS Outage Backup Plan

An FOS outage affects release delivery, crew tracking, and the eAML. It does not suspend operational control. The dispatcher retains it throughout.

26.2.1 Crew Actions

During an FOS outage:

- obtain the release from the dispatcher by ACARS, company radio, or telephone
- record the release number, the fuel required, and any MEL item manually
- use the paper eAML procedure — see section 3m.2 eAML
- confirm crew legality with Crew Scheduling verbally, and record the name of the scheduler and the time

Warning: A verbally issued release is a valid release. A flight departing without one is not dispatched, regardless of the state of the system.

26.2.2 Restoration

On restoration, reconcile the manual records into FOS before the end of the duty period. Advise the dispatcher when reconciliation is complete.

Where a discrepancy is found between the manual record and the reconciled FOS entry, the manual record made at the time governs. Report the discrepancy to the Chief Pilot's office.

26.3 Common Entries

The following entries are provided for reference. The full command set is published in the Controlled Document Library.

Entry	Retrieves
FP*<flight>/<date>	Computer flight plan and dispatch release
WX*<icao>	METAR and TAF for the station
SIG*<region>	SIGMETs and AIRMETs for the region
FD*<region>	Winds and temperatures aloft
AC*<tail>	Aircraft status, open discrepancies, and routing
ML*<tail>	eAML entries for the aircraft
CR*<employee>	Crew assignment and legality
AL*<icao>	Field condition report and NOTAMs
— End —	

Note: Entries are case-insensitive. A malformed entry returns an error rather than incorrect data, but verify the flight number and date on every response.

26.3.1 Reading the Response

Every DECS response is stamped with the time of retrieval, not the time the underlying data was last updated. Where a response is older than expected — particularly a weather or NOTAM query — re-query rather than relying on a cached-looking response.

Where an entry returns no data, confirm the flight number, tail number, or station identifier before concluding the record does not exist. A single transposed character returns "not found," not a warning that the entry was malformed.

26.4 Access and Security

DECS access is tied to the individual crewmember's employee credentials. Do not share a login or leave a DECS session open and unattended on a shared terminal.

A DECS session does not itself authorize a change to a release, a crew assignment, or an eAML entry. Where an entry appears to permit a change, it is a query tool showing what has already been authorized elsewhere — changes to the operation are made by the dispatcher, Crew Scheduling, or Maintenance Control, not by the crewmember's own DECS entry.

Caution: Retrieving a record in DECS is not the same as acting on it. Confirm any change to a release or an assignment through the issuing authority before relying on it.

26.5 Message Traffic and Printer Routing

Operational messages — the dispatch release, an inflight release amendment, an ACARS TPS response, and CCI operational alerts — are routed to the aircraft printer or the CCI display, not through DECS. See section [1c.7 Company to Flight Crew](#) and section [2.2.17 Inflight Release Amendment](#).

DECS is queried by the crew from a company terminal, on the ground, before or after the flight. It is not the channel used for inflight message traffic.

26.6 Reserved

Chapter 27: Glossary

Table of Contents

- [27.1 General](#)
- [27.2 Terms](#)
- [27.3 Abbreviations and Acronyms](#)

27.1 General

A term defined in this chapter carries the stated meaning throughout this manual. Where a term is defined in 14 CFR, the regulatory definition governs and is noted.

27.2 Terms

Term	Definition
AFE	Above field elevation. Height referenced to the elevation of the landing runway.
Alert Height	The height above the runway, established for the fleet and the approach category, above which a CAT III approach is discontinued for a specified failure.
Anti-icing	Application of fluid to a clean surface to prevent the accumulation of contamination.
Cabin Secure	The report from the flight attendant in charge that the cabin is prepared for departure or landing.
Clean Aircraft Concept	The prohibition on takeoff with frost, ice, or snow adhering to a critical surface. See section 3g.7 Deicing .
Company	Northern Airways, Inc.
Contaminated Runway	A runway with more than 25 percent of the required area covered by standing water, slush, snow, or ice.
Controlled Rest	A planned in-seat nap by one pilot at the controls. An ICAO practice permitted by some foreign authorities. Not authorized in 14 CFR Part 121 operations, and not used at Northern Airways. See section 7g.3.
Critical Phase of Flight	All ground operations involving taxi, takeoff, and landing, and all other flight operations below 10,000 feet except cruise. (14 CFR 121.542)

Term	Definition
Critical Surface	A wing, control surface, rotor, propeller, horizontal stabilizer, vertical stabilizer, or any surface where contamination affects performance or control.
Deicing	Removal of contamination from a surface.
Dispatch Release	The document authorizing a flight, signed by the dispatcher and accepted by the captain.
eAML	Electronic Aircraft Maintenance Log. The legal record of discrepancies and their resolution.
Emergency Fuel	The condition in which usable fuel on landing at the nearest suitable airport would be less than final reserve. A declaration of emergency.
ETOPS	Extended Operations. Flight beyond the threshold time from an adequate airport, conducted under OpSpecs authorization.
Final Reserve	The fuel required to be aboard on landing, defined in the release.
Flightdeck	The compartment from which the aircraft is operated. Used throughout in place of <i>cockpit</i> .
FOS	Flight Operating System. The Company operational host. See Chapter 26.
High Minimums Captain	A captain with fewer than 100 hours pilot-in-command in type, subject to increased minimums. (14 CFR 121.652)
Holdover Time	The period anti-icing fluid is expected to prevent the accumulation of contamination, measured from the start of the final application.
Home	The navigation link in the running header returning to the manual index.
Intentional Non-Compliance	A deliberate deviation from an established procedure or policy. See section 22.9.
LAHSO	Land and Hold Short Operations. See section 10g.5.
Mainline	Flights operated by Northern Airways on its own certificate. Excludes Pioneer, Aloha, Aurora, Corvus, Cascadian, Embassy, and Cardinal.
Minimum Fuel	An advisory to ATC that arrival with less than planned reserve is expected. Not a declaration of emergency.
MFF	Maintenance Ferry Flight. See section 14g.3.
NIL	A braking action report indicating no braking effectiveness. Prohibits takeoff and landing.
Non-Normal	A condition addressed by the QRC/QRH or the AOM non-normal chapter.

Term	Definition
OpSpecs	Operations Specifications. The authorizations issued to Northern Airways by the Administrator.
Pilot Flying (PF)	The pilot controlling the flight path, thrust, and configuration.
Pilot Monitoring (PM)	The pilot verifying aircraft state and the actions of the pilot flying, and accomplishing checklists and communications.
Positioning Flight	A flight carrying no revenue passengers, operated to reposition the aircraft. See section 14g.2.
Reserved	A chapter, section, or paragraph number held vacant. Numbering is never compressed and a reserved number is not reissued.
RON	Remain overnight.
Sterile Flightdeck	The period during which activities not required for the safe operation of the aircraft are prohibited. See section 1g.5.
Stabilized Approach	An approach meeting all criteria of section 9g.2 at the applicable gate.
Suitable Airport	An airport meeting the runway, weather, and services criteria for the intended operation.
Threat	An event or condition outside the influence of the crew that increases operational complexity. See section 22.2.
Undesired Aircraft State (UAS)	A position, attitude, condition, or configuration that reduces safety margins. See section 22.4.
— End —	

27.3 Abbreviations and Acronyms

ACARS	Aircraft Communications Addressing and Reporting System	LLWAS	Low-Level Windshear Alert System
ADS-C	Automatic Dependent Surveillance — Contract	LOSA	Line Operations Safety Audit
AFE	Above Field Elevation	MDA	Minimum Descent Altitude
AFM	Aircraft Flight Manual	MEL	Minimum Equipment List
AOM	Aircraft Operating Manual	METAR	Aviation Routine Weather Report

ACARS	Aircraft Communications Addressing and Reporting System	LLWAS	Low-Level Windshear Alert System
AOSSP	Aircraft Operator Standard Security Program	MFF	Maintenance Ferry Flight
AQP	Advanced Qualification Program	MSL	Mean Sea Level
ASAP	Aviation Safety Action Program	NOTAM	Notice to Air Missions
ATC	Air Traffic Control	NTSB	National Transportation Safety Board
ATIS	Automatic Terminal Information Service	PA	Passenger Announcement
ATP	Airline Transport Pilot	PBN	Performance Based Navigation
CDL	Configuration Deviation List	PF	Pilot Flying
CFR	Code of Federal Regulations	PIREP	Pilot Report
CPDLC	Controller Pilot Data Link Communications	PM	Pilot Monitoring
CRM	Crew Resource Management	QNH	Altimeter setting, height above sea level
CVFP	Chartered Visual Flight Procedure	QRC/QRH	Quick Reference Card / Handbook
CVR	Cockpit Voice Recorder	RNAV	Area Navigation
DA/DH	Decision Altitude / Decision Height	RNP	Required Navigation Performance
DECS	Data Entry and Control System	RON	Remain Overnight
DOM	Dispatch Operations Manual	RVR	Runway Visual Range
eAML	Electronic Aircraft Maintenance Log	RVSM	Reduced Vertical Separation Minimum
EFB	Electronic Flight Bag	SELCAL	Selective Calling
ETOPS	Extended Operations	SIGMET	Significant Meteorological Information
FAA	Federal Aviation Administration	SMGCS	Surface Movement Guidance and Control System
FAM	Flight Attendant Manual	SOP	Standard Operating Procedure

ACARS	Aircraft Communications Addressing and Reporting System	LLWAS	Low-Level Windshear Alert System
FCIR	Flight Crew Irregularity Report	TAF	Terminal Aerodrome Forecast
FDP	Flight Duty Period	TCAS	Traffic Alert and Collision Avoidance System
FMA	Flight Mode Annunciator	TDWR	Terminal Doppler Weather Radar
FMS	Flight Management System	TEM	Threat and Error Management
FOM	Flight Operations Manual	TWIP	Terminal Weather Information for Pilots
FOQA	Flight Operations Quality Assurance	UAS	Undesired Aircraft State
FOS	Flight Operating System	VAAC	Volcanic Ash Advisory Center
HIMS	Human Intervention Motivation Study	VMC	Visual Meteorological Conditions
IOC	Integrated Operations Center		
LAHSO	Land and Hold Short Operations	NAPA	Northern Airways Pilots Association
— End —			

Appendix A: Delay Codes

A.1 General

Every delay is coded so that the cause is captured consistently across the system. The delay code is assigned by the station, using the three-character (or shorter) code that most specifically describes the cause. See section [23.7 Delay Coding](#) and section [1c.3.3 Delay Report](#).

Where no code has been assigned by departure, the delay is automatically coded **SD** — **Station Default**. It is the station's responsibility to assign the correct code as soon as possible after departure, by end of day at the latest.

Where more than one cause contributed to a single delay, the delay is split between the applicable codes rather than forced into one.

A.2 Air Traffic Control

Table A.1 A.1 Air Traffic Control Delay Codes

Code	Description
AT1 — ATC Hold Local	Aircraft held at the gate for local ATC conditions, including FAA ground/pushback clearance and local slot-automation departure metering. For ramp congestion, use OP1.
AT2 — ATC Tarmac Delay	Delay incurred to avoid a projected extended onboard taxi delay or another extended tarmac delay caused by an FAA flow control program.
AT3 — ATC Hold Destination	Aircraft held at the gate for enroute or destination ATC conditions — GDP, volume, flow, an international track slot, or oceanic clearance. For IOC-approved Company Arrival Control, use WX5.
AT4 — ATC Equipment Failure	Aircraft held at the gate for a National Airspace System issue — ATC/FAA radar, computer outage, navaid, or a major ATC staffing or equipment failure resulting in a local departure delay.
— End —	

A.3 Weather

Table A.2 A.2 Weather Delay Codes

Code	Description
WX1 — Weather Local	Delay from actual or forecast weather or airport conditions at departure, enroute, or destination — ramp closure, surface condition, ground stop.
WX3 — Weather De-Icing	Delay from snow or ice removal. State in remarks whether de-icing occurred at the gate, in the gate area after push, or away from the gate area.
WX4 — Weather Tarmac Delay	Delay incurred to avoid a projected extended onboard taxi delay caused by WX1, WX3, or WX5 conditions, or to manage another weather-delayed flight.
WX5 — Weather Destination	Aircraft held at the gate for destination weather or airport conditions.
— End —	

A.4 Flight Attendant

Table A.3 A.3 Flight Attendant Delay Codes

Code	Description
FA0 — Completion of Duties/Other	Flight attendant unable to complete scheduled duties before posted departure, or failure to follow published procedure. Items not covered by FA1–FA8.
FA1 — Rest/Legality	Delay to meet legal crew rest requirements.
FA2 — Connection	Awaiting flight attendants from an inbound flight, including deadheads.
FA3 — Broken Thru Flight	Delay caused by a forced aircraft change on a flight attendant through flight.
FA4 — Late	Flight attendant failure to meet report time to the aircraft, except where caused by FA3 or FC8.
FA5 — Sick or No Show	Replacing or accommodating a sick or no-show flight attendant.
FA6 — Boarding Held/Interrupted	Delay directly caused by a flight attendant delaying the start of boarding beyond contractual or policy guidelines.

Code	Description
FA8 — Re-Seating Passengers	Delay from re-seating passengers for passenger convenience. Excludes moving non-qualified passengers from an exit row or seating small children with a parent.
— End —	

A.5 Pilot

Table A.4 A.4 Pilot Delay Codes

Code	Description
FP0 — Completion of Duties/Other	Pilot unable to complete scheduled duties before posted departure — failure to follow published procedure, an unscheduled meal request, not ready for departure, refusal of an operational aircraft, etc. Items not covered by FP1–FP5.
FP1 — Rest/Legality	Delay to meet legal crew rest requirements.
FP2 — Connection	Awaiting pilots from an inbound flight, including deadheads.
FP3 — Broken Thru Flight	Delay caused by a forced aircraft change on a pilot through flight.
FP4 — Late	Pilot failure to meet report time to the aircraft, except where caused by FP3 or FC8.
FP5 — Sick or No Show	Replacing or accommodating a sick or injured pilot.
— End —	

A.6 Crew Scheduling

Table A.5 A.5 Crew Scheduling Delay Codes

Code	Description
FC0 — Crew Scheduling Other	Improper scheduling of pilots or flight attendants. Items not covered by FC1–FC7.
FC1 — Pilot and Flight Attendant Rest	Delay to meet legal crew rest requirements for both groups.

Code	Description
FC2 — Pilot and Flight Attendant Connection	Awaiting both groups from inbound flights, including deadheads.
FC3 — Pilot and Flight Attendant Broken Thru Flight	Delay from a broken through flight affecting both groups.
FC4 — Pilot and Flight Attendant Late to Aircraft	Both groups fail to meet report time, except where caused by FC3 or FC8.
FC5 — Crew Swap/Crew Recovery	Unscheduled crew changes required to mitigate delay length — a lengthy maintenance delay, an up-gauge or down-gauge across fleet types, etc.
FC6 — Scheduled Crew Swaps, In Terminate/Out Originate	Delay from swapping an inbound terminating crew for an outbound originating crew on a minimum-on-ground-time turn. Limited to 5 minutes of delay.
FC7 — Unscheduled Crew Rest	Crew-initiated rest delay not required by 14 CFR.
FC8 — Late Crew, Hotel/Vendor Transportation	Pilot or flight attendant failure to meet report time because hotel or vendor transportation arrived late. Does not include a crewmember missing the scheduled van departure from the hotel.
— End —	

A.7 Passenger Service

Table A.6 A.6 Passenger Service Delay Codes

Code	Description
PR0 — Passenger Service Other	Company-controllable delay from improper passenger processing or late boarding — unauthorized boarding of a late connection, a seat assignment error, slow processing, exceeding the load planning passenger variance, etc. Items not covered by PR1–PR7.
PR1 — Oversale/Over booking	Delay processing passengers during an oversale situation of any size.

Code	Description
PR2 — Customer	Customer accommodation items — a seatbelt extension, a non-English-speaking passenger, a passenger in the restroom at departure time, a cabin-cooling delay after all departments completed cabin preparation on time, etc.
PR3 — Onboard Illness/Injury	Delay from removing or treating an ill or injured passenger. Servicing oxygen bottles or emergency kits used in flight is coded MT0, not PR3, if reported before or on arrival. Cabin cleanup from the event is coded SV2.
PR5 — Special Assistance Passengers	Accommodating passengers using wheelchairs or other special assistance.
PR7 — Passenger Gate-Checked Bags	Delay from Customer Service failing to properly coordinate gate-checked bags for ramp retrieval. Identify Basic Economy baggage delays specifically in the remarks.
— End —	

A.8 Connection

Table A.7 A.7 Connection Delay Codes

Code	Description
CX1 — Hold for Connecting Passengers/Bags, HCC/Tower Authorized	Hub Control Center or tower-initiated gate delay to accommodate connecting passengers or bags.
CX4 — IOC Hold for Connecting Passengers/Bags	IOC-coordinated or authorized gate delay to accommodate connecting passengers or bags.
— End —	

A.9 Baggage and Cargo

Table A.8 A.8 Baggage and Cargo Delay Codes

Code	Description
CG0 — Baggage/Cargo Other	Items not covered by CG1–CG7 — failure to follow published procedure, equipment or personnel not in place, unauthorized loading.
CG1 — Loading Baggage and Cargo	Late delivery of local baggage, extended loading time from baggage volume, late cargo delivered/staged on time, or an inoperative in-plane loading system.
CG2 — Removal Due to Weight and Balance	Removing or reloading baggage or cargo, including after a ground interrupt, to correct an overweight or out-of-balance condition.
CG3 — Cargo Delay	Late delivery or staging of cargo — freight, mail, co-mail — including incorrect or incomplete documentation.
CG4 — Late Connecting Bag Loading, Minimum Connect Not Available	Ramp holds to load connecting baggage with less than minimum connection time available.
CG7 — Gate-Checked Bags	Delay from the ramp agent's late retrieval of properly staged gate-checked baggage.
— End —	

A.10 Servicing

Table A.9 A.9 Servicing Delay Codes

Code	Description
SV0 — Water Service/Other	Late or improper potable water service, including re-servicing. Items not covered by SV1–SV7.
SV1 — Aircraft Cleaning (Controllable)	Untimely or improper cleaning from a failure to follow Company policy or process.
SV2 — Aircraft Cleaning (Non-Controllable)	Non-standard cleaning need — illness, spill, food or beverage — including a required seat cushion change.
SV3 — Lavatory Service	Untimely or improper lavatory servicing, including over-filled fluid, re-servicing, or a crew-requested re-service for odor.

Code	Description
SV4 — Engine Starts (Air/Electric)	Delay from an at-gate engine start with crew and equipment in place on time, or a non-standard start (an air start) known in advance with personnel and equipment in place.
SV5 — Ground Service Equipment (NAC GSE)	Delay from Company-owned ground equipment that is inoperative, fails during use, or is unavailable. Excludes items that are not GSE, such as headsets or wands.
SV6 — Ground Service Equipment, Operator Error	Delay from an operator not following standard GSE procedure — running out of fuel, an uncharged battery, damage, or misuse.
SV7 — Ground Service Equipment, Contract	Delay from contract (non-Company) ground equipment that is inoperative, unavailable, or fails during use. State the equipment that failed in the remarks. For airport-owned facilities, use OP2.
SV8 — NAC Facilities Maintenance	Delay from a Company facilities-maintained item that is inoperative, unavailable, or fails during use. State the equipment that failed in the remarks. For airport-owned facilities, use OP2.
— End —	

A.11 Fuel

Table A.10 A.10 Fuel Delay Codes

Code	Description
FL0 — Fuel Other	Items not covered by FL1–FL5.
FL1 — Fuel Topoff, IOC Requested	IOC-requested fuel boost after correct, on-time initial fueling.
FL2 — Fuel Servicing (Controllable)	Delay from a failure to follow policy or process — late to the aircraft, a spill, an incorrect load.
FL3 — Fuel Servicing (Non-Controllable)	Fuelers called to an aircraft already properly serviced, including a Company-requested defuel.
FL4 — Fuel Topoff, Captain Requested	Captain-requested fuel boost after correct, on-time initial fueling.

Code	Description
FL5 — Fuel Maintenance Required	A maintenance action or inspection required before or during fueling. A mechanical condition identified on the outbound flight is coded MT1, not FL5.
— End —	

A.12 Provisioning and Catering

Table A.11 A.11 Provisioning and Catering Delay Codes

Code	Description
FS1 — Provisioning/Commissary (Controllable)	Company-owned catering's failure to follow policy or process — late to the aircraft, an equipment failure, an inoperative cart.
FS2 — Provisioning/Commissary (Non-Controllable)	Company-owned catering called to an aircraft already properly serviced.
FS3 — Contract Catering (Controllable)	Vendor's failure to follow Company policy or process.
FS4 — Contract Catering (Non-Controllable)	Contract catering called to an aircraft already properly serviced.
— End —	

A.13 Maintenance

Table A.12 A.12 Maintenance Delay Codes

Code	Description
MT0 — Maintenance Mechanical Delay, Inbound	A mechanical condition identified on the inbound flight, before block-in, requiring maintenance action. State the eAML page number in the remarks.
MT1 — Maintenance Mechanical Delay, Outbound	A mechanical condition identified on the outbound flight, after the previous leg blocked in, requiring maintenance action. State the eAML page number.

Code	Description
MT2 — Parts Defective From Stock	A newly installed part failing at installation or on its first flight leg. State the eAML page number.
MT3 — Parts, Materials, Tooling, or Test Equipment Shortage	Delay from a shortage of the item. State the part number of the shortage in the remarks.
MT4 — Delay From Scheduled Maintenance Task	Delay from accomplishing a scheduled maintenance program task. State the task in the remarks.
MT5 — Deferral Management/Technical Data and Personnel Errors	Delay from an MEL, CDL, or deferral awaiting application or approval; a reinspection for a deferral; a logbook entry error; or waiting on technical data from a support organization.
MT6 — Precautionary Inspection	Maintenance responds to a reported mechanical condition and no maintenance action is taken — could not duplicate, found within limits. A delay may be split between MT6 and a crew-side code where the call for assistance could reasonably have come earlier.
MT7 — Late Delivery of Aircraft	A serviceable aircraft delivered late to the gate from maintenance's own preparation or move. Cannot exceed the published minimum-on-gate time. Excludes a tug failure (SV5), gate congestion (OP), or a ramp closure (WX).
MT8 — Onboard Computer/Component Reset	Delay from a maintenance action to reset a computer message, circuit breaker, or component, including a software issue. State the eAML page number.
— End —	

A.14 Damage

Table A.13 A.13 Damage Delay Codes

Code	Description
DA0 — Unspecified Cause, at Arrival/Initial Inspection	Cause or source of damage unknown at arrival. Items not covered by DA1–DA9.
DA1 — Birdstrike or Other Animal Damage	Suspected or confirmed birdstrike or other animal-related damage.

Code	Description
DA2 — Weather/Natural Causes	Damage from lightning, hail, wind, or similar.
DA3 — Flightdeck/Cabin Crew Damage	A broken switch or circuit breaker, jumpseat, interphone equipment, an unauthorized slide deployment, or misuse of oxygen equipment by crew.
DA4 — Passenger Damage	A broken seatback, armrest, or tray table; missing seatbelt; or replacement of medical supplies used for a passenger.
DA6 — Unspecified Cause, After Initial Arrival/Walkaround	Cause or source of damage unknown, found after the initial arrival inspection.
DA8 — Customer Service/Ramp/Catering Damage	A collision with ground or catering equipment, damage from aircraft relocation, or an accidental slide deployment by ramp or catering personnel.
DA9 — Maintenance Damage	A collision with ground equipment, damage from aircraft relocation, or an accidental slide deployment by maintenance personnel.
— End —	

A.15 Load Planning

Table A.14 A.14 Load Planning Delay Codes

Code	Description
OL1 — Late Load Plan From Central Load Planning	Late, incomplete, or incorrect closeout data from Central Load Planning.
OL2 — Adjusting Passengers/Bins After Initial Load	A load adjustment required by a change in weather or runway.
OL3 — Late Delivery of Weight and Balance	Late or incorrect weight and balance information.
OL4 — Moving Passengers/Bags for Weight and Balance	Re-seating passengers or removing bags/cargo to correct an out-of-balance condition.
— End —	

A.16 Operational

Table A.15 A.15 Operational Delay Codes

Code	Description
OP0 — Operational Delays Other	An aircraft change for revenue or seating (not maintenance-driven), a curfew hold, or a schedule adjustment. Items not covered by OP1–OP7. A must-ride part (AOG) or must-ride mechanic delay is coded MT3, not OP0.
OP1 — Ramp Traffic/Pushback	Delay at the gate from aircraft or vehicle congestion on the ramp or taxiway.
OP2 — Airport Facilities	Delay from a non-Company airport resource — a jetway, belt system, or airport power.
OP3 — Late Flight Release	Delay from a late, incomplete, or incorrect release from Dispatch/IOC.
OP6 — VIP Movements	Delay from presidential, governmental, or dignitary movement.
OP7 — Overposted ETDs	Delay attributed to an overposted estimated departure time.
— End —	

A.17 Equipment and Information Technology

Table A.16 A.16 Equipment and IT Delay Codes

Code	Description
EQ — Auto-Generated Code	System-generated; not station-entered.
EQ2 — Incorrect DECS/FOS Turn Delay Calculation	DECS/FOS fails to compute the correct local delay minutes. Applies only to the gap between minimum-on-ground-time and the DECS/FOS time. An overposted ETD is coded OP7, not EQ2.
IT0 — IT Related Issues Other	Items not covered by IT1–IT8.
IT1 — Problems With PSS	PSS/Sabre reservation system issues.
IT2 — Problems With FOS/CLP	Dispatch environment degraded or inoperative — a station FOS issue.

Code	Description
IT3 — Kiosks/Check-In/Ticket Counter IT Infrastructure	Company-controlled IT infrastructure outage, failure, or unavailable equipment.
IT5 — Airport-Owned IT Infrastructure	Non-Company IT infrastructure outage, failure, or unavailable equipment.
IT6 — Network Failure	A network issue — an ARINC outage, for example.
IT7 — Closeout Systems	Delay related to flight closeout systems — weight and balance, bag scanners.
IT8 — Regional Carriers IT Environment	An issue within a regional operator's IT environment. Regional use only.
— End —	

A.18 Security

Table A.17 A.17 Security Delay Codes

Code	Description
SC0 — Security Other	An international security cabin search, a panel or floor removal and search, or a gate security/ID review. Items not covered by SC1–SC5.
SC1 — Bomb/Hijack Threat	Inspecting the aircraft, baggage, freight, mail, or passengers because of a threat.
SC2 — Security/Onboard/PPBM	An onboard security concern, a passenger removal, a ground interrupt, or prohibited passenger-carried baggage processing.
SC3 — Passenger Related Security, TSA/Government	A passenger late to the aircraft because of a TSA or government process, resource, or requirement.
SC4 — Bag Related Security, TSA/Government	Passenger bags late to the aircraft, or cabin baggage processing delayed, because of a TSA or government requirement — an enhanced search, for example. State the specific security item in the remarks.
SC5 — Security Breach	A terminal, concourse, or airfield security breach or evacuation.
— End —	

A.19 Miscellaneous

Table A.18 A.18 Miscellaneous Delay Codes

Code	Description
MI1 — FAA Checks	Delay from accommodating FAA personnel or an inspection.
SD — Station Default	Automatically applied where no code has been assigned. See section A.1 General.
OGD — Objectionable Ground Delay	A delay carried into the turn from an inbound flight's own delay, controllable or non-controllable depending on the inbound cause.
ECD — Equipment Change Delay	An aircraft change at the departure station, most often attributable to maintenance.
ERT — Enroute Delay	A delay from a weather or wind deviation enroute.
— End —	

APPENDICES

Reference material issued with this manual.

Glossary	Alphabetical Index
Appendix A — Delay Codes	

ADDITIONAL CHAPTERS

Chapters that are not tied to a single phase of flight.

Non-Scheduled Flights	Administration
Irregular Operations	Threat & Error Management
International Operations	Reports and Forms
Training and Checking	Meteorology
Passengers and Cargo	Sabre/DECS

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